Dell PowerFlex 4.5.x and PowerFlex Manager 5.1.x

CLI Reference Guide

June 2026
Rev. 1.0

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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Contents

Chapter 1: Introduction **17**
Common PowerFlex terms 17

Chapter 2: Revision history **19**

Part I: PowerFlex Manager CLI **20**

Chapter 3: Accessing CLI **21**
Login command examples 21
Certificate 22
Sessions 22
Filters and operators 22

Chapter 4: Alerts commands **24**
Acknowledge alerts 24
Clear alerts 25
Disacknowledge alerts 25
Query alerts 26

Chapter 5: Alert template commands **27**
Modify alert template 27
Query alert template 28

Chapter 6: PowerFlex CLI version **29**

Chapter 7: Directory service commands **30**
Create directory services 30
Modify directory services 31
Query directory services 31
Sync directory services 32
Delete directory services 32

Chapter 8: Dispatch destination commands **33**
Query dispatch destinations 33
Modify dispatch destinations 34
Delete dispatch destinations 34

Chapter 9: Dispatch policies commands **36**
Query dispatch policies 36
Modify dispatch policies 37
Delete dispatch policies 37

Chapter 10: Dispatch sources commands **39**

Query dispatch sources 39
Modify dispatch sources 40
Delete dispatch sources 40

Chapter 11: Event commands 41
Query events 41

Chapter 12: Group commands 43
Create groups 43
Modify groups 44
Query groups 45
Delete groups 45

Chapter 13: Login banner commands 46
Create login banners 46
Modify login banner 47
Query login banner 47
Delete login banner 48

Chapter 14: Roles commands 49
Query roles 49

Chapter 15: SMTP service commands 50
SMTP services to send test email 50
SMTP services to test connection 51

Chapter 16: Dell Technologies connectivity services configuration commands 52
Dell Technologies connectivity services configuration connection state 52
Dell Technologies connectivity services configuration connectivity 53
Modify Dell Technologies connectivity services configuration 53
Query Dell Technologies connectivity services configuration 54
Send Dell Technologies connectivity services configuration test alert 54
Dell Technologies connectivity services configuration status 55
Dell Technologies connectivity services configuration system mode 55

Chapter 17: Syslogs commands 56
Modify syslogs 56
Query syslogs 57
Delete syslogs 57

Chapter 18: User management commands 58
Create users 58
Modify users 59
Query users 60
Delete users 60

Part II: PowerFlex SCLI 62

Chapter 19: SCLI basics..... 63
Accessing SCLI..... 63
Command format..... 63
Syntax..... 64
Object names..... 65
Using SCLI in non-secure mode..... 65
Logging in..... 66
login..... 66
generate_login_certificate..... 67
logout..... 67
Getting help..... 68
Help..... 68

Chapter 20: Acceleration pool commands..... 70
add_acceleration_pool..... 70
modify_metadata_acceleration..... 70
query_acceleration_pool..... 71
remove_acceleration_pool..... 72
rename_acceleration_pool..... 72

Chapter 21: Device commands..... 74
abort_remove_sds_device..... 74
activate_sds_device..... 74
add_sds_device..... 75
clear_sds_device_error..... 77
modify_sds_device_capacity..... 78
query_all_device_latency_meters..... 79
query_device_latency_meters..... 80
query_device_test..... 81
query_sds_device_info..... 82
remove_sds_device..... 84
rename_device..... 85
set_sds_device_external_acceleration_type..... 86
set_sds_device_led..... 87
set_sds_device_media_type..... 87
start_device_test..... 88
set_sds_device_path..... 90

Chapter 22: Fault set commands..... 91
add_fault_set..... 91
clear_fault_set..... 92
query_all_fault_sets..... 92
query_fault_set..... 93
remove_fault_set..... 94
rename_fault_set..... 94
set_sdt_fault_set..... 95

Chapter 23: General CLI commands..... 96

acknowledge_scanner_error_counters....96
add_trusted_ca....97
extract_root_ca....97
list_trusted_ca....98
query_license....98
query_disconnections....98
query_remote_syslog....99
query_oscillating_failure_counter_parameters....99
remove_trusted_ca....100
reset_oscillating_failure_counters....101
set_cli_login_banner_acceptance_mode....102
set_cli_login_banner_preemptive_acceptance....102
set_login_banner....103
set_oscillating_failure_counter_parameters....103
set_remote_read_only_limit_state....105
set_syslog_facility....105
start_remote_syslog....106
stop_remote_syslog....106
test_remote_system_connectivity....107
add_network_set....107
modify_network_set....108
query_network_set....108

Chapter 24: Protection domain commands....109
activate_protection_domain....109
add_protection_domain....109
disable_fgl_metadata_cache....110
enable_fgl_metadata_cache....110
inactivate_protection_domain....111
query_port_flapping_status....112
query_port_flapping_unsupported_components....113
query_protection_domain....114
remove_protection_domain....116
rename_protection_domain....117
set_default_fgl_metadata_cache_size....117

Chapter 25: Remote system commands....119
add_remote_system....119
query_remote_system....119
query_all_remote_systems....120
rename_remote_system....120
modify_remote_system_bw_limit....120
modify_remote_system_connectivity....121
modify_remote_system_description....121
remove_remote_system....122

Chapter 26: Replication commands....123
abort_replication_consistency_group_snapshots_create....123
activate_replication_consistency_group....123

activate_replication_consistency_group_mappings....124
add_replication_consistency_group....124
add_replication_pair....126
add_replication_peer_system....127
add_sdr....128
add_sdr_ip....129
clear_broken_replication_consistency_group_alerts....130
create_replication_consistency_group_snapshots....131
enter_sdr_maintenance_mode....131
exit_sdr_maintenance_mode....132
failover_replication_consistency_group....133
failover_test_start_replication_consistency_group....133
failover_test_stop_replication_consistency_group....134
freeze_apply_replication_consistency_group....134
inactivate_replication_consistency_group_mappings....135
modify_replication_consistency_group_rpo....135
modify_replication_consistency_group_target_volume_access_mode....136
modify_replication_peer_system_ip....136
modify_replication_peer_system_port....137
modify_sdr_ip_role....138
modify_sdr_port....139
pause_pair_initial_copy....139
pause_replication_consistency_group....140
print_recreate_replication_consistency_group_script....140
query_all_replicated_volumes....141
query_all_replication_consistency_groups....142
query_all_replication_journal_capacity....142
query_all_replication_pairs....143
query_all_replication_peer_system....144
query_all_sdr....144
query_replication_consistency_group....145
query_replication_pair....146
query_replication_peer_system....147
query_replication_peer_system_destination_volume_candidates....148
query_replication_peer_system_protection_domains....148
query_sdr....149
query_sync_now_replication_consistency_group....150
query_replication_destination_sdc_candidates....151
query_replication_destination_storage_pool_candidates....151
remove_replication_consistency_group....152
remove_replication_pair....153
remove_replication_peer_system....153
remove_sdr....154
remove_sdr_ip....154
rename_replication_consistency_group....155
rename_replication_peer_system....155
rename_sdr....156
resume_pair_initial_copy....157
resume_replication_consistency_group....157
restore_replication_consistency_group....158

reverse_replication_consistency_group....158
set_pair_initial_copy_queue_position....159
set_replication_consistency_group_consistent....160
set_replication_consistency_group_inconsistent....160
set_replication_journal_capacity....161
set_remote_replication_journal_capacity....161
switchover_replication_consistency_group....162
sync_now_replication_consistency_group....163
terminate_replication_consistency_group....163
unfreeze_apply_replication_consistency_group....164

Chapter 27: Return messages....165
SDS device name changed....165
Success....165
No resources....166
Timeout....166
Already exists....166
Communication error....167
Wrong reconfiguration mode....167
SDS add device “open” error....167
SDS add device I/O error....168
SDS add device size problem....168
SDS actively attach to this MDM....168
SDS already attached to this MDM....169
SDS active attach to other MDM....169
SDS already attached to other MDM....169
SDS not found....170
Volume not found....170
SDC not found....170
Volume already mapped to SDC....171
Volume already mapped to all SDCs....171
Volume not mapped to SDC....171
Volume mapped to all SDCs....172
Too many mappings....172
Invalid volume ID....172
Invalid SDS ID....173
Invalid SDC ID....173
Host not found....173
Duplicate host name....174
Address and port in use....174
Too many SDSs in protection domain....174
Too many SDSs....175
Too many volumes in protection domain....175
Too many volumes....175
SDS name in use....176
Volume name in use....176
Duplicate SDS device name....176
Duplicate SDC IP address....177
SDS not under removal....177
SDS being removed....177

Volume mapped....178
Use MDM IP address....178
Command only in cluster....178
Command only in single mode....179
Command error when cluster degraded....179
Too many devices....179
SDS device not found....180
License too long....180
License error....180
Unknown license version....181
License not set....181
Configuration exceeds new license....181
License capacity exceeded....182
License expired....182
Installation ID mismatch....182
Customer ID mismatch....183
Unsupported license version....183
Volume decrease not supported....183
SDS device being removed....184
Cannot remove last SDS device....184
Device not under removal....184
SDS already adding device....185
Invalid device ID....185
Too many protection domains....185
Protection domain name in use....186
Protection domain not found....186
Protection domain has SDSs....186
Not enough SDSs in storage pool....187
Remove in progress....187
No network test results....187
Network test in progress....188
SDS device already active....188
SDS device wrong test mode....188
SDS device wrong test state....189
SDS device test in progress....189
Volume allocation “busy” error....189
Capacity highly unbalanced....190
Volume creation failed....190
Invalid password....190
No permissions....191
Too many storage pools in protection domain....191
Storage pool already exists....191
Storage pool not found....192
Storage pool has devices....192
Storage pool name already exists....192
Invalid storage pool ID....193
Protection domain has storage pools....193
SDC has mappings....193
Too many volumes in V-Tree....194
Too many snapshots in V-Tree....194

No volume to delete....194
Too many devices in storage pool....195
Volume not found for consistency group....195
Too many IP addresses....195
Cannot remove last SDS IP address....196
SDS IP address not found....196
Volume size too large....196
Capacity too low for snapshot....197
IP address belongs to other SDS....197
Invalid port....197
LIA package not installed....198
LIA multiple packages installed....198
LIA command failed....199
LIA invalid password....199
LIA operation not permitted....199
LIA file is too large....200
LIA wrong ID....200
LIA disk does not exist....200
SDS IP address not found....201
Bad threshold value....201
Storage pool has volumes....201
Protection domain and storage pool mismatch....202
SDS IP address already exists....202
IP role already set....202
Cluster ID mismatch....203
Device already exists....203
Volume size is illegal....203
SDC must be disconnected....204
Invalid percentage....204
Wrong receive group....204
Mismatched SW ID....205
Invalid license issuer....205
Number of parallel messages is too low....205
Number of parallel messages is too high....206
Network test size is too high....206
Network test length too long....206
Network test must limit either time or size....207
Not enough SDSs in protection domain for test....207
Not allowed in storage pool with devices....207
License capacity mismatch....208
Exceeds system capacity limitations....208
SDC mapped to too many volumes....208
Not enough devices in storage pool for removal....209
Invalid network limits....209
A device is currently being removed....209
A device is already active....210
RAM cache wrong state....210
More than one mapping not allowed....210
At least one SDS failed....211
A device is in incorrect test state....211

A device test is in progress....211
Enterprise features not enabled....212
Volume already mapped to an SDC....212
SDS already exists and is being removed....212
Invalid spare percentage....213
Invalid background device scanner bandwidth limit....213
Zero padding with background device scanner data comparison mode....213
IP address already assigned to another SDS....214
Capacity too low for thin volumes....214
Too many storage pools in the system....214
Remote access to the MDM is blocked....215
Cannot add a partitioned device....215
Cannot add a mounted device....215
Device header mismatch....216
Device has user content....216
Missing storage device media type....216
Invalid media type....217
Media type mismatch....217
Invalid external acceleration type....217
External acceleration type mismatch....218
Metadata feature cannot be configured on the acceleration device....218
Metadata feature is not enabled on the storage pool....218
Metadata feature is already enabled on the storage pool....219
Metadata feature is not enabled on the devices....219
Metadata is already enabled on devices....219
Insufficient capacity for accelerating metadata....220

Chapter 28: NVMe host and SDC commands....221
Understanding the effect of network latency on NVMe over TCP latency counters....221
Limitations of measuring latency between the host and SDT....222
add_sdc....223
remove_host....223
rename_host....224
check_sdc_authentication_status....224
generate_sdc_password....225
query_all_approved_sdc....226
query_host....226
query_all_hosts....226
query_potential_nvme_hosts....227
query_all_sdc....227
query_restricted_sdc_mode....228
query_sdc....228
query_sdc_volume_limits....229
reset_sdc_sds_connections....230
set_restricted_sdc_mode....230
set_sdc_authentication....231
set_sdc_volume_limits....232
set_approved_sdc_ips....233
add_nvme_host....233
modify_nvme_host_max_paths....234

modify_nvme_host_max_sys_ports....235
modify_nvme_host_os_full_type....235
set_manual_nvme_host_connectivity....236
reset_manual_nvme_host_connectivity....236
set_manual_nvme_mapping_connectivity....237
reset_manual_nvme_mapping_connectivity....238
rebalance_nvme_host_connectivity....238
query_nvme_discovery_info....238
clear_nvme_reservation....239
add_host_group....240
assign_host_to_host_group....240
query_all_host_groups....241
query_host_group....241
modify_host_group....241
rename_host_group....242
remove_host_from_host_group....242
remove_host_group....242
set_auto_update_sdc_with_mdm_ip_addresses....243
restart_connections_to_realign_net_path....243

Chapter 29: SDC operations....245
Add an external SDC to an existing PowerFlex system....245
Install the SDC on an ESXi-based server and connect it to PowerFlex using esxcli....245
Install SDC on a Linux-based server and connect it to PowerFlex....246
Install SDC on an AIX-based server and connect it to PowerFlex....248
Install SDC on a Windows-based server and connect it to PowerFlex....248
Performing SDC operations....249
Update the SDC driver with IP address changes....249
Detecting new volumes....250
Query volumes using drv_cfg....250
Query tgt objects using drv_cfg....251
Query GUID using drv_cfg....251
Query MDMs using drv_cfg....252
Loading a configuration file using drv_cfg....252
Adding an MDM using drv_cfg....253
Modifying an MDM IP address using drv_cfg....254
Permanent device loss state....255
Run the SDC driver on Red Hat Enterprise Linux or SUSE Enterprise Linux Enterprise Server with multipath enabled....256
SDC authentication....256
Manual SDC configuration/verification....258

Chapter 30: SDS commands....259
abort_remove_sds....259
add_sds....259
add_sds_ip....262
disable_sds_rmcache....263
enter_maintenance_mode....264
exit_maintenance_mode....265
enable_sds_rmcache....266

enter_protected_maintenance_mode....267
exit_protected_maintenance_mode....267
modify_sds_ip_role....268
modify_sds_port....269
query_all_sds....270
query_sds_connectivity_status....270
query_sds....271
query_sds_connectivity_status....273
query_sds_network_test_results....274
query_sdc_to_sds_disconnections....274
remove_sds....275
remove_sds_ip....275
rename_sds....276
set_drl_properties....277
set_fgl_metadata_cache_size....278
set_sds_network_limits....278
set_sds_rmcache_size....279
show_certificate....280
start_sds_network_test....281
update_sds_original_paths....282
set_sds_device_path....283

Chapter 31: SDT commands....284
add_sdt....284
remove_sdt....285
rename_sdt....285
add_sdt_ip....286
remove_sdt_ip....286
modify_sdt_ip_role....287
modify_sdt_port....288
query_sdt....288
query_all_sdt....289
enter_sdt_maintenance_mode....289
exit_sdt_maintenance_mode....290

Chapter 32: Snapshot commands....292
clear_snapshot_security....292
remove_snapshots_consistency_group....292
set_snapshot_security....293
snapshot_volume....294
add_snapshot_copy_task....295
query_snapshot_copy_task....296
query_all_snapshot_copy_tasks....296
pause_snapshot_copy_task....296
resume_snapshot_copy_task....297
remove_snapshot_copy_task....297

Chapter 33: Storage pool commands....299
add_storage_pool....299

disable_background_device_scanner....301
disable_persistent_checksum....301
enable_background_device_scanner....302
enable_persistent_checksum....303
modify_compression_method....304
modify_persistent_checksum....305
modify_zero_padding_policy....305
query_inaccessible_user_address_space....306
query_storage_pool....307
remove_storage_pool....309
rename_storage_pool....309
reset_scanner_error_counters....310
set_capacity_alerts_threshold....311
set_rebalance_mode....312
set_rebalance_policy....313
set_rebuild_mode....314
set_rebuild_policy....315
set_rebuild_rebalance_parallelism....316
set_rmcache_usage....316
set_rmcache_write_handling_mode....317
set_storage_pool_external_acceleration_type....318
set_storage_pool_media_type....319
set_checksum_mode....320
set_protected_maintenance_mode_policy....320
test_inaccessible_data....321

Chapter 34: System commands....323
abort_upgrade....323
add_certificate....323
add_standby_mdm....324
allow_commands_during_upgrade....325
create_mdm_cluster....325
finalize_upgrade....326
generate_certificate....326
generate_mdm_certificate....327
generate_mdm_csr_file....328
modify_cluster_virtual_ips....329
modify_management_ip....329
modify_spare_policy....330
modify_virtual_ip_interfaces....331
query_all....332
Output description....335
query_cluster....339
query_performance_parameters....341
query_properties....344
Object presets....346
query_remote_read_only_limit_state....346
query_system_limits....347
query_upgrade....348
refresh_mdm_cluster_capabilities....348

remove_standby_mdm....349
rename_mdm....349
rename_system....350
query_all_system_networks....350
query_system_network....350
add_system_network....351
rename_system_network....351
remove_system_network....352
replace_cluster_mdm....352
replace_mdm_security_files....353
set_component_authentication_properties....354
set_management_client_communication....355
set_performance_parameters....355
start_upgrade....357
switch_cluster_mode....357
switch_mdm_ownership....359
generate_mdm_mno_csr....359
replace_mdm_mno_certificate....360
query_all_network_sets....360
query_network_set....360
modify_network_set....361
rename_network_set....361
remove_network_set....362

Chapter 35: Volume commands....363
add_volume....363
clear_scsi2_reservation....364
clear_scsi3_reservation....364
map_volume_to_host....365
unmap_volume_from_host....366
modify_volume_capacity....367
overwrite_volume_content....367
query_all_volumes....368
query_volume....369
query_volume_sdc_meters....370
query_volume_sdt_meters....371
remove_volume....371
rename_volume....372
set_volume_rmcache_usage....372
set_volume_mapping_access_mode....373
set_volume_access_mode_limit....374
query_auto_provisioned_destination_volume_suffix....374
modify_auto_provisioned_destination_volume_suffix....374
add_snapshot_policy....375
add_source_volume_to_snapshot_policy....376
align_snapshot_policy_to_current_time....376
lock_auto_snapshot....377
modify_snapshot_policy....377
pause_snapshot_policy....378
query_all_snapshot_policies....378

query_snapshot_policy....379
remove_snapshot_policy....380
remove_source_volume_from_snapshot_policy....381
rename_snapshot_policy....381
resume_snapshot_policy....382
unlock_auto_snapshot....382

Chapter 36: V-Tree commands....384
migrate_vtree....384
pause_vtree_migration....385
query_all_vtree_migrations....386
query_vtree....387
query_vtree_migration....388
resume_vtree_migration....388
set_vtree_migration_policy....389
set_vtree_migration_queue_position....390

1

Introduction

This document provides a reference for using the PowerFlex command-line interface (CLI) to work with your system.

The PowerFlex CLI is intended for users who want to interactively manage a PowerFlex system, or to use commands in scripts for automating routine tasks.

PowerFlex file and file services are supported only for PowerFlex systems with PowerFlex Manager 4.6.x and earlier.

Integrated PowerFlex management (non-KVM) was previously known as co-resident. Customer-provided infrastructure for PowerFlex management (Linux/KVM) was previously known as bring your own Linux. References in the documentation will be updated over time.

SupportAssist is now known as Dell Technologies connectivity services. References in the documentation will be updated over time.

For additional PowerFlex software documentation, go to Dell PowerFlex Software: Technical Documentation.

Common PowerFlex terms

The following table identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms

Term	Acronym
management virtual machine	MVM
management data store	MDS
metadata manager	MDM
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data server	SDS
storage virtual machine	SVM
storage data replicator	SDR
storage data target	SDT
system configuration reporter	SCR

The following table summarizes key terminology changes. References in the documentation will be updated over time.

Table 2. Terminology changes

Term	Formerly known as
PowerFlex management cluster (ESXi)	PowerFlex management controller 2.0
PowerFlex management cluster (vSAN)	PowerFlex management controller 1.0
PowerFlex management cluster	PowerFlex management controller
Single ESXi PowerFlex management node	PowerFlex management node
PowerFlex management node	PowerFlex controller node
Dell Technologies connectivity services	SupportAssist

Table 2. Terminology changes (continued)

Term	Formerly known as
Enterprise Project Services (EPS) form	Logical Configuration Survey (LCS)
Dell Code and Compatibility Knowledge Hub	Release Certification Matrix (RCM) Portal

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

2

Revision history

Table 3. Revisions

Date	Document revision	Description of changes
June 2026	1.0	Initial release

I

PowerFlex Manager CLI

This section explains the basic concepts of using the PowerFlex CLI, including how to access the SCLI and log in.

3

Accessing CLI

The `powerflex_cli` file is available in the Installer node in the folder `/opt/dell/pfmp`. You must download the bundle from the Dell Technologies support site and open it to locate the file. The PowerFlex management platform bundle includes meta data manager and other components. PowerFlex CLI is a tool that connects to the MDM, which in turn, sends commands or requests information about the system. You can use the PowerFlex CLI to run commands that are related to the management and operations cluster—the management layer, that is, events and alerts, and services in the management and operations cluster.

Logging in to CLI

To log in to CLI, it is mandatory to have a server address and enter the username and password. The command options are:

- `server_address`—(Mandatory) For other commands, the `server_address` is taken from the session file.
- `server_port`—(Optional) Default value: PowerFlex Default port: 443
- `session_name`—(Optional) Default value is "default". It is a user-provided name for the session.
- `cert_file`—(Mandatory) when you are logging in for the first time. However, it is possible to log in using a certificate that exists from a prior PowerFlex CLI session.

The CLI commands are derived from the RESTful API. For example:

- Accessing help

```
powerflex_cli -help
```

- Running a query

```
powerflex_cli alerts query --session_token de98deb6b9927fc5 --id "3af3ab8277783029"
```

Login command examples

```
./powerflex_cli login --username scaleio --password scaleio --cert_file /tmp/
my_ca_cert.crt --server_address dellpowerflex.com
```

Try to log in to dellpowerflex.com using the certificate, `/tmp/my_ca_cert.crt`. The certificate is copied automatically to `~/.powerflex_cli/certificates/dellpowerflex.com/ca.crt`. For details about how to get the certificate from PowerFlex Manager server, see the section, Certificate. If the login succeeds, an encrypted session file `~/.powerflex_cli/cli/default.token` is generated.

A session token is returned, which you can use for subsequent CLI sessions.

NOTE: You must have a session token to use the CLI.

Login command with session name

An example of a login command with a session name is given below.

```
./powerflex_cli login --username scaleio --password scaleio --cert_file /tmp/
my_ca_cert.crt --server_address dellpowerflex.com -session_name some_script_name
```

1. Copy `/tmp/my_ca_cert.crt` to `~/.powerflex_cli/certificates/dellpowerflex.com/ca.crt`
2. Try to log in to dellpowerflex.com using the certificate. If the login succeeds:
   - An encrypted session file `~/.powerflex_cli/cli/some_script_name.token` is generated.

A session token is returned.

Login command without certificate

An example of a login command with a session name is given below.

```
./powerflex_cli login --username scaleio --password scaleio --server_address
dellpowerflex.com
```

1. Try to log in to dellpowerflex.com using the certificate `~/.powerflex_cli/certificates/dellpowerflex.com/ca.crt`.
2. If the certificate does not exist, the login without the certificate.

Certificate

The CLI must provide a certificate at least one time per server. The certificates can be provided more than once too. When the certificate is provided, the CLI saves it in `~/.powerflex_cli/certificates/<server>/ca.crt`. The certificate is used again if a user logs in to the same server without a certificate.

You can set the certificate using a flag in the login command.

To get the certificate from the PowerFlex Manager server, run the command:

```
kubectl get secret platform-ca -o json | jq -r '.data["ca-bundle"]' | base64 --decode
```

Sessions

If the login succeeds, the CLI generates an encrypted session file and saves it in `~/.powerflex/cli/`. The name of the session token file is in the format, `<session_name>.token` if the session name is given or `default.token` if the session name is not given. The session name allows you to run more than once script that calls login command simultaneously and to be logged in to more than one server at the same time. The session file contains the name of the server, server port, and keycloak access token in an encrypted format.

Session token

The CLI returns a user session token, which is the key to open the session file. The session token is mandatory in any call to the CLI, except login. It expires if the session is idle for 30 minutes (default value). While in use, the session token is valid for 10 hours (default value). The CLI refreshes the session automatically. If a session name other than `default`, is passed in the log in command, the same session name must be passed in every command with the session token.

Filters and operators

You can use filters for querying alerts, alert templates, and events.

You can use filters for a command many times. This is applicable to filters that have a logical AND between them.

The following are some examples of filter formats:

- query to get all alerts filtered by name=HEALTH_CHECK_FAILED -

  ```
  ./powerflex_cli alerts query --session_token 0541629989f63e34 --filter 'name eq
  HEALTH_CHECK_FAILED'
  ```

- query alerts on multiple filters - name = HEALTH_CHECK_FAILED and severity=CRITICAL

  ```
  ./powerflex_cli alerts query --session_token 0541629989f63e34 --filter 'name eq
  HEALTH_CHECK_FAILED' --filter 'severity eq CRITICAL'
  ```

- query events -

```
./powerflex_cli events query --session_token a909dcc30ddca7a9
```

- query events with filters -

```
./run_cli.sh events query --session_token a909dcc30ddca7a9 --filter 'code eq 20010001'
```

You can use the following with the filters:

- eq—
- ne—means 'not equal' - can be used in the same way of "eq"
- like—when you want to query all the items that their description contains a substring. and you can use "%" as a wildcard.

  For example:

  filter=description like "%this is %"

  It returns all the items with the substring "this is " anywhere in the string. The usage and behavior of the like operator and "%" is exactly as in SQL.

- in—to retrieve elements that the field value is one of that lists. Filter=name in (SDS, PROTECTION_DOMAIN)
- gt—greater than (>)
- lt—less then (<)
- ge—greater then equal (>=)
- le—less then or equal (<=)

The greater than, less than, greater than or equal to, and less than or equal to operators behave the same as the corresponding SQL operator in brackets. They are used in the syntax similarly to equal to or not equal to operators.

4

Alerts commands

This section contains commands for managing alerts.

Syntax

```
powerflex_cli alerts [command options]
```

Parameters

-h, --help
Help for alerts.

--session_name [string]
Session name.

--session_token [string]
Secret session token.

Command options

acknowledge
Acknowledges an alert template.

clear
Clears an alert.

disacknowledge
Disacknowledges an alert.

query
Returns a list of alerts.

Acknowledge alerts

Acknowledges an alert.

Syntax

```
powerflex_cli alerts acknowledge [parameters]
```

Parameters

-h, --help
Help for acknowledge

--id [string]

ID of an alert that needs to be acknowledged

--session_name [string]

Session name.

--session_token [string]

Secret session token.

Clear alerts

Clears an alert.

Syntax

```
powerflex_cli alerts clear [parameters]
```

Parameters

-h, --help

Help for clear

--id [string]

ID of an alert that needs to be cleared

--session_name [string]

Session name.

--session_token [string]

Secret session token.

Disacknowledge alerts

Disacknowledges an alert.

Syntax

```
powerflex_cli alerts disacknowledge [parameters]
```

Parameters

-h, --help

Help for disacknowledge.

--id [string]

ID of an alert that needs to be disacknowledged.

--session_name [string]

Session name.

--session_token [string]

Secret session token.

Query alerts

Fetches a list of alerts.

Syntax

```
powerflex_cli alerts query [parameters]
```

Parameters

-h, --help

Help for querying alerts.

--filter [string]

Conditions for filtering alerts.

--id [string]

ID of alert to return.

--limit [int32]

Limit number of returned objects.

--offset [int]

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

--order [string]

Sorts the result set by ascending or descending order. Ascending order is default if not specified. The syntax to sort the result is:

```
<fieldname>.asc
```

Or

```
<fieldname>.dsc
```

Example: To fetch records ordered by timestamp in ascending order:

```
--order timestamp.asc
```

--session_name [string]

Session name.

--session_token [string]

Secret session token.

5

Alert template commands

This section contains commands for managing alert templates.

Syntax

```
powerflex_cli alert-templates [command]
```

Parameters

-h, --help
Help for querying dispatch sources.

--session name [string]
Session name.

--session token [string]
Secret session token.

Command options

modify
Updates an alert template.

query
Returns a list of alert templates

Modify alert template

Updates an alert template.

Syntax

```
powerflex_cli alert-templates modify [parameters]
```

Parameters

-h, --help
Help for modifying alert templates.

--id [string]
ID of alert template that needs to be updated.

--severity [int32]
Indicates the severity of the alert. The options are:

- INFORMATION
- MINOR

- MAJOR
- CRITICAL

--session_name [string]

Session name. The default value is "default".

--session_token [string]

Token of the secret session.

Query alert template

Fetches a list of alert templates.

Syntax

```
powerflex_cli alert-templates query [parameters]
```

Parameters

-h, --help

Help for query

--filter [string]

Conditions for filtering alerts

--id [string]

ID of alert template to return

--limit [int32]

Limit number of returned objects

--offset [int]

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

--order [string]

Sorts the result set by ascending or descending order. Ascending order is default if not specified. The syntax to sort the result is:

```
<fieldname>.asc
```

Or

```
<fieldname>.dsc
```

Example: To fetch records ordered by timestamp in ascending order:

```
--order timestamp.asc
```

--session_name string

Session name. The default value is "default".

--session_token [string]

Token of the secret session.

6

PowerFlex CLI version

This command displays the version of PowerFlex CLI.

Syntax

```
powerflex_cli version [parameters]
```

Parameters

-h, --help

Help for PowerFlex CLI.

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

7

Directory service commands

This section contains commands for managing directory services.

Syntax

```
powerflex_cli directory-services [command option]
```

Command options

create
: Create a directory service.

delete
: Delete a directory service by ID

modify
: Modify a directory service by ID

query
: Queries a directory service.

sync
: Synchronizes a directory service.

Create directory services

Creates a directory service.

Syntax

```
powerflex_cli directory-services create [parameters]
```

Parameters

-h, --help
: Help for creating directory services

--configuration [string]

--directory_service_type [string]
: Options are:
- ad
- other

--group_configuration [string]

--name [string]
: Name of the directory service

--state [string]

--user_search_setting [string]

--session_name [string]
Session name. The default name is "default".

--session_token [string]
Secret token of the session.

Modify directory services

Modifies a directory service by ID.

Syntax

```
powerflex_cli directory-services modify [parameters]
```

Parameters

-h, --help
Help for modifying directory services

--configuration [string]

--group_configuration [string]

--id [string]
ID of an existing directory service

--name [string]
Name of the directory service

--state [string]

--user_search_setting [string]

--session_name [string]
Session name. The default name is "default".

--session_token [string]
Secret token of the session.

Query directory services

Query directory services.

Syntax

```
powerflex_cli directory-services query [parameters]
```

Parameters

-h, --help
Help for querying directory services.

--id [string]

ID of an existing directory service.

--session_name [string]

Session name. The default name is "default".

--session_token [string]

Secret token of the session.

Sync directory services

Synchronizes directory services.

Syntax

```
powerflex_cli directory-services sync [parameters]
```

Parameters

-h, --help

Help for synchronizing directory services

--id [string]

--session_name [string]

Session name. The default name is "default".

--session_token [string]

Secret token of the session.

Delete directory services

Deletes a directory service by ID.

Syntax

```
powerflex_cli directory-services delete [parameters]
```

Parameters

-h, --help

Help for deleting directory services

--id [string]

Id of an existing directory service

--session_name [string]

Session name. The default name is "default".

--session_token [string]

Secret token of the session.

8

Dispatch destination commands

This section contains commands for managing dispatch destinations.

Syntax

```
powerflex_cli dispatch-destinations [command]
```

Parameters

-h, --help
Help for dispatch destinations.

--session_name [string]
Session name.

--session_token [string]
Secret session token.

Command options

modify
Modifies a dispatch destination.

query
Queries dispatch destination.

delete
Deletes a dispatch destination.

Query dispatch destinations

Returns a list of dispatch destinations.

Syntax

```
powerflex_cli dispatch-destinations query [parameters]
```

Parameters

-h, --help
Help for querying dispatch destinations

--id [string]
ID of dispatch destination

--session_name [string]

Session name.

--session_token [string]

Secret session token.

Modify dispatch destinations

Modifies dispatch destinations.

Syntax

```
powerflex_cli dispatch-destinations modify [parameters]
```

Parameters

-h, --help

Help for modifying dispatch destinations

--id [string]

--data [string]

--description [string]

--name [string]

--type [string]

Options:

- SNMPV2C
- SNMPV3
- SYSLOG
- WEBHOOKS
- Dell Technologies connectivity services
- EMAIL

--session_name [string]

Session name.

--session_token [string]

Secret session token.

Delete dispatch destinations

Deletes dispatch destinations.

Syntax

```
powerflex_cli dispatch-destinations delete [parameters]
```

Parameters

-h, --help

Help for deleting dispatch destinations

--id [string]

Unique identifier of the dispatch destination.

--session_name [string]

Session name.

--session_token [string]

Secret session token.

9

Dispatch policies commands

This section contains commands for managing dispatch policies.

Syntax

```
powerflex_cli dispatch-policies [command options]
```

Parameters

-h, --help
Help for querying dispatch policies.

--session_name [string]
Session name.

--session_token [string]
Secret session token.

Command options

modify
Modifies a dispatch policies.

query
Queries a list of dispatch policies.

delete
Deletes dispatch policies

Query dispatch policies

Syntax

```
powerflex_cli dispatch-policies query [parameters]
```

Parameters

-h, --help
Help for querying dispatch policies.

--id [string]
Unique identifier of dispatch policy.

--session_name [string]
Session name.

--session_token [string]

Secret session token.

Modify dispatch policies

Syntax

```
powerflex_cli dispatch-policies modify [parameters]
```

Parameters

-h, --help

Help for modifying dispatch policies.

--id [string]

Unique identifier of dispatch policy.

--description [string]

--domain [string]

Values are:

- MANAGEMENT
- BLOCK
- NETWORK
- PLATFORM
- COMPUTE
- SECURITY

--name [string]

--severity [string]

--source [string]

Values are:

- SNMPV2C
- SNMPV3
- SYSLOG

--session_name [string]

Session name.

--session_token [string]

Secret session token.

Delete dispatch policies

Syntax

```
powerflex_cli dispatch-policies delete [parameters]
```

Parameters

-h, --help

Help for deleting dispatch policies.

--id [string]

Unique identifier of dispatch policy.

--session_name [string]

Session name.

--session_token [string]

Secret session token.

10

Dispatch sources commands

This section contains commands for managing dispatch sources.

Syntax

```
powerflex_cli dispatch-sources [command options]
```

Parameters

-h, --help
Help for querying dispatch sources.

--session_name [string]
Session name.

--session_token [string]
Secret session token.

Command options

modify
Modifies a dispatch sources.

query
Queries a list of dispatch sources.

delete
Deletes dispatch sources

Query dispatch sources

Syntax

```
powerflex_cli dispatch-sources query [parameters]
```

Parameters

-h, --help
Help for querying dispatch sources.

--id [string]
Unique identifier of dispatch sources.

--session_name [string]
Session name.

--session_token [string]

Secret session token.

Modify dispatch sources

Syntax

```
powerflex_cli dispatch-sources modify [parameters]
```

Parameters

-h, --help

Help for modifying dispatch sources.

--id [string]

Unique identifier of dispatch sources.

--data [string]

--description [string]

--name [string]

--type [string]

Values are:

- SNMPV2C
- SNMPV3
- SYSLOG

--session_name [string]

Session name.

--session_token [string]

Secret session token.

Delete dispatch sources

Syntax

```
powerflex_cli dispatch-sources delete [parameters]
```

Parameters

-h, --help

Help for deleting dispatch sources.

--id [string]

Unique identifier of dispatch sources.

--session name [string]

Session name.

--session token [string]

Secret session token.

11

Event commands

This section contains commands for managing events.

Syntax

```
powerflex_cli events [command options]
```

Parameters

-h, --help
Help for events.

--session name [string]
Session name.

--session token [string]
Secret session token.

Command options

query
Returns a list of events

Query events

Returns a list of events.

Syntax

```
powerflex_cli events query [flags]
```

Parameters

-h, --help
Help for querying events

--id [string]

--filter [string]
Conditions for filtering alerts

--limit [int32]
Limit number of returned objects

--offset [int]

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

--order [string]

Sorts the result set by ascending or descending order. Ascending order is default if not specified. The syntax to sort the result is:

```
<fieldname>.asc
```

Or

```
<fieldname>.dsc
```

Example: To fetch records ordered by timestamp in ascending order:

```
--order timestamp.asc
```

--session name [string]

Session name.

--session token [string]

Secret session token.

12

Group commands

This section contains commands to manage groups.

Syntax

```
powerflex_cli groups [command options]
```

Parameters

-h, --help
Help for groups.

--session name [string]
Session name.

--session token [string]
Secret session token.

Command options

create
Creates a group.

delete
Deletes a group by ID.

modify
Modifies a group by ID.

query
Queries a group.

Create groups

Creates groups.

Syntax

```
powerflex_cli groups create [parameters]
```

Parameters

-h, --help
Help for creating groups.

--name [string]

--role [string]

Options are:

- SuperUser
- SystemAdmin
- StorageAdmin
- LifecycleAdmin
- ReplicationManager
- SnapshotManager
- SecurityAdmin
- DriveReplacer
- Technician
- Monitor
- Support
- NA

--session name [string]

Session name.

--session token [string]

Secret session token.

Modify groups

Modify a group by ID.

Syntax

```
powerflex_cli groups modify [parameters]
```

Parameters

-h, --help

Help for modifying groups.

--id [string]

ID of an existing group.

--role [string]

Options are:

- SuperUser
- SystemAdmin
- StorageAdmin
- LifecycleAdmin
- ReplicationManager
- SnapshotManager
- SecurityAdmin
- DriveReplacer
- Technician
- Monitor
- Support
- NA

--session name [string]

Session name.

--session token [string]

Secret session token.

Query groups

Get a group by ID.

Syntax

```
powerflex_cli groups query [parameters]
```

Parameters

-h, --help
Help for querying groups.

--id [string]
ID of an existing group.

--limit [int32]
Limit number of returned objects.

--offset [int]
Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

--session name [string]
Session name.

--session token [string]
Secret session token.

Delete groups

Delete a group by ID.

Syntax

```
powerflex_cli groups delete [parameters]
```

Parameters

-h, --help
Help for deleting groups.

--id [string]
ID of an existing group.

--session name [string]
Session name.

--session token [string]
Secret session token.

13

Login banner commands

This section contains commands to manage login banners.

Syntax

```
powerflex_cli login-banners [command options]
```

Parameters

-h, --help
Help for login-banners.

--session name [string]
Session name.

--session token [string]
Secret session token.

Command options

create
Creates a group.

delete
Delete a login banner by ID.

modify
Modify a login banner by ID.

query
Queries a login banner.

text
Text for a login banner.

Create login banners

Creates login banners.

Syntax

```
powerflex_cli login-banners create [parameters]
```

Parameters

-h, --help

Help for creating login banners.

--login_banner [string]

--session name [string]

Session name.

--session token [string]

Secret session token.

Modify login banner

Modify a login banner by ID.

Syntax

```
powerflex_cli login-banners modify [parameters]
```

Parameters

-h, --help

Help for modifying login banners.

--id [string]

ID of an existing login banner.

--login_banner [string]

--session name [string]

Session name.

--session token [string]

Secret session token.

Query login banner

Queries login banner.

Syntax

```
powerflex_cli login-banners query [parameters]
```

Parameters

-h, --help

Help for querying login banners.

--id [string]

ID of an existing login banner.

--session name [string]

Session name.

--session token [string]

Secret session token.

Delete login banner

Delete a login banner by ID.

Syntax

```
powerflex_cli login-banners delete [parameters]
```

Parameters

-h, --help
Help for deleting login banners.

--id [string]
ID of an existing login banner.

--session name [string]
Session name.

--session token [string]
Secret session token.

14

Roles commands

This section contains commands for managing roles.

Syntax

```
powerflex_cli roles [command options]
```

Parameters

-h, --help
Help for roles.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Command options

query
Query roles.

Query roles

Queries all roles.

Syntax

```
powerflex_cli roles query [parameters]
```

Parameters

-h, --help
Help for querying roles.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

15

SMTP service commands

This section contains commands to manage SMTP services.

Syntax

```
powerflex_cli smtp-services [command options]
```

Parameters

-h, --help
Help for file interfaces.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Command options

send-test-email
Sends test email to the destination

test-connection
Tests SMTP connection.

SMTP services to send test email

Syntax

```
powerflex_cli smtp-services send-test-email [parameters]
```

Parameters

-h, --help
Help for sending test emails.

--password [string]

--recipient_1 [string]

--recipient_2 [string]

--recipient_3 [string]

--recipient_4 [string]

--recipient_5 [string]

--sender_address [string]

--server_type [string]

--username [string]

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

SMTP services to test connection

Syntax

```
powerflex_cli smtp-services test-connection [parameters]
```

Parameters

-h, --help

Help for testing connections.

--password [string]

--recipient_1 [string]

--recipient_2 [string]

--recipient_3 [string]

--recipient_4 [string]

--recipient_5 [string]

--sender_address [string]

--server_type [string]

--username [string]

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

16

Dell Technologies connectivity services configuration commands

This section contains commands to manage Dell Technologies connectivity services configuration.

Syntax

```
powerflex_cli support-assist-config [command options]
```

Parameters

-h, --help
Help for file interfaces.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Command options

connection_state

connectivity

modify
Modifies SupportAssist commands.

query
Queries all SupportAssist commands.

send-test-alert
Sends test alerts.

status

system_mode

Dell Technologies connectivity services configuration connection state

Syntax

```
powerflex_cli support-assist-config connection_state [parameters]
```

Parameters

-h, --help
Help for connection state.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Dell Technologies connectivity services configuration connectivity

Syntax

```
powerflex_cli support-assist-config connectivity [parameters]
```

Parameters

-h, --help
Help for connectivity.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Modify Dell Technologies connectivity services configuration

Syntax

```
powerflex_cli support-assist-config modify [parameters]
```

Parameters

-h, --help
Help for modifying SupportAssist configuration.

--contacts [string]

--enabled [string]

--eventMuted [string]

--systemMode [string]

--session name [string]
Session name. The default name is "default".

--session token [string]

Secret session token.

Query Dell Technologies connectivity services configuration

Syntax

```
powerflex_cli support-assist-config query [parameters]
```

Parameters

-h, --help
Help for querying SupportAssist configuration.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Send Dell Technologies connectivity services configuration test alert

Syntax

```
powerflex_cli support-assist-config send-test-alert [parameters]
```

Parameters

-h, --help
Help for sending SupportAssist configuration test alert.

--message [string]

--symptomid [string]

--type [string]

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Dell Technologies connectivity services configuration status

Syntax

```
powerflex_cli support-assist-config status [parameters]
```

Parameters

-h, --help
Help for Dell Technologies connectivity services configuration status.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Dell Technologies connectivity services configuration system mode

Syntax

```
powerflex_cli support-assist-config system_mode [parameters]
```

Parameters

-h, --help
Help for Dell Technologies connectivity services configuration system mode.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

17

Syslogs commands

This section contains commands to manage syslogs.

Syntax

```
powerflex_cli syslogs [command options]
```

Parameters

-h, --help
Help for syslogs.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Command options

modify
Modify the syslogs commands.

query
Query syslogs.

delete
Deletes syslogs.

Modify syslogs

Modifies the syslogs.

Syntax

```
powerflex_cli syslogs modify [parameters]
```

Parameters

-h, --help
Help for modifying the syslogs.

--id [string]
Unique Identifier of the syslogs.

--description [string]

--destination_address [string]

--name [string]

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Query syslogs

Queries syslogs.

Syntax

```
powerflex_cli syslogs query [parameters]
```

Parameters

-h, --help
Help for modifying the syslog.

--id [string]
Unique Identifier of the syslog.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Delete syslogs

Deletes syslogs.

Syntax

```
powerflex_cli syslogs delete [parameters]
```

Parameters

-h, --help
Help for deleting the syslog.

--id [string]
Unique Identifier of the syslog.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

18

User management commands

This section contains commands to manage users.

Syntax

```
powerflex_cli users [command options]
```

Parameters

-h, --help
Help for users commands.

--session name [string]
Session name. The default name is "default".

--session token [string]
Secret session token.

Command options

create
Creates a user.

modify
Modify user settings.

query
Query all users.

delete
Delete a user.

reset-password
Resets the password.

Create users

Syntax

```
powerflex_cli users create [parameters]
```

Parameters

-h, --help
Help for creating users.

--email_address [string]

--first_name [string]

--last_name [string]

--is_enabled [string]

--is_temporary [string]

--username [string]

--password [string]

--role [string]

Values are:

- SuperUser
- SystemAdmin
- StorageAdmin
- LifecycleAdmin
- ReplicationManager
- SnapshotManager
- SecurityAdmin
- DriveReplacer
- Technician,Monitor
- Support
- NA

--type [string]

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

Modify users

Modify a user by ID.

Syntax

```
powerflex_cli users modify [parameters]
```

Parameters

-h, --help

Help for modifying users.

--id [string]

Unique Identifier of an existing user.

--is_enabled [string]

--first_name [string]

--last_name [string]

--role [string]

Values are:

- SuperUser
- SystemAdmin
- StorageAdmi

- LifecycleAdmin
- ReplicationManager
- SnapshotManager
- SecurityAdmin
- DriveReplacer
- Technician
- Monitor
- Support
- NA

--username [string]

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

Query users

Syntax

```
powerflex_cli users query [parameters]
```

Parameters

-h, --help

Help for querying users.

--id [string]

Unique Identifier of an existing user.

--filter [string]

Conditions for filtering users (username, first_name, last_name, type, is_enabled, email_address).

--limit [int32]

Limit number of returned objects.

--offset [string]

Starting row of the result set, used with limit for paging. Value is zero based (offset=0 starts the response from the first row of results).

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

Delete users

Delete a user by ID.

Syntax

```
powerflex_cli users delete [parameters]
```

Parameters

-h, --help

Help for deleting users.

--id [string]

Unique identifier of an existing user.

--session name [string]

Session name. The default name is "default".

--session token [string]

Secret session token.

II

PowerFlex SCLI

This section explains the basic concepts of using the PowerFlex SCLI, including how to access the SCLI and log in and lists the SCLI commands.

19

SCLI basics

This section explains the basic concepts of using the PowerFlex SCLI, including how to access the SCLI and log in.

Accessing SCLI

PowerFlex CLI (SCLI) is installed as part of the MDM component. It can be found in the following path:

Table 4. Operating systems and CLI paths

Operating system	Path
Linux	`scli`
ESXi	`scli`
Citrix Hypervisor (XenServer)	`siocli`

Before using most SCLI commands, you must log in using the following command:

```
scli --login --management_system_ip <MNO IP> --username <username> --password <password>
```

`--management_system_ip <MNO_IP>`—IP address of the management system.

`--username <NAME>`—User name to log in.

`--password <PASSWORD>`—User password. If the password is missing, the user is prompted to enter the password.

Command format

All SCLI commands use the following format:

Table 5. Operating systems and command format

Operating system	Command format
Linux	`scli [--mdm_ip <IP>] --<cmd_name>`
ESXi	`scli [--mdm_ip <IP>] --<cmd_name>`
Windows	`scli [--mdm_ip <IP>] --<cmd_name>`
Xen	`siocli [--mdm_ip <IP>] --<cmd_name>`

Where:

- `--mdm_ip <IP>`

  is one or more IP addresses of the servers running the primary MDM and secondary MDM. In a non-clustered environment, use the MDM IP address.

  If the CLI does not reside on the MDM, the `--mdm_ip` parameter must be added to every CLI command.

- `--<cmd_name>`

  Is the command to be run.

Example:

```
scli --mdm_ip 10.10.10.3,10.10.10.4 --query_all
```

Additional rules:

- All parameters are preceded by --
- The order of the parameters in the command is insignificant.
- SCLI commands are lowercase and case-sensitive.

--mdm_ip

The `mdm_ip` parameter indicates the MDM designated to receive and run the command. If the command is run from the primary MDM, this switch may be omitted.

Example:

```
scli --query_all
```

To avoid using the `--mdm_ip` parameter in every command, or avoid having to install the CLI on other servers, use SSH, or RDM to log in to the shell running on the management server.

You cannot run SCLI commands on the secondary MDM. However, you can send a command from the secondary MDM by adding the `--mdm_ip` parameter with the primary MDM IP address to the command.

Syntax

SCLI syntax follows these rules:

Format

The following command syntax is mandatory:

```
scli --cmd_name
```

Parentheses ( ) indicate a group of required parameters, from which you must choose one.

Angle brackets <> indicate an argument.

Braces { } indicate a series of options for parameter values, from which you must choose one.

A pipe | is used to separate items in a series of options (parameters or parameter values). Select one of them.

Square brackets [ ] indicate an optional parameter.

Usage

Each command entry uses the above syntax and looks similar to the following example:

```
scli --cmd_example --r1 (--r2 | --r3 <V1>) [o1 <V2> | o2 {--so1 <V3> | --so2 }] [Options]
```

Parameter description

--r1

The `r1` parameter description

--r2

The `r2` parameter description

--r3<*V1*>

The `r3` parameter description with possible <*V1*> input values

--o1<*V2*>

The `o1` description with possible <*V2*> input values

--o2 {--so1 <V3> | --so2 }

The `o2` description

[Options]

Additional optional parameters:

so1 <V3>

The `so1` description with possible *V3* input values

so2

The `so2` description

Interpretation

- The text `scli --cmd_example r1` is mandatory. In commands that have no parameters, just `scli --cmd_example` is necessary.
- `(--r2 | --r3 <V1>)` indicates that you must choose one of the options separated by |. Selecting an option is required, as indicated by ( ).
- `[o1 <V2>| o2 {--so1 <V3> | --so2}]` indicates that you may choose one of the options separated by |. Selecting one of these options is optional, as indicated by [ ].
- `[Options]` indicates that you may choose from the following additional parameters. It is optional, as indicated by [ ].

Object names

Define object names according to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID using a query or the property sheet of the object, available in the PowerFlex Manager user interface.

It is recommended that you give each object a meaningful name that is associated with its operational role.

NOTE: The system has a unique name as it is a single object.

Using SCLI in non-secure mode

If PowerFlex is running in non-secure mode, you must disable secure communications on every MDM server to enable execution of commands.

- To disable secure communications in Windows, on each MDM open the SCLI `conf.txt` file, and add the following line:

```
cli_use_secure_communication=0
```

- To disable secure communications in Linux, run the following on each MDM:

```
echo cli_use_secure_communication=0 >> ~/.scli/conf.txt
```

NOTE: Changing from secure mode to non-secure mode after renewing certificates is not supported.

Logging in

To access the CLI, you must first log in to the management system using a terminal application.

If the CLI and the MDM do not reside on the same server, add the `--mdm_ip` parameter to all CLI commands.

In a non-clustered environment, use the MDM IP address. In a clustered environment, use the IP addresses of the primary and secondary MDMs, separated by a comma. For example:

```
scli --mdm_ip 10.10.10.3,10.10.10.4 --login --username supervisor1 --password password1
--management_system_ip 10.11.12.13
```

If you do not enter the password, you will be prompted to enter the password.

login

Log in to the system by either specifying management credentials or using a pre-generated p12 certificate. It is recommended to avoid the password options. If not used, the user is prompted to enter the passwords in a secure manner.

Once you are authenticated by the system, all commands can be run with the respective role until a logout is performed, or until the session expires by reaching one of the following timeouts:

- Maximum session length (default: 8 hours)
- Session idle time (default: 10 minutes)

Before you log in for the first time, run the following commands:

`cp/opt/emc/scaleio/mdm/cfg/mgmt_ca.pem/`—Copy the management certificate to the root location.

`scli --add_certificate --certificate_file /opt/emc/scaleio/mdm/cfg/mgmt_ca.pem`—Add the management certificate to the trusted certificate.

Syntax

```
scli --login ( (--management_system_ip <MNO_IP> --username <NAME> [--password
<PASSWORD>] ) | ( [--p12_path <P12_PATH>] [--p12_password <P12_PASSWORD>] ) ) [--
accept_banner_by_scripts_only]
```

Parameters

`--management_system_ip <MNO_IP>`

IP address of the management system.

`--username <NAME>`

User name to log in

`--password <PASSWORD>`

User password. If the password is missing, you are prompted to enter the password.

NOTE: In Linux, to prevent the password from being recorded in the history log, type the command without password parameter and then enter the password in the new line when it is prompted.

`--p12_path <P12_PATH>`

Path to p12 certificate file. If not specified, uses same default path as in generate_login_certificate

`p12_password <P12_PASSWORD>`

Password for p12 certificate. If not specified, you are prompted to enter the password.

`--accept_banner_by_scripts_only`

Preemptive approval of login banner

Examples

```
scli --login --management_system_ip dellpowerflex.com --username myuser --password 2@3#4A
```

If a login banner is configured and enabled on your system, press any key to display the banner. Enter **q** to quit the login banner, and then enter **y** to approve the banner.

generate_login_certificate

Generate a P12 certificate for CLI.

Syntax

```
scli --generate_login_certificate --management_system_ip <MNO_IP> --username <NAME> [--
password <PASSWORD>] [--p12_password <P12_PASSWORD>] [--p12_path <P12_CERT_PATH>]
```

Parameters

--management_system_ip <MNO_IP>

IP address of the management system.

--username <NAME>

User name to log in

--password <PASSWORD>

User password. If the password is missing, the you are prompted to enter the password.

NOTE: In Linux, to prevent the password from being recorded in the history log, leave out the `password` flag and enter the password interactively.

--p12_password <P12_PASSWORD> (Optional)

The P12 password is used to encrypt the certificate. If missing, you are prompted to enter the password.

--p12_path <P12_CERT_PATH> (Optional)

The path where the p12 certificate is stored. If not specified, use the same default path as in login.

Examples

```
scli --login --management_system_ip dellpowerflex.com --username myuser --password
2@3#4A --p12_path /home/myuser/my_certificate.p12 --p12_password "password"
```

logout

Log out the current user of the system.

Syntax

```
scli --logout
```

Example

```
scli --logout
```

Getting help

The PowerFlex CLI supports auto-completion. To complete a command or parameters, press the Tab key while typing CLI commands.

NOTE:

In Windows, PowerFlex does not support auto-completion.

Alternately, you can run the `help` command.

Help

Display help for the CLI commands.

Syntax

```
scli --help
  [--mdm]
  [--sds]
  [--vol]
  [--sdc]
  [--general]
  [--all]
  [--user]
```

NOTE: To view all help options, run the command without any additional options:

```
scli --help
```

Options

--mdm

Displays help for MDM commands

--sds

Displays help for SDS commands

--vol

Displays help for volume commands

--sdc

Displays help for SDC commands

--general

Displays help for general commands

--all

Displays help for all commands

--user

Displays help for commands related to user management

Example

```
scli --help --mdm
```

CLI command help syntax

- `Text`—Required
- <>—Argument
- `()`—Required element
- `[]`—Optional element
- `|`—Select from options A | B | C

Example command:

```
scli --cmd_example --req1 (--req2 | --req3 <VAL1>) [opt1 <VAL2> | opt2] Options
```

Some description of what `cmd_example` does.

Parameters:

`--req1`
req1 description

`--req2`
req2 description

`--req3 <V1>`
req3 description with possible VAL1 input values

`--opt1 <VAL2>`
opt1 description with possible VAL2 input values

`--opt2`
opt2 description

Options:
CHOOSE ONE/SEVERAL

`--spec_opt1 <VAL3>`
spec_opt1 description with possible VAL3 input values

`--spec_opt2`
spec_opt2 description

20

Acceleration pool commands

This section contains commands for managing acceleration pools.

add_acceleration_pool

Add an acceleration pool to a protection domain.

Syntax

```
scli --add_acceleration_pool (--protection_domain_id <ID> | --protection_domain_name
<NAME>) (--media_type <TYPE> {SSD | NVDIMM})
  [--acceleration_pool_name <NAME>]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--media_type {SSD | NVDIMM}
Type of devices in the acceleration pool: SSD or NVDIMM

--acceleration_pool_name *<NAME>*
Name to assign the acceleration pool you are creating

Example

```
scli --add_acceleration_pool --protection_domain_name pd1 --media_type SSD --
acceleration_pool_name RfcAccPool
```

modify_metadata_acceleration

Modify the acceleration pool that holds the metadata for the specified storage pool.

Syntax

```
scli --modify_metadata_acceleration (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id
<ID>) (((--protection_domain_id <ID> | --protection_domain_name <NAME>) --
acceleration_pool_name <NAME>) | --acceleration_pool_id <ID>)
  [--on_device_metadata_on_missing_capacity]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

--acceleration_pool_name *<NAME>*
Acceleration pool name

--acceleration_pool_id *<ID>*
Acceleration pool ID

--on_device_metadata_on_missing_capacity
When there is insufficient capacity in the acceleration pool, the metadata of some of the devices resides on the devices themselves.

Example

```
scli --modify_metadata_acceleration --protection_domain_name pd2 --
storage_pool_name sp4 --protection_domain_name pd2 --acceleration_name ap4 --
on_device_metadata_missing_capacity
```

query_acceleration_pool

Retrieve information about a protection domain's acceleration pool.

Syntax

```
scli --query_acceleration_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--acceleration_pool_name *<NAME>*
Acceleration pool name

--acceleration_pool_id *ID>*
Acceleration pool ID

Example

```
scli --query_acceleration_pool --protection_domain_name pd2 --acceleration_pool_name
NVDIMM_pool
```

Sample output

```
Acceleration Pool RFcache_Pool (Id: 6794008e00000000)
        Media Type: SSD
        Used for RFcache: yes
```

remove_acceleration_pool

Remove an acceleration pool from a protection domain.

Syntax

```
scli --remove_acceleration_pool (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id
<ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--acceleration_pool_name *<NAME>*
Acceleration pool name

--acceleration_pool_id *ID>*
Acceleration pool ID

Example

```
scli --remove_acceleration_pool --protection_domain_name pd1 --acceleration_pool_name
ap_flash1
```

rename_acceleration_pool

Name or rename an acceleration pool.

Syntax

```
scli --rename_acceleration_pool (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id
<ID>) --new_name <NAME>
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--acceleration_pool_name *<NAME>*

Acceleration pool name

--acceleration_pool_id *ID>*

Acceleration pool ID

--new_name *<NAME>*

New name to be assigned to the acceleration pool

Example

```
scli --rename_accleration_pool --protection_domain_name pd1 --acceleration_pool_name
ap_1 --new_name ap_flash1
```

21

Device commands

This section contains commands for managing devices.

abort_remove_sds_device

Stop the process of removing a storage data server (SDS) device from an SDS. If it is performed at too late a stage in the removal process, the command may fail and the device is removed.

NOTE: Removal of acceleration devices cannot be aborted.

Syntax

```
scli --abort_remove_sds_device (--device_id <ID> ((--sds_id <ID> | sds_name <NAME> |
sds_ip [sds_port <PORT>]) (device_name <NAME>  | device_path <PATH>)))
```

Parameters

--device_id *<ID>*

The ID of the storage device that was requested to be removed from the SDS

--sds_id *<ID>*

The ID of the SDS from which the device was to be removed

--sds_name *<NAME>*

The name of the SDS from which the device was to be removed

--sds_ip *<IP>*

The IP address of the SDS from which the device was to be removed

--sds_port *<PORT>*

The port associated with the SDS

--device_name *<NAME>*

The name of the storage device or file that was requested to be removed from the SDS

--device_path *<PATH>*

The path of the device that was requested to be removed

Example

```
scli --abort_remove_sds_device --sds_ip 192.168.1.6 --device_name /dev/sdb
```

activate_sds_device

Activate SDS devices that have completed their tests.

The system does not use SDS devices that are added using the `add_sds` or `add_sds_device` command, with the `test_only` parameter, until activated by this command.

This command does not apply to acceleration devices which are activated automatically.

NOTE: You can activate one or all SDS devices on an SDS with a single command.

Syntax

```
scli --activate_sds_device (--sds_id <ID> |
--sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) (--device_id <ID> | --device_name
<NAME>  | --device_path <PATH> | --activate_all_devices)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--device_id *<ID>*
ID of the SDS device to activate

--device_name *<NAME>*
Storage device name or file name

--device_path *<PATH>*
Path of the SDS device to activate

--activate_all_devices
Activate all of the specified SDS devices. If all the devices cannot be activated, this command fails.

Example

```
scli --mdm_ip 192.168.1.200 --activate_sds_device
--sds_ip 192.168.1.5 --activate_all_devices
```

add_sds_device

Add a storage or acceleration device to an SDS.

CAUTION: During execution of the command, PowerFlex checks that the device is clear before adding it. If the device is not clear, an error message is returned, and the command fails for that device. If you want to overwrite existing data on the device by forcing the command, use the `force_device_takeover` flag. This flag destroys all existing data on the device.

Syntax

```
scli --add_sds_device (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --device_path <PATH> --device_name <NAME>] (--storage_pool_name <NAME> | --
storage_pool_id <ID> | --acceleration_pool_name <NAME> | --acceleration_pool_id <ID>)
  [--force_device_takeover]
  [--media_type {HDD | SSD}]
  [--external_acceleration {none | read | write | read_and_write}]
```

```
   [--test_time <TIME>]
   [--test_only | --no_test | <BLANK> ]
```

Parameters

--sds_id *<ID>*

ID of the SDS

--sds_name *<NAME>*

Name of the SDS

--sds_ip *<IP>*

SDS IP address

-sds_port *<PORT>*

Port associated with the SDS

--device_path *<PATH>*

Full path of the device to be added. A device can be a disk, an unmounted partition, or a file that represents free space on a mounted device.

--device_name *<NAME>*

Name to assign to the device

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--acceleration_pool_name *<NAME>*

Acceleration pool name

--acceleration_pool_id *<ID>*

Acceleration pool ID

--force_device_takeover

Add the device, ignoring data from other SDSs on the device.

CAUTION: Use this flag with caution, because all data on the device is destroyed.

--media_type {HDD | SSD}

Media type of the device, where the device will be HDD or SSD.

--external_acceleration *<TYPE>*

External acceleration for the device where *<TYPE>* can be one of the following:

none

The device is not accelerated by non-PowerFlex cache.

read

The device is accelerated by a non-PowerFlex read cache.

write

The device is accelerated by a non-PowerFlex write cache.

read_and_write

The device is accelerated by both non-PowerFlex read and write cache.

--test_time *<TIME>*

The maximum test run-time in seconds. Default: 10. The test will stop when it reaches either this limit, or the time it takes to complete 128 MB of data read/write, whichever is first. When `no_test` is selected, this switch is ignored.

Test options (choose one):

By default, PowerFlex tests the performance of the device being added before its capacity can be used, and saves the results. Two tests are performed: random writes and random reads. When the tests are

complete, the device capacity is added automatically to the storage pool used by the MDM. To modify this behavior, specify one of the test options:

--test_only

Devices will be tested, but not used. To start using their capacity, run the `--activate_sds_device` command.

--no_test

The device capacity will be used without any device testing.

<BLANK>

Read and write test will be run on devices before their capacity is used.

Example

```
scli --mdm_ip 192.168.1.200 --add_sds_device
--sds_ip 192.168.1.6 --device_path /dev/sdb --device_name sd02
```

Devices

The following devices are considered to be not "clear," and cannot be added to an SDS:

- Linux: A complete device with either a file system or partition, or a partitioned device with a file system.
- Windows: A complete device with a partition, or a partitioned device with a file system.
- ESXi: The same as the above, depending on the operating system of the SVM where the SDS is installed.

Limitations:

- SAN devices are not prevented from being added to an SDS.
- Devices in an LVM group cannot be added to an SDS.
- Within the database devices, only Oracle ASM devices can be detected and blocked.

SDS device names

Assigning each SDS device a meaningful name facilitates future object identification, because the defined name remains constant even if the path changes. When a name has not been defined, the system may display default system-defined names that use the first IP address of the SDS.

Each name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

clear_sds_device_error

Clear an SDS device error.

NOTE: The command does not check whether the device error was fixed. If the error persists, the device returns to an error state when it is accessed. You can verify that the device error is fixed by running an SDS query.

Syntax

```
scli --clear_sds_device_error (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) (--device_id <ID> | --device_name <NAME> | --device_path <PATH> | --
clear_all)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--device_id *<ID>*
ID of the device

--device_name *<NAME>*
Storage device name or file name

--device_path *<PATH>*
Path of the device

--clear_all
Clear errors on all SDS devices.

Example (single device)

```
scli --mdm_ip 192.168.1.200 --clear_sds_device_error
--device_name /dev/sdb --sds_ip 192.168.1.7
```

Example (all devices)

```
scli --mdm_ip 192.168.1.200 --clear_sds_device_error
--clear_all --sds_ip 192.168.1.7
```

modify_sds_device_capacity

Modify the amount of an SDS device's storage capacity that is made available to PowerFlex.

Use this command to reduce (or increase) the device capacity that is available to PowerFlex, in order to assist in balancing storage, without affecting the physical disk size. Ensure that the modified capacity is less than the total physical size of the device. If you reduce capacity, data will be moved to other devices in an asynchronous fashion.

When reducing capacity, the excess device capacity is not accessible to the user.

Syntax

```
scli --modify_sds_device_capacity (--device_id <ID> | ((--sds_id <ID> | --sds_name
<NAME> | --sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path
<PATH>))) --size_gb <SIZE>
```

Parameters

--device_id *<ID>*
ID of the SDS device

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--device_name *<NAME>*
SDS storage device name or file

--device_path *<PATH>*
Path of the device whose capacity is to be modified

--size_gb *<SIZE>*
Size in GB of the device to use. The size must be within the device size limits and slightly less than the full device size.

Example

```
scli --mdm_ip 192.168.1.200 --modify_sds_device_capacity --sds_ip 192.168.1.6 --
device_name /dev/sdb --size_gb 150
```

query_all_device_latency_meters

Retrieve device latency meters for all the devices, or for devices in the specified protection domain.

NOTE: To query the devices in a specific SDS, use the `query_device_latency_meters` command.

Syntax

```
scli --query_all_device_latency_meters
  [(--protection_domain_id <ID> | --protection_domain_name <NAME>)]
```

NOTE: If the background device scanner is enabled, several device read statistics are dramatically affected.

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

Example

```
scli --query_all_device_latency_meters --protection_domain_name pd10
```

Sample output

```
Protection Domain: fcc9af5500000000, Name: pd1 contains 4 SDS
Storage Pool: 3df1cd9600000000, Name: sp1 contains 8 devices
  SDS sds_idd_1:
    Device: /dev/sdb Average latency read: 836 write: 921 (microseconds)
    Device: /dev/sdc Average latency read: 755 write: 843 (microseconds)

  SDS sds_idd_2:
    Device: /dev/sdb Average latency read: 658 write: 815 (microseconds)
    Device: /dev/sdc Average latency read: 689 write: 732 (microseconds)

  SDS sds_idd_4:
    Device: /dev/sdb Average latency read: 842 write: 811 (microseconds)
    Device: /dev/sdc Average latency read: 852 write: 918 (microseconds)

  SDS sds_idd_5:
    Device: /dev/sdb Average latency read: 844 write: 698 (microseconds)
    Device: /dev/sdc Average latency read: 795 write: 801 (microseconds)

  Devices with maximal read latency:
    SDS: sds_idd_4 Device: /dev/sdc  Latency: 852 (microseconds)
    SDS: sds_idd_5 Device: /dev/sdb  Latency: 844 (microseconds)
    SDS: sds_idd_4 Device: /dev/sdb  Latency: 842 (microseconds)
  Devices with maximal write latency:
    SDS: sds_idd_1 Device: /dev/sdb  Latency: 921 (microseconds)
    SDS: sds_idd_4 Device: /dev/sdc  Latency: 918 (microseconds)
    SDS: sds_idd_1 Device: /dev/sdc  Latency: 843 (microseconds)

Protection Domain: fcc9af5600000001, Name: pd2 contains 1 SDS
Storage Pool: 3df1cd9700000001, Name: sp2 contains 1 devices
  SDS sds_idd_3:
    Device: /dev/sdb Average latency read: 0 write: 0 (microseconds)


Protection Domain: fcc9d66500000002, Name: rrr contains 0 SDS
```

query_device_latency_meters

Retrieve the latency meters for all the devices of the specified SDS.

The SDS maintains statistics about reads and writes to its devices. This command returns the average I/O size and latency for both reads and writes.

NOTE: To query all devices in the system, or all devices in a specific protection domain, use the `query_all_device_latency_meters` command.

Syntax

```
scli --query_device_latency_meters (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>]) [(--storage_pool_name <NAME> | --storage_pool_id <ID>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

NOTE: If the background device scanner is enabled, several device read statistics are dramatically affected.

Example

```
scli --mdm_ip 192.168.1.200 --query_device_latency_meters --sds_name sds_idd_2 --
storage_pool_name sp1
```

Sample output

```
SDS sds_idd_2 returned information on 2 devices
Device: /dev/sdc
  Average read size: 277.1 KB (283763 Bytes)
  Average read latency: 873 (microseconds)
  Average write size: 355.6 KB (364140 Bytes)
  Average write latency: 1144 (microseconds)
Device: /dev/sdb
  Average read size: 292.8 KB (299816 Bytes)
  Average read latency: 928 (microseconds)
  Average write size: 362.0 KB (370731 Bytes)
  Average write latency: 1157 (microseconds)
```

query_device_test

Retrieve the SDS device test results. Every SDS saves results for the last two tests per device.

Syntax

```
scli --query_device_test ((--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
| (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) | (((--
protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>)
| --storage_pool_id <ID>) | (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id <ID>))
```

Parameters

--device_id *<ID>*

Device ID

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_id *<ID>*

Device ID

--device_name *<NAME>*

SDS storage device name or file name

--device_path *<PATH>*

File path to storage device

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--acceleration_pool_name *<NAME>*

Storage pool name

--acceleration_pool_id *<ID>*

Storage pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --query_device_test
--sds_ip 192.168.1.6 --device_name /dev/sdb
```

query_sds_device_info

Retrieve detailed information for the specified SDS device, or for all devices in the specified SDS.

Syntax

```
scli --query_sds_device_info (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
  [--all_devices]
```

Parameters

--device_id *<ID>*

ID assigned to the storage device

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

Name assigned to the storage device

--device_path *<PATH>*

Device path or file path of the storage device

--all_devices

Return detailed information for all devices in the specified SDS.

Example

```
scli --query_sds_device_info --sds_id 06b108a400000004 --all_devices
```

Sample output

```
Storage Device Information (total 2 devices):
        Device ID: c8bea95800040000 Name: N/A Path: /dev/sdb
                ScaleIO Device Configuration:
                        Original Path: /dev/sdb
                        Storage Pool Name: sp1
                        Capacity Limit: 148.9 GB (152500 MB)
                        Device State: Normal
                        Data Layout: Medium Granularity
                        MG Persistent checksum state: PROTECTED
                Physical Device Information:
                        Device Type: UNKNOWN
                        Media Type: HDD
                        Auto Detected Media Type: SSD
                        External Acceleration Type: NONE
                        Queue Depth: 16
                        Preset Queue Depth: HDD
                        Vendor Name: N/A
                        Model Name: N/A
                        Serial Number: N/A
                        Slot Number: N/A
                        Firmware Version: N/A
                        Cache Look-ahead: not Active
                        Write Cache: not Active
                        ATA Security: not Active
                        Logical Sector Size: 0 B
                        Physical Sector Size: 0 B
                        Capacity: 0 GB
                        LED Setting: OFF
                Background Device Scanner Information:
                        Scanned: 1873793 MB
                        Unresolved Read Errors: 0/0
                        Unresolved Compare Errors: 0/0
                SMART Information:
                        Aggregated State: NEVER_FAILED
                        Temperature State: NEVER_FAILED
                                Current Value: 0 Worst Value: 0 Threshold: 0
                        Media Wearout Indicator State: NEVER_FAILED
                                Current Value: 0 Worst Value: 0 Threshold: 0
                RAID Controller Information:
                        Serial Number: N/A
                        RAID vDisk Status: N/A
                        RAID vDisk Type: N/A
                        RAID vDisk Cache: N/A

        Device ID: c8bea95900040001 Name: N/A Path: /dev/sdc
                ScaleIO Device Configuration:
                        Original Path: /dev/sdc
                        Storage Pool Name: sp1
                        Capacity Limit: 148.9 GB (152500 MB)
```

```
                        Device State: Normal
                        Data Layout: Medium Granularity
                        MG Persistent checksum state: PROTECTED
                Physical Device Information:
                        Device Type: UNKNOWN
                        Media Type: HDD
                        Auto Detected Media Type: SSD
                        External Acceleration Type: NONE
                        Queue Depth: 16
                        Preset Queue Depth: HDD
                        Vendor Name: N/A
                        Model Name: N/A
                        Serial Number: N/A
                        Slot Number: N/A
                        Firmware Version: N/A
                        Cache Look-ahead: not Active
                        Write Cache: not Active
                        ATA Security: not Active
                        Logical Sector Size: 0 B
                        Physical Sector Size: 0 B
                        Capacity: 0 GB
                        LED Setting: OFF
                Background Device Scanner Information:
                        Scanned: 1873886 MB
                        Unresolved Read Errors: 0/0
                        Unresolved Compare Errors: 0/0
                SMART Information:
                        Aggregated State: NEVER_FAILED
                        Temperature State: NEVER_FAILED
                                Current Value: 0 Worst Value: 0 Threshold: 0
                        Media Wearout Indicator State: NEVER_FAILED
                                Current Value: 0 Worst Value: 0 Threshold: 0
                RAID Controller Information:
                        Serial Number: N/A
                        RAID vDisk Status: N/A
                        RAID vDisk Type: N/A
                        RAID vDisk Cache: N/A


Acceleration Device Information (total 0 devices):

RAID Controllers Information (total 0 controllers):
```

remove_sds_device

Initiate removal of a storage device from an SDS. Note that this process is asynchronous, and will run in the background.

An SDS device can be removed at any time, and no downtime is required. The removal process is asynchronous and runs in the background.

During execution of this command, the associated data is replicated to different nodes. Therefore, the process may take a long time.

When removing an SDS device on a VMware server, use the vSphere client.

NOTE: If the capacity of this SDS is still used by volumes, and cannot be replaced, the command will fail.

Syntax

```
scli --remove_sds_device (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> | --
sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
```

Parameters

--device_id *<ID>*

ID of the device to be removed

--sds_id *<ID>*

ID of the SDS from which the device should be removed

--sds_name *<NAME>*

Name of the SDS from which the device should be removed

--sds_ip *<IP>*

IP address of the SDS from which the device should be removed

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

Name of the storage device or file to be removed from the SDS

--device_path *<PATH>*

Path of the storage device to be removed from the SDS

Example

```
scli --mdm_ip 192.168.1.200 --remove_sds_device
--sds_ip 192.168.1.6 --device_name /dev/sdb
```

Related operations

To abort the removal, run the command `scli --abort_remove_sds_device`.

To track progress of the removal operation, use either of the following queries:

- `scli --query_sds_device_info`
- `scli --query_sds`

rename_device

Assign a name to, or rename, an SDS device.

Syntax

```
scli --rename_device (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) --new_name
<NAME>
```

Parameters

--device_id *<ID>*

ID of the SDS device

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

Name of the SDS device

--device_path *<PATH>*

Path of the SDS device

--new_name *<NAME>*

New name to be assigned to the SDS device

Example

```
scli --rename_device --sds_ip 8.8.8.100 --device_name flash_18 --new_name flash_10018
```

set_sds_device_external_acceleration_type

Set the external acceleration type of an SDS device.

Syntax

```
scli --set_sds_device_external_acceleration_type (--device_id <ID> | ((--sds_id <ID>
| --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --
device_path <PATH>))) --external_acceleration <TYPE>
```

Parameters

--device_id *<ID>*

Device ID

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

Device name

--device_path *<PATH>*

SDS storage device path or file path

--external_acceleration *<TYPE>*

External acceleration for the device, where TYPE is one of the following:

none

The device is not accelerated by a non-PowerFlex read or write cache.

read

The device is accelerated by a non-PowerFlex read cache.

write

The device is accelerated by a non-PowerFlex write cache.

read_and_write

The device is accelerated by both non-PowerFlex read and write cache.

Example

```
scli --set_sds_device_external_acceleration_type --sds_name sds_23 --device_name
storage_2310 --external_acceleration read
```

set_sds_device_led

Turn the LED on a storage or cache device on or off.

Syntax

```
scli --set_sds_device_led (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) (--
led_on | --led_off)
```

Parameters

--device_id *<ID>*
: ID assigned to the storage device

--sds_id *<ID>*
: SDS ID

--sds_name *<NAME>*
: SDS name

--sds_ip *<IP>*
: SDS IP address

--sds_port *<PORT>*
: Port associated with the SDS

--device_id *<ID>*
: Device ID.

--device_name *<NAME>*
: Name assigned to the storage device

--device_path *<PATH>*
: Device path or file path of a storage device

--led_on
: Turn the LED on.

--led_off
: Turn the LED off.

Example

```
scli --set_sds_device_led --sds_ip  192.168.1.5 --sds_port 7072 --device_name slot5 --
led_on
```

set_sds_device_media_type

Set the media type of an SDS device.

Syntax

```
scli --set_sds_device_media_type (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME>
| --sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) --
media_type <TYPE>
```

Parameters

--device_id *<ID>*
Device ID

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or file path

--media_type *<TYPE>*
Media type of the device where TYPE is one of the following:

HDD
The device is an HDD device.

SSD
The device is an SSD device.

Example

```
scli --set_sds_device_media_type --sds_name sds22 --device_name storage_210 --media_type
HDD
```

start_device_test

Initiate a device test for one or all SDS devices.

Syntax

```
cli --start_device_test ((--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
| (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) | (((--
protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>)
|--storage_pool_id <ID>)) | (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id <ID>)) (Options)
(Control)
  [--i_am_sure]
```

Parameters

--device_id *<ID>*

Device ID

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

IP address associated with the SDS

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

SDS storage device name or file

--device_path *<PATH>*

SDS storage device path or file path

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--acceleration_pool_name *<NAME>*

Acceleration pool name

--acceleration_pool_id *<ID>*

Acceleration pool ID

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Options

Choose several:

--io_type_write

Use write I/O type (default is read). If the storage device is in use, only read testing can be performed.

--io_pattern_squential

Use sequential I/O pattern (default is random).

--io_size_kb *<SIZE>*

The read or write I/O size in KB. Range is 4-512 KB. Default: 8 KB.

Control

Choose several:

--test_time *<TIME>*

Maximum test run time in seconds. If not specified, the test will be limited by I/O size only.

--io_total_size_mb *<SIZE>*

The total size of the test reads or writes in MB. Range is 1-32768 MB, which specifies the total I/O size in MB. Default: 128 MB

<blank>

Use defaults

Example

```
scli --mdm_ip 192.168.1.200 --start_device_test
--sds_ip 192.168.1.6 --device_path /dev/sdb --io_type_write --io_size_kb 8
```

set_sds_device_path

Configure the path of the given storage device.

Syntax

```
scli --set_sds_device_path (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) --
new_device_path <PATH>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--device_id *<ID>*
Device ID

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or file path. The file path is used only when the device is available at this path

--new_device_path *<PATH>*
The new path to configure

22

Fault set commands

This section contains commands for managing fault sets.

add_fault_set

Add an empty fault set to a protection domain.

Fault sets are subsets of a given protection domain. They provide additional safeguards for protecting your data against hardware failure. When you define SDSs later on, you can add them to a fault set.

Fault sets are enabled by default.

Syntax

```
scli --add_fault_set (--protection_domain_id <ID> | --protection_domain_name <NAME>)
  [--fault_set_name <NAME>]
```

Parameters

--protection_domain_id *<ID>*

ID of the protection domain to which the fault set will be added

--protection_domain_name *<NAME>*

Name of the protection domain to which the fault set will be added

--fault_set_name *<NAME>*

Name of the new fault set

Example

```
scli --add_fault_set --protection_domain_name pd1 --fault_set_name fs1-mg
```

Fault set names

Assign each fault set a meaningful name. When a name has not been defined, the system may display default system-defined names that use the ID of the fault set.

Each fault set name must conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

clear_fault_set

Remove all SDSs from the specified fault set.

Syntax

```
scli --clear_fault_set (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>))
  [--i_am_sure]
```

Parameters

--fault_set_id *<ID>*
Fault set ID

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_name *<NAME>*
Fault set name

--i_am_sure
Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --clear_fault_set --protection_domain_name pd18 --fault_set_name sio-fs1
```

query_all_fault_sets

Retrieve information for all the fault sets in the specified protection domain.

Syntax

```
scli --query_all_fault_sets (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

Example

```
scli --query_all_fault_sets --protection_domain_name pd1
```

Sample output

```
Protection Domain has 1 Fault Sets:
Protection Domain ID: fcc9af5500000000 Fault Set: ID: 36c9d76800000000 Name: fs1_mg SDS
Count: 2
  SDS ID: 06b108a400000004 Name: sds_idd_1
  SDS ID: 06b108a300000003 Name: sds_idd_2
```

query_fault_set

Retrieve information about the specified fault set.

Syntax

```
scli --query_fault_set (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>))
```

Parameters

--fault_set_id *<ID>*
: Fault set ID

--protection_domain_id *<ID>*
: Protection domain ID

--protection_domain_name *<NAME>*
: Protection domain name

--fault_set_name *<NAME>*
: Name of the new fault set

Example

```
scli --query_fault_set --protection_domain_name pd18 --fault_set_name fs1_mg
```

Sample output

```
Protection Domain ID: fcc9af5500000000 Fault Set: ID: 36c9d76800000000 Name: fs1_mg SDS
Count: 2
  SDS ID: 06b108a400000004 Name: sds_idd_1
  SDS ID: 06b108a300000003 Name: sds_idd_2
```

remove_fault_set

Remove the specified fault set.

Ensure that the fault set is empty before executing this command. You can remove SDSs from the fault set using the `clear_fault_set` command.

Syntax

```
scli --remove_fault_set (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>))
```

Parameters

--fault_set_id *<ID>*
Fault set ID

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_name *<NAME>*
Name of the new fault set

Example

```
scli --remove_fault_set --protection_domain_name pd18 --fault_set_name sio-fs1
```

rename_fault_set

Rename the specified fault set.

Syntax

```
scli --rename_fault_set (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>)) --new_name <NAME>
```

Parameters

--fault_set_id *<ID>*
Fault set ID

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_name *<NAME>*
Fault set name

--new_name *<NAME>*

New name to assign to the fault set

Example

```
scli --rename_fault_set --protection_domain_name pd18 --fault_set_name sio-fs-test --
new_name sio-fs1
```

set_sdt_fault_set

Set the fault set of an SDT.

Syntax

```
scli --set_sdt_fault_set (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) (--fault_set_id <ID> | ((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --fault_set_name <NAME>))
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_id *<ID>*
Fault set ID

--fault_set_name *<NAME>*
Fault set name

23

General CLI commands

This section contains general CLI commands, including commands relating to licenses, certificates, login banner, and syslog.

acknowledge_scanner_error_counters

Acknowledge read and compare errors so that alerts are no longer be shown for the current counter value. Acknowledgment can be performed at storage pool level or at device level, and for counter type.

Syntax

```
scli --acknowledge_scanner_error_counters (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id
<ID>) | (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) ) [--
acknowledge_compare_errors] [--acknowledge_read_errors]
```

Parameters

--protection_domain_id *<ID>*
: Protection domain ID

--protection_domain_name *<NAME>*
: Protection domain name

--storage_pool_name *<NAME>*
: Storage pool name

--storage_pool_id *<ID>*
: Storage pool ID

--sds_id *<ID>*
: SDS ID

--sds_name *<NAME>*
: SDS name

--sds_ip *<IP>*
: SDS IP address

--sds_port *<PORT>*
: Port assigned to the SDS

--device_id *<ID>*
: Device ID

--device_name *<NAME>*
: Device name

--device_path *<PATH>*
: SDS storage device path or file path

--acknowledge_read_errors
: Acknowledge read error counter

--acknowledge_compare_errors

Acknowledge compare error counter

Example

```
scli --acknowledge_scanner_error_counters --sds_name sds25 --device_name abc123 --
acknowledge_compare_errors
```

add_trusted_ca

Upload the certificate from the specified file as a trusted certificate.

Syntax

```
scli --add_trusted_ca --certificate_file <FILE> --comment <COMMENT>
```

Parameters

--certificate_file *<FILE>*

The certificate file to upload as trusted

--comment *<COMMENT>*

Optional free text to describe the certificate

Example

```
scli --mdm_ip 192.168.1.200 --add_trusted_ca --certificate_file system2_ca.cer --comment
"Certificate for system 2"
```

extract_root_ca

Extract the MDM root certificate to the specified file. This is the first step in the process of exchanging certificates between replication peer systems.

Syntax

```
scli --extract_root_ca --certificate_file <FILE>
```

Parameters

--certificate_file *<FILE>*

The file to write the certificate to

Example

```
scli --mdm_ip 192.168.1.200 --extract_root_ca --certificate_file system1_ca.cer
```

list_trusted_ca

Display a list of the currently trusted certificates in the system.

Syntax

```
scli --list_trusted_ca
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --list_trusted_ca
```

query_license

Retrieve information about the current license and restrictions.

Syntax

```
scli --query_license
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_license
```

query_disconnections

Query various connection statuses.

Syntax

```
scli --query_disconnections [--disconnection_type <TYPE>] [--print_all] [--print_proxy]
```

Parameters

--disconnection_type <TYPE>

Print disconnections of the given type only. The following options are available:

sdc_sds

SDC-SDS connections

sdc_sdr

SDC-SDR connections

sdr_sds

SDR-SDS connections

sdr_sdr

SDR-SDR connections

--print_all

Use this flag to print all disconnections

--print_proxy

When proxy access mode is active, print SDSs that can not serve as a proxy.

Example

```
scli --mdm_ip 192.168.1.200 --query_disconnections --disconnection_type sdr_sdr
```

query_remote_syslog

Retrieve the remote syslog module state and configuration.

Syntax

```
scli --query_remote_syslog
```

Parameters

None.

Example

```
scli --query_remote_syslog
```

query_oscillating_failure_counter_parameters

Retrieve oscillating failure counter parameters.

Syntax

```
scli --query_oscillating_failure_counter_parameters --failure_counter (<COUNTER FROM
GROUP A> (--protection_domain_id <ID> | --protection_domain_name <NAME>) | <COUNTER FROM
```

```
GROUP B> | <COUNTER FROM GROUP C> | --all_counters(((----protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>))
```

Parameters

--failure_counter *<COUNTER>*

The oscillating failure counter to be queried. Counters are grouped as follows:

Counter Group A

MDM and SDS-related counters:

- mdm_sds_network_disconnections
- sds_sds_network_disconnections
- sds_decoupled
- sds_configuration_failures
- sds_receive_buffer_allocation_failures

Counter Group B

SDC-related counters:

- sdc_mdm_network_disconnections
- sdc_sds_network_disconnections
- sdc_long_operations
- sdc_memory_allocation_failures
- sdc_socket_allocation_failures

Counter Group C

SDS device-related counters:

- sds_device_long_successful_ios

--all_counters

Query all oscillating failure counters parameters of the selected group

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

Example

```
scli --query_oscillating_failure_counter_parameters --failure_counter
sds_device_long_successful_ios --protection_domain_name pd10 --storage_pool_name sp18
```

remove_trusted_ca

Remove the certificate with the given fingerprint.

Syntax

```
scli --remove_trusted_ca --fingerprint <FINGERPRINT>
```

Parameters

--fingerprint <FINGERPRINT>

The fingerprint of the certificate to be removed

Example

```
scli --mdm_ip 192.168.1.200 --remove_trusted_ca --fingerprint
B5:F0:43:1E:9B:7B:55:7F:05:98:B7:D9:7E:50:D4:9F:6D:1B:A7:5F
```

reset_oscillating_failure_counters

Reset the specified oscillating failure counters to zero. This command is useful when you have fixed a problem and want to ensure that an alert is no longer active in the system.

Syntax

```
scli --reset_oscillating_failure_counters (--failure_counter (<COUNTER FROM GROUP A> ((--
protection_domain_id <ID> | --protection_domain_name <NAME>) | --all_sds) | <COUNTER
FROM GROUP B> --all_sdc | <COUNTER FROM GROUP C> (((----protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) |
--all_devices) | --all_counters)
```

Parameters

--failure_counter *<COUNTER>*

The oscillating failure counter to be reset to zero. Counters are grouped as follows:

Counter Group A

MDM and SDS-related counters:

- `mdm_sds_network_disconnections`
- `sds_sds_network_disconnections`
- `sds_decoupled`
- `sds_configuration_failures`
- `sds_receive_buffer_allocation_failures`

Counter Group B

SDC-related counters:

- `sdc_mdm_network_disconnections`
- `sdc_sds_network_disconnections`
- `sdc_long_operations`
- `sdc_memory_allocation_failures`
- `sdc_socket_allocation_failures`

Counter Group C

SDS device-related counters:

- `sds_device_long_successful_ios`

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

--all_sds
Reset the counter parameters on all SDSs.

--all_sdc
Reset the counter parameters on all SDCs.

--all_devices
Reset the counter parameters on all SDS devices.

--all_counters
Reset all counters to zero.

Example

```
scli --reset_oscillating_failure_counters --failure_counter
mdm_sds_network_disconnections --all_sds
```

set_cli_login_banner_acceptance_mode

Set the login banner acceptance mode.

Syntax

```
scli --set_cli_login_banner_acceptance_mode (--password_only | --accept_banner)
```

Parameters

--password_only
Allow CLI users to accept the login banner by entering a valid password.

--accept_banner
Require CLI users to accept the login banner by pressing 'y'.

Example

```
scli --set_cli_login_banner_acceptance_mode --password_only
```

set_cli_login_banner_preemptive_acceptance

Enable or prevent CLI users from preemptively accepting the login banner.

Syntax

```
scli --set_cli_login_banner_preemptive_acceptance (--enable | --disable)
```

Parameters

--enable

Allow CLI users to preemptively accept the login banner

--disable

Prevent CLI users from preemptively accepting the login banner

Example

```
scli --set_cli_login_banner_preemptive_acceptance --enable
```

set_login_banner

Set the login banner that is displayed at every login to the system, or remove an existing login banner.

The maximum file size for the banner is 16 KB. Only ASCII text files are supported.

Syntax

```
--set_login_banner (--filename <FILE> | --remove_banner)
```

Parameters

--filename *<FILE>*

File containing the login banner to set

--remove_banner

Remove the login banner.

Example

```
scli --set_login_banner --filename login_banner.txt
```

set_oscillating_failure_counter_parameters

Modify oscillating failure counter parameters. You can use this command to configure the time interval associated with each window type and the number of failures allowed before reporting commences for each window type, per counter.

Syntax

```
scli --set_oscillating_failure_counter_parameters --failure_counter (<COUNTER FROM GROUP
A> ((--protection_domain_id <ID> | --protection_domain_name <NAME>) | --all_sds) |
<COUNTER FROM GROUP B> --all_sdc | <COUNTER FROM GROUP C> (((----protection_domain_id
<ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id
<ID>) | --all_devices) --window_type {SHORT | MEDIUM | LONG} --window_interval
<INTERVAL> --failures_threshold<ID> <NUM OF FAILURES IN INTERVAL>
```

Parameters

--failure_counter *<COUNTER>*

The oscillating failure counter to be reset to zero. Counters are grouped as follows:

Counter Group A

MDM and SDS-related counters:

- `mdm_sds_network_disconnections`
- `sds_sds_network_disconnections`
- `sds_decoupled`
- `sds_configuration_failures`
- `sds_receive_buffer_allocation_failures`

Counter Group B

SDC-related counters:

- `sdc_mdm_network_disconnections`
- `sdc_sds_network_disconnections`
- `sdc_long_operations`
- `sdc_memory_allocation_failures`
- `sdc_socket_allocation_failures`

Counter Group C

SDS device-related counters:

- `sds_device_long_successful_ios`

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--all_sds

Apply the counter parameters to all SDSs.

--all_sdc

Apply the counter parameters to all SDCs.

--all_devices

Apply the counter parameters to all SDS devices.

--window_type {SHORT | MEDIUM | LONG}

Window type to be modified: short, medium, or long

--window_interval *<INTERVAL>*

New window time interval in seconds. The valid interval range is 1-86400. If the interval is set to 0, the window type is disabled.

--failures_threshold*<ID>* *<NUM OF FAILURES IN INTERVAL>*

Maximum number of failures per window type to be ignored before error reporting begins

Example

```
scli --set_oscillating_failure_counter_parameters --failure_counter
mdm_sds_network_disconnections --window_interval 120 --failures_threshold 10 --
window_type short --all_sds
```

set_remote_read_only_limit_state

Configure the PowerFlex system to restrict access from remote clients to read-only operations on the MDM. When this restriction is enabled, remote client users cannot change any configurations on the MDM. However, users on a local host may still make configuration changes while the restriction is enforced.

NOTE: A local user is one that successfully communicates with the MDM using the IP address 127.0.0.1 (the default IP address used by SCLI). If you try to issue commands from the local machine using other local IP addresses, you will not be able configure the system.

Syntax

```
scli --set_remote_read_only_limit_state --remote_read_only_limit_state {enabled |
disabled}
```

Parameters

--remote_read_only_limit_state {enabled | disabled}

Enable or disable restricted state. Default: disabled.

Example

```
scli --set_remote_read_only_limit_state --remote_read_only_limit_state enabled
```

set_syslog_facility

Set the facility field of the syslog events. Legal values are 0 to 23.

Syntax

```
scli --set_syslog_facility --remote_syslog_server_ip <IP> --syslog_facility <FACILITY>
```

Parameters

--remote_syslog_server_ip *<IP>*

Comma-separated list of IP addresses or hostnames of syslog servers. Omit the space after each comma.

--syslog_facility *<FACILITY>*

Control the facility field of the event. Default is 16.

Example

```
scli --set_syslog_facility --remote_syslog_server_ip 192.168.1.201 --syslog_facility 16
```

start_remote_syslog

Start posting events to a remote syslog server.

Syntax

```
scli --start_remote_syslog --remote_syslog_server_ip <IP>
  [--remote_syslog_server_port <PORT>]
  [--syslog_facility <FACILITY>]
  [--attach_event_code]
```

Parameters

--remote_syslog_server_ip *<IP>*

Comma-separated list of IP addresses or hostnames of syslog servers. Omit the space after each comma.

--remote_syslog_server_port *<PORT>*

Syslog server port. Default is 1468.

--syslog_facility *<FACILITY>*

Control the facility field of the event. Default is 16.

--attach_event_code

Add the posted event code to the event message. This parameter is disabled by default.

Example

```
scli --start_remote_syslog --remote_syslog_server_ip 192.168.1.201 --syslog_facility 16
```

stop_remote_syslog

Stop posting events to a remote syslog server.

Syntax

```
scli --stop_remote_syslog (--remote_syslog_server_ip <IP> | --all_remote_syslog_servers)
```

Parameters

--remote_syslog_server_ip *<IP>*

Comma-separated list of IP addresses or hostnames of syslog servers. Omit the space after each comma.

--all_remote_syslog_servers

Stop posting events to all remote syslog servers.

Example

```
scli --stop_remote_syslog --remote_syslog_server_ip 192.168.1.201
```

test_remote_system_connectivity

Perform a connectivity test between the specified SDTs of the protection domain and remote system.

Syntax

```
scli --test_remote_system_connectivity (--remote_system_id <ID> | --remote_system_name
<NAME>) (--protection_domain_id <ID> | --protection_domain_name <NAME>) [--
destination_volume_nsid <NSID>] [--full_output]
```

Parameters

--remote_system_id *<ID>*

The ID of the remote system

--remote_system_name *<NAME>*

The name of the remote system

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--destination_volume_nsid *<NSID>*

The destination volume NSID. If not specified, only an indication of the success of discovery information retrieval is returned.

--full_output

Print a list of IO connections to the remote system, If not specified, only the number of healthy connections is returned.

add_network_set

Add a network set to the PowerFlex system.

Syntax

```
scli --add_network_set [--network_set_name <NAME>] [--network_id <ID> | --network_name
<NAME>]
```

Parameters

--network_set_name *<NAME>*

A name to be assigned to the network set

--network_id *<ID>*

A comma separated list of system network IDs to be associated with the network set

--network_name *<NAME>*

A comma separated list of system network names to be associated with the network set

modify_network_set

Modify the list of system networks of an existing network set.

Syntax

```
scli --modify_network_set (--network_set_id <ID> | --network_set_name <NAME>) [--
network_id <ID> | --network_name <NAME>]
```

Parameters

--network_set_id *<ID>*
: Network set ID

--network_set_name *<NAME>*
: Network set name

--network_id *<ID>*
: A comma separated list of system network IDs to be replaced with the old one

--network_name *<NAME>*
: A comma separated list of system network names to be replaced with the old one

query_network_set

Retrieve information about a network set.

Syntax

```
scli --query_network_set (--network_set_id <ID> | --network_set_name <NAME>)
```

Parameters

--network_set_id *<ID>*
: Network set ID

--network_set_name *<NAME>*
: Network set name

24

Protection domain commands

This section contains commands for managing protection domains.

activate_protection_domain

Activate a protection domain.

Protection domains are activated by default. Use this command to activate a protection domain that was inactivated.

Syntax

```
scli --activate_protection_domain (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
  [--force_activate]
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--force_activate

Activates the protection domain, ignoring current system errors

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --activate_protection_domain --protection_domain_name pd1
```

add_protection_domain

Add a protection domain.

Syntax

```
scli --add_protection_domain
  [--protection_domain_name <NAME>]
```

Parameters

--protection_domain_name *<NAME>*

Unique name to assign the protection domain

Example

```
Example
scli --mdm_ip 192.168.1.200 --add_protection_domain
--protection_domain_name rack_1.1
```

Protection domain names

Assign each protection domain a meaningful name associated with its operational role. When a name has not been defined, the system may display default system-defined names that use the ID of the volume.

Each protection domain name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

disable_fgl_metadata_cache

Disable fine granularity metadata caching for the specified protection domain.

Syntax

```
scli --disable_fgl_metadata_cache (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --mdm_ip 192.168.1.200 --disable_fgl_metadata_cache --protection_domain_name PD1
```

enable_fgl_metadata_cache

Enable fine granularity metadata caching for the specified protection domain. Fine granularity metadata caching helps improve the performance of the system when using the fine granularity layout.

Syntax

```
scli --enable_fgl_metadata_cache (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --mdm_ip 192.168.1.200 --enable_fgl_metadata_cache --protection_domain_name PD1
```

inactivate_protection_domain

Inactivate the specified protection domain. Using this command is an effective way to shut down nodes and is preferable to shutting them down manually.

NOTE: When you inactivate a protection domain, the data remains on the SDSs. It is preferable to remove a protection domain if you no longer need it.

When this feature is in effect, the system may perform the following activities in the background:

- Determine whether there are any current rebuild/rebalance activities taking place. If so, the shutdown is delayed (unless it is forced) until they are finished.
- Block future rebuild/rebalance activities.
- Pause application I/O and disable access to volumes.
- Move the DRL mode of all SDSs to harden in preparation for rebooting the server.
- Reload all SDSs before re-enabling data access.

Syntax

```
scli --inactivate_protection_domain (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
  [--force_inactivate]
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--force_inactivate

Inactivate the protection domain, ignoring current system errors.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --inactivate_protection_domain --protection_domain_name pd1
```

query_port_flapping_status

Query the system port flapping status.

Syntax

```
scli --query_port_flapping_status [--weak] (--all | --protection_domain_id <ID> | --
protection_domain_name <NAME> | --sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>] | --sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --
sdc_ip<IP> | --sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>] |--
apply_to_mdm | --peer_system_id <ID> | --peer_system_name <NAME> | --peer_system_ip <IP>
[--peer_system_port <PORT>] | --peer_system_object_id <ID> | --sdt_id <ID> | --sdt_name
<NAME> | --sdt_ip <IP> [--sdt_port <PORT>])
```

Parameters

--weak
Include reported paths with weak ERT violations.

--all
Query all protection domains in the system.

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--sds_id *<ID>*
SDS ID

--sds_ip *<IP>*
SDS IP address

--sds_name *<NAME>*
SDS name

--sds_port *<PORT>*
SDS port

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC global unique identifier

--sdc_ip *<IP>*
SDC IP address

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
SDR IP address

--sdr_port *<PORT>*
Port assigned to the SDR

--apply_to_mdm
Display MDM only

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_ip *<IP>*
Replication peer system IP

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

query_port_flapping_unsupported_components

Query components that do not support the port flapping feature, and/or have it disabled.

Syntax

```
scli --query_port_flapping_unsupported_components [(--full | --protection_domain_id <ID>
| --protection_domain_name <NAME>) [--raw]] [(--only_disabled | --only_unsupported)]
```

Parameters

--full
Display a full unsummarized report

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--raw
Outputs results in raw format with one component displayed per line

--only_disabled
Query only the the components that have the port flapping feature disabled

--only_unsupported

Query only the the components that do not support the port flapping feature. Omit components that have the feature disabled.

Example

```
scli --mdm_ip 192.168.1.200 --query_port_flapping_unsupported_components --full
```

query_protection_domain

Retrieve aggregated information about all objects in a single protection domain.

Syntax

```
scli --query_protection_domain (--protection_domain_id <ID> | --protection_domain_name
<NAME>
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --mdm_ip 192.168.1.200 --query_protection_domain --protection_domain_name pd2
```

Sample output

```
Protection Domain pd2 (Id: fcc9af5600000001) has 1 storage pools, 1 acceleration pools,
0 Fault Sets, 1 SDS nodes and 0 volumes
        Free net capacity: 22.3 GB (22875 MB), Volume allocation limit: 0 Bytes
        Operational state is Active
        I/O flow control is enabled
        Fine granularity default number of concurrent writes: 1000
        FGL metadata cache is disabled
        FGL default metadata cache size: 0 MB
        Replication capacity utilization is in Normal level

Storage Pool sp2 (Id: 3df1cd9700000001) has 0 volumes and 22.3 GB (22875 MB) free net
capacity. 0 Bytes volume allocation limit.
        The number of parallel rebuild/rebalance jobs: 2
        Rebuild is enabled and using Limit-Concurrent-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1
        Rebalance is enabled and using Favor-Application-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        VTree Migration is enabled and using Favor-Application-IO policy with the
following parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        Protected Maintenance Mode is enabled and using Limit-Concurrent-IO policy with
```

```
the following parameters:
          Number of concurrent IOs per device: 1
        Fragmentation: Enabled
        Background device scanner: Enabled, Read Error Action: report and fix, Compare
Error Action: report and fix, Bandwidth Limit 3072 KBps per device
        Zero padding: Enabled
        Spare policy: 70% out of total
        Inflight Checksum: Disabled
        Doesn't use RAM Read Cache
        Capacity alert thresholds: High: 80, Critical: 90
        Media Type: SSD, External Acceleration Type: NONE, Inflight Requests Factor:
115, Inflight Bandwidth Factor: 115
        Data layout: Medium granularity
        Replication Capacity Max Allowed Ratio: 10%
        Replication Capacity Max Allowed: 3.0 GB (3072 MB)
        Replication Allocated Capacity: 0 Bytes
        Replication Used Capacity: 0 Bytes
        Number of devices with inaccessible capacity: 0
        Medium granularity persistent checksum:
                Persistent checksum is enabled
                State is PROTECTED
                Checksum builder limit is: 3072 KBps
                Validation on read is disabled

SDR-SDS connectivity status: All connected

SDS Summary:
        Total 1 SDS Nodes
        1 SDS nodes have membership state 'Joined'
        1 SDS nodes have connection state 'Connected'

Physical data:
        148.9 GB (152500 MB) total capacity
                0 Bytes in-use capacity
                        0 Bytes user data
                        0 Bytes metadata
                        0 Bytes fine granularity spares
                44.7 GB (45750 MB) unused capacity
                104.2 GB (106750 MB) spare capacity
                0 Bytes decreased capacity
        Capacity Health:
                0 Bytes protected capacity
                0 Bytes failed capacity
                0 Bytes degraded-failed capacity
                0 Bytes degraded-healthy capacity
                0 Bytes unreachable-unused capacity
                0 Bytes semi-protected capacity
                0 Bytes in-maintenance capacity
                0 Bytes temporary capacity

User data:
        0 Bytes total user data
                0 Bytes medium granularity user data
                0 Bytes fine granularity user data
                        0 Bytes compressed data (Compression ratio: Not applicable)
                        0 Bytes uncompressed data
        0 Bytes trimmed capacity
        0 Bytes user data without trimmed capacity
        Total compression ratio: Not applicable
        Thin and snapshot saving: Not applicable
        Overall saving: Not applicable
        Data Movement:
                0 Bytes rebalance capacity
                0 Bytes forward-rebuild capacity
                0 Bytes backward-rebuild capacity
                0 Bytes enter protected maintenance mode capacity
                0 Bytes exit protected maintenance mode capacity
                0 Bytes at-rest capacity
                0 Bytes total moving capacity
                        0 Bytes active moving capacity
                                0 Bytes active rebalance capacity
                                0 Bytes active forward-rebuild capacity
                                0 Bytes active backward-rebuild capacity
```

```
                                0 Bytes active enter protected maintenance mode capacity
                                0 Bytes active exit protected maintenance mode capacity
                        0 Bytes pending moving capacity
                                0 Bytes pending rebalance capacity
                                0 Bytes pending forward-rebuild capacity
                                0 Bytes pending backward-rebuild capacity
                                0 Bytes pending enter protected maintenance mode capacity
                                0 Bytes pending exit protected maintenance mode capacity

Bandwidth:
        Primary-reads                               0 IOPS 0 Bytes per second
        Primary-writes                              0 IOPS 0 Bytes per second
        Secondary-reads                             0 IOPS 0 Bytes per second
        Secondary-writes                            0 IOPS 0 Bytes per second
        Backward-rebuild-reads                      0 IOPS 0 Bytes per second
        Backward-rebuild-writes                     0 IOPS 0 Bytes per second
        Forward-rebuild-reads                       0 IOPS 0 Bytes per second
        Forward-rebuild-writes                      0 IOPS 0 Bytes per second
        Rebalance-reads                             0 IOPS 0 Bytes per second
        Rebalance-writes                            0 IOPS 0 Bytes per second
        Volume-migration-reads                      0 IOPS 0 Bytes per second
        Volume-migration-writes                     0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-reads      0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-writes     0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-reads       0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-writes      0 IOPS 0 Bytes per second

Volumes summary:
        0 Bytes all volumes size
        0 Bytes base volumes size
        0 Volumes.
```

remove_protection_domain

Remove a protection domain from the PowerFlex system. You can only remove a protection domain if no SDS and no storage pool are associated with it.

Syntax

```
scli --remove_protection_domain (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

Example

```
scli --mdm_ip 192.168.1.200 --remove_protection_domain
--protection_domain_name my_protection_domain
```

rename_protection_domain

Assign a name to, or rename, a protection domain.

Syntax

```
scli --rename_protection_domain (--protection_domain_id <ID> | --protection_domain_name
<NAME>) --new_name <NAME>
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--new_name *<NAME>*

Example

```
scli --mdm_ip 192.168.1.200 --rename_protection_domain --protection_domain_name pd1 --
new_name protection_domain_1.
```

Protection domain names

Assign the protection domain a meaningful name associated with its operational role. The protection domain name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

set_default_fgl_metadata_cache_size

Configure the default fine granularity metadata cache size for a protection domain. New SDSs in the protection domain inherit this value when created.

Syntax

```
scli --set_default_fgl_metadata_cache_size (--protection_domain_id <ID> | --
protection_domain_name <NAME>) --metadata_cache_size_mb <SIZE>
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--metadata_cache_size_mb *<SIZE>*

Define the default size of metadata cache for the protection domain in MB

Example

```
scli --mdm_ip 192.168.1.200 --set_default_fgl_metadata_cache_size --
protection_domain_name PD1 --metadata_cache_size_mb 10248
```

25

Remote system commands

This section contains commands for managing remote systems.

add_remote_system

Add a remote system that can be used for data mobility.

Syntax

```
scli --add_remote_system --discovery_ip <IPs> [--remote_system_name <NAME>] [--
remote_system_type <SYSTEM_TYPE>] [--discovery_port <PORT>] [--remote_system_description
<DESCRIPTION>] [--transport <TRANSPORT_TYPE>] [--bw_limit_mbps <Mbps>]
```

Parameters

--discovery_ip *<IP>*
A list of IP addresses seperated by commas

--remote_system_name *<NAME>*
The remote system name

--remote_system_type *<SYSTEM_TYPE>*
The remote system type. Default is NVMe

--discovery_port *<PORT>*
NVMe discovery port. Default port is 8009

--remote_system_description *<DESCRIPTION>*
Description of the remote system

--transport *<TRANSPORT_TYPE>*
Currently, only TCP is supported

--bw_limit_mbps *<Mbps>*
The remote system bandwidth limit

query_remote_system

Retrieve information about the remote system.

Syntax

```
scli --query_remote_system (--remote_system_id <ID> | --remote_system_name <NAME>)
```

Parameters

--remote_system_id *<ID>*
The ID of the remote system

--remote_system_name *<NAME>*

The name of the remote system

query_all_remote_systems

Retrieve information about all the snapshot copy tasks.

rename_remote_system

Modify the remote system name.

Syntax

```
scli --rename_remote_system (--remote_system_id <ID> | --remote_system_name <NAME>) --
new_remote_system_name <NAME>
```

Parameters

--remote_system_id *<ID>*

The ID of the remote target system

--remote_system_name *<NAME>*

The name of the remote target system

--new_remote_system_name *<NAME>*

The new name for the remote system

modify_remote_system_bw_limit

Modify a remote system bandwidth limit.

Syntax

```
scli --modify_remote_system_bw_limit (--remote_system_id <ID> | --remote_system_name
<NAME>) --new_bw_limit_mbps <Mbps>
```

Parameters

--remote_system_id *<ID>*

The ID of the remote target system

The ID of the remote target system *<NAME>*

The name of the remote target system

--new_bw_limit_mbps *<Mbps>*

New bandwidth limit of the system

modify_remote_system_connectivity

Modify the IP list or port for the remote system.

Syntax

```
scli --modify_remote_system_connectivity (--remote_system_id <ID> | --remote_system_name
<NAME>) --discovery_ip <IP> --discovery_port <PORT>
```

Parameters

--remote_system_id *<ID>*

The ID of the remote target system

--remote_system_name *<NAME>*

The name of the remote target system.

Select one of the following options:

--discovery_ip *<IPs>*

A list of IP addresses seperated by commas

--discovery_port *<PORT>*

The new port of the system

modify_remote_system_description

Modify a remote system description.

Syntax

```
scli --modify_remote_system_description (--remote_system_id <ID> | --remote_system_name
<NAME>) --remote_system_description <DESCRIPTION>
```

Parameters

--remote_system_id *<ID>*

The ID of the remote target system

--remote_system_name *<NAME>*

The name of the remote target system

--remote_system_description *<DESCRIPTION>*

The description of the new system

remove_remote_system

Remove the remote system.

Syntax

```
scli --remove_remote_system (--remote_system_id <ID> | --remote_system_name <NAME>)
```

Parameters

--remote_system_id *<ID>*
The ID of the remote system

--remote_system_name *<NAME>*
The name of the remote system

26

Replication commands

This section contains commands for managing replication.

abort_replication_consistency_group_snapshots_create

Cancel a previous request to create a snapshot of a replication consistency group (RCG).

NOTE: This command cannot be performed when the specified RCG is inactive.

Syntax

```
scli --abort_replication_consistency_group_snapshots_create (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --abort_replication_consistency_group_snapshots_create --
replication_consistency_group_name rcg27
```

activate_replication_consistency_group

Activate the specified replication consistency group (RCG).

Notes

Replication is enabled while activating the RCG. The initial copy process is started when there are sufficient system resources.

Syntax

```
scli --activate_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --activate_replication_consistency_group --
replication_consistency_group_name rcg27
```

activate_replication_consistency_group_mappings

Activate replication consistency group mappings on the peer system that is addressed by the command.

Notes

In general, the disaster recovery processes of failover, switchover, recover, and restore perform all the required operations, including activating mappings. This command is useful in situations when you choose to perform each of the individual operations directly. You can only activate mappings when the replication consistency group (RCG) role is neutral or source. You cannot activate mappings if the RCG role is target.

Syntax

```
scli --activate_replication_consistency_group_mappings (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --activate_replication_consistency_group_mappings --
replication_consistency_group_name rcg27
```

add_replication_consistency_group

Add a replication consistency group (RCG).

Notes

You may only add an RCG after the replication peer system has been added using `add_replication_peer_system`.

Syntax

```
scli --add_replication_consistency_group (--destination_system_id <ID> |
--destination_system_name <NAME> | --destination_system_ip <IP> [--
destination_system_port <PORT>] | --destination_system_object_id <ID>) --rpo
<PERIOD> [--replication_consistency_group_name <NAME>] (--protection_domain_id
<ID> | --protection_domain_name <NAME>) (--remote_protection_domain_id
<ID> | --remote_protection_domain_name <NAME>) [--target_volume_access_mode
<TARGET_VOLUME_ACCESS_MODE>] [--replication_consistency_group_activity_mode
<ACTIVITY_MODE>]
```

Parameters

--destination_system_id *<ID>*

Destination replication system ID

--destination_system_name *<NAME>*

Destination replication system Name

--destination_system_ip *<IP>*

Destination replication system IP address

--destination_system_port *<PORT>*

Port assigned to the destination replication system

--destination_system_object_id *<ID>*

Destination replication system internal object ID

--rpo *<PERIOD>*

Desired RPO in units of time. To specify the time period, use s|m|h respectively indicating seconds, minutes, or hours. Examples: 30s (30 seconds), 15m (15 minutes), or 1h (one hour).

--replication_consistency_group_name *<NAME>*

A unique name to be assigned to the replication consistency group

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--remote_protection_domain_id *<ID>*

Remote protection domain ID

--remote_protection_domain_name *<NAME>*

Remote protection domain name

--target_volume_access_mode *<TARGET_VOLUME_ACCESS_MODE>*

Target volume access mode. Can be one of the following: `no_access` (default) or `read_only`

--replication_consistency_group_activity_mode *<ACTIVITY_MODE>*

Replication consistency group activity mode. Can be one of the following:

active

(Default). When the RCG is active, replication is enabled and initial copy begins as soon as there are sufficient system resources.

inactive

The RCG is created in inactive state and data is not copied to the target. The RCG can be activated later using the `activate_replication_consistency_group` command.

Example

```
scli --mdm_ip 192.168.1.200 --add_replication_consistency_group --destination_system_id
459865f0774b81e6 --rpo 15m --protection_domain_id 1caf743100000000 --
remote_protection_domain_id dc65bd9900000000 --target_volume_access_mode read_only
```

add_replication_pair

Add a replication pair consisting of a volume from the source system and a volume from the target system.

A volume can be a member of at most one replication pair. When the replication pair is created, both the source volume and the target volume must be the same size. It is recommended, but not mandatory, that the volumes in the volume pair have the same attributes (including zero padding and granularity). Not doing so can impact performance and capacity.

Syntax

Syntax for adding pair without destination volume auto-provisioning (destination volume must be pre-created):

```
scli --add_replication_pair (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>) (--source_volume_id <ID> | --
source_volume_name <NAME>) (--destination_volume_id <ID> | --destination_volume_name
<NAME>) [--replication_pair_name <NAME>]
```

Syntax for adding pair with destination volume auto-provisioning:

```
scli --add_replication_pair  [--replication_pair_name <NAME>]
(--replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
(--source_volume_id <ID> | --source_volume_name <NAME>) (--destination_storage_pool_name
<NAME> | --destination_storage_pool_id <ID>) [--destination_sdc_id <ID> | --
destination_sdc_name <NAME> | --destination_sdc_guid <GUID> | --destination_sdc_ip <IP>]
 [--compression_method <METHOD>] [Provisioning options] [Use RAM Read Cache options]
[--access_mode <ACCESS_MODE>] [--allow_multi_map]
```

Parameters

--replication_pair_name *<NAME>*

Replication pair name

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--source_volume_id *<ID>*

Source volume ID

--source_volume_name *<NAME>*

Source volume name

--destination_sdc_ip *<IP>*

Destination SDC IP

Provisioning options:

If auto provision option is selected, provide the following:

Compression (optional):

--compression_method *<METHOD>*

Indicates the compression method to be used, relevant only for fine granularity data layouts.

`none`— No compression applied

`normal`—Normal compression applied

Destination storage pool option: Select any one of the following:

--destination_storage_pool_name *<NAME>*

Destination storage pool name

--destination_storage_pool_id *<ID>*

Destination storage pool ID

Provisioning options: Select any one of the following:

--thin_provisioned

This volume will be thin-provisioned

--thick_provisioned

This volume will be thick-provisioned

<BLANK>

Use storage pool default (thick for medium granularity, thin for fine granularity)

Use RAM read cache: Select any one of the following:

--use_rmcache

The specified volume uses caching

--dont_use_rmcache

The specified volume does not use caching

<BLANK>

Use storage pool default value

Indicates what dst (destination) SDC exposes the dst volume in case of disaster recovery:

--destination_sdc_id *<ID>*

SDC ID

--destination_sdc_name *<NAME>*

SDC name

--destination_sdc_guid *<GUID>*

SDC global unique identifier

--destination_sdc_ip *<IP>*

SDC IP address

Destination volume mapping to SDC (optional):

--access_mode <ACCESS_MODE>

Mapping access mode when not in disaster recovery. Can be one of the following: no_access (default), read_only

--allow_multi_map

Allow this volume to be mapped to more than one SDC

add_replication_peer_system

The peer system must be added to each of the systems involved in replication. There must be network connectivity between the two systems.

Syntax

```
scli --add_replication_peer_system --peer_system_ip <IP> [--peer_system_port <PORT>] --
peer_system_id <ID> [--peer_system_name <NAME>]
```

Parameters

--peer_system_ip *<IP>*

A comma-separated list of IP addresses assigned to the peer system. These are necessary for the replication to survive failures.

--peer_system_port *<PORT>*

Port assigned to the peer system. If this is omitted, the port used is the default peer listening port.

--peer_system_id *<ID>*

Peer system ID

--peer_system_name *<NAME>*

A name to be assigned to the peer system. The name should be unique.

Example

```
scli --mdm_ip 192.168.1.200 --add_replication_peer_system --peer_system_ip 10.55.105.21
--peer_system_id 7d897dd0269f5b0f --peer_system_name Durham1
```

add_sdr

Add an SDR to the specified protection domain. The SDR is one of the main components that must be installed on each of the peer protection domains for replication. It transfers data from the source volume to the target volume.

Syntax

```
scli --add_sdr --sdr_ip <IP> [--sdr_ip_role <ROLE>] [--sdr_name <NAME>] (--
protection_domain_id <ID> | --protection_domain_name <NAME>) [--sdr_port <PORT>] [--
profile <PROFILE>] [--force_clean [--i_am_sure]]
```

Parameters

--sdr_ip *<IP>*

A comma-separated list of IP addresses assigned to the SDR

--sdr_ip_role *<ROLE>*

The role assigned to each SDR IP address where *<ROLE>* can be one of the following:

application_only

Application only, for communicating with SDC

storage_only

Storage only, for communicating with SDS

external_only

External only, for communicating with remote SDR

application_and_storage

Application and storage, for communicating with SDC and SDS

application_and_external

Application and external, for communicating with SDC and remote SDR

storage_and_external

Storage and external, for communicating with SDS and remote SDR

all

Application, storage, and external, for communicating with SDC, SDS, and remote SDR

--sdr_name *<NAME>*

A name to be assigned to the SDR

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--sdr_port *<PORT>*

Port assigned to the SDR

--profile {compact | high_performance}

Set the performance profile as compact or high performance. The high performance profile configures a predefined set of parameters for very high performance use cases.

--force_clean

Clean a previous SDR configuration. Use this if the SDR was previously part of a PowerFlex system.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --add_sdr --sdr_ip 10.55.105.41,10.76.30.14 --sdr_name SDR2
--protection_domain_id 2c3071a200000001
```

SDR names

Assign each SDR a meaningful name. When a name has not been defined, the system may display default system-defined names that use IDs of the SDRs.

Each name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object’s property sheet in the PowerFlex Manager.

add_sdr_ip

Add an IP address to the specified SDR.

Syntax

```
scli --add_sdr_ip (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --new_sdr_ip <IP> [--sdr_ip_role <ROLE>]
```

Parameters

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

IP address assigned to the SDR

--sdr_port *<PORT>*

Port assigned to the SDR

--new_sdr_ip *<IP>*

New IP address to be assigned to the SDR

--sdr_ip_role *<ROLE>*

The role assigned to each SDR IP address where *<ROLE>* can be one of the following:

application_only

Application only, for communicating with SDC

storage_only

Storage only, for communicating with SDS

external_only

External only, for communicating with remote SDR

application_and_storage

Application and storage, for communicating with SDC and SDS

application_and_external

Application and external, for communicating with SDC and remote SDR

storage_and_external

Storage and external, for communicating with SDS and remote SDR

all

Application, storage, and external, for communicating with SDC, SDS, and remote SDR

Example

```
scli --mdm_ip 192.168.1.200 --add_sdr_ip --sdr_name SDR27 --new_sdr_ip 10.76.30.14 --
sdr_ip_role application_only
```

clear_broken_replication_consistency_group_alerts

Clear broken replication consistency group (RCG) indications for all volumes associated with the specified protection domain. The clearing process is useful in situations when the SDRs run out of journal capacity and RCG autotermination cannot be applied (such as when the remote MDM does not support it). At that point, an alert is produced for every replicated volume in the affected protection domain. These alerts can be cleared using this command.

Syntax

```
scli --clear_broken_replication_consistency_group_alerts (--protection_domain_id <ID> |
--protection_domain_name <NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --mdm_ip 192.168.1.200 --clear_broken_replication_consistency_group_alerts --
protection_domain_name pd1
```

create_replication_consistency_group_snapshots

Create replication consistency group snapshots on the target.

Notes

The resulting snapshots contain the latest consistent image. If a consistent image cannot be created, the command fails. The `--force` option can be used to create a snapshot even if a consistent image is not available. Use this option carefully as the resulting snapshot may be crash inconsistent and may not be usable by applications.

Syntax

```
scli --create_replication_consistency_group_snapshots (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>) [--
force_during_partial_consistency]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--force_during_partial_consistency

Run the command even if some volumes would be excluded because of the uncompleted initial copy.

enter_sdr_maintenance_mode

Enter the SDR into maintenance mode to allow for maintenance of the SDR. Note that there must be at least two other active SDRs joined to the system for replication to continue. When maintenance mode is activated, the MDM changes the mapping and distributes the I/Os to the remaining SDRs.

Syntax

```
scli --enter_sdr_maintenance_mode (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--
sdr_port <PORT>])
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
IP address assigned to the SDR

--sdr_port *<PORT>*
Port assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --enter_sdr_maintenance_mode --sdr_name SDR27
```

exit_sdr_maintenance_mode

Move the SDR out of maintenance mode.

Notes

An SDR exiting maintenance mode requires free journal capacity in order to become active in the replication.

Syntax

```
scli --exit_sdr_maintenance_mode (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--
sdr_port <PORT>])
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
IP address assigned to the SDR

--sdr_port *<PORT>*
Port assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --exit_sdr_maintenance_mode --sdr_name SDR27
```

failover_replication_consistency_group

Failover the specified replication consistency group. During failover, the I/O applications are stopped at the source and the access mode of the source volumes is changed to inaccessible. The target volumes are moved to the newest consistent image available on the destination and their access mode is changed to read-write. Recovery from a failover state can be achieved by either restoring the replication or reversing it.

Syntax

```
scli --failover_replication_consistency_group (--replication_consistency_group_id <ID>
| --replication_consistency_group_name <NAME>) [--force_during_partial_consistency] [--
i_am_sure]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID.

--replication_consistency_group_name *<NAME>*

Replication consistency group name.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

--force_during_partial_consistency

Run the command even if some volumes would be excluded because of the uncompleted initial copy.

failover_test_start_replication_consistency_group

Start a test failover of the specified replication consistency group (RCG). This test allows you to demonstrate failover mode without interrupting data synchronization.

Syntax

```
scli --failover_test_start_replication_consistency_group (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>) [--
force_during_partial_consistency]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--force_during_partial_consistency

Perform the command despite some volumes would be excluded due to uncompleted initial copy.

failover_test_stop_replication_consistency_group

Stop a failover test for the specified replication consistency group (RCG).

Syntax

```
scli --failover_test_stop_replication_consistency_group (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID.

--replication_consistency_group_name *<NAME>*

Replication consistency group name.

Example

```
scli --mdm_ip 192.168.1.200 --failover_test_stop_replication_consistency_group --
replication_consistency_group_name rcg27
```

freeze_apply_replication_consistency_group

Apply the latest consistent image available at the destination to the target volumes and freeze the replication consistency group (RCG). While the RCG is frozen, replication remains active and I/O operations are replicated to the target journal. The I/Os are stored in the target journal and are not applied to the target volumes. Retaining in a frozen state for a long time runs the risk of running out of space in the target journal.

Syntax

```
scli --freeze_apply_replication_consistency_group (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>)[--force_during_partial_consistency]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--force_during_partial_consistency

Run the command even if some volumes would be excluded because of the uncompleted initial copy.

inactivate_replication_consistency_group_mappings

Inactivate replication consistency group mappings on the peer to which the command is applied.

Notes

In general, the disaster recovery processes of failover, switchover, recover, and restore perform all the required operations, including inactivating the replication consistency group mappings. This command is useful in situations when you choose to perform each of the individual operations directly.

Syntax

```
scli --inactivate_replication_consistency_group_mappings (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --inactivate_replication_consistency_group_mappings --
replication_consistency_group_name rcg27w
```

modify_replication_consistency_group_rpo

Modify the recovery point objective (RPO) of the replication consistency group.

The RPO is the maximum data loss (in units of time) that you are willing to lose. When you set the RPO, you are setting the goal of replication.

Syntax

```
scli --modify_replication_consistency_group_rpo (--replication_consistency_group_id <ID>
| --replication_consistency_group_name <NAME>) --rpo <PERIOD>
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--rpo *<PERIOD>*

Desired RPO in units of time, between 30 seconds and 1 hour. To specify the time period, use s|m|h respectively indicating seconds, minutes, or hours. Examples: 30s (30 seconds), 15m (15 minutes), or 1h (one hour).

Example

```
scli --mdm_ip 192.168.1.200 --modify_replication_consistency_group_rpo --
replication_consistency_group_name rcg27 --rpo 20m
```

modify_replication_consistency_group_target_volume_access_mode

Modify the target volume access mode of a replication consistency group.

Syntax

```
scli --modify_replication_consistency_group_target_volume_access_mode (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>) --
target_volume_access_mode <TARGET_VOLUME_ACCESS_MODE>
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--target_volume_access_mode *<TARGET_VOLUME_ACCESS_MODE>*

Target volume access mode. Can be one of the following: `no_access` (default), `read_only`

Example

```
scli --mdm_ip 192.168.1.200
--modify_replication_consistency_group_target_volume_access_mode --
replication_consistency_group_name rcg27 --target_volume_access_mode read_only
```

modify_replication_peer_system_ip

Modify an IP address of a replication peer system.

Syntax

```
scli --modify_replication_peer_system_ip (--peer_system_id <ID> | --peer_system_name
<NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id
<ID>) --new_peer_system_ip <IP>
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
Replication peer system IP address

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

--new_peer_system_ip *<IP>*
A comma-separated list of new IP addresses assigned to the peer system

Example

```
scli --mdm_ip 192.168.1.200 --modify_replication_peer_system_ip --peer_system_name RPS1
--new_peer_system_ip 10.55.105.21
```

modify_replication_peer_system_port

Modify the port of the specified replication peer system.

Notes

Before running this SCLI command to modify port assigned to the replication peer system, you must first edit the `/opt/emc/scaleio/mdm/cfg/conf.txt` configuration file and set the `mdm_external_port` field to the new port. After you've saved the configuration file, restart the MDM service. Then run this command to modify the port assigned to the replication peer system.

Syntax

```
scli --modify_replication_peer_system_port (--peer_system_id <ID> | --peer_system_name
<NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id
<ID>) --new_peer_system_port <PORT>
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
IP address of the replication peer system

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*

Replication peer system internal object ID

--new_peer_system_port *<PORT>*

New port assigned to the peer system. When this parameter is omitted, the default listening port is used.

Example

```
scli --mdm_ip 192.168.1.200 --modify_replication_peer_system_port --peer_system_name RP1
--new_peer_system_port 7611
```

modify_sdr_ip_role

Modify the role of the specified SDR's IP address.

Syntax

```
scli --modify_sdr_ip_role (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP>  [--
sdr_port <PORT>]) [--sdr_ip_to_modify <IP>] --new_sdr_ip_role <ROLE>
```

Parameters

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

IP address assigned to the SDR

--sdr_port *<PORT>*

Port assigned to the SDR

--sdr_ip_to_modify *<IP>*

(Optional) The IP address of the SDR that you wish to modify

--new_sdr_ip_role *<ROLE>*

The new role assigned to the modified SDR IP address where *<ROLE>* can be one of the following:

application_only

Application only, for communicating with SDC

storage_only

Storage only, for communicating with SDS

external_only

External only, for communicating with remote SDR

application_and_storage

Application and storage, for communicating with SDC and SDS

application_and_external

Application and external, for communicating with SDC and remote SDR

storage_and_external

Storage and external, for communicating with SDS and remote SDR

all

Application, storage, and external, for communicating with SDC, SDS, and remote SDR

Example

```
scli --mdm_ip 192.168.1.200 --modify_sdr_ip_role --sdr_name SDR27  --sdr_ip_to_modify
10.76.30.14 --new_sdr_ip_role application_and_storage
```

modify_sdr_port

Modify the port assigned to the specified SDR.

Syntax

```
scli --modify_sdr_port (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --new_sdr_port <PORT>
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
IP address of the SDR

--sdr_port *<PORT>*
Port assigned to the SDR

--new_sdr_port *<PORT>*
New port to be assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --modify_sdr_port --sdr_id ffa0be3900000000 --new_sdr_port
11088
```

pause_pair_initial_copy

Pause the initial copy process of the specified replication pair.

Syntax

```
scli --pause_pair_initial_copy (--replication_pair_id <ID> | --replication_pair_name
<NAME>)
```

Parameters

--replication_pair_id *<ID>*
Replication pair ID

--replication_pair_name *<NAME>*

Replication pair name

Example

```
scli --mdm_ip 192.168.1.200 --pause_pair_initial_copy --replication_pair_id
ca77eb5e00000000
```

pause_replication_consistency_group

Pause the replication data transfer of a replication consistency group.

Syntax

```
scli --pause_replication_consistency_group (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --pause_replication_consistency_group --
replication_consistency_group_name rcg27
```

print_recreate_replication_consistency_group_script

Display a list of SCLI commands required to recreate the specified replication consistency group.

Syntax

```
scli --print_recreate_replication_consistency_group_script (--
replication_consistency_group_name <NAME>| --replication_consistency_group_id <ID>)
```

Parameters

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--replication_consistency_group_id *<ID>*

Replication consistency group ID

Example

```
scli --mdm_ip 192.168.1.200 --print_recreate_replication_consistency_group_script --
replication_consistency_group_name rcg27
```

Sample output

```
scli --add_replication_consistency_group --replication_consistency_group_name RCG1 --
destination_system_object_id c876517d00000000 --protection_domain_id 036a797000000000
--remote_protection_domain_id 073b95a000000000 --rpo 3600 --target_volume_access_mode
no_access
scli --add_replication_pair --replication_consistency_group_name RCG1 --source_volume_id
6730b3a800000003 --destination_volume_id 2d3d6cd500000003 --copy_type full
```

query_all_replicated_volumes

Retrieve information about all replicated volumes. The query result can be optionally filtered by protection domain or by storage pool.

Syntax

```
scli --query_all_replicated_volumes [(((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replicated_volumes
```

Sample output

```
Protection Domain f11d784d00000000 Name: pd1
Storage Pool bf83a76700000000 Name: sp1
 Volume ID: c3a62eb200000003 Name: vol1 Size: 16.0 GB (16384 MB)
   Pair ID: ef199ea900000000 (CG ID: 4607a08e00000000)

Protection Domain f11d784e00000001 Name: pd2
Storage Pool bf83a76800000001 Name: sp2
 <No volumes defined>
```

```
Protection Domain f11d9f5d00000002 Name: dfg
 <No volumes defined>
Query-all-replicated-volumes returned 1 volumes
```

query_all_replication_consistency_groups

Retrieve information about all the replication consistency groups (RCG).

Syntax

```
scli --query_all_replication_consistency_groups
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replication_consistency_groups
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
 ID: 399c609c00000000 Name: rcg1 Activity: INACTIVE Peer System: peer_to_118 RPO: 1m
Compliance: 0% Status: Stopped by User

Protection Domain fcc9af5600000001 Name: pd2

Protection Domain fcc9d66500000002 Name: rrr

query_all_replication_consistency_groups returned 1 Replication Consistency Groups.
```

query_all_replication_journal_capacity

Retrieve information about all replication journal volumes. The results can be filtered by protection domain or by storage pool.

Syntax

```
scli --query_all_replication_journal_capacity [(((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replication_journal_capacity --storage_pool_name
SP1
```

Sample output

```
Protection Domain a86cde9d00000000 Name: pd1
Storage Pool 6d1751fe00000000 Name: sp1
 Volume ID: f7d341ef00000000 Name: N/A Size: 8.0 GB (8192 MB) SDR ID: 3179c35600000000
Thin-provisioned  SdrSessionIndex 0
 Volume ID: f7d341f100000002 Name: N/A Size: 8.0 GB (8192 MB) SDR ID: 3179c35800000002
Thin-provisioned  SdrSessionIndex 2
 Volume ID: f7d341f400000001 Name: N/A Size: 8.0 GB (8192 MB) SDR ID: 3179c35a00000001
Thin-provisioned  SdrSessionIndex 1

Journal Capacity - Allocated: 24.0 GB (24576 MB) Max Allowed: 26.0 GB (26624 MB) (10% of
total)

Query-all-volumes returned 3 volumes
```

query_all_replication_pairs

Retrieve information about all the replication pairs in the system.

Syntax

```
scli --query_all_replication_pairs
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replication_pairs
```

Sample output

```
CG Id: 399c609c00000000 CG Name: rcg1
Peer System Name: peer_to_118 Direction: LOCAL_TO_REMOTE
    Local ID: bee0361f00000000 Remote ID: c557f02500000000 Copy Type: IDENTICAL Initial
Copy State: DONE Name: (null)
    Local Volume ID f6351e3700000003 Remote Volume ID: dbf2341e00000003 Local Volume
Name: vol1_114 Remote Volume Name: vol1_118
```

```
    Initial Copy progress: 100% Local Activity State: INACTIVE
query_all_replication_pairs returned 1 Replication Pairs.
```

query_all_replication_peer_system

Retrieve information about all the replication peer systems.

Syntax

```
scli --query_all_replication_peer_system
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replication_peer_system
```

Sample output

```
Query-all-Replication Peer System returned 1 Replication Peer System nodes.

Replication Peer System ID: 7c3e2613207b310f Replication Peer System
internal ID: 3ea1845400000000 Name: peer_to_118 State: Joined, SUCCESS IP:
10.55.137.118,10.55.137.120 Port: 7611 Version: 3.6.0

SDR-SDR connectivity status: All connected
```

query_all_sdr

Retrieve information about all SDRs.

Syntax

```
scli --query_all_sdr
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_sdr
```

Sample output

```
Query-all-SDR returned 3 SDR nodes.

Protection Domain ID: 036a797000000000
SDR ID: 5747c3f800000000 Name: SDR2 State: Connected, Joined IP:
10.55.105.40,10.76.30.11,192.168.177.28 Port: 11088 Version: 3.5.0
SDR ID: 5747c3f900000001 Name: SDR3 State: Connected, Joined IP:
10.55.105.41,10.76.30.14 Port: 11088 Version: 3.5.0
SDR ID: 5747c3fa00000002 Name: SDR1 State: Connected, Joined IP:
10.55.105.39,10.76.30.10 Port: 11088 Version: 3.5.0
```

query_replication_consistency_group

Retrieve information about the specified replication consistency group.

Syntax

```
scli --query_replication_consistency_group (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --query_replication_consistency_group --
replication_consistency_group_id 399c609c00000000
```

Sample output

```
Replication Consistency Group ID: 399c609c00000000 Name: rcg1
  Protection Domain ID: fcc9af5500000000 Name: pd1

Configuration:
  RPO: 1m
  Target volume access mode: NO_ACCESS
  Number of volume pairs: 1, Capacity: 16.0 GB (16384 MB)

Remote info:
  Peer System ID: 7c3e2613207b310f Name: peer_to_118
  Protection Domain ID: e6bce87900000000 Name: pd1
  Remote CG Id: 2215ba5800000000

State:
  Lifetime State: NORMAL
  Transmit State: NOT_PAUSED
  Freeze State: UNFROZEN
  Consistency Mode: PARTIALLY_CONSISTENT
  Direction: LOCAL_TO_REMOTE
  Disaster recovery flow: NONE
```

```
  Disaster recovery state: None

Consistency state:
  RPO Compliance: 0%
  Lag Received: 0s (at 10:51:55)
  Lag Journalled: 0s (at 10:51:55)
  Lag Applied: 0s (at 10:51:55)
  Last Barrier: 0s (at 10:51:55)
  Next Planned Barrier: 0s (at 10:51:55)

Bandwidth:
  User Read: 479 IOPS 239.2 MB (244941 KB) per-second
  User Write: 514 IOPS 256.8 MB (262960 KB) per-second
  Transmit: 0 IOPS 0 Bytes per-second
  Apply: 0 IOPS 0 Bytes per-second

Latency:
  User Read: 240 IOPS 20.917 ms average latency
  User Write: 257 IOPS 221.345 ms average latency
  Transmit: 0 IOPS 0 us average latency
  Apply: 0 IOPS 0 us average latency
  Journaler Write: 0 IOPS 0 us average latency
  Journaler Read: 0 IOPS 0 us average latency

Journalled data:
  Journalled on Source: 0 Bytes
  Ready For Transmit: 0 Bytes
  Journalled on Target: 0 Bytes
  Ready For Apply: 0 Bytes

Activity State:
  Local Activity State: ACTIVE
  Remote Activity State: ACTIVE
```

query_replication_pair

Retrieve information about the specified replication pair.

Syntax

```
scli --query_replication_pair --replication_pair_id <ID> --replication_pair_name <NAME>
```

Parameters

--replication_pair_id *<ID>*
Replication pair ID

--replication_pair_name *<NAME>*
Replication pair name

Example

```
scli --mdm_ip 192.168.1.200 --query_replication_pair --replication_pair_id
bee0361f00000000
```

Sample output

```
Local ID: bee0361f00000000 Remote ID: c557f02500000000 Name: N/A Initial Copy State:
DONE Initial Copy Priority: -1 Initial Copy paused by user: FALSE
Peer System Name: peer_to_118 Direction: LOCAL_TO_REMOTE CG ID: 399c609c00000000 CG
Name: rcg1
Local Volume ID: f6351e3700000003 Remote Volume ID: dbf2341e00000003 Local Volume Name:
vol1_114 Remote Volume Name: vol1_118 Remote Volume Capacity (MB): 16384 Copy Type:
IDENTICAL Lifetime State: NORMAL
Initial Copy progress: 100%
Local Activity State: INACTIVE Remote Activity State: INACTIVE Last EnableRpl Failure
Reason: INVALID
```

query_replication_peer_system

Retrieve information about the specified replication peer system.

Syntax

```
scli --query_replication_peer_system (--peer_system_id <ID> | --peer_system_name <NAME>
| --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id <ID>)
```

Parameters

--peer_system_id *<ID>*

Replication peer system ID

--peer_system_name *<NAME>*

Replication peer system name

--peer_system_ip *<IP>*

Replication peer system IP address

--peer_system_port *<PORT>*

Port assigned to the replication peer system port

--peer_system_object_id *<ID>*

Replication peer system internal object ID

Example

```
scli --mdm_ip 192.168.1.200 --query_replication_peer_system --peer_system_id
7c3e2613207b310f
```

Sample output

```
Replication Peer System 7c3e2613207b310f Internal ID 3ea1845400000000 Name: peer_to_118
Version: 3.6.0
IP information (total 2 IP addresses):
         1: 10.55.137.118
         2: 10.55.137.120
         Port: 7611
Membership-state: Joined; Coupling_rc: SUCCESS;
```

query_replication_peer_system_destination_volume_candidates

Retrieve information about destination volume candidates in the replication peer system. This command returns available remote volumes in the specified protection domain of the peer system with the specified capacity.

Syntax

```
scli --query_replication_peer_system_destination_volume_candidates (--peer_system_id
<ID> | --peer_system_name <NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>]
| --peer_system_object_id <ID>) (--protection_domain_id <ID> | --protection_domain_name
<NAME>) --size_gb <SIZE>
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
Replication peer system IP address

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--size_gb *<SIZE>*
Size of volume in GBs

Example

```
scli --mdm_ip 192.168.1.200 --
query_replication_peer_system_destination_volume_candidates --peer_system_id
1820896a6b53a70f --protection_domain_name pd1 --size_gb 540
```

query_replication_peer_system_protection_domains

Retrieve information about the available protection domains in the target replication peer system.

Syntax

```
Usage: scli --query_replication_peer_system_protection_domains (--peer_system_id <ID>
| --peer_system_name <NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>] | --
peer_system_object_id <ID>)
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
IP address assigned to the replication peer system

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

Example

```
scli --mdm_ip 192.168.1.200 --query_replication_peer_system_protection_domains --
peer_system_id 1820896a6b53a70f
```

Sample output

```
Received 1 PDs
PD Id 382dca8200000000
PD Name pd1
```

query_sdr

Retrieve information about the specified SDR.

Syntax

```
scli --query_sdr (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>])
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
IP address assigned to the SDR

--sdr_port *<PORT>*
Port assigned to the SDR

Example

```
scli --query_sdr --sdr_id 82dc5ccd00000000
```

Sample output

```
SDR 82dc5ccd00000000 Name: SDR3 Version: 3.6.0
 Protection Domain ID: fcc9af5500000000
IP information (total 1 IP addresses):
         1: 10.55.137.116    Role: all (application, storage, external)
         Port: 11088
Membership-state: Joined; Connection-state: Connected; Maintenance-state: Normal; SDR-
state: Normal;
```

query_sync_now_replication_consistency_group

Retrieve the status of a synchronization process for the specified replication consistency group (RCG).

Syntax

```
scli --query_sync_now_replication_consistency_group (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>) --sync_now_key <SYNC_NOW_KEY>
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--sync_now_key *<SYNC_NOW_KEY>*

The key returned from running the `--sync_now_replication_consistency_group` command.

Example

```
scli --mdm_ip 192.168.1.200 --query_sync_now_replication_consistency_group --
replication_consistency_group_name rcg27 --sync_now_key 667432
```

Sample output

```
Successfully queried SyncNowKey 190531 of Replication Consistency Group RCG1: The data
was applied to the destination volumes
```

Return codes

Table 6. Return codes

Return code	Message
MOS_RC__SUCCESS	Success.
MOS_RC__RPL_CG_INVALID_ID	Invalid replication consistency group ID.
MOS_RC__RPL_CG_INVALID_KEY	Invalid replication consistency group key.
MOS_RC__RPL_CG_NOT_FOUND	Replication consistency group not found.

Table 6. Return codes (continued)

Return code	Message
MOS_RC__RPL_CG_IS_INACTIVE	The RCG is not active due to deactivated protection domain.
MOS_RC__RPL_CG_COMMAND_IS_NOT_ALLOWED_ON_SYSTEM_RCG	A user operation is not allowed on a system RCG.
MOS_RC__RPL_CG_COMMAND_IS_NOT_ALLOWED_DURING_DISASTER_RECOVERY	The command is not allowed during disaster recovery.
MOS_RC__RPL_CG_IS_NOT_IN_NORMAL_STATE	Replication consistency group is not in its normal state.

query_replication_destination_sdc_candidates

Query a replication peer-system for available SDCs. These SDCs can be used in destination volumes auto-provisioning.

Syntax

```
scli --query_replication_destination_sdc_candidates (--peer_system_id <ID> | --
peer_system_name <NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>] | --
peer_system_object_id <ID>)
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
Replication peer system IP address

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

query_replication_destination_storage_pool_candidates

Query a replication peer-system for storage pools on which a volume of specified size and provisioning mode can be created. These storage pools can be used in destination volumes auto-provisioning.

Syntax

```
scli --query_replication_destination_storage_pool_candidates (--peer_system_id
<ID> | --peer_system_name <NAME> | --peer_system_ip <IP> [--peer_system_port
<PORT>] | --peer_system_object_id <ID>) (--remote_protection_domain_id <ID> | --
remote_protection_domain_name <NAME>) --size_gb <SIZE> [Provisioning options]
```

Parameters

--peer_system_id *<ID>*

Replication peer system ID

--peer_system_name *<NAME>*

Replication peer system name

--peer_system_ip *<IP>*

Replication peer system IP address

--peer_system_port *<PORT>*

Port assigned to the replication peer system

--peer_system_object_id *<ID>*

Replication peer system internal object ID

--remote_protection_domain_id *<ID>*

Remote protection domain ID

--remote_protection_domain_name *<NAME>*

Remote protection domain name

--size_gb *<SIZE>*

The required volume size. The query results show only storage pools on which a volume of this size can be created.

Select any one of the following provisoning options:

--thin_provisioned

Volume is thin-provisioned

--thick_provisioned

Volume is thick-provisioned

<BLANK>

Use storage pool default (thick for medium granularity, thin for fine granularity)

remove_replication_consistency_group

Initiate removal of a replication consistency group.

Notes

Removal of a replication consistency group is a lengthy process and it may complete some time after the command completion.

Syntax

```
Usage: scli --remove_replication_consistency_group (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --remove_replication_consistency_group --
replication_consistency_group_name rcg27
```

remove_replication_pair

Remove the specified replication pair.

NOTE: The target volume image may not be consistent after the pair removal. To obtain a consistent image on the target volume: Run `freeze_apply_replication_consistency_group` on the the replication consistency group, remove the replication pair, and then run `unfreeze_apply_replication_consistency_group` on the replication consistency group.

Syntax

```
scli --remove_replication_pair (--replication_pair_id <ID> | --replication_pair_name
<NAME>) [--i_am_sure]
```

Parameters

--replication_pair_id *<ID>*
Replication pair ID

--replication_pair_name *<NAME>*
Replication pair name

--i_am_sure
Preemptive approval

Example

```
scli --mdm_ip 192.168.1.200 --remove_replication_pair --replication_pair_name rp1 --
i_am_sure
```

remove_replication_peer_system

Remove a replication peer system.

Syntax

```
scli --remove_replication_peer_system (--peer_system_id <ID> | --peer_system_name <NAME>
| --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id <ID>)
```

Parameters

--peer_system_id *<ID>*
Peer system ID

--peer_system_name *<NAME>*
Peer system name

--peer_system_ip *<IP>*
A comma-separated list of IP addresses assigned to the peer system

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

Example

```
scli --mdm_ip 192.168.1.200 --remove_replication_peer_system --peer_system_name RPS1
```

remove_sdr

Initiate removal of the specified SDR.

Syntax

```
scli --remove_sdr (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>])
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
SDR IP address

--sdr_port *<PORT>*
Port assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --remove_sdr --sdr_name SDR2
```

remove_sdr_ip

Remove an IP address from an SDR.

Syntax

```
scli --remove_sdr_ip (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --sdr_ip_to_remove <IP>
```

Parameters

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

IP address assigned to the SDR

--sdr_port *<PORT>*

Port assigned to the SDR

--sdr_ip_to_remove *<IP>*

IP address to remove from the SDR

Example

```
scli --mdm_ip 192.168.1.200 --remove_sdr_ip --sdr_name SDR27 --sdr_ip_to_remove
10.76.30.14
```

rename_replication_consistency_group

Name or rename the specified replication consistency group.

Syntax

```
scli --rename_replication_consistency_group (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>) (--new_name <NAME> | --remove_name)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--new_name *<NAME>*

New name to be assigned to the replication consistency group

--remove_name

Use this flag to delete the replication consistency group name without giving it a new name.

Example

```
scli --mdm_ip 192.168.1.200 --rename_replication_consistency_group --
replication_consistency_group_name rcg27 --new_name RCG27
```

rename_replication_peer_system

Name or rename the replication peer system. The name should be unique.

Syntax

```
scli --rename_replication_peer_system (--peer_system_id <ID> | --peer_system_name <NAME>
| --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id <ID>) --
new_peer_system_name <NAME>
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
IP address assigned to the replication peer system

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

--new_peer_system_name *<NAME>*
A new unique name to be assigned to the peer system.

Example

```
scli --mdm_ip 192.168.1.200 --rename_replication_peer_system --peer_system_id
1820896a6b53a70f --new_peer_system_name RPS1
```

rename_sdr

Name or rename an SDR.

Syntax

```
scli --rename_sdr (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --new_name <NAME>
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
SDR IP address

--sdr_port *<PORT>*
Port assigned to the SDR

--new_name *<NAME>*
New name to be assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --rename_sdr --sdr_name SDR2 --new_name SDR27
```

resume_pair_initial_copy

Resume the paused initial copy process of the specified replication pair.

Syntax

```
scli --resume_pair_initial_copy (--replication_pair_id <ID> | --replication_pair_name
<NAME>)
```

Parameters

--replication_pair_id <ID>

Replication pair ID

--replication_pair_name <NAME>

Replication pair name

Example

```
scli --mdm_ip 192.168.1.200 --resume_pair_initial_copy --replication_pair_name RP1
```

resume_replication_consistency_group

Resume a paused replication consistency group (RCG). Once the RCG is resumed, the source will resume transmission of the collected write data to the target.

Syntax

```
scli --resume_replication_consistency_group (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --resume_replication_consistency_group --
replication_consistency_group_name rcg27
```

restore_replication_consistency_group

Once the replication consistency group is in failover or switchover mode, you can reverse or restore the replication. Restoring replication maintains the replication direction from the original source and overwrites all data at the target.

Notes

During a restore operation, data movement from the source to the target is carried out without maintaining consistency of the data during the process. Consistency is only achieved at the target once the source and target synchronized.

Prior to executing the restore operation, you consider creating a snapshot on the new target volumes in order to preserve the consistent image during the resynchronization process.

Syntax

```
scli --restore_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --restore_replication_consistency_group --
replication_consistency_group_name rcg27
```

reverse_replication_consistency_group

Once the replication consistency group is in failover or switchover mode, you can reverse or restore the replication. Reversing replication changes the direction so that the original target becomes the source. All data at the original source is overwritten by the data at the target.

Notes

During a reverse operation, data movement from the new source to the new target is carried out without maintaining consistency of the data during the process. Consistency is only achieved at the new target once the new source and the new target synchronized.

Prior to executing the reverse operation, you consider creating a snapshot on the new target volumes in order to preserve the consistent image during the resynchronization process.

Syntax

```
scli --reverse_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --reverse_replication_consistency_group --
replication_consistency_group_name rcg27
```

set_pair_initial_copy_queue_position

Set the priority of the initial copy process of the specified replication pair to be the highest or lowest priority in the initial copy queue.

NOTE: This command cannot be performed when the specified RCG is inactive.

Syntax

```
scli --set_pair_initial_copy_queue_position (--replication_pair_id <ID> | --
replication_pair_name <NAME>) (--initial_copy_priority <TYPE>)
```

Parameters

--replication_pair_id *<ID>*

Replication pair ID

--replication_pair_name *<NAME>*

Replication pair name

--initial_copy_priority *<TYPE>*

Desired initial copy priority. Can be one of the following:

highest

Move the initial copy process to the front of the queue

lowest

Move the initial copy process to the end of the queue

Example

```
scli --mdm_ip 192.168.1.200 --set_pair_initial_copy_queue_position --
replication_pair_name RP1 --initial_copy_priority highest
```

set_replication_consistency_group_consistent

Set the specified replication consistency group (RCG) as consistent. While in consistent mode, the journal at the target applies its content to the target volumes in a way that guarantees that a consistent image of the target volumes can be achieved.

NOTE: This command cannot be performed when the specified RCG is inactive. It cannot be performed when in failover or switchover mode. Similarly, it cannot be performed on a disabled protection domain or during freeze apply.

Syntax

```
scli --set_replication_consistency_group_consistent (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --set_replication_consistency_group_consistent --
replication_consistency_group_name rcg27
```

set_replication_consistency_group_inconsistent

Set the specified replication consistency group (RCG) as inconsistent. While in inconsistent mode, the contents of the target journal are applied to the target volumes without guarantee of consistency. This means that the target volume image may not be consistent when the RCG is inconsistent. The RCG automatically regains consistency when possible.

It is recommended to create a snapshot of the application volumes at the target before running this command. The snapshot captures the last consistent image before the RCG is inconsistent.

NOTE: This command cannot be performed when the specified RCG is inactive or while in failover or switchover mode. It also cannot be performed on a disabled protection domain, during freeze apply, while creating snapshots, or during failover test.

Syntax

```
scli --set_replication_consistency_group_inconsistent (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --set_replication_consistency_group_inconsistent --
replication_consistency_group_name rcg27
```

set_replication_journal_capacity

Set the replication journal capacity ratio for the specified storage pool. This defines the maximum ratio of a storage pool capacity that can be used by the replication system for journaling.

Notes

Journal capacity is defined as a percentage of the total storage capacity in the storage pool and must equal at least 28 GB per SDR. In general, journal capacity should be at least 5% of replicated usable capacity in the protection domain, including volumes used as source and targets. It is important to assign enough storage capacity for the replication journal.

Syntax

```
scli --set_replication_journal_capacity (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) --
replication_journal_capacity_max_ratio <RATIO>)
```

Parameters

--protection_domain_id *<ID>*
: Protection domain ID

--protection_domain_name *<NAME>*
: Protection domain name

--storage_pool_name *<NAME>*
: Storage pool name

--storage_pool_id *<ID>*
: Storage pool ID

--replication_journal_capacity_max_ratio *<RATIO>*
: Maximum capacity percentage to be allocated for journal capacity (from 0 to 100).

Example

```
scli --mdm_ip 192.168.1.200 --set_replication_journal_capacity --protection_domain_name
PD1 --storage_pool_name SP1 --replication_journal_capacity_max_ratio 25
```

set_remote_replication_journal_capacity

Define the maximum capacity allowed for the remote system to use for replication consistency groups (RCG) sending to this protection domain.

Syntax

```
scli --set_remote_replication_journal_capacity (--protection_domain_id <ID>
| --protection_domain_name <NAME>) (--remote_capacity_max_ratio <RATIO>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--remote_capacity_max_ratio *<RATIO>*

New capacity ratio to be assigned to the remote Protection domain.

Valid range: 10-100

Example

```
scli --mdm_ip 192.168.1.200 --set_remote_replication_journal_capacity --
protection_domain_id fcc9af5500000000 --remote_capacity_max_ratio 50
```

switchover_replication_consistency_group

Switch over the specified replication consistency group (RCG). Switchover is a planned failover. The application I/Os are stopped at the source, and the data is synchronized so that the target volume is consistent with the source volume. The access mode of the original source volumes is changed to unavailable or read-only. The access mode of the original target volumes is changed so that the hosts can read and write to the volumes.

NOTE: This command cannot be performed when the specified RCG is inactive.

Syntax

```
scli --switchover_replication_consistency_group (--replication_consistency_group_id <ID>
| --replication_consistency_group_name <NAME>)[--i_am_sure]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --switchover_replication_consistency_group --
replication_consistency_group_name rcg27 --i_am_sure
```

sync_now_replication_consistency_group

Synchronize the source and target to ensure that all new data that has been written to the source is also written to the target volumes. A sync key is returned that can be used to query the status of the synchronization process.

Notes

In general, the disaster recovery processes of failover, switchover, recover, and restore perform all the required operations, including synchronizing the source and target volumes. This command is useful in situations when you choose to perform each of the individual operations directly.

Syntax

```
scli --sync_now_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>) [--force_during_partial_consistency]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--force_during_partial_consistency

Perform the command despite some volumes would be excluded due to uncompleted initial copy.

terminate_replication_consistency_group

Terminate the specified replication consistency group (RCG).

Notes

Terminating the RCG moves it into an inactive state. Replication is stopped, and the journal capacity is released. From this point on, the I/Os are handled by the SDS and not the SDR and no journal capacity is consumed. All RCG configurations are maintained, including source and target roles, pair configuration, and volume access modes.

Terminating an RCG does not require connectivity between the sites. If the RCG is terminated on a peer when there is no connectivity, then the other peer terminates once connectivity is restored.

NOTE: Activating a terminated RCG requires full initialization. The initial copy process may take a long time.

Syntax

```
scli --terminate_replication_consistency_group (--replication_consistency_group_id <ID>
| --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --terminate_replication_consistency_group --
replication_consistency_group_name rcg27
```

unfreeze_apply_replication_consistency_group

Unfreeze the replication consistency group (RCG) and apply the I/Os stored in the target journal to the target volumes.

Syntax

```
scli --unfreeze_apply_replication_consistency_group (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --unfreeze_apply_replication_consistency_group --
replication_consistency_group_name rcg27
```

27

Return messages

This section describes the contents and the types of return messages in the PowerFlex CLI.

Each return message contains the following:

- Name of the return message (return code)
- Message presented with the return code
- Commands that may return the relevant code
- Action or actions that may resolve the issue (if relevant)

SDS device name changed

Parameters

Name

SDS_DEVICE_NAME_CHANGED

Message

The SDS detected that the device name has changed. Re-run the command using the device ID.

message

The SDS detected that the device name has changed. Re-run the command using the device ID.

Commands

`remove_sds_device`, `abort_remove_sds_device`

Description

In some situations (usually following a reboot), the operating system changes the device name. The SDS detects this change. `Remove_sds_device` is a time consuming operation. To ensure that you have chosen the correct device, PowerFlex alerts you to the name change.

action

Identify the device in the output of `query_sds` and use the device ID to run the command again. You can also resynchronize the device name.

Success

Parameters

Name

SUCCESS

Message

Success

commands

All

Description

Command was executed successfully

Action

None

No resources

Parameters

Name

NO_RESOURCES

Message

Insufficient resources

Commands

`add_volume`

Description

There is insufficient space to create a volume of the requested size.

Action

Add capacity to the system by adding another SDS/device, or create a smaller volume

Timeout

Parameters

Name

TIMEOUT

Message

A timeout occurred

Commands

All

Description

No reply received from the MDM within the specified timeout period

Action

Validate the requested operational state using queries

Already exists

Parameters

Name

ALREADY_EXISTS

Message

Already Exists

Commands

All object addition commands

Description

The added object already exists

Action

Check command parameters, and use queries to view existing objects' properties

Communication error

Parameters

Name

COMMUNICATION_ERROR

Message

Communication error

Commands

SDS related commands

Description

SDS has communication errors

Action

Use `query_all` to determine which SDS is disconnected

Wrong reconfiguration mode

Parameters

Name

WRONG_RECONF_MODE

Message

SDS being configured, retry command

Commands

All SDS related commands

Description

SDS is being configured by the MDM

Action

Retry the command

SDS add device “open” error

Parameters

Name

TGT_ADD_DEV_OPEN_ERROR

Message

Could not open SDS device

Commands

`add_sds,add_sds_device`

Description

SDS cannot open one of the specified devices.

Action

Validate the command parameters. If correct, validate that the device can be accessed on the SDS.

SDS add device I/O error

Parameters

Name

TGT_ADD_DEV_IO_ERROR

Message

Add SDS device I/O error

Commands

`add_sdsadd_sds_device`

Description

The SDS was unable to write to one of the specified devices

Action

Validate the command parameters, and check the specified devices

SDS add device size problem

Parameters

Name

TGT_ADD_DEV_SIZE_PROBLEM

Message

SDS device size error

Commands

`add_sds,add_sds_device`

Description

The device size is smaller or larger than the allowed size.

Action

Validate the command parameters, and set the device size appropriately.

SDS actively attach to this MDM

Parameters

Name

TGT_ACTIVE_ATTACH_TO_THIS_MDM

Message

SDS is actively attached to this MDM

Commands

`add_sds`

Description

SDS was already configured by this MDM. This usually implies that the provided IP address resolves to an already configured SDS that is currently connected. “Actively” implies that the SDS is configured and communicating with the MDM.

Action

Check the provided IP address. Contact Dell Technologies Support.

SDS already attached to this MDM

Parameters

Name

TGT_ALREADY_ATTACH_TO_THIS_MDM

Message

SDS is already attached to this MDM

Commands

`add_sds`

Description

SDS was previously configured by this MDM. This usually indicates that the SDS was previously removed using `remove_sds`, but was not cleaned up properly.

Action

Validate the SDS IP address and port, and check the MDM configuration to validate that the SDS is not already configured under a different IP address. If the SDS is not found in the MDM configuration, this might indicate that it was not cleaned up properly when it was removed. To add the SDS forcefully, thus overriding its configuration and data, use the `force_clean` flag.

SDS active attach to other MDM

Parameters

Name

TGT_ACTIVE_ATTACH_TO_OTHER_MDM

Message

SDS is actively attached to another MDM

Commands

`add_sds`

Description

SDS is already managed by another MDM. "Actively" implies that the SDS is configured and communicating with the MDM.

Action

Check the command parameters to validate the SDS IP address and port. Check the configuration of any other MDMs installed that might be connected to this SDS. Contact Dell Technologies Support.

SDS already attached to other MDM

Parameters

Name

TGT_ALREADY_ATTACH_TO_OTHER_MDM

Message

SDS is already attached to another MDM

Commands

`add_sds`

Description

SDS was previously configured by another MDM

Action

Check your command parameters to validate the SDS IP address and port. Check for other PowerFlex installations of MDMs. If the SDS is not found in any MDM configuration, this might indicate that it was not cleaned up properly when it was removed. To add the SDS, use the `force_clean` flag.

NOTE: This command erases the existing SDS configuration and data.

SDS not found

Parameters

Name

TGT_NOT_FOUND

Message

Could not find SDS

Commands

All SDS related commands

Description

The SDS was not found

Action

Check the provided parameters. Use `query_all_sds` to view the configured SDSs.

Volume not found

Parameters

Name

VOL_NOT_FOUND

Message

Could not find the volume

Commands

All volume related commands

Description

The volume was not found

Action

Check the provided parameters. Use `query_all_volumes` to view the configured volumes.

SDC not found

Parameters

Name

INI_NOT_FOUND

Message

Could not find the SDC

Commands

All SDC related commands

Description

SDC was not found

Action

Use `query_all_sdc` to view the configured SDCs

Volume already mapped to SDC

Parameters

Name

VOL_ALREADY_MAPPED_TO_THIS_INI

Message

The volume is already mapped to this SDC

Commands

`map_volume_to_sdc` (when mapping to a single SDC)

Description

The specified volume is already mapped to the specified SDC

Action

Check the command parameters, and use `query_all_volumes` to validate the current mapping

Volume already mapped to all SDCs

Parameters

Name

VOL_ALREADY_MAPPED_TO_ALL_INIS

Message

The volume is already mapped to all SDCs

Commands

`map_volume_to_sdc` (when mapping to all SDCs)

Description

The specified volume is already mapped to all SDCs

Action

Check the command parameters, and use `query_all_volumes` to validate the current mapping

Volume not mapped to SDC

Parameters

Name

VOL_NOT_MAPPED_TO_INI

Message

The volume is not mapped to SDC

Commands

`unmap_volume_from_sdc,query_sdc_volume_limits,set_sdc_volume_limits`

Description

The specified volume is not mapped to the specified SDC

Action

Check the command parameters, and use `query_all_sdc` to validate the current mapping

Volume mapped to all SDCs

Parameters

Name

VOL_MAPPED_TO_ALL_INIS

Message

The volume is mapped to all SDCs

Commands

`map_volume_to_sdc, unmap_volume_from_sdc, set_sdc_volume_limits`

Description

The volume is mapped to all SDCs

Action

Check the command parameters, and use `query_all_volumes` to validate the current mapping

Too many mappings

Parameters

Name

TOO_MANY_MAPPINGS

Message

There are too many mappings

Commands

`map_volume_to_sdc`

Description

Mapping of an additional volume will exceed the system limit

Action

Validate if all existing volumes that are mapped are relevant. Contact Dell Technologies Support.

Invalid volume ID

Parameters

Name

INVALID_VOL_ID

Message

Invalid volume ID. Try again with a valid ID.

Commands

All volume-related commands.

Description

The specified volume ID is invalid.

Action

Check the specified volume ID. Use `query_all_volumes` to find the correct ID, or use the volume name.

Invalid SDS ID

Parameters

Name

INVALID_TGT_ID

Message

Invalid SDS ID. Try again with a valid ID.

Commands

All SDS commands.

Description

The specified SDS ID is invalid.

Action

Check the specified SDS ID. Use `query_all_sds` to find the correct ID, or use the SDS name or IP address.

Invalid SDC ID

Parameters

Name

INVALID_INI_ID

Message

Invalid SDC ID. Try again with a valid ID.

Commands

All SDC commands.

Description

The specified SDC ID is invalid.

Action

Check the specified SDC ID. Use `query_all_sdc` to find the correct ID, or use the SDC IP address.

Host not found

Parameters

Name

HOST_NOT_FOUND

Message

Could not find the host

Commands

Secret session token. All SDS commands

Description

The specified host name was not found

Action

Check the specified host name

Duplicate host name

Parameters

Name

DUP_HOST_NAME

Message

Duplicate host name

Commands

`add_sds`

Description

The list of host names includes two identical host names or IP addresses, or two host names are resolved to the same IP address

Action

Check the specified host name list

Address and port in use

Parameters

Name

TGT_ADDR_AND_PORT_IN_USE

Message

The SDS IP address and port already in use. Try with different ones.

Commands

`add_sds`

Description

The specified IP address and port are already in use by another defined SDS.

Action

Check the specified IP address and port.

Too many SDSs in protection domain

Parameters

Name

TOO_MANY_TGTS_IN_FDOM

Message

There are too many SDSs in a protection domain.

Commands

`add_sds`

Description

Adding the SDS exceeds the system limit of allowed SDSs in a protection domain.

Action

Check the configuration.

Too many SDSs

Parameters

Name

TOO_MANY_TGTS

Message

There are too many SDSs

Commands

`add_sds`

Description

Adding the SDS will exceed the system limit of allowed SDSs in the system

Action

Check configuration

Too many volumes in protection domain

Parameters

Name

TOO_MANY_VOLUMES_IN_FD

Message

Too many volumes in a protection domain.

Commands

`add_volume`

Description

Adding a volume exceeds the limit for the number of volumes allowed in a protection domain.

Action

Check the configuration.

Too many volumes

Parameters

Name

TOO_MANY_VOLS

Message

There are too many volumes

Commands

`add_volume`

Description

Adding a volume exceeds the limit for the number of volumes allowed in a system

Action

Check configuration

SDS name in use

Parameters

Name

TGT_NAME_IN_USE

Message

SDS name already in use. Use a different name.

Commands

`add_sds`, `rename_sds`

Description

The specified SDS name is already in use by another SDS.

Action

Use `query_all_sds` to validate existing names, and issue the command with another name.

Volume name in use

Parameters

Name

VOL_NAME_IN_USE

Message

Volume name already in use. Use a different name.

Commands

`add_volume`, `rename_volume`

Description

The specified volume name is already in use by another volume.

Action

Use `query_all_volumes` to validate existing names, and issue the command with another name.

Duplicate SDS device name

Parameters

Name

DUPLICATE_TGT_DEVICE_NAME

Message

Duplicate SDS device name

Commands

`add_sds`

Description

A device appears more than once in the `add_sds` command

Action

Validate command parameters

Duplicate SDC IP address

Parameters

Name

DUPLICATE_SDC_IP

Message

More than one SDC exists for the same IP address. Use name or ID instead, or remove the duplicate SDC.

Commands

`map_volume_to_sdc`, `unmap_volume_from_sdc`, `set_sdc_volume_limits` , `query_sdc_volume_limits`

Description

The specified SDC IP address is not unique. It is associated with more than one SDC.

Action

Use `query_all_sdc` to find the relevant ID. Use the SDC ID in the command instead of the SDC IP address.

SDS not under removal

Parameters

Name

TGT_NOT_UNDER_REMOVAL

Message

SDS is not being removed

Commands

`abort_remove_sds`

Description

The SDS is not being removed

Action

Check command parameters. If the parameters are correct, it is still possible that the SDS has already been removed.

SDS being removed

Parameters

Name

TGT_BEING_REMOVED

Message

SDS is being removed

Commands

SDS related commands

Description

The SDS is currently being removed

Action

None

Volume mapped

Parameters

Name

VOL_MAPPED

Message

Volume is mapped

Commands

`remove_volume`, `remove_consistency_group`

Description

The volume (for `remove_volume`) or any volume in the group (`remove_consistency_group`) is mapped to an SDC

Action

Unmap before removing the volume or consistency group. Use `query_all_volumes` to check current mapping.

Use MDM IP address

Parameters

Name

USE_MDM_IP

Message

Use cluster MDM physical IP to access the MDM.

Commands

All cluster commands.

Description

The command reached the secondary MDM. Use the physical IP address to access the MDM.

Action

Add `--mdm_ip` to the commands.

Command only in cluster

Parameters

Name

CMD_ONLY_IN_CLUSTER

Message

Command can only be used in cluster mode.

Commands

`switch_mdm_ownership`

Description

This command can be issued only when in cluster mode.

Action

Use `query_cluster` to check the current configuration.

Command only in single mode

Parameters

Name

CMD_ONLY_IN_SINGLE

Message

Command can only be used in a single mode.

Commands

MDM cluster commands

Description

This command can be issued in a single mode only.

Action

Use `switch_mdm_ownership` to switch to single mode before issuing this command.

Command error when cluster degraded

Parameters

Name

CMD_ERROR_CLUSTER_DEGRADED

Message

Command cannot be used when a cluster is degraded

Commands

`switch_mdm_ownership`

Description

This command cannot be issued when the cluster is degraded

Action

Check the cluster state and hardware configuration to determine the reason for cluster degradation

Too many devices

Parameters

Name

TOO_MANY_DEVICES

Message

There are too many devices for SDS

Commands

`add_sds,add_sds_device`

Description

Too many devices specified (`add_sds`), or the SDS already has the maximum number of allowed devices

Action

None

SDS device not found

Parameters

Name

TGT_DEVICE_NOT_FOUND

Message

Could not find the SDS device

Commands

SDS device related commands

Description

SDS device was not found

Action

Check the command parameters. Use `query_sds` to view the device details.

License too long

Parameters

Name

LICENSE_TOO_LONG

Message

The license key is too long. Check your key and enter it again.

Commands

`set_license`

Description

The specified license key is too long.

Action

Validate the specified license key with the one received from Dell.

License error

Parameters

Name

LICENSE_ERROR

Message

License key is corrupted.

Commands

`set_license`

Description

The specified license key is invalid or does not match this version. Contact Dell Technologies Support.

Action

Validate the specified license key with the one received from Dell.

Unknown license version

Parameters

Name

LICENSE_UNKNOWN_VERSION

Message

Unknown license version

Commands

`set_license`

Description

The specified license key is corrupted

Action

Validate the specified license key with the one received from Dell.

License not set

Parameters

Name

LICENSE_NOT_SET

LICENSE_NOT_SET

License has not been set

Commands

`query_license`, `add_sds`, `add_sds_device`, `set_data_copy_limit`

Description

License was not set. Product cannot be used.

Action

Set the license before continuing to work

Configuration exceeds new license

Parameters

Name

CONFIG_EXCEEDS_NEW_LICENSE

Message

The current system configuration exceeds the license entitlements.

Commands

`set_license`

Description

The specified license decreases the amount of licensed capacity. The current system capacity already exceeds the specified license.

Action

Contact Dell Technologies support to receive a license for a larger amount of capacity.

License capacity exceeded

Parameters

Name

CAPACITY_LICENSE_EXCEEDED

Message

The operation could not be completed. The license capacity has been exceeded.

Commands

`add_sds`, `add_sds_device`

Description

Adding the SDS or device exceeds the current licensed capacity

Action

Contact Dell Technologies support to receive a license for a larger amount of capacity

License expired

Parameters

Name

LICENSE_TIME_EXCEEDED

Message

The license has expired

Commands

Most commands

Description

The system time-based license has expired.

Action

Contact Dell Technologies support to receive a new license.

Installation ID mismatch

Parameters

Name

INSTALL_ID_MISMATCH

Message

The license installation ID does not match the ID of this system

Commands

`set_license`

Description

The provided license key does not match the current installation.

Action

Contact Dell Technologies support

Customer ID mismatch

Parameters

Name

CUSTOMER_ID_MISMATCH

Message

License customer ID does not match this system.

Commands

`set_license`

Description

The provided license key does not match the current installation.

Action

Contact Dell Technologies support.

Unsupported license version

Parameters

Name

LICENSE_UNSUPPORTED_VERSION

Message

Unsupported license version.

Commands

`set_license`

Description

The provided license key does not match the current installation.

Action

Contact Dell Technologies support.

Volume decrease not supported

Parameters

Name

VOL_DECREASE_NOT_SUPPORTED

Message

Volume capacity can only be increased.

Commands

`modify_volume_capacity`

Description

The specified volume capacity is smaller than the existing volume capacity. Decreasing volume capacity is not supported.

Action

Use `query_all_volumes` to validate your current configuration.

SDS device being removed

Parameters

Name

TGT_DEV_BEING_REMOVED

Message

The SDS device is being removed.

Commands

`remove_sds`, `remove_sds_device`, `activate_sds_device`, `start_sds_device_test`

Description

The specified SDS device is currently being removed. The command cannot be executed.

Action

Wait for the device to be removed

Cannot remove last SDS device

Parameters

Name

CANNOT_REMOVE_LAST_SDS_DEVICE

Message

Last remaining SDS device cannot be removed.

Commands

`remove_sds_device`

Description

The command attempts to remove the last remaining SDS device. This is an illegal operation.

Action

To remove the last remaining device, remove the SDS.

Device not under removal

Parameters

Name

DEV_NOT_UNDER_REMOVAL

Message

The SDS device is not being removed.

Commands

`abort_remove_sds`

Description

The specified device is not in the process of being removed.

Action

Check the command parameters.

SDS already adding device

Parameters

Name

TGT_ALREADY_ADDING_DEV

Message

A device is being added to this SDS.

Commands

`add_sds_device`, `remove_sds`

Description

This operation cannot be performed while a device is being added to the SDS.

Action

Wait for the device to be added.

Invalid device ID

Parameters

Name

INVALID_DEV_ID

Message

The device ID is invalid. Use a valid device ID.

Commands

SDS device-related commands.

Description

The specified SDS device ID is invalid.

Action

Use `query_sds` to determine the correct ID.

Too many protection domains

Parameters

Name

TOO_MANY_FDs.

Message

There are too many protection domains.

Commands

`add_protection_domain`

Description

Adding a protection domain exceeds the system limit of the allowed protection domains.

Action

None

Protection domain name in use

Parameters

Name

FD_NAME_IN_USE

Message

Protection domain name already in use. Try a different name.

Commands

`add_protection_domain`, `rename_protection_domain`

Description

The specified protection domain name is already in use by another protection domain.

Action

Use `query_all` to validate the existing names, and issue the command with another name.

Protection domain not found

Parameters

Name

FD_NOT_FOUND

Message

Could not find protection domain.

Commands

All protection domain-related commands.

Description

The specified protection domain name or ID do not match any existing protection domains.

Action

Check the command parameters. Use `query_all` to validate the existing protection domain names.

Protection domain has SDSs

Parameters

Name

FD_HAS_TGTS

Message

Protection domain has SDSs

Commands

`remove_protection_domain`

Description

The protection domain contains SDSs and cannot be removed.

Action

Remove the SDSs before attempting to remove the protection domain.

Not enough SDSs in storage pool

Parameters

Name

NOT_ENOUGH_TGTS_IN_STORAGE_POOL

Message

Fewer than three devices from different SDSs are defined in this storage pool. Add devices to the storage pool from additional SDSs.

Commands

`add_volume`

Description

The storage pool must contain devices from at least three different SDSs in order to create a volume.

Action

Add more devices to the storage pool before creating a volume.

Remove in progress

Parameters

Name

REMOVE_IN_PROGRESS

Message

Removal of the object is in progress. Please wait.

Commands

Multiple commands

Description

The operation cannot be performed, because the volume is being removed.

Action

Wait for the volume to be removed.

No network test results

Parameters

Name

NET_TEST_NOT_FOUND

Message

Could not find network test results

Commands

`query_sds_network_test_results`

Description

The specified SDS has no network test results.

Action

Check the command parameters to make sure that a `start_sds_network_test` was sent to the specified SDS. If this is the correct SDS, resend the `start_sds_network_test` command.

Network test in progress

Parameters

Name

NET_TEST_IN_PROGRESS

Message

Network test in progress.

Commands

`query_sds_network_test_results`

Description

The SDS network test has not been completed.

Action

Wait for the test to be completed. Use the same query periodically until results are displayed.

SDS device already active

Parameters

Name

TGT_DEV_ALREADY_ACTIVE

Message

The SDS device is already active.

Commands

`activate_sds_device`

Description

At least one specified device is already active.

Action

Use `query_sds` to view the device status.

SDS device wrong test mode

Parameters

Name

TGT_DEV_WRONG_TEST_MODE

Message

The SDS device is in incorrect test mode.

Commands

`activate_sds_device`

Description

At least one specified device is automatically activated when its test is completed.

Action

Use `query_sds` to view the device status.

SDS device wrong test state

Parameters

Name

TGT_DEV_WRONG_TEST_STATE

Message

The SDS device test is not complete.

Commands

`activate_sds_device`

Description

At least one specified device test is not yet complete. This device cannot be activated.

Action

Use `query_sds` to view the devices' status.

SDS device test in progress

Parameters

Name

TGT_DEV_TEST_IN_PROGRESS

Message

The SDS device test is in progress.

Commands

`start_sds_device_test`

Description

At least one specified device test is in-progress. A new test for this device cannot be started now.

Action

Use `query_sds` to view the devices' test status.

Volume allocation "busy" error

Parameters

Name

VOL_ALLOC_ERROR_BUSY

Message

The system is busy. Retry later.

Commands

`add_volume`

Description

The MDM is attempting to recover a connection with an SDS. You cannot create a volume now.

Action

Use `query_all` to view the system status. Retry the command when the situation has returned to normal.

Capacity highly unbalanced

Parameters

Name

CAPACITY_HIGHLY_UNBALANCED

Message

System capacity is unbalanced.

Commands

`add_volume`

Description

System capacity is unbalanced. This can be a temporary or permanent state.

- Temporary—Due to adding or removing capacity, or due to disconnections. In this case, the system automatically works in the background to correct the situation.
- Permanent—The capacity of the devices added to PowerFlex is highly unbalanced. This requires user intervention to correct the situation.

Action

If a rebalance is in progress, wait for it to finish and resubmit the command. If the situation persists, contact support.

Volume creation failed

Parameters

Name

VOL_CREATE_FAIL

Message

Could not create the volume.

Commands

`add_volume`

Description

The volume creation failed.

Action

Use `query_all` to validate the system state, and retry the command.

Invalid password

Parameters

Name

INVALID_PASSWORD

Message

Invalid password. Try again.

Commands

Most commands.

Description

The specified password is incorrect.

Action

Check that you are using the correct password in the command.

No permissions

Parameters

Name

NO_PERMISSIONS

Message

You do not have permission to perform this operation. Contact your system administrator for assistance.

Commands

Most commands.

Description

The specified user does not have permission to issue this command.

Action

Contact the system administrator.

Too many storage pools in protection domain

Parameters

Name

TOO_MANY_STORAGE_POOLS_IN_PROTECTION_DOMAIN

Message

There are too many storage pools in protection domain.

Commands

`add_storage_pool`

Description

Addition of another storage pool exceeds the system limit of allowed storage pools in a protection domain.

Action

Use `query_protection_domain` to view the storage pools in the protection domain.

Storage pool already exists

Parameters

Name

STORAGE_POOL_ALREADY_EXISTS

Message

Storage pool exists.

Commands

`add_storage_pool`

Description

A storage pool with the specified name exists in the specified protection domain.

Action

Use `query_protection_domain` to view the storage pools in the protection domain, and choose a different name.

Storage pool not found

Parameters

Name

STORAGE_POOL_NOT_FOUND

Message

Could not find a storage pool.

Commands

All storage pool–related commands.

Description

The specified storage pool was not found.

Action

Use `query_protection_domain` to view the storage pools in the protection domain, and check the issued parameters.

Storage pool has devices

Parameters

Name

STORAGE_POOL_HAS_DEVS

Message

The storage pool has SDS devices.

Commands

`remove_storage_pool`

Description

The specified storage pool has SDS devices that are associated with it.

Action

Remove the SDS devices before removing the storage pool.

Storage pool name already exists

Parameters

Name

STORAGE_POOL_NAME_ALREADY_EXISTS

Message

Storage pool name exists.

Commands

`rename_storage_pool`

Description

A storage pool with the specified new name exists in the protection domain.

Action

Use `query_protection_domain` to view the storage pools in the protection domain. Choose another name for the storage pool.

Invalid storage pool ID

Parameters

Name

INVALID_STORAGE_POOL_ID

Message

Invalid storage pool ID.

Commands

All storage pool-related commands.

Description

The specified storage pool ID is invalid.

Action

Check the specified storage pool ID. Use `query_protection_domain` to find the correct ID, or use the storage pool name.

Protection domain has storage pools

Parameters

Name

FD_HAS_STORAGE_POOLS

Message

Protection domain has storage pools.

Commands

`remove_protection_domain`

Description

The protection domain has more than one storage pool associated with it.

Action

Remove the storage pools before removing the protection domain.

SDC has mappings

Parameters

Name

INI_HAS_MAPPINGS

Message

SDC has mapped volume(s).

Commands

`remove_sdc`

Description

At least one volume is mapped to the specified SDC.

Action

Remove the mapping before removing the SDC.

Too many volumes in V-Tree

Parameters

Name

TOO_MANY_VOLS_IN_VTREE

Message

There are too many volumes in the V-Tree.

Commands

`snapshot_volume`

Description

Taking a snapshot of the specified volume exceeds the allowed system limit of snapshots per volume.

Action

Use `query_volume` to view the current snapshots. Consider deleting some snapshots.

Too many snapshots in V-Tree

Parameters

Name

TOO_MANY_SNAPS_IN_VTREE

Message

There are too many snapshots in the V-Tree.

Commands

`snapshot_volume`

Description

Taking a snapshot of the specified volume exceeds the allowed system limit of snapshots per volume.

Action

Contact Dell Technologies support.

No volume to delete

Parameters

Name

NOTHING_TO_DO

Message

No volume to delete.

Commands

`remove_volume`

Description

The combination of parameters resulted in an empty set of volumes to delete.

Action

Use `query_volume` to view the current state, and review the command parameters accordingly.

Too many devices in storage pool

Parameters

Name

TOO_MANY_DEVICES_IN_STORAGE_POOL

Message

There are too many devices in storage pool.

Commands

`add_sds`, `add_sds_device`

Description

Adding a device to the specified storage pool exceeds the allowed system limit of SDS devices per storage pool.

Action

Use `query_sds` to find the association of devices to storage pools.

Volume not found for consistency group

Parameters

Name

VOL_NOT_FOUND_FOR_SNAP_GROUP

Message

No volumes were found for the given consistency group.

Commands

`remove_consistency_group`

Description

No volume found for the specified consistency group.

Action

Use `query_all_volumes` to view the existing consistency groups, and review command parameters accordingly.

Too many IP addresses

Parameters

Name

TOO_MANY_IPS

Message

There are too many IP addresses for SDS.

Commands

`add_sds_ip`

Description

Addition of an IP address to the SDS exceeds the allowed system limit of IP addresses per SDS.

Action

Use `query_sds` to view the SDS IP addresses.

Cannot remove last SDS IP address

Parameters

Name

LAST_IP

Message

SDS must have at least one IP address.

Commands

`remove_sds_ip`

Description

An SDS must have at least one IP address. Cannot remove the last remaining IP address.

Action

Use `query_sds` to view the SDS IP addresses.

SDS IP address not found

Parameters

Name

SDS_IP_NOT_FOUND

Message

SDS IP not found.

Commands

`remove_sds_ip`

Description

The specified SDS IP address was not found.

Action

Use `query_sds` to view the SDS IP addresses.

Volume size too large

Parameters

Name

VOL_SIZE_TOO_LARGE

Message

The volume size too large.

Commands

`add_volume`, `modify_volume_capacity`

Description

The specified volume size is too large.

Action

Use a smaller volume size.

Capacity too low for snapshot

Parameters

Name

CAPACITY_TOO_LOW_FOR_SNAP

Message

The capacity of the storage pool is too small to create new snapshots.

Commands

`snapshot_volume`

Description

Snapshots are disabled when the free capacity in the storage pool is very low. The threshold is defined using the `set_capacity_alerts_threshold` command. The default threshold is 10% below the predefined spare.

Action

Add storage capacity, or free up space by removing some snapshots.

IP address belongs to other SDS

Parameters

Name

IP_BELONGS_TO_OTHER_TGT

Message

The specified IP address belongs to another SDS. Use the correct IP address.

Commands

`add_sds`, `add_sds_ip`

Description

The specified IP address is already assigned to another SDS.

Action

Check command parameters.

Invalid port

Parameters

Name

INVALID_PORT

Message

The specified SDS port is not in the valid range. Try again with a valid port number.

Commands

`add_sds`, `add_sds_ip`

Description

The specified port must be greater than 1024.

Action

Check command parameters.

LIA package not installed

Parameters

Name

LIA_PACKAGE_NOT_INSTALLED

Message

The package is not installed.

Description

You are trying to uninstall the LIA package, but the package is not installed.

Action

Check why you are trying to uninstall a component that is not installed. Verify that you have the correct package name for the package that you want to uninstall.

LIA multiple packages installed

Parameters

Name

LIA_MULTIPLE_PACKAGES_INSTALLED

Message

More than one package is installed.

Description

More than one package of the specified component is installed.

Action

Uninstall the packages manually:

Linux:

1. Find the package name:

   ```
   rpm -qa | grep -i lia
   ```

   The LIA package name is displayed. For example:

   ```
   EMC-ScaleIO-lia-2.50-0.30.el6.x86_64
   ```

2. Uninstall the package:

   ```
   rpm -e <PACKAGE_NAME>
   ```

Windows:

- **Windows Control Panel > Add/Remove programs** > **EMC-ScaleIO-lia**

LIA command failed

Parameters

Name

LIA_COMMAND_FAILED

Message

The command failed. Try again.

Description

his is a general error message.

Action

Look for more information about the error in the LIA log:

Linux: `/opt/emc/scaleio/lia/logs`

Windows: `C:\Program Files\emc\scaleio\lia\logs`

LIA invalid password

Parameters

Name

LIA_INVALID_PASSWORD

Message

Token authentication was not successful.

Description

The wrong LIA token was used, and authentication failed.

Action

Verify that you have the correct LIA password (token).

LIA operation not permitted

Parameters

Name

LIA_OPERATION_NOT_PERMITTED

Message

The operation is not permitted by current configuration.

Description

LIA was configured to disallow the operation that you tried to perform.

Action

Contact your system administrator; change the LIA configuration in your system.

LIA file is too large

Parameters

Name

LIA_FILE_IS_TOO_LARGE

Message

The file is too large to send back to the user.

Description

The file returned by the `get_info` command is larger than 500 MB, and therefore cannot be sent back to you.

Action

Contact Dell Technologies support.

LIA wrong ID

Parameters

Name

LIA_WRONG_ID

Message

LIA installation ID verification failed.

Description

The LIA ID that was passed in the installation/uninstall package is different to the installation ID that LIA is currently using.

Action

Edit the LIA's configuration file with LIA token and restart the LIA. The configuration file is found at:

- Linux: `/opt/emc/scaleio/lia/cfg/cont.txt`
- Windows: `C:\Program Files\emc\scaleio\lia\cfg\conf.txt`

LIA disk does not exist

Parameters

Name

LIA_DISK_DOES_NOT_EXIST

Message

The disk does not exist in the system.

Description

The disk name that was passed with the `prepare_disk` command does not exist in the system.

Action

Check that you are using the correct the disk name.

SDS IP address not found

Parameters

Name

TGT_IP_NOT_FOUND

Message

The SDS IP address cannot be found.

Commands

`modify_sds_ip`, `remove_sds_ip`

Description

You tried to remove or modify an SDS IP address that does not exist in the system.

Action

Verify that you are specifying the correct IP address in the command.

Bad threshold value

Parameters

Name

BAD_THRESHOLD_VALUE

Message

The threshold value is out of range. Enter a valid percentage value.

Commands

`set_capacity_alert_threshold`

Description

The legal range for the capacity alert threshold is a percentage value between 0 and 100. The value that is specified for critical alerts must be less than the value for high alerts.

Action

Specify a value between 0 and 100. Ensure that the value for critical alerts is less than the value for high alerts.

Storage pool has volumes

Parameters

Name

STORAGE_POOL_HAS_VOLS

Message

The storage pool has volumes. Remove all the volumes before removing the storage pool.

Commands

`remove_storage_pool remove_protection_domain`

Description

Only empty storage pools can be removed.

Action

Remove all volumes before removing the storage pool or protection domain.

Protection domain and storage pool mismatch

Parameters

Name

FD_AND_SP_MISMATCH

Message

The specified storage pool is not in the specified protection domain.

Commands

Any command which uses the combination of storage pool and protection domain.

Description

The operation could not be performed because the combination of storage pool and protection domain is incorrect and the storage pool could not be found.

Action

Use the correct combination of storage pool and protection domain in the command. Use the command `query_all` to list all the storage pools to find this information.

SDS IP address already exists

Parameters

Name

TGT_IP_ALREADY_EXISTS

Message

This SDS IP exists. Use a different IP address.

Commands

`add_sds_ip`

Description

The IP address that is used in the command is already in use in the system.

Action

Verify that you are using the correct IP address in the command, and that the address has not been assigned to any other SDSs.

IP role already set

Parameters

Name

ROLE_ALREADY_SET

Message

The IP role for this SDS has already been configured.

Commands

`modify_sds_ip`

Description

The IP role for the specified SDS has already been configured.

Action

Verify that you are configuring the intended SDS, and that you are specifying the correct IP address.

Cluster ID mismatch

Parameters

Name

CLUSTER_ID_MISMATCH

Message

The system ID number that you provided does not match the actual system ID.

Commands

`query_properties`, `query_system`

Description

The system ID number has been modified, and does not match the ID specified in the command.

Action

No action required - this code is provided for information purposes for advanced users.

Device already exists

Parameters

Name

DEV_ALREADY_EXISTS

Message

A device with the given name already exists in the SDS.

Commands

`add_sds_device`

Description

You tried to add a device to the system, using a name that is already defined in the SDS.

Action

Verify that the specified device has not already been added to the system.

Volume size is illegal

Parameters

Name

VOL_SIZE_ILLEGAL

Message

The specified volume size is illegal. Volume size must be in 8 GB blocks.

Commands

`add_volume`, `modify_volume_size`

Description

You tried to define a volume size that is not a multiple of 8 GB. Volume size can only be defined in multiples of 8 GB.

Action

Specify volume size in blocks of 8 GB.

SDC must be disconnected

Parameters

Name

INI_MUST_BE_DISCONNECTED

Message

The SDC must be disconnected.

Commands

`remove_sdc`

Description

Only SDCs that are currently inactive (disconnected) can be removed.

Action

This command is intended to remove old, disconnected SDCs. Do not remove active SDCs.

Invalid percentage

Parameters

Name

INVALID_PERCENTAGE

Message

The percentage value that you provided is invalid. Use a value between 0—100.

Commands

`set_thick_volume_percentage`

Description

The value that you entered was less than 0 or greater than 100.

Action

Use a value between 1 and 100 in the command.

Wrong receive group

Parameters

Name

WRONG_RECEIVE_GROUP

Message

The command was sent to the wrong receive group due to component software mismatch.

Commands

All commands

Description

There are different software versions installed on some components in your system, and this is causing a software mismatch error.

Action

Verify that the same software version is installed on all components, and fix, if necessary.

Mismatched SW ID

Parameters

Name

MISMATCHING_SWID

Message

The license contains a mismatch of the SWID number. Contact Dell Technologies support.

Commands

`set_license`

Description

The SWID number that you entered does not match your license.

Action

Contact Dell Technologies support.

Invalid license issuer

Parameters

Name

INVALID_LICENSE_ISSUER

Message

The issuer of the license you are attempting to add does not match that of the product.

Commands

`set_license`

Description

There is a problem with your license.

Action

Contact Dell Technologies support.

Number of parallel messages is too low

Parameters

Name

NUM_PARALLEL_MSG_TOO_LOW

Message

The number of parallel messages is too low. Use a number in the range 1–6.

Commands

`start_sds_network_test`

Description

You used a number less than 1 in the command.

Action

Use a number in the range of 1 to 6 for parallel messages in the command.

Number of parallel messages is too high

Parameters

Name

NUM_PARALLEL_MSG_TOO_HIGH

Message

The number of parallel messages is too high. Use a number in the range 1- 6.

Commands

`start_sds_network_test`

Description

You used a number greater than six in the command.

Action

Use a number in the range of 1-6 for parallel messages in the command.

Network test size is too high

Parameters

Name

NETWORK_TEST_SIZE_TOO_HIGH

Message

The network test size is too high. Use a test size less than 10.

Commands

`start_sds_network_test`

Description

You used a number greater than 10 in the command.

Action

Use a number less than 10 for the test size in the command.

Network test length too long

Parameters

Name

NETWORK_TEST_LENGTH_TOO_HIGH

Message

The network test length is too long. Use a test length less than 43200 seconds.

Commands

`start_sds_network_test`

Description

You used a number greater than 43200 for the test length in the command.

Action

Use a number less than 43200 for test length in the command.

Network test must limit either time or size

Parameters

Name

NETWORK_TEST_MUST_LIMIT_EITHER_TIME_OR_SIZE

Message

The network test must limit either time or size.

Commands

`start_sds_network_test`

Description

Your command did not contain a value for time or size.

Action

Include a value for either time or size in the command.

Not enough SDSs in protection domain for test

Parameters

Name

NOT_ENOUGH_TGTS_IN_FD_FOR_TEST

Message

To start the test, there must be at least two SDSs in the protection domain.

Commands

`start_sds_network_test`

Description

There are fewer than two SDSs in the protection domain, and therefore the test operation cannot be performed.

Action

Ensure that there are at least two SDSs in the protection domain before attempting to test it.

Not allowed in storage pool with devices

Parameters

Name

NOT_ALLOWED_IN_SP_WITH_DEVS

Message

This operation is only allowed when there are no devices in the storage pool. Remove all devices from the storage pool.

Commands

`modify_zero_padding_policy`

Description

You can only modify the zero padding policy of empty storage pools.

Action

Ensure that all devices are removed from the storage pool before you attempt to modify its zero padding policy.

License capacity mismatch

Parameters

Name

LICENSE_CAPACITY_MISMATCH

Message

The license contains a mismatch of the capacity values for basic and advanced features. Contact Dell Technologies support.

Commands

`set_license`

Description

The capacity allowed for basic and advanced features in your license do not match your installation.

Action

Contact Dell Technologies support.

Exceeds system capacity limitations

Parameters

Name

EXCEED_SYSTEM_CAPACITY_LIMITATIONS

Message

The request exceeds system limitations.

Commands

`add_volume`, `modify_volume_capacity`

Description

The command specifies a volume larger in size than the maximum system capacity.

Action

Remove one or more volumes in the system, to make room for the new one.

SDC mapped to too many volumes

Parameters

Name

INI_MAPPED_TO_TOO_MANY_VOLS

Message

The SDC is mapped to too many volumes. Remove some mapping.

Commands

`map_volume_to_sdc`

Description

You attempted to map more than the maximum allowed number of volumes (8192) to one or more SDCs.

Action

Avoid mapping more than the maximum allowed volumes to a single SDC.

Not enough devices in storage pool for removal

Parameters

Name

NOT_ENOUGH_DEVICES_IN_STORAGE_POOL_FOR_REMOVAL

Message

Not enough devices in the storage pool.

Commands

`remove_sds`, `remove_sds_device`

Description

Each storage pool must contain at least two devices. If there are two or less devices in the storage pool and you try to remove one, the command fails.

Action

Add devices to the storage pool before trying to remove this one.

Invalid network limits

Parameters

Name

INVALID_TGT_NETWORK_LIMITS

Message

Bad limits given. The overall must be larger than both rebuild and rebalance and all limits must be over 5MB.

Commands

`sds_network_limits`

Description

The total (overall) limit must be greater than the sum of the rebuild limit and the rebalance limit. All limits must be greater than 5 MB.

Action

Ensure that limit values are greater than 5 MB, and that the overall limit value is greater than rebuild limit plus rebalance limit.

A device is currently being removed

Parameters

Name

ONE_TGT_DEV_BEING_REMOVED

Message

This command cannot be completed. An SDS device is currently being removed.

Commands

`activate_sds_device` with the parameter `activate_all_devices`.

Description

If a device is in the process of being removed, you cannot use the `activate_all_devices` parameter to activate the remaining ones.

Action

Wait until the device removal is complete, and try again.

A device is already active

Parameters

Name

ONE_TGT_DEV_ALREADY_ACTIVE

Message

This command cannot be completed; An SDS device is currently active.

Commands

`activate_sds_device` with the parameter `activate_all_devices`

Description

A device is already active.

Action

Do not use the `activate_all_devices` option in the command. Instead, specify each device individually in the command.

RAM cache wrong state

Parameters

Name

RMCACHE_WRONG_STATE

Message

Wrong RAM cache state.

Commands

`enable_sds_rmcache` , `disable_sds_rmcache`, `set_sds_rmcache_size`

Description

The cache is in transient state, so the requested operation cannot be executed.

Action

Wait for a few seconds and try again.

More than one mapping not allowed

Parameters

Name

NOT_ALLOWED_MORE_THAN_ONE_MAPPING

Message

Only a single SDC may be mapped to this volume at a time.

Commands

`map_volume_to_sdc`

Description

The volume is already mapped, and the `allow_multi_map` flag was not used in the command.

Action

The volume is already mapped. Use the `allow_multi_map` flag to map the volume to additional SDCs.

At least one SDS failed

Parameters

Name

AT_LEAST_ONE_TGT_FAILED

Message

At least one SDS failed.

Commands

`enable_sds_rmcache`, `disable_sds_rmcache`, `set_sds_rmcache_size`

Description

You tried to enable or disable, or change the size of cache for all SDSs in the protection domain, and this failed for at least one of the SDSs.

Action

Check the state of the cache in all the SDSs, and try again for the specific SDS with the problem.

A device is in incorrect test state

Parameters

Name

ONE_TGT_DEV_WRONG_TEST_STATE

Message

This command cannot be completed; An SDS device is currently in incorrect test mode.

Commands

`activate_sds_device` with `activate_all_devices` parameter.

Description

A device was not added with the `test_only` option, so it cannot be activated with this command.

Action

It will activate itself on its own. No action required.

A device test is in progress

Parameters

Name

ONE_TGT_DEV_WRONG_TEST_MODE

Message

This command cannot be completed; An SDS device test is currently in progress.

Commands

`activate_sds_device` with `activate_all_devices` parameter.

Description

A device test is currently in progress, and the device cannot be activated until the test is finished.

Action

Wait for the device test to finish, and then try again.

Enterprise features not enabled

Parameters

Name

ENTERPRISE_FEATURES_NOT_ENABLED

Message

Enterprise features are not enabled for this system.

Commands

Any command that requires a special license.

Description

The command that you tried to use cannot be used with the type of license that you have activated in your system.

Action

Verify that you are using a license with enterprise features enabled. If problems persist, contact Dell Technologies support.

Volume already mapped to an SDC

Parameters

Name

VOL_ALREADY_MAPPED_TO_AN_INI

Message

The volume is already mapped to a SDC.

Commands

`map_volume_to_sdc` with the `all_sdcs` option.

Description

If the volume is already mapped to one or more SDCs, you cannot use the `all_sdcs` option.

Action

Remove the volume's mapping to the SDCs, and then try again.

SDS already exists and is being removed

Parameters

Name

TGT_ALREADY_EXISTS_AND_BEING_REMOVED

Message

The SDS exists and is being removed. Wait for the operation to finish.

Commands

`add_sds`

Description

If you try to add an SDS that exists, and is in the process of being removed, this operation fails.

NOTE: If an `add_sds` command failed, the system tries to remove it automatically, without user intervention. Therefore, if you issue the same `add_sds` command twice in a row, and the first time failed, this return code may be generated (depending on why adding the SDS failed).

Action

Verify that you are adding the correct SDS. If you are adding an SDS which is in the process of being removed, wait for the removal process to finish before trying to add the SDS back to the system.

Invalid spare percentage

Parameters

Name

INVALID_SPARE_PERCENTAGE

Message

The spare percentage value that you provided is invalid. Use a value between 0 - 99.

Commands

`modify_spare_policy`

Description

The spare percentage that is required in the command is invalid.

Action

Verify that the correct input parameters (a value between 0-99) are used with the command.

Invalid background device scanner bandwidth limit

Parameters

Name

INVALID_SCANNER_BW_LIMIT

Message

The scanner bandwidth limit is invalid. Please use a value between 10KB - 1GB.

Commands

`enable_background_device_scanner`

Description

The bandwidth value that you provided for the bandwidth limit parameter is invalid.

Action

Verify that correct input parameters (a value between 10KB–1GB) are used with the command.

Zero padding with background device scanner data comparison mode

Parameters

Name

ZEROPAD_AND_SCANNER

Message

Zero padding must be enabled in order to set scanner to data comparison mode.

Commands

`enable_background_device_scanner`

Description

The background device scanner has two modes: device only, and data comparison mode. The storage pool must have zero padding enabled before you attempt to enable background device scanning in data comparison mode.

Action

Enable zero padding in the storage pool (you must do this before adding devices to the storage pool).

IP address already assigned to another SDS

Parameters

Name

ADDRESS_BELONGS_TO_DIFFERENT_TGT

Message

The given address belongs to a different SDS. Check the system configuration.

Commands

`add_sds`, `add_sds_ip`

Description

This message is returned if the specified IP address has already been assigned to another SDS in the system.

Action

Check SDS network configuration and IP address assignments.

Capacity too low for thin volumes

Parameters

Name

CAPACITY_TOO_LOW_FOR_THIN_VOL

Message

Storage pool reached critical capacity utilization. Unable to create new thin volumes.

Commands

`add_volume`, `modify_volume_capacity`

Description

Thin volumes cannot be allocated if the specified storage pool has reached the critical capacity level.

Action

Add more devices to the storage pool.

Too many storage pools in the system

Parameters

Name

TOO_MANY_STORAGE_POOLS

Message

There are too many storage pools in the system.

Commands

`add_storage_pool`

Description

Addition of another storage pool exceeds the system limit of allowed storage pools.

Action

Remove unused storage pools. If the problem is not solved, install a new system.

Remote access to the MDM is blocked

Parameters

Name

REMOTE_PERMISSION_DENIED

Message

Permission denied. Remote read-only limit state is enabled and is blocking this operation from a remote location. Run this command from the primary MDM host.

Commands

All

Description

Remote read-only limit state is enabled and is blocking this operation from a remote location.

Action

Run the command from the primary MDM host.

Cannot add a partitioned device

Parameters

Name

TGT_ADD_DEV_IS_PARTITIONED

Message

Unable to add a device that was already partitioned.

Commands

`add_sds`, `add_sds_device`

Description

The added device is partitioned and cannot be used by the system.

Action

Remove partitions from the device, or add the partitioned device to the system.

Cannot add a mounted device

Parameters

Name

TGT_ADD_DEV_IS_MOUNTED

Message

Unable to add a device that was already mounted.

Commands

`add_sds`, `add_sds_device`

Description

The added device is mounted and cannot be used by the system.

Action

Unmount the device.

Device header mismatch

Parameters

Name

INVALID_DEVICE_HEADER_SIGNATURE

Message

Found an invalid device header signature.

Commands

`add_sds`, `add_sds_device`

Description

The device was previously added to the system, or after SDS reboot the signature of the header does not match the expected signature.

Action

Check the specific SDS devices configuration.

Device has user content

Parameters

Name

USER_CONTENT_DETECTED

Message

User content is present on the device.

Commands

`add_sds`, `add_sds_device`

Action

Clear the device from all content or partitions before adding the device.

Missing storage device media type

Parameters

Name

MISSING_MEDIA_TYPE

Message

The media type is not configured.

Commands

`add_storage_pool`, `add_sds`, `add_sds_device`, `set_storage_pool_media_type`

Action

Specify the media type of the storage devices when issuing the command. Storage pools must contain devices of the same media type.

Invalid media type

Parameters

Name

INVALID_MEDIA_TYPE

Message

The media type is invalid.

Commands

`add_storage_pool`, `add_sds`, `add_sds_device`, `set_storage_pool_media_type` , `set_sds_device_media_type`

Action

Issue the command with an appropriate media type.

Media type mismatch

Parameters

Name

MEDIA_TYPE_MISMATCH

Message

The device media type is not compatible with the storage pool media type.

Commands

`add_sds`, `add_sds_device`, `set_storage_pool_media_type`, `set_sds_device_media_type`

Action

Ensure that the storage device and the storage pool are the same media type. For example, storage pool defined for HDD can only contain HDD devices.

Invalid external acceleration type

Parameters

Name

INVALID_EXTERNAL_ACCELERATION_TYPE

Message

The external acceleration type is invalid.

Commands

`add_sds_device`, `set_storage_pool_external_acceleration_type` , `set_sds_device_external_acceleration_type`

Action

Issue the command with an appropriate external acceleration type.

External acceleration type mismatch

Parameters

Name

EXTERNAL_ACCELERATION_TYPE_MISMATCH

Message

The device external acceleration type is not compatible with the storage pool external acceleration type.

Commands

`set_storage_pool_external_acceleration_type`

Action

Ensure that the external acceleration type assigned to the storage pool and its storage devices are set to the same type.

Metadata feature cannot be configured on the acceleration device

Parameters

Name

TGT_DEV_METADATA_NOT_CONFIGURABLE

Message

The SDS device metadata feature cannot be configured for the selected devices because they are still undergoing previous configuration changes.

Commands

`enable_metadata_acceleration` , `disable_metadata_acceleration`, `modify_metadata_acceleration`, `enable_metadata_acceleration_per_device`, `disable_metadata_acceleration_per_device`, `modify_metadata_acceleration_per_device`

Action

Wait until the current configuration changes have been applied, and then try again.

Metadata feature is not enabled on the storage pool

Parameters

Name

TGT_DEV_METADATA_NOT_ENABLED_ON_SP

Message

The SDS device metadata feature is not enabled on the selected storage pool.

Commands

`modify_metadata_acceleration`

Action

Enable the feature at storage pool level and then issue the command again.

Metadata feature is already enabled on the storage pool

Parameters

Name

TGT_DEV_METADATA_ALREADY_ENABLED_ON_SP

Message

The SDS device metadata feature is already enabled on the selected storage pool. To change the feature settings, use the modify command.

Commands

`enable_metadata_acceration`

Action

None required

Metadata feature is not enabled on the devices

Parameters

Name

TGT_DEV_METADATA_NOT_ENABLED_ON_DEVICES

Message

The device metadata feature is not enabled on some or all of the selected devices.

Commands

`modify_metadata_acceleration_per_device`

Action

Enable the metadata feature on all of the selected devices.

Metadata is already enabled on devices

Parameters

Name

TGT_DEV_METADATA_ALREADY_ENABLED_ON_DEVICES

Message

The SDS device metadata feature is already enabled on some or all of the selected devices. To change the feature settings for those devices, use the modify command.

Commands

`modify_metadata_acceleration_per_device`

Action

None required.

Insufficient capacity for accelerating metadata

Parameters

Name

INSUFFICIENT_ACC_METADATA_CAPACITY

Message

The required acceleration capacity for metadata exceeds the amount of available capacity on the acceleration device(s).

Commands

`modify_metadata_acceleration_per_device`, `add_sds`, `add_sds_device`, `enable_metadata_acceleration_per_device`

Action

Free capacity in the acceleration pool, or add one or more acceleration devices to the pool.

28

NVMe host and SDC commands

This section contains commands for managing NVMe host and SDC.

Understanding the effect of network latency on NVMe over TCP latency counters

This section describes the possible effects of network latency between the host and storage data target (SDT) in an NVMe over TCP environment.

To access the performance counters, run the following commands:

- For storage data target

```
scli --query_properties --object_type SDT --all_objects --properties
SDT_DATA_READ_LATENCY,SDT_DATA_WRITE_LATENCY,SDT_HOST_READ_LATENCY,SDT_HOST_WRITE_LATE
NCY
```

Output sample:

```
754135e800000000:
          SDT_DATA_READ_LATENCY             3595 IOPS 0.823 ms average latency
          SDT_DATA_WRITE_LATENCY          0 IOPS 0 us average latency
          SDT_HOST_READ_LATENCY             899 IOPS 82 microseconds average latency
          SDT_HOST_WRITE_LATENCY          899 IOPS 12 microseconds average latency
```

- For host:

```
scli --query_properties --object_type HOST --all_objects --properties
USER_DATA_SDC_READ_LATENCY,USER_DATA_SDC_WRITE_LATENCY
```

Output sample:

```
f3e742f300010000:
          USER_DATA_SDC_READ_LATENCY        930 IOPS 69 microseconds average latency
          USER_DATA_SDC_WRITE_LATENCY       930 IOPS 12 microseconds average latency
```

Table 7. I/O latency measurement by counters

PowerFlex object	Performance counter	Measures I/O latency		
		Type	From	To
storage data target	SDT_DATA_**READ**_LATENCY	Read	SDS	storage data target
	SDT_DATA_**WRITE**_LATENCY	Write	storage data target	SDS
	SDT_HOST_**READ**_LATENCY	Read	storage data target	Host
	SDT_HOST_**WRITE**_LATENCY	Write	Host	storage data target
HOST	USER_DATA_SDC_**READ**_LATENCY	Read	SDS	storage data target
	USER_DATA_SDC_**WRITE**_LATENCY	Write	storage data target	SDS

NOTE: The term storage data client (SDC) used in the counter name is for backward compatibility purposes only.

Limitations of measuring latency between the host and SDT

In an NVMe over TCP setup, PowerFlex does not have a component that is installed on the host. As a result of the above fact there is no way of knowing when the host initiates the I/O request or when it receives requested data or a response about data it sent. Therefore, it is impossible to measure latency between the host and storage data target. The effect this has on the host-related counters are described below. Counters for traffic between storage data server (SDS) and storage data target (SDT) are reliable and reflect any network latency between those nodes.

Exception

Latency can be measured in one scenario. The following table describes how network latency affects the counters that measure latency between the host and the storage data target.

Table 8. Network latency effect on counters measuring latency between host and storage data target

I/O type	I/O size	I/O direction	Result
READ	All	Host to and from the storage data target	Values are estimates based on the I/O size. Network latency does not affect the values, and the values cannot provide any information about such issues.
WRITE	Less than or equal to 4 KB		
	More than 4 KB	Storage data target to the host	PowerFlex can accurately measure the network latency between the host and the storage data target. The network latency affects the counter value **twice**. That is, if network latency is increased by 100 microseconds then the counter value increases by 200 microseconds.
		Host to the storage data target	

Significance of large write requests

In all the host to storage data target flow cases, the host sends a read or write request to the storage data target.

The flow is complicated while writing large IOs in the host to the storage data target:

1. Host sends a write request to the storage data target.
2. Storage data target sends the host a request to send the write data.
3. Host sends the write data to the storage data target.

Since the system knows when it initiated step 2 and it knows when step 3 is completed it can measure the time of these steps. Any network latency affects both steps 2 and 3, and this is the reason that the latency affects twice as described in the table above.

Summary

To check for possible effects of network latency between the host and the storage data target in an NVMe over TCP setup, complete the following:

1. Run a write I/O load with I/O size larger than 4 KB and examine the **SDT_HOST_WRITE_LATENCY** counter.
2. If a baseline value is available to compare with, divide the delta of the two counter values by two for an estimate of the network latency.

add_sdc

Approve an SDC to map volumes and allow an SDC to access the MDM.

Syntax

```
scli --add_sdc --sdc_ip <ip> | --sdc_guid <guid>
  [--sdc_name <name>]
```

Parameters

--sdc_ip *<ip>*
SDC IP address

--sdc_guid *<guid>*
SDC GUID

--sdc_name *<name>*
Name to assign to the SDC

NOTE: You can find the GUID of the SDC using any of these methods:

- From the command line, run:

  ```
  scli --query_all_sdc
  ```

- View the properties of the SDC in PowerFlex Manager.
- Locally in `/opt/emc/scaleio/sdc/bin/drv_cfg`, with root access permission, run:

  ```
  drv_cfg --query_guid
  ```

Example

```
scli --add_sdc --sdc_ip 10.125.120.53 --sdc_name sdc53
```

remove_host

Remove a host from the system.

Syntax

```
scli --remove_host (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --
sdc_guid <GUID> | --sdc_ip <IP>)
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--sdc_guid *<GUID>*

SDC global unique identifier

--sdc_ip *<IP>*

SDC IP address

Example

```
scli --remove_host --host_name host1
```

rename_host

Name or rename an existing host.

Syntax

```
scli --rename_host (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --
sdc_guid <GUID> | --sdc_ip <IP>) --new_name <NAME>
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--sdc_guid *<GUID>*

SDC global unique identifier

---sdc_ip *<IP>*

SDC IP address

- --new_name *<NAME>*

New name assigned to the host.

Example

```
scli --rename_host --host_name host1 --new_name new_host_name
```

check_sdc_authentication_status

Print the SDC authentication password status for all SDCs.

Syntax

```
scli --check_sdc_authentication_status [--run_test] [--filename <FILENAME>]
```

Parameters

--run_test

Run a connectivity test to check whether the SDCs can successfully authenticate using CHAP.

--filename *<FILENAME>*

The filename and filepath where detailed SDC authentication information will be stored. This parameter is mandatory when using the `--run_test` option.

Example

```
scli --mdm_ip 192.168.1.200 --check_sdc_authentication_status --run_test --filename
report.txt
```

generate_sdc_password

Generate a new SDC authentication password for the specified SDC.

Syntax

```
scli --generate_sdc_password (--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> |
--sdc_ip <IP>) [--reason <REASON>]
```

Parameters

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC GUID

--sdc_ip *<IP>*

SDC IP address

--reason *<REASON>*

Free text to describe the reason for the SDC password reset

Example

```
scli --mdm_ip 192.168.1.200 --generate_sdc_password --sdc_name SDC1 --reason
MonthlyUpdate
```

query_all_approved_sdc

Query the system when it is operating in restricted SDC mode for all SDCs that have been approved by the system.

NOTE: This query is useful when restricted SDC mode is enabled and you want to determine which SDCs can be used for volume mapping.

Syntax

```
scli  --query_all_approved_sdc
```

Parameters

None.

Example

```
scli --query_all_approved_sdc
```

query_host

Retrieve information about a host.

Syntax

```
scli --query_host (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --
sdc_guid <GUID> | --sdc_ip <IP>)
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe Host NQN

--sdc_guid *<GUID>*
SDC Global Unique Identifier

--sdc_ip *<IP>*
SDC IP address

query_all_hosts

Retrieve information about all SDCs and NVMe hosts.

Syntax

```
scli --query_all_hosts
```

query_potential_nvme_hosts

Retrieve information about all potential NVMe hosts. The query result can be optionally filtered by protection domain.

Syntax

```
scli --query_potential_nvme_hosts (--all | --protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

Parameters

--all

Retrieve information about all potential hosts.

--protection_domain_id *<ID>*

Protection domain ID

-protection_domain_name *<NAME>*

Protection domain name

query_all_sdc

Retrieve information about all SDCs in the system.

Syntax

```
scli --query_all_sdc [--agent_status] [--mdm_ip_addrs_current] [--mdm_ip_addrs_hardened]
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_sdc
```

Sample output

```
MDM restricted SDC mode: NONE
Query all SDC returned 5 SDC nodes.
SDC ID: 30b5c18600000000 Name: N/A IP: 10.55.137.119 State: Connected GUID:
96CCE42A-6ADB-45EA-8E3F-B71C1F67808C  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
```

```
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18700000001 Name: N/A IP: 10.55.137.114 State: Connected GUID:
C08C87B5-7E28-421A-A82C-C9C5B3070FF2  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18800000002 Name: N/A IP: 10.55.137.115 State: Connected GUID:
89947FF3-6E84-4DF9-B869-481BCB8E3B3D  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18900000003 Name: N/A IP: 10.55.137.116 State: Connected GUID: 7F5BC40E-
A1FB-44EC-A8B0-76C63A7B1123  OS Type: LINUX Loaded Version: 3.6.0 Installed Version:
3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18a00000004 Name: N/A IP: 10.55.137.117 State: Connected GUID:
34EE9476-65D3-44C1-83DC-3CF6912CCB86  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
```

query_restricted_sdc_mode

Query whether restricted SDC mode is enabled or disabled in the PowerFlex system.

If restricted SDC mode is enabled, you must register to the system each SDC to which you want to map volumes using the `--add_sdc` command.

Syntax

```
scli --query_restricted_sdc_mode
```

Parameters

None.

Example

```
scli --query_restricted_sdc_mode
```

query_sdc

Retrieve information about the specified SDC.

Syntax

```
scli --query_sdc --sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>
  [--show_oscillating_failures]
```

Parameters

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC global unique Identifier

--sdc_ip *<IP>*

SDC IP address

--show_oscillating_failures

Show extended oscillating failure information if there are one or more failures.

Example

```
scli --mdm_ip 192.168.1.200 --query_sdc --sdc_id c27ab52600000001
```

Sample output

```
SDC ID: c27ab52600000001 Name: N/A IP: 10.55.137.119 State: Connected GUID:
C6533636-224B-44E3-A9EB-3CD651116CF5  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  525 IOPS 262.4 MB (268697 KB) per-second
    Writes: 503 IOPS 251.2 MB (257222 KB) per-second

Volume information (total 2 volumes):
         1: Volume ID: f6351e3900000005 Name: vol3_pol1 Provisioning: Thin Size: 8.0 GB
(8192 MB) Access mode: READ_WRITE
         2: Volume ID: f6351e3700000003 Name: vol1_114 Provisioning: Thin Size: 16.0 GB
(16384 MB) Access mode: READ_WRITE
```

query_sdc_volume_limits

Retrieve the IOP and bandwidth limits that one SDC generates for the specified volume.

Syntax

```
scli --query_sdc_volume_limits (--volume_id <ID> | --volume_name <NAME>) (--sdc_id <ID>
| --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>)
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC global unique Identifier

--sdc_ip *<IP>*

SDC IP address

Example

```
scli --mdm_ip 192.168.1.200 --query_sdc_volume_limits --volume_id f6351e3700000003 --
sdc_id c27ab52600000001 --mdm_port 6611
```

Sample output

```
Successfully queried limits from SDC with ID c27ab52600000001 to volume with ID
f6351e3700000003: IOPS limit is unlimited. Bandwidth limit is unlimited.
```

reset_sdc_sds_connections

Reset SDC-SDS connections.

Syntax

```
scli --reset_sdc_sds_connections --i_am_sure
```

Parameters

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --reset_sdc_sds_connections --i_am_sure
```

set_restricted_sdc_mode

Enable or disable restricted SDC mode. When enabled, SDCs must be approved (by GUID or IP address) before volumes can be mapped to them.

NOTE: In a system that is upgraded and already has volumes mapped to SDCs, if you want to enable restricted SDC mode, you must first approve the SDCs and only then enable restricted SDC mode. Use the `add_sdc` command to approve SDCs. After approving SDCs, add all the SDCs to the approved list before enabling the restricted mode.

Syntax

```
scli --set_restricted_sdc_mode --restricted_sdc_mode {approved_ip | guid | none}
```

Parameters

--restricted_sdc_mode {approved_ip | guid | none

Set the restricted SDC mode to one of the following:

- `approved_ip` — Volumes can be mapped only to approved SDCs that are also approved by IP address.
- `guid` — Volumes can be mapped only to approved SDCs.
- `none` — Restricted SDC mode is disabled and volumes can be mapped to any SDC.

Example

```
scli --add_sdc --sdc_ip approved_ip | guid | none
```

```
scli --set_approved_sdc_ips --sdc_id --sdc_ips approved_ip | guid | none
```

```
scli --set_restricted_sdc_mode --restricted_sdc_mode approved_ip | guid | none
```

set_sdc_authentication

Enable or disable SDC authentication. This feature ensures security by enforcing password-based authentication of the SDC to the MDM for access to mapped volumes. This prevents the SDC from accessing unauthorized volumes.

Syntax

```
scli --set_sdc_authentication (--enable | --disable) [--allow_old_sdc] [--
allow_no_sdc_password]
```

Parameters

--enable

Enable SDC authentication and authorization

--disable

Disable SDC authentication and authorization

--allow_old_sdc

Allow enabling SDC authentication and authorization when there is an old SDC. All old SDCs will be disconnected

--allow_no_sdc_password

Allow enabling SDC authentication and authorization when there is an SDC without a password. All SDCs without a password will be disconnected.

Example

```
scli --mdm_ip 192.168.1.200 --set_sdc_authentication --enable --allow_old_sdc
```

set_sdc_volume_limits

Set limits to the IOPS and bandwidth that one SDC generates for the specified volume. This enables you to control the quality of service (QoS).

Syntax

```
scli --set_sdc_volume_limits (--volume_id <ID> | --volume_name <NAME>) (--sdc_id <ID> |
--sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>) (Options)
  [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC global unique Identifier

--sdc_ip *<IP>*

SDC IP address

Options

Choose at least one:

--limit_iops *<NUMBER>*

Limit the volume IOPS. The number of IOPS must be larger than 10. 0 is unlimited.

--limit_bandwidth *<NUMBER>*

Limits the volume network bandwidth. The bandwidth is in MB/s. 0 is unlimited

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --set_sdc_volume_limit
--volume_name vol1 --sdc_ip 192.168.1.3 --limit_iops 100
```

set_approved_sdc_ips

Set a list of up to four IP addresses that are approved for the given SDC.

Syntax

```
scli --set_approved_sdc_ips (--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --
sdc_ip <IP>) --sdc_ips <IPs>
```

Parameters

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC GUID

--sdc_ip *<IP>*
IP address of an SDC

--sdc_ips *<IPS>*
A comma-separated list of SDC IP addresses. Omit the space after each comma.

Example

```
scli --set_approved_sdc_ips --sdc_ips 10.76.60.10,10.76.60.11,10.76.60.12,10.76.60.13
```

add_nvme_host

Add an NVMe host to the system.

Syntax

```
scli --add_nvme_host --nvme_host_nqn <NQN> [--host_name <NAME>] [--max_number_paths
<VALUE>] [--max_number_sys_ports <VALUE>] [--host_os_full_type <OS>] [--host_group_name
<ID> | --host_group_id <NAME>] | --force
```

Parameters

--nvme_host_nqn *<NQN>*
NVMe host NQN

--host_name *<NAME>*
Host name

--max_number_paths *<VALUE>*
Maximal number of paths allowed per mapped volume

Valid range: 2 to 8

Dafault: 4

-max_number_sys_ports *<VALUE>*-

Maximal number of system ports allowed per protection domain

Valid range: 2 to 16

Dafault: 10

--host_os_full_type *<OS>*

OS type. Used for OS with known NVMe host issues

Options:

- generic (default)
- powerflex

--host_group_id *<ID>*

Host group ID.

--host_group_name *<NAME>*

Host group name.

--force

Force NQN, ignore all NQN formating rules.

Example

```
scli --add_nvme_host --host_name host1 --nvme_host_nqn
nqn.2014-08.org.nvmexpress:uuid:11111111-1111-2222-3333-444455556666
```

modify_nvme_host_max_paths

Modify the maximum number of paths per mapped volume for an existing NVMe host.

Syntax

```
scli --modify_nvme_host_max_paths (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn
<NQN>) --max_number_paths <VALUE>
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--volume_id *<ID>*

Volume ID

--max_number_paths *<VALUE>*

Maximal number of paths allowed per mapped volume

Valid range: 2-8

Default: 4

Example

```
scli --modify_nvme_host_max_paths --host_name host1 --max_number_paths 6
```

modify_nvme_host_max_sys_ports

Modify the maximum number of system ports per protection domain for an existing NVMe host.

Syntax

```
scli --modify_nvme_host_max_sys_ports (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) --max_number_sys_ports <VALUE>
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--max_number_sys_ports *<VALUE>*

Maximal number of system ports allowed per protection domain

Valid range: 2-128

Default: 10

Example

```
scli --modify_nvme_host_max_sys_ports --host_name host1 --max_number_sys_ports 6
```

modify_nvme_host_os_full_type

Modify the operating system (OS) type and version for an existing NVMe host.

Syntax

```
scli --modify_nvme_host_os_full_type (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) --host_os_full_type <OS>
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe Host NQN

--host_os_full_type *<OS>*

OS type. Used for OS with known NVMe host issues

Options:

- generic (default)
- powerflex

set_manual_nvme_host_connectivity

Set a manual connectivity goal for NVMe host to a protection domain.

Syntax

```
scli --set_manual_nvme_host_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--protection_domain_id <ID> | --protection_domain_name <NAME>) (--
system_port_ip <IP> | --system_port_id <ID>) --force
```

Parameters

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--system_port_ip *<IP>*

A comma separated list of system port IP addresses.

--system_port_id *<ID>*

A comma separated list of system port IDs (4 hexadecimal digits).

--force

Force non-standard connectivity settings

reset_manual_nvme_host_connectivity

Reset manual connectivity goal for NVMe host to a protection domain.

Syntax

```
scli --reset_manual_nvme_host_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--protection_domain_id <ID> | --protection_domain_name <NAME>)
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

set_manual_nvme_mapping_connectivity

Set manual connectivity for mapping of a volume to an NVMe host.

Syntax

```
scli --set_manual_nvme_mapping_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--volume_id <ID> | --volume_name <NAME>) [--protection_domain_id
<ID> | --protection_domain_name <NAME>] (--system_port_ip <IP> | --system_port_id <ID>)
--force
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--system_port_ip *<IP>*
A comma separated list of system port IP addresses.

--system_port_id *<ID>*
A comma separated list of system port IDs (4 hexadecimal digits)

--force
Force non standard connectivity settings

reset_manual_nvme_mapping_connectivity

Reset manual connectivity of volume mapping to an NVMe host.

Syntax

```
scli --reset_manual_nvme_mapping_connectivity (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>) (--volume_id <ID> | --volume_name <NAME>) [--protection_domain_id
<ID> | --protection_domain_name <NAME>]
```

Parameters

--host_id *<ID>*
Host ID

--host_name *<NAME>*
Host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

rebalance_nvme_host_connectivity

Set connectivity goal and system ports balancing on all system NVMe Hosts.

Syntax

```
scli --rebalance_nvme_host_connectivity
```

query_nvme_discovery_info

Get the information required to set up a discovery service on hosts.

Syntax

```
scli --query_nvme_discovery_info (--protection_domain_id <ID> | --protection_domain_name
<NAME>) --service <TYPE> --transport <TYPE>
```

Parameters

--protection_domain_id *<ID>*

The failure domain ID

--protection_domain_name *<NAME>*

The failure domain name

--service *<TYPE>*

The service type

Option: discovery

--transport *<TYPE>*

The transport type

Option: TCP

--remote_system_description *<DESCRIPTION>*

Description of the remote system

--transport *<TRANSPORT_TYPE>*

Currently only TCP is supported

clear_nvme_reservation

Clear NVMe reservation and registration from the given volume. If there is a host that is a registrant and has an active controller, the command fails unless the `--force` flag is used.

Syntax

```
scli --clear_nvme_reservation (--volume_id <ID> | --volume_name <NAME>) [--wait] [--
force] [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--wait

Wait for the command to complete. If specified, the return code is the actual return code of the command

--force

Force clearing NVMe reservation

--i_am_sure

Preemptive approval

add_host_group

Add a host group to the system.

Syntax

```
scli --add_host_group [--host_group_name <NAME>] [--network_id <ID> | --network_name
<NAME>] [--host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN>] [--force]
```

Parameters

--host_group_name *<NAME>*
: A name to be assigned to the host group

--network_id *<ID>*
: A comma separated list of system network IDs to be associated with the group

--network_name *<NAME>*
: A comma separated list of system network names to be associated with the group

--host_id *<ID>*
: A comma separated list of NVMe host IDs to be added to the group

--host_name *<NAME>*
: A comma separated list of NVMe host names to be added to the group

--nvme_host_nqn *<NQN>*
: A comma separated list of NVMe host NQNs to be added to the group

--force
: Force assigning NVMe hosts to group regardless of system networks accessibility

assign_host_to_host_group

Assign a host to a host group.

Syntax

```
scli --assign_host_to_host_group (--host_id <ID> | --host_name <NAME> | --nvme_host_nqn
<NQN>) (--host_group_id <ID> | --host_group_name <NAME>) [--force]
```

Parameters

--host_id *<ID>*
: Host ID

--host_name *<NAME>*
: Host name

--nvme_host_nqn *<NQN>*
: NVMe host NQN

--host_group_id *<ID>*
: Host group ID

--host_group_name *<NAME>*
: Host group name

--force *<NAME>*

Force assigning NVMe host to host group regardless of system networks accessibility

query_all_host_groups

Retrieve information about all host groups.

query_host_group

Retrieve information about a host group.

Syntax

```
scli --query_host_group (--host_group_id <ID> | --host_group_name <NAME>)
```

Parameters

--host_group_id *<ID>*

Host group ID

--host_group_name *<NAME>*

Host group name

modify_host_group

Modify the list of system networks of an existing host group.

Syntax

```
scli --modify_host_group (--host_group_id <ID> | --host_group_name <NAME>) [--network_id
<ID> | --network_name <NAME>]
```

Parameters

--host_group_id *<ID>*

Host group ID

--host_group_name *<NAME>*

Host group name

--network_id *<ID>*

A comma separated list of system network IDs to be replaced with the old one

--network_name *<NAME>*

A comma separated list of system network names to be replaced with the old one

rename_host_group

Name or rename a host group.

Syntax

```
scli --rename_host_group (--host_group_id <ID> | --host_group_name <NAME>) --new_name
<NAME>
```

Parameters

--host_group_id *<ID>*
: Host group ID

--host_group_name *<NAME>*
: Host group name

--new_name *<NAME>*
: A new name to be assigned to the host group

remove_host_from_host_group

Remove a host from a host group.

Syntax

```
scli --remove_host_from_host_group (--host_id <ID> | --host_name <NAME> | --
nvme_host_nqn <NQN>)
```

Parameters

--host_id *<ID>*
: Host ID

--host_name *<NAME>*
: Host name

--nvme_host_nqn *<NQN>*
: NVMe host NQN

remove_host_group

Remove a host group from the system.

Syntax

```
scli --remove_host_group (--host_group_id <ID> | --host_group_name <NAME>)
```

Parameters

--host_group_id *<ID>*

Host group ID

--host_group_name *<NAME>*

Host group name

set_auto_update_sdc_with_mdm_ip_addresses

Enable or disable SDC automatic update with MDM IP addresses.

Syntax

```
scli --set_auto_update_sdc_with_mdm_ip_addresses ((--enable [--force] (--
mdm_control_ip_addrs | --mdm_management_ip_addrs) | disable)
```

Parameters

--enable

Enable SDC automatic update with MDM IP addresses.

--disable

Disable SDC automatic update with MDM IP addresses.

--force

Enable update with MDM IP addresses though some SDCs are disconnected

--mdm_control_ip_addrs

Set auto update of SDC with MDM IP addresses to control IPs

--mdm_management_ip_addrs

Set auto update of SDC with MDM IP addresses to management IPs

NOTE: To enable this command, you must deploy the SDC hardening agent. For more information, see the Dell PowerFlex Manager MDM IP Address Dynamic Updates TechNote.

restart_connections_to_realign_net_path

Set a restart connection to realign the network path interval.

Syntax

```
scli --restart_connections_to_realign_net_path [--disable | --enable | --force | --
interval_hour <hours> | --enable --interval_hour <hours>]
```

Parameters

--enable

Restart connections to realign network path. The default interval is 24 hours.

--disable

Disable periodic connections realignment.

--interval_hour *<INTERVAL_HOUR>*

Set the restart connection interval, 1-168 hours.

--force

Force on time refresh, even if periodic connections realignment has been disabled.

29

SDC operations

PowerFlex uses different methods to add external SDCs to an existing PowerFlex system, depending on the operation system of the server. For existing SDCs, many operations use the drv_cfg command line, which is a local utility that affects only the client on which the SDC is running.

Add an external SDC to an existing PowerFlex system

Add an external SDC to an existing PowerFlex system.

During manual installation of an SDC, you can install the SDC according to the operating system-specific instructions in the following section, and it will be connected to the existing PowerFlex system.

Install the SDC on an ESXi-based server and connect it to PowerFlex using esxcli

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this task, to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Customer Support site)
- A GUID string for the SDC. These strings can be generated by tools that are freely available online. The GUID needs to conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

About this task

The following procedure explains how to manually install an external SDC on an ESXi-based using esxcli in command line.

NOTE: This procedure requires two server reboots.

Steps

1. On the ESXi-based server on which you are installing the SDC, set the acceptance level:

```
esxcli software acceptance set --level PartnerSupported
```

2. Install the SDC:

Option	Description
ESXi 6.x	`esxcli software vib install -d <full_path_to_offline_bundle>`

Option	Description
	where *<full_path_to_offline_bundle>* is the absolute path to the SDC VIB file. **NOTE:** The SDC offline ESXi 6.0 bundle is the same for all ESXi 6.x versions.
ESXi 7.x	`esxcli software component apply -d <path_to_zip> -n scaleio-sdc-esx7` where *<path_to_zip>* is the path to the SDC zip file.

3. Get permission to reboot the ESXi-based server, and perform a graceful reboot cycle to load the SDC driver on the server.

   In a first-time install, the SDC will not automatically boot up at this point. Further configuration and an additional reboot are required, as described in this procedure.
4. Set the IP address of the MDM:

   ```
   esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
   IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
   ```

   where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
     - Separate the IP addresses of the same MDM cluster with a "," symbol.
     - Separate multiple MDM clusters with the "+" symbol.

       In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one: `10.20.30.40,50.60.70.80+11.22.33.44`
   - *<XXXXXX>* is the user-generated GUID string. Example GUID: 12345678-90AB-CDEF-1234-567890ABCDEF
   - Optional: `IoctlMdmPasswordStr` is used for SDC authentication to the MDM for access to the PowerFlex system.
5. Back up the parameters:

   ```
   /bin/auto-backup.sh
   ```
6. Perform another graceful reboot cycle.
7. Ensure that the driver is loaded:

   ```
   vmkload_mod -l | grep scini
   ```

   Output similar to the following indicates that the driver is loaded:

   ```
   scini 2 808
   ```

   The SDC component is installed on the ESXi-based server.

Results

The SDC is installed on the ESXi-based server and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Install SDC on a Linux-based server and connect it to PowerFlex

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC operating system (available from the zipped software packages that can be downloaded from the Dell Technologies Support site).
- Secure Boot is not supported; ensure that Secure Boot is disabled in the BIOS.

About this task

The following procedure explains how to manually install an external SDC on a Linux-based server. External SDCs can be deployed by the PowerFlex Installer.

The steps in the procedure below are relevant for Linux Red Hat Enterprise Linux/CentOS/Oracle Linux operating systems.

Table 9. Modifications required for deployment on other operating systems

Operating system	Modifications required
Ubuntu/OL	Before installing PowerFlex, ensure that you have followed the required preparation procedures relating to various types of Linux operating systems. ● PowerFlex component packages are delivered as TAR files. Before installing, perform the following: 1. Untar all the packages: `tar -xvf <tar_file>`. This yields SIOB files. 2. Extract the DEB from the SIOB files: `siob_extract <siob_file>` 3. The DEB file is in the following format: `EMC-ScaleIO-sdc-.<operating_system>.X.X.x86_64.deb` Use the extracted DEB files for the installation. ● Some commands are a bit different, noted where applicable.

Steps

1. Install the GPG key on every server on which SDC must be installed. From the PowerFlex installation folder, run the following command on every server:

```
rpm --import RPM-GPG-KEY-ScaleIO
```

2. In the command line, install the SDC:
   - Red Hat Enterprise Linux/CentOS /Oracle Linux/SLES

```
MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
```

   - Ubuntu (These files must be extracted before use, as described above.)

```
MDM_IP=<LIST_VIP_MDM_IPS> dpkg -i <SDC_PATH>.deb
```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located.
3. You can use the following command to verify if the driver is loaded:
   - SLES 15:

```
service sdc status
```

   - All other Linux flavors:

```
service scini status
```

Results

The SDC is installed on the Linux-based server, and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks that are described in this guide.

In existing systems, map volumes to the new SDCs that were added to the system.

Install SDC on an AIX-based server and connect it to PowerFlex

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. These instructions are relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC operating system (available from the zipped software packages that can be downloaded from the Dell Technologies Support site).

About this task

The following procedure explains how to manually install an external SDC on an AIX-based server. The PowerFlex Installer cannot be used.

Steps

1. In the command line, install the SDC:

   ```
   MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
   ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located.

     The SDC package is in a format similar to this:
     `EMC-ScaleIO-sdc-.aix7.aix7.2.ppc.rpm`
2. Optional: For SDC authentication to the MDM for access, in the command line, enter:

   ```
   chdev -l scini -a "mdm_password=<ip>-<password>"
   ```

Results

The SDC is installed on the AIX-based server and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the recommended post-deployment tasks that are described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Install SDC on a Windows-based server and connect it to PowerFlex

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC operating system (available from the zipped software packages that can be downloaded from the Dell Technologies Support site).

About this task

The following procedure explains how to manually install an external SDC on a Windows-based server. Alternatively, you can install the external SDC using the PowerFlex Installer. For more information about adding system components to an existing system, see the *Dell PowerFlex and PowerFlex Manager Install and Upgrade Guide*.

NOTE: This procedure requires a server reboot.

Steps

1. On the Windows-based server on which you are installing the SDC, run the following command in the command line:

   ```
   msiexec /i <SDC_PATH>.msi MDM_IP=<LIST_VIP_MDM_IPS>
   ```

   Where
   - *<SDC_PATH>* is the path where the SDC installation package is located.
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not add tiebreaker addresses here.
2. Get permission to reboot the Windows-based server, and perform a reboot to load the SDC driver on the server.

Results

The SDC is installed on the Windows-based server, and is connected to PowerFlex.

Next steps

In newly deployed systems, perform the post-deployment tasks that are described in this guide.

In existing systems, map volumes to the new SDCs that you added to the system.

Performing SDC operations

Many SDC operations use `drv_cfg`. The `drv_cfg` command line is a local CLI utility that affects only the client on which the SDC is running.

Possible SDC operations include updating the SDC driver with IP changes, detecting new volumes, querying volumes, loading a configuration file, adding an MDM, modifying an MDM IP address, and enabling support of PDL state.

Update the SDC driver with IP address changes

Use the drv_cfg to update the IP address.

Steps

1. Edit `drv_cfg.txt` and change the IP address in the last line to the new IP address.

   Location of `drv_cfg.txt`:
   - Linux: `/etc/emc/scaleio/`
   - Windows: In the following registry key - `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\services\scini\Parameters\mdms`

   NOTE:

   On ESXi, GUID and MDM lists are stored as module parameters, and not in a `drv_cfg.txt` file. To modify these parameters, use `esxcli` commands.

2. Save and close the file.
3. Type the following command:

```
/etc/init.d/scini restart
```

NOTE: ESXi guidelines are described in the PowerFlex plug-in chapter.

Detecting new volumes

Volumes are always exposed to the operating system as devices with the prefix `scini` (such as `/dev/scinia`, `/dev/scinib` and so on). Unique names can be found under `/dev/disk/by-id/`.

PowerFlex periodically scans the system to detect new volumes. You can initiate a scan for the most up-to-date status on a particular SDC node. This command is unique because it is not a CLI command, but rather a command issued on the specific SDC.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`
- ESXi: See the vSphere Client VMware guidelines on how to detect new storage. If troubleshooting is needed, contact Customer support.

Command

drv_cfg --rescan

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --rescan
```

Query volumes using drv_cfg

This utility retrieves information about all known active volume objects in KVM mode. You can use this utility to determine which volumes are mapped, and the ID of each volume in PowerFlex.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

Command

drv_cfg --query_vols

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_vols
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_vols
```

Query tgt objects using drv_cfg

This utility retrieves information about all known active tgt objects (SDSs) in KVM mode.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

Command

`drv_cfg --query_tgts`

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_tgts
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_tgts
```

Query GUID using drv_cfg

This utility retrieves the unique ID of the KVM module. The utility can be used to verify that all SDC GUIDs in the system are unique.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

NOTE:

If the SDC was removed and reinstalled, the GUID of the SDC is different than its original GUID. In such a case, you may have to remove the SDC, if two SDCs now have the same GUID.

Command

`drv_cfg --query_guid`

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_guid
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_guid
```

Query MDMs using drv_cfg

This utility retrieves information about all known MDM objects in KVM mode. This utility is typically used to determine to which MDM an SDC is connected.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

Command

`drv_cfg --query_mdms`

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_mdms
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --query_mdms
```

Loading a configuration file using drv_cfg

This utility reads a configuration file containing MDM IP addresses, and calls the KVM to connect to them.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

The configuration file that is loaded when using the `drv_cfg --load_cfg_file` utility is not persistent; when you restart the SDC, the changes will be lost.

To make the changes persistent, perform either of the following:

- Install the SDC on every server that exposes volumes to the application running, by running the following command:

  ```
  MDM_IP=<IP of the MDM> rpm -i <full rpm file path>
  ```

- Use the following `drv_cfg` command:

```
/opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
--ip <EXISTING_MDM_IP_ADDRESS> --new_mdm_ip
<NEW_MDM_IP_ADDRESSES>
```

Command

drv_cfg --load_cfg_file

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

This command can not be used on ESXi servers. Instead, follow the steps described in "Modify parameters on ESXi servers".

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg
--load_cfg_file <FILE_NAME>
```

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg
--load_cfg_file /etc/emc/scaleio/drv_cfg.txt
```

Adding an MDM using drv_cfg

This utility calls the KVM module to connect to an MDM. This command is typically used in cases where an SDC is connected to more than one PowerFlex system.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

NOTE: Extending your PowerFlex system with another MDM requires that you update all SDCs in your system with the new MDM IP address. Run the `drv_cfg` utility with the `--add_mdm` option, and to make the change persistent, use the `--file` parameter. In addition, any additional objects or systems which interface with the MDM must also be updated. For more information, see Modifying an MDM IP address using drv_cfg.

Command

drv_cfg --add_mdm

NOTE:

This is not a CLI command, but rather an executable that is run on the SDC server.

This command cannot be used on ESXi servers.

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --add_mdm
--ip <MDM_IP_ADDRESS_LIST>
```

Parameters

Parameters

--ip <MDM_IP_ADDRESS_LIST>

List of IP addresses (comma delimited) for this primary or secondary MDM.

--file <CONFIG_FILE_NAME> (optional)

Name of the configuration file to which the MDM information should be written.

--only_cfg (optional)

Do not call the KVM to actually connect.

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --add_mdm
--ip 10.100.22.20,10.100.22.30
--file /etc/emc/scaleio/drv_cfg.txt
```

Modifying an MDM IP address using drv_cfg

This utility calls the KVM to modify the IP address list of an MDM. It is typically used in cases when an MDM IP address has changed, or when MDMs are added/removed from/to the system. The command must be run on every SDC in the system. To bring the changes into effect, a server restart is required.

Location of `drv_cfg` command:

- Linux: `/opt/emc/scaleio/sdc/bin/drv_cfg`
- Windows: `C:\Program Files\emc\scaleio\sdc\bin\drv_cfg`

NOTE: Extending your PowerFlex system with another MDM requires that you update all SDCs in your system with the new MDM IP address. Run the `drv_cfg` utility with the `--add_mdm` option, and to make the change persistent, use the `--file` parameter. In addition, any additional objects or systems which interface with the MDM must also be updated.

Command

drv_cfg --mod_mdm_ip

NOTE: This is not a CLI command, but rather an executable that is run on the SDC server. This command cannot be used on ESXi servers. Instead, follow the steps that are described in "Modify parameters in ESXi servers".

Syntax

```
/opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
--ip <EXISTING_MDM_IP_ADDRESS>
--new_mdm_ip <NEW_MDM_IP_ADDRESSES> [--file <CONFIG_FILE_NAME>]["-""-"only_cfg]
```

Parameters

Parameters

--ip <EXISTING_MDM_IP_ADDRESS>

One of the existing MDM IP addresses.

--new_mdm_ip <NEW_MDM_IP_ADDRESSES>

The new IP address list (comma delimited) for this MDM. If you want to retain the existing address(es), include them in this list.

--file <CONFIG_FILE_NAME> (optional)

The name of the configuration file to which the MDM information should be written.

--only_cfg (optional)

Do not call the KVM to actually connect.

Example

```
/opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
--ip 10.100.20.20
--new_mdm_ip 10.100.20.20,10.100.20.30,10.100.20.40
```

Permanent device loss state

When the MDM has disconnected from the SDC and a volume that is mapped to this SDC has experienced an I/O error, the ESXi host continuously sends I/Os to the device to determine if the device has become accessible. This can subsequently cause a high I/O-error load that can lead to the host freezing. In cases where the device is disconnected long-term, such as when the entire MDM cluster is down during a network upgrade, the SDC can change the volume state to permanent device loss (PDL) to prevent more I/O errors coming from the ESXi.

PDL is an ESXi state, which, once enabled, is supported in PowerFlex. Once the ESXi host loses connectivity with a device, if a timeout value is reached, the ESXi is notified that the device is in a PDL state. The timeout value can be manually set. Once a device is in a PDL state, the ESXi host no longer attempts to re-establish connectivity or issue commands to the device.

Recovering a device from PDL state is described in the VMware documentation for your operating system version.

Use the instructions that match your environment.

Enable support of PDL state on the ESXi

Enable support of permanent device loss (PDL) state on the ESXi host.

Steps

1. On the ESXi host:

   ```
   esxcli system module parameters set -m scini "<<PREVIOUS_MODULE_PARAMS>>
   bBlkDevIsPdlActive=1 blkDevPdlTimeoutMillis=<TIMEOUT_VALUE>"
   ```

   where:
   - *<<PREVIOUS_MODULE_PARAMS>>* is any previous module parameters being used for this ESXi host.
   - *<TIMEOUT_VALUE>* is the timeout time in milliseconds. Its value can be 1000-3600000 (default is 60000) and including it in the command is optional.
2. Reboot the host.

Disable support of PDL state on the ESXi

Disable support of permanent device loss (PDL) state on the ESXi host.

Steps

1. On the ESXi host:

   ```
   esxcli system module parameters set -m scini "<<PREVIOUS_MODULE_PARAMS>>
   bBlkDevIsPdlActive=0"
   ```

   Where *<<PREVIOUS_MODULE_PARAMS>>* is any previous module parameters being used for this ESXi host.

2. Reboot the host.

Run the SDC driver on Red Hat Enterprise Linux or SUSE Enterprise Linux Enterprise Server with multipath enabled

Run the SDC driver on Red Hat Enterprise Linux or SUSE Enterprise Linux Enterprise Server with multipath enabled

Steps

1. On the machine where the SDC is running, open `/etc/udev/rules.d/20-scini.rules` for editing.
2. In the first line at the end add:

```
ENV{ID_WWN}="%c"
```

   The line should read like this:

```
KERNEL=="scini*[!0-9]", SUBSYSTEM=="block", PROGRAM="/bin/emc/scaleio/drv_cfg --
query_block_device_id --block_device $tempnode", SYMLINK+="disk/by-id/emc-vol-%c",
ENV{ID_BUS}="scsi", ENV{ID_SERIAL}="%c", ENV{ID_WWN}="%c"
```

3. Open `/etc/multipath.conf` and add the following additional settings to the configuration file:

```
defaults {
          retain_attached_hw_handler "no"
                find_multipaths no
}
```

```
blacklist_exceptions {
                    devnode "scini"
}
```

   NOTE: From the command line, run `dracut -f.` This includes the changes in initrd.
   `[retain_attached_hw_handler no]` is the default setting in Red Hat Enterprise Linux 7.x or 8.x.

4. Restart the PowerFlex service and multipathd service:

```
systemctl restart scini
```

```
systemctl restart multipathd.service
```

   NOTE: If LUNs are in use and a restart is not possible, restart the node.

SDC authentication

This feature ensures secured connectivity between the SDC and the storage cluster over the data path by applying Challenge-Handshake Authentication Protocol (CHAP) based authentication of the SDC to the MDM for access to the system in general and to mapped volumes in particular. This prevents the SDC from accessing the cluster if it is not admitted as well as unauthorized volumes. Once enabled, the SDC internally performs mutual CHAP authentication with the SDSs and the SDRs with no manual intervention.

Prerequisites

Enable SDC authentication according to the following rules:

- For each SDC, a CHAP authentication password is generated by the MDM.
- All SDCs must be configured with their generated passwords
- Run the `--check_sdc_authentication_status` command, to check the status of the SDCs and whether they are ready to authenticate

About this task

This procedure describes how to enable SDC authentication.

NOTE: Using CHAP authentication with SDC also means that an SDC can only perform I/O operations on volumes explicitly mapped to it. The SDS blocks SDC I/O operations on unmapped volumes. CHAP authentication is also used internally for I/O authentication to the SDS and SDR, however it is always enabled and not controlled by the user.

Steps

1. Get the shared generated password for SDC from the MDM using the command:

   ```
   scli --generate_sdc_password --(sdc_id <ID> | sdc_name <NAME) | sdc_guid <GUID> |
   sdc_ip <IP>) [--reason <REASON>]
   ```

   Use the reason parameter to verify that the SDC password is being reset and not changed by accident. The reason is stored in the MDM events log.

   NOTE: SDCs not configured with a password are disconnected after the feature is enabled in step 3.

   NOTE: You must comply with organization password protection policy rules when performing this operation.

   Copy the password that was generated in <SDC_PASSWORD_STRING>, used in the next step.

2. On the SDC, run the following command:
   - Linux:

     ```
     /opt/emc/scaleio/sdc/bin/drv_cfg --set_mdm_password --ip <MDM_IP> --password
     <SDC_PASSWORD_STRING> --file/etc/emc/scaleio/drv_cfg.txt
     ```

     NOTE: The file option is required for password persistency, for cases such as service scini restart or SDC reboot. Open the file to verify the <SDC_PASSWORD_STRING> is logged at the end of the MDM line.

   - ESXi:

     a. `cat /etc/vmware/esx.conf | grep scini | grep options`

     A string is returned representing all the ESXi configuration parameters currently set. Copy the string with the enclosing quotation marks and paste in a text editor for editing.

     b. At the end of the string, add the following text, within the quotation marks:

     ```
     IoctlMdmPasswordStr=<MDM_IP>-<MDM_PASSWORD>
     ```

     where:
     - *<MDM_IP>* is the MDM IP address.
     - *<MDM_PASSWORD>* is the MDM password.

     For example:

     ```
     "IoctlIniGuidStr=cd069ce3-bf2a-5dea-b50a-1a5ebc8ef3de
     IoctlMdmIPStr=192.169.217.165,172.17.217.165,192.169.217.166,172.17.217.166,192.1
     69.217.167,172.17.217.167 IoctlMdmPasswordStr=192.169.217.165-AQAAAAAAAADu/
     10fXW3BS1wPBDgnkR06tdneGoUK7VQ"
     ```

     c. Run the following command with the string that is appended to the end:

     ```
     esxcli system module parameters set -m scini -p <STRING>
     ```

     For example:

     ```
     esxcli system module parameters set -m scini -p "IoctlIniGuidStr=cd069ce3-
     bf2a-5dea-b50a-1a5ebc8ef3de
     IoctlMdmIPStr=192.169.217.165,172.17.217.165,192.169.217.166,172.17.217.166,192.1
     69.217.167,172.17.217.167 IoctlMdmPasswordStr=192.169.217.165-AQAAAAAAAADu/
     10fXW3BS1wPBDgnkR06tdneGoUK7VQ"
     ```

3. To check SDC readiness for all SDCs in the system, before enabling SDC authentication, run the following command:

   NOTE: It is important to complete the previous steps for all SDCs before running the command.

   ```
   scli --check_sdc_authentication_status [--run_test] [--file_name <FILENAME>]
   ```

   Where:

   - `--run_test` runs a connectivity test to check whether the SDCs can successfully authenticate using CHAP
   - `--filename <FILENAME>` is the full file name and path for the generated report.

   The command sends a report that includes the SDCs authentication password status.

   NOTE: When running this command, the SDCs are disconnected for a short period from the MDM. This does not interrupt running I/Os or have any impact on MDM/SDC activity. It is recommended to run the command when the system is in a healthy state and not during rebalancing or rebuilding operations.

4. To enable SDC authentication, run the following command:

   ```
   scli --set_sdc_authentication --enable
   ```

5. To disable SDC authentication, run the following command:

   ```
   scli --set_sdc_authentication --disable
   ```

6. Reboot the ESXi for the configuration to take effect. For more information, see *Gracefully shut down or reboot a node* in the Dell PowerFlex and PowerFlex Manager Administration Guide.

Results

SDC authentication is enabled or disabled.

Manual SDC configuration/verification

If there are problems connecting to a PowerFlex system after a reboot, users can modify the `drv_cfg.txt` file or use the drv_cfg utility to ensure that the SDC reconnects to the MDM automatically upon node restart/reboot.

To do this, perform one of the following procedures:

- Edit `/etc/emc/scaleio/drv_cfg.txt` and change the #MDM line to include each IP address for any node(s) containing an MDM in your cluster.

  For example, change the line:

  From:

  #mdm 10.20.30.40

  To:

  mdm 10.108.158.48,10.108.158.49

  where .48 is the primary node, and .49 is the secondary node. The tiebreaker node should not be included.

- Use the drv_cfg binary or drv_cfg.exe on Windows to re-attach the MDM.
- Rescan all volumes using the drv_cfg utility.

30

SDS commands

This section contains commands for managing storage data severs (SDS).

abort_remove_sds

Abort the removal of an SDS from a PowerFlex system. If it is performed at too late a stage in the removal process, the command may fail and the SDS will be removed.

Syntax

```
scli --abort_remove_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>])
```

Parameters

--sds_id *<ID>*
: The ID of the SDS whose removal you want to stop

--sds_name *<NAME>*
: The name of the SDS whose removal you want to stop

--sds_ip *<IP>*
: The IP address of the SDS whose removal you want to stop

--sds_port *<PORT>*
: The port associated with the SDS

Example

```
scli --mdm_ip 192.168.1.200 --abort_remove_sds --sds_ip 192.168.1.4
```

add_sds

Add an SDS to a protection domain.

This command creates an SDS component and can be used to apply many configuration options at the time of creation. It informs the SDS about which devices to use, and for what purpose to use them (storage or acceleration). By default, the command also performs tests on the devices it adds, and saves the results.

NOTE: During execution of the command, PowerFlex checks that the devices are clear before adding them. If a device is not clear, an error message is returned, and the command fails for that device. If you would like to overwrite existing data on the devices by forcing the command, use the `force_device_takeover` flag. Note that this flag destroys all existing data on the device!

Syntax

```
scli --add_sds --sds_ip <IP> [--sds_ip_role <ROLE> {sdc_only | sds_only
| all}] (--protection_domain_id <ID> | --protection_domain_name <NAME>) [(--
storage_pool_name <NAME> | --storage_pool_id <ID>) --device_path <PATHS> [--
device_name <NAMES>] [--media_type <TYPES> {HDD | SSD}]] [(--acceleration_pool_name
<NAME> | --acceleration_pool_id <ID>) --acceleration_device_path <PATHS> [--
acceleration_device_name <NAMES>]] [--sds_name <NAME>] [--fault_set_id <ID> | --
fault_set_name <NAME>]
   [--rmcache_size_mb <SIZE>]
   [{--enable_rmcache | --disable_rmcache}]
   [--sds_port <PORT>]
   [--force_clean]
   [--force_device_takeover]
   [--test_time <TIME>]
   [{--test_only | --no_test | <BLANK>}]
   [--profile <PROFILE>]
   [--i_am_sure]
```

Parameters

--sds_ip *<IP>*

Comma-separated list of one or more IP addresses associated with the SDS over which the data will be transferred. See "SDS IP addresses and roles" below. Omit the space after each comma.

--sds_ip_role {sdc_only | sds_only | all}

Comma-separated list of IP roles associated with the SDS IP addresses. Omit the space after each comma. See "SDS IP addresses and roles" below.

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Comma-separated list of storage pool names to be associated with the storage devices, respectively. Omit the space after each comma. If only one storage pool is specified in the list, all devices are associated with it. Otherwise, each device is associated with the storage pool situated in the same position in the list.

--storage_pool_id *<ID>*

Comma-separated list of storage pool IDs to be associated with the storage devices, respectively. Omit the space after each comma. If only one storage pool is specified in the list, all devices are associated with it. Otherwise, each device is associated with the storage pool situated in the same position in the list.

--device_path *<PATHS>*

Comma-separated list of the full path to each devices to be added. Omit the space after each comma. A device can be a disk, an unmounted partition, or a file that represents free space on a mounted device. To force device take-over, use this command with the `force_device_takeover` flag, proceeding with caution.

--device_name *<NAMES>*

Comma-separated list of device names. Omit the space after each comma. When using this flag to assign names to the SDS devices. The number of names must equal the number of devices, and the names will be appropriated to the devices in the same order as they appear in the `device_path` list.

--media_type {HDD | SSD}

If one media type is specified in the list, all devices are assigned with this type. Otherwise, each device is assigned the media type located in the same position as the list.

--acceleration_pool_name *<NAME>*

Comma-separated list of acceleration pool names to be associated with the acceleration devices. Omit the space after each comma. If one acceleration pool is specified, all acceleration devices are assigned to it. Otherwise, each device is assigned to the acceleration pool located in the same position in the list.

--acceleration_pool_id *<ID>*

Comma-separated list of acceleration pool IDs to be associated with the acceleration devices. Omit the space after each comma. If one Acceleration Pool is specified, all acceleration devices are assigned to it. Otherwise, each device is assigned to the acceleration pool located in the same position in the list.

--acceleration_device_path *<PATHS>*

Comma-separated list of SDS acceleration device paths. Omit the space after each comma. A device can be a disk, an unmounted partition, or a file that represents free space on a mounted device. To force device take-over, use the command with the `force_device_takeover` flag, proceeding with caution.

--acceleration_device_name *<NAMES>*

Comma-separated list of acceleration device names. Omit the space after each comma.

--sds_name *<NAME>*

Name assigned to the SDS

--fault_set_id *<ID>*

Fault set ID

--fault_set_name *<NAME>*

Fault set name

--rmcache_size_mb *<SIZE>*

Size of the SDS RAM cache in MB. Range: 128 MB to 300GB

--enable_rmcache | --disable_rmcache

Enable/disable RAM cache on the SDS

--sds_port *<PORT>*

Port associated with the SDS. Default: 7072

--force_clean

Clean a previous SDS configuration. Use this if the SDS was previously part of a PowerFlex system.

--force_device_takeover

Force adding the devices, ignoring data from other SDSs that are stored on the devices.

CAUTION: Use this flag with caution. All data saved on the devices will be destroyed.

--test_time *<TIME>*

The maximum test run-time in seconds. Default: 10. The test will stop when it reaches either this limit, or the time it takes to complete 128 MB of data read/write, whichever is first.

Test options (choose one):

By default, PowerFlex tests the performance of each device being added before before its capacity can be used. Two tests are performed on the given devices: random writes and random reads. When the tests are complete, the device capacity is added automatically to the protection domain storage used by the MDM. To modify this behavior, specify one of the test options:

--test_only

Devices are tested, but not used. To start using their capacity, run the `--activate_sds_device` command.

--no_test

The device capacity is used without any device testing.

<BLANK>

Read and write test is run on devices before their capacity is used.

--profile {compact | high_performance}

Set the performance profile. The default is `high_performance`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --add_sds --sds_ip 192.168.1.9,192.168.1.29 -sds_ip_role
sds_only,sdc_only --protection_domain_name rack_1.1 --device_path /dev/cciss/c0d1p3,/dev/
cciss/c0d1p4 --device_name sds01,sds02 --sds_name sds_1
```

SDS names

Assign each SDS and SDS device a meaningful name in order to facilitate future object identification. This can be helpful for SDS devices, because the defined name remains constant even if the path changes. When a name has not been defined, the system may display default system-defined names that use the first IP address of the SDS.

Each name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

SDS IP addresses and roles

Configure up to eight IP addresses per SDS. Define each one with one of the following roles:

- sdc_only: Communications are enabled only between this SDS and SDCs. The MDM is not aware of this IP; therefore, internal error-checking is not performed. If an incorrect IP address is supplied, throughput may be affected.
- sds_only: Communications are enabled only between this SDS and other SDS and MDM components.
- all (default): Communications are enabled with all components:

You can define a role for every IP address, or use the default for all of them. A situation where all IP addresses are either sdc_only or sds_only is not valid. Each SDS must have one of the following:

- At least one IP address with the role all

  or

- Two IP addresses with the roles sdc_only and sds_only

IP roles can be configured when adding an SDS, by adding a new IP address to an SDS, or by editing a current IP role.

add_sds_ip

Add an IP address to an existing SDS.

Syntax

```
scli --add_sds_ip (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --new_sds_ip <IP>
  [--sds_ip_role {sdc_only | sds_only | all}] [--force_add_ip]
```

Parameters

`--sds_id <ID>`

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port number on the server to be associated with the SDS

--new_sds_ip *<IP>*

New IP address to be associated with the SDS

--sds_ip_role {sdc_only | sds_only | all}

The IP role assigned to each SDS IP address: SDC only, SDS only, or all (the default). When adding multiple IP addresses, use a comma-separated list for IP roles, with no space after the commas.

--force_add_ip

Force the addition of the new IP address even if it could not be verified.

Example

```
scli --mdm_ip 192.168.1.200 --add_sds_ip --sds_ip 192.168.1.4 --new_sds_ip 192.168.1.24
```

disable_sds_rmcache

Disable read RAM cache on the specified SDS.

Syntax

```
scli --disable_sds_rmcache ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | (--protection_domain_id <ID> | --protection_domain_name <NAME>))
  [--i_am_sure]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --disable_sds_rmcache --sds_name sds23 --protection_domain_name pd01
```

NOTE: To disable the read RAM cache for all SDSs in a protection domain using a single command, add the relevant protection domain name or ID to the `--disable_sds_rmcache` command. For example:

```
scli --disable_sds_rmcache --sds_ip 10.100.5.25 --protection_domain pd25
```

enter_maintenance_mode

Place one or more SDSs in maintenance mode.

Syntax

```
scli --enter_maintenance_mode ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>]) | (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>)))
  [--force_insufficient_spare_or_free_space]
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--fault_set_id *<ID>*
Fault set ID

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_name *<NAME>*
Fault set name

--force_insufficient_spare_or_free_space
Allow entry into maintenance mode, even without enough available capacity.

Example

```
scli --enter_maintenance_mode --sds_name sds10 --force_insufficient_spare_or_free_space
```

exit_maintenance_mode

Cancel maintenance mode for one or more SDSs.

NOTE: The system displays an error message if you attempt to remove an SDS in Maintenance Mode, ("The task failed since there is an SDS in the Protection Domain currently in Maintenance Mode"). However, you can force an SDS out of Maintenance Mode by using one of the flags, and then removing the SDS.

Syntax

```
scli --exit_maintenance_mode ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>]) | (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>)))
  [--force_failed_device_id <IDs>]
  [--force_failed_sds]
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--fault_set_id *<ID>*
Fault set ID

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_name *<NAME>*
Fault set name

--force_failed_device_id *<IDs>*
Allow exit from maintenance mode, even when the specified devices are in an error state.

--force_failed_sds
Allow exit from maintenance mode, even when an SDS is down.

NOTE: If you attempt to remove an SDS in maintenance mode, the system displays an error message: `The task failed since there is an SDS in the Protection Domain currently in Maintenance Mode`. Using the `force_failed_device_id` or `force_failed_sds` flag forces the SDS out of maintenance mode.

Example

```
scli --exit_maintenance_mode --sds_ip 10.76.1.10 --sds_port 7072
```

enable_sds_rmcache

Enable read RAM cache on the specified SDS.

For a read to be stored in the RAM of a specific SDS, the read RAM cache feature on that SDS must be enabled, and the relevant storage pool and the relevant volume must both be configured to use read RAM cache. Caching only begins after one or more devices are added to the SDS.

Syntax

```
scli --enable_sds_rmcache ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | (--protection_domain_id <ID> | --protection_domain_name <NAME>))
  [--i_am_sure]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --enable_sds_rmcache --sds_ip 10.100.5.25 --protection_domain_name pd01
```

NOTE: To enable read RAM cache for all SDSs in a protection domain using a single command, add the relevant protection domain name or ID to the `--enable_sds_rmcache` command. For example:

```
scli --enable_sds_rmcache --sds_ip 10.100.5.25 --protection_domain pd25
```

enter_protected_maintenance_mode

When entering protected maintenance mode (PMM), a third copy of the SDS is created, which ensures that if there is a node failure where the second copy of SDS is required, there is still a full backup of the SDS. Move one or more SDSs to protected maintenance mode.

Syntax

```
scli --enter_protected_maintenance_mode ((--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>]) | (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>)))
```

Parameters

None.

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--fault_set_id *<ID>*
Fault set ID

--fault_set_name *<NAME>*
Fault set name

Example

```
scli --enter_protected_maintenance_mode --mdm_ip 192.168.1.200 --sds_name sds120
```

exit_protected_maintenance_mode

Move one or more SDSs out of protected maintenance mode.

Syntax

```
scli --exit_protected_maintenance_mode ((--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>]) | (--fault_set_id <ID> | ((--protection_domain_id <ID> |
--protection_domain_name <NAME>) --fault_set_name <NAME>))) [--force_failed_device_id
<IDs>] [--force_failed_sds] [--abort_entering_maintenance_mode]
```

Parameters

None.

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--fault_set_id *<ID>*

Fault set ID

--fault_set_name *<NAME>*

Fault set name

--force_failed_device_id *<IDs>*

Allow exiting protected maintenance mode when the specified devices are in an error state.

--force_failed_sds

Allow exiting protected maintenance mode when the SDS is down.

--abort_entering_maintenance_mode

Acknowledge in case the process of entering protected maintenance mode is not yet complete.

Example

```
scli --exit_protected_maintenance_mode --mdm_ip 192.168.1.200 --sds_name sds120 --
force_failed_sds
```

modify_sds_ip_role

Modify the role of an SDS's IP address. The IP address can be used for SDC only, SDS only, or both.

Syntax

```
scli --modify_sds_ip_role (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) [--sds_ip_to_modify <IP>] --new_sds_ip_role {sdc_only | sds_only | all} [--
force_role_modification]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--sds_ip_to_modify *<IP>*

IP address of the SDS being modified

--new_sds_ip_role {sdc_only | sds_only | all}

New role to assign the SDS IP: SDC only, SDS only, or all

--force_role_modification

Force the modification of the SDS's role even if the IP address could not be verified.

Example

```
scli --modify_sds_ip_role --sds_name sds01 --sds_ip_to_modify 192.168.1.4 --
new_sds_ip_role all
```

modify_sds_port

Modify the port used by the SDS for communication purposes.

Syntax

```
scli --modify_sds_port (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --new_sds_port <PORT>
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--new_sds_port *<PORT>*

New port to be associated with the SDS

Example

```
scli --mdm_ip 192.168.1.200 --modify_sds_port --sds_ip 192.168.1.4 --new_sds_port 6801
```

query_all_sds

Retrieve information about all SDSs in the system.

Syntax

```
scli --query_all_sds
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_sds
```

Sample output

```
Query-all-SDS returned 5 SDS nodes.

Protection Domain f11d784d00000000 Name: pd1
SDS ID: 7a3cf71e00000004 Name: sds_idd_1 State: Connected, Joined IP: 10.55.137.114
Port: 7072 Version: 3.6.0
SDS ID: 7a3cf71c00000003 Name: sds_idd_2 State: Connected, Joined IP: 10.55.137.115
Port: 7072 Version: 3.6.0
SDS ID: 7a3cf71d00000002 Name: sds_idd_4 State: Connected, Joined IP: 10.55.137.116
Port: 7072 Version: 3.6.0
SDS ID: 7a3cf71b00000001 Name: sds_idd_5 State: Connected, Joined IP: 10.55.137.117
Port: 7072 Version: 3.6.0
Protection Domain f11d784e00000001 Name: pd2
SDS ID: 7a3cf71a00000000 Name: sds_idd_3 State: Connected, Joined IP: 10.55.137.119
Port: 7072 Version: 3.6.0
Protection Domain f11d9f5d00000002 Name: dfg
```

query_sds_connectivity_status

Retrieve information regarding the connectivity status of the SDSs in the specified protection domain.

Syntax

```
scli --query_sds_connectivity_status (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

Example

```
scli --query_sds_connectivity_status --protection_domain_name pd_18
```

Sample output

```
Successfully queried protection domain pd1
Got responses from 4 SDSs:
        ID: 06b108a100000001 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117  : Connected
        ID: 06b108a100000001 <===> ID: 06b108a100000001  Connected Count 8
                10.55.137.116  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a100000001  Connected Count 4
                10.55.137.116  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a300000003  Connected Count 8
                10.55.137.115  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a200000002  Connected Count 4
                10.55.137.119  : Connected
        ID: 06b108a000000000 <===> ID: 06b108a000000000  Connected Count 8
                10.55.137.117  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a100000001  Connected Count 4
                10.55.137.116  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a300000003  Connected Count 4
                10.55.137.115  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a400000004  Connected Count 8
                10.55.137.114  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a200000002  Connected Count 4
                10.55.137.119  : Connected
```

query_sds

Retrieve detailed information about the specified SDS.

Syntax

```
scli --query_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>])
[--show_oscillating_failures]
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--show_oscillating_failures
Show extended oscillating failure information, if such failures have occurred

Example

```
scli --mdm_ip 192.168.1.200 --query_sds --sds_id 06b108a400000004
```

Sample output

```
SDS 06b108a400000004 Name: sds_idd_1 Version: 3.6.0
Protection Domain: fcc9af5500000000, Name: pd1
Fault Set: 36c9d76800000000, Name: fs1_mg
Medium granularity DRL mode: Volatile
Authentication error: NONE
IP information (total 1 IP Addresses):
         1: 10.55.137.114    Role: All (SDS and SDC)
        Port: 7072
I/O flow control is enabled
Fine granularity number of concurrent writes: 1000
RAM Read Cache information:
        128.0 MB (131072 KB) total size
        Cache is enabled
        RAM Read Cache memory allocation state is pending
FGL metadata cache information:
        FGL metadata cache is disabled
        FGL metadata cache size: 0 MB

Storage Pool information (total of 1 Storage Pools):
         1: Storage Pool: sp1  inflight requests factor: 115, inflight bandwidth factor
115

Device information (total 2 devices):
         1: Name: N/A  Path: /dev/sdb  Original-path: /dev/sdb  ID: c8bea95800040000
                Storage Pool: sp1, Capacity: 148 GB, State: Normal
                Scanned 936040 MB, Unresolved Read errors: 0/0, Unresolved Compare
errors: 0/0
         2: Name: N/A  Path: /dev/sdc  Original-path: /dev/sdc  ID: c8bea95900040001
                Storage Pool: sp1, Capacity: 148 GB, State: Normal
                Scanned 936134 MB, Unresolved Read errors: 0/0, Unresolved Compare
errors: 0/0

Acceleration device information (total 0 devices):

Membership-state: Joined, Connection-state: Connected, SDS-state: Normal, 0 devices with
error,

Physical data:
        297.9 GB (305000 MB) total capacity
        280.5 GB (287184 MB) unused capacity
        17.4 GB (17816 MB) in-use capacity
        0 Bytes unreachable-unused capacity
        0 active moving-in forward-rebuild jobs
        0 active moving-in backward-rebuild jobs
        0 active moving-in rebalance jobs
        0 active moving-in enter protected maintenance mode jobs
        0 active moving-in exit protected maintenance mode jobs
        0 active moving-out forward-rebuild jobs
        0 active moving-out backward-rebuild jobs
        0 active moving-out enter protected maintenance mode jobs
        0 active moving-out exit protected maintenance mode jobs
        0 active moving-out rebalance jobs
        0 pending moving-in forward-rebuild jobs
        0 pending moving-in backward-rebuild jobs
        0 pending moving-in rebalance jobs
        0 pending moving-in enter protected maintenance mode jobs
        0 pending moving-in exit protected maintenance mode jobs
        0 pending moving-out forward-rebuild jobs
        0 pending moving-out backward-rebuild jobs
        0 pending moving-out rebalance jobs
        0 pending moving-out enter protected maintenance mode jobs
        0 pending moving-out exit protected maintenance mode jobs
```

```
        0 Bytes decreased capacity

Bandwidth:
        Primary-reads                               114 IOPS 81.2 MB (83148 KB) per-second
        Primary-writes                              150 IOPS 75.6 MB (77450 KB) per-second
        Secondary-reads                             0 IOPS 0 Bytes per second
        Secondary-writes                            142 IOPS 69.8 MB (71516 KB) per-second
        Backward-rebuild-reads                      0 IOPS 0 Bytes per second
        Backward-rebuild-writes                     0 IOPS 0 Bytes per second
        Forward-rebuild-reads                       0 IOPS 0 Bytes per second
        Forward-rebuild-writes                      0 IOPS 0 Bytes per second
        Rebalance-reads                             0 IOPS 0 Bytes per second
        Rebalance-writes                            0 IOPS 0 Bytes per second
        Volume-migration-reads                      0 IOPS 0 Bytes per second
        Volume-migration-writes                     0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-reads      0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-writes     0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-reads       0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-writes      0 IOPS 0 Bytes per second
```

query_sds_connectivity_status

Retrieve information regarding the connectivity status of the SDSs in the specified protection domain.

Syntax

```
scli --query_sds_connectivity_status (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

Example

```
scli --query_sds_connectivity_status --protection_domain_name pd_18
```

Sample output

```
Successfully queried protection domain pd1
Got responses from 4 SDSs:
        ID: 06b108a100000001 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117   : Connected
        ID: 06b108a100000001 <===> ID: 06b108a100000001  Connected Count 8
                10.55.137.116   : Connected
        ID: 06b108a300000003 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117   : Connected
        ID: 06b108a300000003 <===> ID: 06b108a100000001  Connected Count 4
                10.55.137.116   : Connected
        ID: 06b108a300000003 <===> ID: 06b108a300000003  Connected Count 8
                10.55.137.115   : Connected
        ID: 06b108a300000003 <===> ID: 06b108a200000002  Connected Count 4
                10.55.137.119   : Connected
        ID: 06b108a000000000 <===> ID: 06b108a000000000  Connected Count 8
```

```
                10.55.137.117   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a100000001  Connected Count 4
                10.55.137.116   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a300000003  Connected Count 4
                10.55.137.115   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a400000004  Connected Count 8
                10.55.137.114   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a200000002  Connected Count 4
                10.55.137.119   : Connected
```

query_sds_network_test_results

Retrieve the results of the last network test performed between an SDS and its peers.

Syntax

```
scli --query_sds_network_test_results (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>])
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

Example

```
scli --mdm_ip 192.168.1.200 --query_sds_network_test_results --sds_ip 192.168.1.4
```

query_sdc_to_sds_disconnections

Retrieve details on any disconnections between SDCs and SDSs currently in the system.

Syntax

```
scli --query_sdc_to_sds_disconnections
```

Parameters

None.

Example

```
scli --query_sdc_to_sds_disconnections
```

remove_sds

Initiate the removal of an SDS. The command removes an SDS object and disconnects the SDS from the MDM.

You can remove an SDS at any time, with no downtime required. During execution of this command, the associated data is replicated to different nodes. Therefore, the removal process is asynchronous and might take a long time.

NOTE: If the capacity of this SDS is still used by volumes, and the capacity cannot be replaced due to lack of available free space, the command will fail.

Syntax

```
scli --remove_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>])
```

Parameters

--sds_id *<ID>*
ID of the SDS to remove

--sds_name *<NAME>*
Name of the SDS to remove

--sds_ip *<IP>*
IP address of the SDS to remove

--sds_port *<PORT>*
Port associated with the SDS

Example

```
scli --mdm_ip 192.168.1.200 --remove_sds --sds_ip 192.168.1.4
```

Related operations

The removal operation runs in the background. To track its progress, run the `query_all_sds` command periodically.

To abort the removal, run the `abort_remove_sds` command.

remove_sds_ip

Remove an IP address from an SDS.

Possible scenarios for using this command are:

- In PowerFlex setups that include an SDS installed on a server with more than one NIC (network interface card), you might want to divert the usage of one of the NICs to a different purpose.
- An IP range may have changed. For example, an SDS may be using an IP address in a certain range, but the IP range has been changed to a different one in order to free up the previously used addresses.

Syntax

```
scli --remove_sds_ip (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --sds_ip_to_remove <IP>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--sds_ip_to_remove *<IP>*
IP address to remove from the SDS

Example

```
scli --mdm_ip 192.168.1.200 --remove_sds_ip
 --sds_ip 192.168.1.4 --sds_ip_to_remove 192.168.1.24
```

rename_sds

Assign a new name to the specified SDS.

Syntax

```
scli --rename_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --new_name <NAME>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--new_name *<NAME>*
New, unique name to assign to the specified SDS

Example

```
scli --mdm_ip 192.168.1.200 --rename_sds --sds_ip 192.168.1.4 --new_name sds_new_name
```

SDS names

Assign the SDS a meaningful name in order to facilitate future object identification. The name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

set_drl_properties

Set the properties of the dirty-region-logging used by one or all SDSs.

NOTE: The settings apply to medium granularity layout devices only.

Syntax

```
scli --set_drl_properties ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | --set_all) (--volatile | --hardened)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--set_all
Apply the properties to all SDSs. Overrides previous settings

--volatile
Dirty-region-logging information will be stored in memory only. This is the default.

--hardened
Dirty-region-logging information will be stored both in memory and on disk.

Example

```
scli --set_drl_properties --sds_ip 192.168.1.4 --hardened
```

set_fgl_metadata_cache_size

Configure the fine granularity metadata cache size for the specified SDS.

Syntax

```
scli --set_fgl_metadata_cache_size (--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--metadata_cache_size_mb <SIZE> | --
use_protection_domain_default_size)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--metadata_cache_size_mb *<SIZE>*
Define the default size of the metadata cache for the SDS in MB

--use_protection_domain_default_size
Use the default size of the protection domain

Example

```
scli --mdm_ip 192.168.1.200 --set_fgl_metadata_cache_size --sds_name SDS1 --
metadata_cache_size_mb 10248
```

set_sds_network_limits

Limit the network bandwidth used by all the SDSs in the specified protection domain for various traffic types, including I/O.

SDS nodes transfer data among themselves. This data consists of user-data being replicated as part of the RAID protection, and data copied for internal rebalancing and recovery from failures.

You can modify the balance between these types of data loads by limiting the data copy bandwidth. This change affects all SDSs in the specified protection domain.

CAUTION: Contact Dell Technologies Support before you modify this configuration.

Syntax

```
scli --set_sds_network_limits (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
[--rebuild_limit <LIMIT>] [--rebalance_limit <LIMIT>] [--volume_migration_limit <LIMIT>]
[--protected_maintenance_mode_limit <LIMIT>] [--overall_limit <LIMIT>]
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--rebuild_limit *<LIMIT>*

Limit the network bandwidth for rebuild in MB/s. 0 is unlimited

--rebalance_limit *<LIMIT>*

Limit the network bandwidth for rebalance in MB/s. 0 is unlimited

--volume_migration_limit *<LIMIT>*

Limit the network bandwidth for V-Tree migration in MB/s. 0 is unlimited

--protected_maintenance_mode_limit *<LIMIT>*

Limit the network bandwidth for protected maintenance mode in MB/s. 0 means unlimited bandwidth.

--overall_limit *<LIMIT>*

Limit the overall network bandwidth in MB/s. This includes the rebuild, rebalance, and application I/O bandwidth. 0 is unlimited

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.68.1.200 --set_sds_network_limits --protection_domain_name pd1 --
rebuild_limit 100
```

set_sds_rmcache_size

Set the RMcache size of the specified SDS.

By default, the RMcache size is set to 128 MB in all SDSs. The amount of RAM you can allocate for cache is limited by the amount of RAM on the SDS server:

- If the RAM is less than 32 GB, 50 percent of memory can be used for cache.
- If the RAM is more than 32 GB, 75 percent of memory can be used for cache.

The maximum amount of RAM cache is 128 GB.

NOTE: In cases of Non-Uniform Memory Access (NUMA), the total RAM refers to the first node only.

Syntax

```
scli --set_sds_rmcache_size ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | (--protection_domain_id <ID> | --protection_domain_name <NAME>)) --
rmcache_size_mb <SIZE>
  [--i_am_sure]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--rmcache_size_mb *<SIZE>*

New size of RMcache in MB. Range: 128 MB-300 GB.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_sds_rmcache_size --protection_domain_name pd01 --sds_name sds10 --
rmcache_size_mb 256
```

show_certificate

Display the certificate properties for the specified SDS or SDR.

Syntax

```
scli --show_certificate ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) | --all_sds | (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) | --all_sdr | (--protection_domain_id <ID> | --protection_domain_name <NAME>))
  [--days_until_expiration <DAYS>]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--all_sds

Show certificates for all SDSs.

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

SDR IP address

--sdr_port *<PORT>*

Port associated with the SDR

--all_sdr

Show certificates for all SDRs.

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--days_until_expiration *<DAYS>*

Only show certificates expiring within a given number of days.

Example

```
scli --show_certificate --all_sds --days_until_expiration 7
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
Protection Domain fcc9af5600000001 Name: pd2
Protection Domain fcc9d66500000002 Name: rrr
```

start_sds_network_test

Initiate a network test between an SDS and all of its peers.

Syntax

```
scli --start_sds_network_test (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>])
  [--parallel_messages <NUMBER>]
  [--network_test_size_gb <SIZE>]
  [--network_test_length_secs <SECONDS>]
  [--test_associated_protection_domains]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--parallel_messages *<NUMBER>*

Number of parallel messages sent during the test. Range is 1 - 16. Default: 4.

--network_test_size_gb *<SIZE>*

Amount of data sent between the tested SDS and every other SDS. Default size: 1 GB.

--network_test_length_secs *<SECONDS>*

Maximum amount of time spent testing the SDS, regardless of the number of SDSs in the system. Default value: 0 (unlimited).

--test_associated_protection_domains

Test associated protection domains. Protection domains can be connected during V-Tree migration between different protection domains.

Example

```
scli --mdm_ip 192.168.1.200 --start_sds_network_test
--sds_ip 192.168.1.4 --parallel_messages 8 --network_test_size_gb 2 --
network_test_length_secs 120
```

update_sds_original_paths

Change the original path configuration of all devices to the path currently assigned to the devices in the SDS.

Syntax

```
scli --update_sds_original_paths (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) [--force_failed_devices]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

SDS port

--force_failed_devices

Force path update even for devices in error

Example

```
scli --update_sds_original_paths --sds_name sds1 --force_failed_devices
```

set_sds_device_path

Configure the path of the given storage device.

Syntax

```
scli --set_sds_device_path (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) --
new_device_path <PATH>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--device_id *<ID>*
Device ID

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or file path. The file path is used only when the device is available at this path

--new_device_path *<PATH>*
The new path to configure

31

SDT commands

This section contains commands for managing storage data targets (SDT).

add_sdt

Add an SDT IP address.

Syntax

```
scli --add_sdt --sdt_ip <IP> [--sdt_ip_role <ROLE>] [--storage_port <PORT>] [--nvme_port
<PORT>] [--discovery_port <PORT>] [--sdt_name <NAME>] (--protection_domain_id <ID> | --
protection_domain_name <NAME>) [--profile <PROFILE>] [--force_clean [--i_am_sure]]
```

Parameters

--sdt_ip *<IP>*

A comma separated list of IP addresses assigned to the SDT

--sdt_ip_role *<ROLE>*

A comma separated list of roles assigned to each SDT IP address

Role options: storage_only, host_only, or storage_and_host

--storage_port *<PORT>*

Port assigned to the SDT (default: 12200)

--nvme_port *<PORT>*

Port used by the NVMe hosts (default: 4420)

--discovery_port *<PORT>*

Port used by the NVMe hosts for discovery (default: 8009). Set to 1 in order to indicate no use of discovery port.

--sdt_name *<NAME>*

A name assigned to the SDT

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--fault_set_id *<ID>*

Fault set ID

--fault_set_name *<NAME>*

Fault set name

--profile *<PROFILE>*

Set the performance profile from the following options: compact | high_performance

The default is high_performance

--force_clean

Clean a previous SDT configuration

--i_am_sure

Preemptive approval

Example

```
scli --add_sdt --sdt_name sdt1 --sdt_ip 10.0.0.1,10.0.1.1 --sdt_ip_role
storage_and_host, storage_and_host --protection_domain_name pd1
```

remove_sdt

Start the removal of an SDT.

Syntax

```
scli --remove_sdt (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) --force
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

--force
Force removing IP regardless the number of SDTs left on protection domains.

Example

```
scli --remove_sdt --sdt_name sdt1
```

rename_sdt

Name or rename an SDT.

Syntax

```
scli --rename_sdt (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) --new_name <NAME>
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--new_name *<NAME>*

A new name to be assigned to the SDT

Example

```
scli --rename_sdt sdt1 --new_name someSdt
```

add_sdt_ip

Add an IP address to an SDT.

Syntax

```
scli --add_sdt_ip (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) --new_sdt_ip <IP> [--sdt_ip_role <ROLE>]
```

Parameters

--sdt_id *<ID>*

SDT ID

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--new_sdt_ip *<IP>*

New IP address to be assigned to the SDT

--sdt_ip_role *<ROLE>*

The role assigned to the new SDT IP address

Role options: storage_only, host_only, or storage_and_host

Example

```
scli --add_sdt_ip  --sdt_name sdt1 --new_sdt_ip 10.0.3.1 --sdt_ip_role storage_and_host
```

remove_sdt_ip

Remove an IP address from an SDT.

Syntax

```
scli --remove_sdt_ip (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) --sdt_ip_to_remove <IP> --force
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

--sdt_ip_to_remove *<IP>*
IP address to remove from the SDT

--force
Force removing IP regardless the number of hosts connections

Example

```
scli --remove_sdt_ip --sdt_name sdt1 --sdt_ip_to_remove 10.0.3.1
```

modify_sdt_ip_role

Modify the role of an SDT IP address.

Syntax

```
scli --modify_sdt_ip_role (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) [--sdt_ip_to_modify <IP>] --sdt_ip_role <ROLE> --force
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

--sdt_ip_to_modify *<IP>*
IP address for which to modify the role. Required if --sdt_ip is not specified

--sdt_ip_role *<ROLE>*

The new role assigned to the modified SDT IP address

Role options: storage_only, host_only, or storage_and_host

--force

Force modifying IP role regardless the number of hosts connections

Example

```
scli --modify_sdt_ip_role --sdt_name sdt1 --sdt_ip_to_modify 10.0.3.1 -sdt_ip_role
host_only
```

modify_sdt_port

Modify the port of an SDT.

Syntax

```
scli --modify_sdt_port (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) [--storage_port <PORT>] [--nvme_port <PORT>] [--discovery_port <PORT>]
```

Parameters

--sdt_id *<ID>*

SDT ID

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--storage_port *<PORT>*

Port assigned to the SDT

--nvme_port *<PORT>*

Port used by the NVMe hosts

--discovery_port *<PORT>*

Port used by the NVMe hosts for discovery.

Example

```
scli --modify_sdt_port (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port
<PORT>]) [--storage_port <PORT>] [--nvme_port <PORT>] [--discovery_port <PORT>]
```

query_sdt

Retrieve information about an SDT.

Syntax

```
scli --query_sdt (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port <PORT>])
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*
Port assigned to the SDT

Example

```
scli --quert_sdt --sdt_name sdt1
```

query_all_sdt

Retrieve information about all SDTs.

Syntax

```
scli --query_all_sdt
```

enter_sdt_maintenance_mode

Move the SDT into maintenance mode.

Syntax

```
scli --enter_sdt_maintenance_mode (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--
sdt_port <PORT>]) --force
```

Parameters

--sdt_id *<ID>*
SDT ID

--sdt_name *<NAME>*
SDT name

--sdt_ip *<IP>*
SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--force

Force entering maintenance mode regardless the number of SDTs in a protection domain.

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--fault_set_id *<ID>*

Fault set ID

--fault_set_name *<NAME>*

Fault set name

Example

```
scli --enter_sdt_maintenance_mode --sdt_name sdt1
```

exit_sdt_maintenance_mode

Move the SDT out of maintenance mode.

Syntax

```
scli --exit_sdt_maintenance_mode (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--
sdt_port <PORT>])
```

Parameters

--sdt_id *<ID>*

SDT ID

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--fault_set_id *<ID>*

Fault set ID

--fault_set_name *<NAME>*

Fault set name

Example

```
scli --exit_sdt_maintenance_mode --sdt_name sdt1
```

32

Snapshot commands

This section contains commands for managing snapshots.

clear_snapshot_security

Clear the snapshot security for the specified volume to enable editing or removing the volume. This command can only be run with a signature file obtained from customer support. If you need to clear the snapshot security, contact Dell Technologies Support to obtain the signature file.

Syntax

```
scli --clear_snapshot_security (--volume_id <ID> | --volume_name <NAME>) --
signature_file <FILE> [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--signature_file *<FILE>*

Signature file provided by customer support.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --clear_snapshot_security --volume_name vol1 --
signature_file /opt/vol-dd6e34001.sig --i_am_sure
```

remove_snapshots_consistency_group

Remove all snapshots that belong to a specific consistency group. Depending on the size of the consistency group, this command may take a long time.

NOTE: You cannot remove the consistency group if it contains auto snapshots.

Syntax

```
scli --remove_snapshots_consistency_group --scg_id <ID>
  [--i_am_sure]
```

Parameters

--scg_id *<ID>*

The consistency group's ID number.

NOTE: The consistency group ID is displayed when you create the consistency group. To find this ID at a later time, use the `query_all_volumes` command.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli  --remove_snapshots_consistency_group
--scg_id 783c7f00000000 --i_am_sure
```

set_snapshot_security

Set the specified snapshot as secure. The snapshot is not editable or removable for the specified retention period. If the specified snapshot is already secure, the security expiration date may be postponed, but not configured to an earlier point in time.

Syntax

```
scli --set_snapshot_security (--volume_id <ID> | --volume_name <NAME>) --
retention_period <PERIOD> [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--retention_period *<PERIOD>*

The snapshots will not be editable or removable for the specified time period. To specify the time period, use `m` | `h` | `D` | `W` | `M` | `Y` respectively indicating minutes, hours, days, weeks, months, or years. Examples: 30m (thirty minutes), 5D (five days), 1M (one month, which equals 31 days).

--i_am_sure

Preemptive approval

Example

```
scli --mdm_ip 192.168.1.200 --set_snapshot_security --volume_id 022beb2500000006 --
retention_period 2Y --i_am_sure
```

snapshot_volume

Creates a snapshot of one or more volumes.

When you specify more than one volume (a list), a consistency group is generated and the ID associated with it is displayed. The snapshots under the consistency group are taken simultaneously for all listed volumes, thus ensuring their consistency.

Secure snapshots, created using the `--retention_period` parameter, prohibit anyone from deleting the snapshot until the expiration time that is defined.

NOTE: The consistency group is for convenience purposes only. There are no protective measures to conserve the consistency group, and you can delete members from it.

Syntax

```
scli --snapshot_volume (--volume_id <ID>| --volume_name <NAME> | --volume_id_from_file
<FILE> | --volume_name_from_file <FILE> | --volume_class <CLASS>)
  [--snapshot_name <NAME> | --snapshot_name_from_file <FILE>] [--read_only] [--
retention_period <PERIOD>] [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Comma-separated list of volume IDs. Omit the space after each comma.

--volume_name *<NAME>*

Comma-separated list of volume names. Omit the space after each comma.

--volume_id_from_file *<FILE>*

File containing new, line-separated list of volume IDs

--volume_name_from_file *<FILE>*

File containing a new, line-separated list of volume names

--snapshot_name *<NAME>*

Comma-separated list of names to be associated with the created snapshots. The names in this list correspond to the names in the volume name or volume ID list. Omit the space after each comma.

--snapshot_volume_class *<Class>*

Allows creating snapshot in special cases

--snapshot_name_from_file *<FILE>*

File containing a new, line-separated list of names to be associated with the created snapshots

--read_only

All mappings of the created volume will be limited to read-only access

--retention_period *<PERIOD>*

Use this parameter to create the snapshots as secure. The snapshots will not be editable or removable for the specified time period. To specify the time period, use `m|h|D|W|M|Y` respectively indicating minutes, hours, days, weeks, months, or years. Examples: 30m (thirty minutes), 5D (five days), 1M (one month, which equals 31 days)

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --snapshot_volume
--volume_name vol_1 --snapshot_name snap_1 --retention_period 2Y
```

Snapshot names

Assign the snapshot a meaningful name. When a name has not been defined, the system may display default system-defined names that use the ID of the volume.

Assign snapshot names according to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

add_snapshot_copy_task

Adds an asynchronous task to copy a snapshot image to a target volume.

Syntax

```
scli --add_snapshot_copy_task (--remote_system_id <ID> | --remote_system_name
<NAME>) (--source_volume_id <ID> | --source_volume_name <NAME> | --
source_volume_id_from_file <FILE> | --source_volume_name_from_file <FILE>) [--
base_volume_id <ID> | --base_volume_name <NAME> | --base_volume_id_from_file
<FILE> | --base_volume_name_from_file <FILE>]  (--destination_volume_nsid <NSID>
| --destination_volume_nsid_from_file <FILE>) [--snapshot_copy_task_name <NAME> | --
snapshot_copy_task_name_from_file <FILE>] [--copy_allocated_blocks_only]
```

Parameters

--snapshot_copy_task_name *<NAME>*

The snapshot copy task name

--remote_system_id *<ID>*

The ID of the remote system

--remote_system_name *<NAME>*

The name of the remote target system

--source_volume_id *<ID>*

A comma-separated list of the source volume IDs

--source_volume_name *<NAME>*

A comma-separated list of the source volume names

--source_volume_id_from_file *<FILE>*

A file containing new line separated list of the source volume IDs

--source_volume_name_from_file *<FILE>*

A file containing new line separated list of the source volume names

--base_volume_id *<ID>*

A comma-separated list of the base volume IDs

--base_volume_name *<NAME>*

A comma-separated list of the base volume names

--base_volume_id_from_file *<FILE>*

A file containing new line separated list of the base volume IDs

--base_volume_name_from_file *<FILE>*

A file containing new line separated list of the base volume names

--destination_volume_nsid *<NSID>*

A comma-separated list of the destination volume NSIDs

--destination_volume_nsid_from_file *<FILE>*

A file containing new line separated list of the destination volume NSIDs

--snapshot_copy_task_name *<NAME>*

A comma-separated list of the snapshot copy task names

--snapshot_copy_task_name_from_file *<FILE>*

A file containing new line separated list of the snapshot copy task names

--copy_allocated_blocks_only

Skip the copy of unallocated blocks (in full snapshot copy tasks only)

query_snapshot_copy_task

Retrieve information about the snapshot copy task.

Syntax

```
scli --query_snapshot_copy_task (--snapshot_copy_task_id <ID> | --
snapshot_copy_task_name <NAME>)
```

Parameters

--snapshot_copy_task_id *<ID>*

The snapshot copy task ID

--snapshot_copy_task_name *<NAME>*

The snapshot copy task name

query_all_snapshot_copy_tasks

Retrieve information about all snapshot copy tasks.

Syntax

```
cli --query_all_snapshot_copy_tasks
```

pause_snapshot_copy_task

Pause the snapshot copy task.

Syntax

```
scli --pause_snapshot_copy_task (--snapshot_copy_task_id <ID> | --
snapshot_copy_task_name <NAME> | --snapshot_copy_task_id_from_file <FILE> | --
snapshot_copy_task_name_from_file <FILE>)
```

Parameters

--snapshot_copy_task_id *<ID>*

A comma-separated list of the task IDs

--snapshot_copy_task_name *<NAME>*

A comma-separated list of the task names

--snapshot_copy_task_id_from_file *<FILE>*

A file containing new line separated list of the task IDs

--snapshot_copy_task_name_from_file *<FILE>*

A file containing new line separated list of the task names

resume_snapshot_copy_task

Resume the snapshot copy task.

Syntax

```
scli --resume_snapshot_copy_task (--snapshot_copy_task_id <ID> | --
snapshot_copy_task_name <NAME> | --snapshot_copy_task_id_from_file <FILE> | --
snapshot_copy_task_name_from_file <FILE>)
```

Parameters

--snapshot_copy_task_id *<ID>*

A comma-separated list of the task IDs

--snapshot_copy_task_name *<NAME>*

A comma-separated list of the task names

--snapshot_copy_task_id_from_file *<FILE>*

A file containing new line separated list of the task IDs

--snapshot_copy_task_name_from_file *<FILE>*

A file containing new line separated list of the task names

remove_snapshot_copy_task

Remove the snapshot copy task.

Syntax

```
scli --remove_snapshot_copy_task (--snapshot_copy_task_id <ID> | --
snapshot_copy_task_name <NAME> | --snapshot_copy_task_id_from_file <FILE> | --
snapshot_copy_task_name_from_file <FILE>) [--force]
```

Parameters

--snapshot_copy_task_id *<ID>*

A comma-separated list of the task IDs

--snapshot_copy_task_name *<NAME>*

Protection domain name

--snapshot_copy_task_id_from_file *<FILE>*

A file containing new line separated list of the task IDs

--snapshot_copy_task_name_from_file *<FILE>*

A file containing new line separated list of the task names

--force

Allow removal of non-completed tasks

33

Storage pool commands

This section contains commands for managing storage pools.

add_storage_pool

Add a storage pool to a protection domain.

A storage pool is a group of storage devices within a protection domain. Each time that you add devices to the system, you must map them to either storage pools or acceleration pools. Create storage pools before you start adding SDSs and storage devices to the system. You can modify storage pools post-installation using all management clients, except for OpenStack.

Syntax

```
scli --add_storage_pool (--protection_domain_id <ID> | --protection_domain_name <NAME>)
  [--storage_pool_name <NAME>]
  [--use_rmcache | --dont_use_rmcache]
  [--rmcache_write_handling_mode  {cached | passthrough}]
  [--media_type {HDD | SSD}]
  [--external_acceleration <TYPE>] {none | read | write | read_and_write}]
  [--data_layout <LAYOUT> {medium_granularity |fine_granularity}
  [--compression_method <METHOD> {none | normal}]
  (--fine_granularity_acceleration_pool_name <NAME> | --
fine_granularity_acceleration_pool_id <ID>)]]
  [--enable_inflight_checksum | --disable_inflight_checksum]
  [--persistent_checksum_enabled | --persistent_checksum_disabled]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Name to be associated with the new storage pool

--use_rmcache

Use RMcache on all SDSs in the storage pool. You can change this setting later on specific SDSs (they can be individually configured not to use RMcache).

--dont_use_rmcache

Do not use read RAM cache on any SDSs in the storage pool (default)

--rmcache_write_handling_mode *<MODE>* **{cached | passthrough}**

Determines whether the system stores the data of this storage pool writes in the SDS read RAM cache, or not. The default is to store the write data in cache (`cached`). You can change this setting later, using the command `set_rmcache_write_handling_mode`.

NOTE: Only I/Os that are multiples of 4k bytes can be cached.

--media_type *<TYPE>* **{HDD | SSD}**

Media type of the devices in the storage pool, where the devices will all be HDD or SSD.

--external_acceleration *<TYPE>*

External acceleration for the device where *<TYPE>* can be one of the following:

none

All devices are not accelerated by a non-PowerFlex read or write cache (default).

read

All devices are accelerated by a non-PowerFlex read cache.

write

All devices are accelerated by a non-PowerFlex write cache.

read_and_write

All devices are accelerated by both non-PowerFlex read and write cache.

--data_layout *<LAYOUT>* **{medium_granularity | fine_granularity}**

Indicates the layout of the data for devices in the storage pool, where *<LAYOUT>* can be medium granularity or fine granularity. The default is medium granularity.

--compression_method *<METHOD>*

When using fine granularity data layout, this Indicates the compression method to be used. *<METHOD>* can be one of the following:

none

No compression applied.

normal

Normal compression applied.

--fine_granularity_acceleration_pool_name *<NAME>*

Name of the NVDIMM acceleration pool to be used for fine granularity.

--fine_granularity_acceleration_pool_id *<ID>*

ID of the NVDIMM acceleration pool to be used for fine granularity.

--enable_inflight_checksum

Enable inflight checksum protection.

--disable_inflight_checksum

Disable inflight checksum protection.

--persistent_checksum_enabled

Enable persistent checksum protection.

--persistent_checksum_disabled

Disable persistent checksum protection.

Example

```
scli --add_storage_pool --protection_domain_name pd1 --storage_pool_name sp2
--dont_use_rmcache --media_type ssd --external_acceleration read_and_write --
persistent_checksum_enabled
```

Storage pool names

Assign each storage pool a meaningful name. When a name has not been defined, the system may display default system-defined names that use the ID of the storage pool.

Each volume name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

disable_background_device_scanner

Disable the background device scanner on the specified storage pool.

Syntax

```
scli --disable_background_device_scanner (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --disable_background_device_scanner --protection_domain_name pd120 --
storage_pool_name sp180
```

disable_persistent_checksum

Disable persistent checksum for the specified storage pool.

Syntax

```
scli --disable_persistent_checksum (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

Example

```
scli --mdm_ip 192.168.1.200 --disable_persistent_checksum --storage_pool_id
d089bdc100000000
```

enable_background_device_scanner

Enable the background device scanner on the devices in the specified storage pool to check for errors (not supported in fine granularity storage pools).

Syntax

```
scli --enable_background_device_scanner (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) --
read_error_action <ACTION>  --compare_error_action <ACTION>
  [--scanner_bandwidth_limit <LIMIT>]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--read_error_action

Read error handling. Can be one of the following:

--report_and_fix

Perform read operations and automatically fix from peer on errors.

--report_only

Perform read operations and report errors without fixing them.

--compare_error_action

Compare error handling between primary and secondary copies of data.

NOTE: Zero padding must be enabled in order to set the background device scanner to data compare mode.

Can be one of the following:

--no_compare

Perform read operations without comparing data content with peer.

--report_and_fix

Perform read operations, compare data content with peer, and automatically fix.

--report_only

Perform read operations, compare data content with peer, and report mismatches without fixing them.

--scanner_bandwidth_limit *<LIMIT>*

Bandwidth limit in KB per second per device. The value should be in the range of 10 KB to 10 MB. Default: 3 MB.

NOTE: High bandwidth should be used very carefully for extreme cases only (such as an urgent need to check some devices), as it may create negative impact on the system performance. Setting the background device scanner bandwidth should take into account maximum bandwidth of the devices.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --enable_background_device_scanner --protection_domain_name pd120 --
storage_pool_name sp18 --read_error_action --report_only --compare_error_action --
report_only --scanner_bandwidth_limit 2000
```

Errors

To check for errors detected by the background device scanner, query SDSs using the `--query_sds` command. Error information is provided in event reports.

Bandwidth limits

High bandwidth may create a negative impact on system performance and should be used carefully and in extreme cases only—for example, when there is an urgent need to check certain devices. When setting the background device scanner bandwidth, you should consider the maximum bandwidth of the devices.

enable_persistent_checksum

Enable a persistent checksum for the specified storage pool. Persistent checksum can be used to support the medium granularity layout in protecting the storage device from data corruption.

Syntax

```
scli --enable_persistent_checksum (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) [--
validate_on_read <MODE>] [--builder_bandwidth_limit <LIMIT>]
```

Parameters

`--protection_domain_id <ID>`
Protection domain ID

`--protection_domain_name <NAME>`
Protection domain name

`--storage_pool_name <NAME>`
Storage pool name

`--storage_pool_id <ID>`
Storage pool ID

`--validate_on_read <MODE>`
Validate checksum upon reading data: `enabled` or `disabled`

`--builder_bandwidth_limit <LIMIT>`

Bandwidth limit in KB/s for the checksum building process. Valid range: 1024-10240. Default: 3072

Example

```
scli --mdm_ip 192.168.1.200 --enable_persistent_checksum --storage_pool_id
d089bdc100000000 --validate_on_read enabled --builder_bandwidth_limit 6144
```

modify_compression_method

Modify the compression method of a storage pool or a V-Tree. Compression can only be used for fine granularity layouts.

NOTE: When fine granularity layouts are not used, use the `none` compression option.

Syntax

```
scli --modify_compression_method ( (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) | --
vtree_id <ID> | --volume_id <ID> | --volume_name <NAME> ) --compression_method <METHOD>
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

--storage_pool_id *<ID>*

Storage pool ID

--vtree_id *<ID>*

V-Tree ID

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--compression_method *<NAME>*

Indicates the compression method to be used on the storage pool or V-Tree. Compression is only relevant for fine granularity data layouts. Set to `none` if fine granularity is not used. Options include:

`none`- No compression is applied

`normal`- Normal compression is applied

Example

```
scli --modify_compression_method --protection_domain_name pd2 --storage_pool_name sp21 --
compression_method normal
```

modify_persistent_checksum

Modify the persistent checksum configuration for the specified storage pool.

Syntax

```
scli --modify_persistent_checksum (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) [--
validate_on_read <MODE>] [--builder_bandwidth_limit <LIMIT>]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

--validate_on_read *<MODE>*
Validate checksum upon reading data: `enabled` or `disabled`

--builder_bandwidth_limit *<LIMIT>*
Bandwidth limit in KB/s for the checksum building process. Valid range: 1024-10240. Default: 3072

Example

```
scli --mdm_ip 192.168.1.200 --modify_persistent_checksum --storage_pool_id
d089bdc100000000 --validate_on_read disabled --builder_bandwidth_limit 1024
```

modify_zero_padding_policy

Modify the zero padding policy for the storage pool. Use the zero-padded policy when the storage pool data layout is fine granularity (FG). The zero-padded policy ensures that every read from an area that is previously not written to, returns zeros.

Syntax

```
scli --modify_zero_padding_policy (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) (--
enable_zero_padding | --disable_zero_padding)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--enable_zero_padding

Enable zero padding in this storage pool

--disable_zero_padding

Disable zero padding in this storage pool

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --modify_zero_padding_policy --protection_domain_name pd10 --storage_pool_name sp1
--zero_padding_disabled
```

query_inaccessible_user_address_space

List all inaccessible address spaces for the specified storage pool.

Syntax

```
scli --query_inaccessible_user_address_space (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id <ID>

Storage pool ID

Example

```
scli --mdm_ip 192.168.1.200 --query_inaccessible_user_address_space --storage_pool_id
d089bdc100000000
```

query_storage_pool

Retrieve aggregated information about all objects in the specified storage pool.

Syntax

```
scli --query_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --query_storage_pool --protection_domain_name pd2 --
storage_pool_name sp2
```

Sample output

```
Storage Pool sp2 (Id: 3df1cd9700000001) has 0 volumes and 22.3 GB (22875 MB) free net
capacity. 0 Bytes volume allocation limit.
        The number of parallel rebuild/rebalance jobs: 2
        Rebuild is enabled and using Limit-Concurrent-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1
        Rebalance is enabled and using Favor-Application-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        VTree Migration is enabled and using Favor-Application-IO policy with the
following parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        Protected Maintenance Mode is enabled and using Limit-Concurrent-IO policy with
the following parameters:
          Number of concurrent IOs per device: 1
        Fragmentation: Enabled
        Background device scanner: Enabled, Read Error Action: report and fix, Compare
Error Action: report and fix, Bandwidth Limit 3072 KBps per device
        Zero padding: Enabled
        Spare policy: 70% out of total
        Inflight Checksum: Disabled
        Doesn't use RAM Read Cache
        Capacity alert thresholds: High: 80, Critical: 90
        Media Type: SSD, External Acceleration Type: NONE, Inflight Requests Factor:
```

```
115, Inflight Bandwidth Factor: 115
        Data layout: Medium granularity
        Replication Capacity Max Allowed Ratio: 10%
        Replication Capacity Max Allowed: 3.0 GB (3072 MB)
        Replication Allocated Capacity: 0 Bytes
        Replication Used Capacity: 0 Bytes
        Number of devices with inaccessible capacity: 0
        Medium granularity persistent checksum:
                Persistent checksum is enabled
                State is PROTECTED
                Checksum builder limit is: 3072 KBps
                Validation on read is disabled


Physical data:
        148.9 GB (152500 MB) total capacity
                0 Bytes in-use capacity
                44.7 GB (45750 MB) unused capacity
                104.2 GB (106750 MB) spare capacity
                0 Bytes decreased capacity
        Capacity Health:
                0 Bytes protected capacity
                0 Bytes failed capacity
                0 Bytes degraded-failed capacity
                0 Bytes degraded-healthy capacity
                0 Bytes unreachable-unused capacity
                0 Bytes semi-protected capacity
                0 Bytes in-maintenance capacity
                0 Bytes temporary capacity
                0 Bytes inaccessible capacity

User data:
        0 Bytes total user data
        0 Bytes total thin user data
        Thin and snapshot saving: Not applicable
        Overall saving: Not applicable
        Data Movement:
                0 Bytes rebalance capacity
                0 Bytes forward-rebuild capacity
                0 Bytes backward-rebuild capacity
                0 Bytes enter protected maintenance mode capacity
                0 Bytes exit protected maintenance mode capacity
                0 Bytes at-rest capacity
                0 Bytes total moving capacity
                        0 Bytes active moving capacity
                                0 Bytes active rebalance capacity
                                0 Bytes active forward-rebuild capacity
                                0 Bytes active backward-rebuild capacity
                                0 Bytes active enter protected maintenance mode capacity
                                0 Bytes active exit protected maintenance mode capacity
                        0 Bytes pending moving capacity
                                0 Bytes pending rebalance capacity
                                0 Bytes pending forward-rebuild capacity
                                0 Bytes pending backward-rebuild capacity
                                0 Bytes pending enter protected maintenance mode capacity
                                0 Bytes pending exit protected maintenance mode capacity

Bandwidth:
        Primary-reads                             0 IOPS 0 Bytes per second
        Primary-writes                            0 IOPS 0 Bytes per second
        Secondary-reads                           0 IOPS 0 Bytes per second
        Secondary-writes                          0 IOPS 0 Bytes per second
        Backward-rebuild-reads                    0 IOPS 0 Bytes per second
        Backward-rebuild-writes                   0 IOPS 0 Bytes per second
        Forward-rebuild-reads                     0 IOPS 0 Bytes per second
        Forward-rebuild-writes                    0 IOPS 0 Bytes per second
        Rebalance-reads                           0 IOPS 0 Bytes per second
        Rebalance-writes                          0 IOPS 0 Bytes per second
        Volume-migration-reads                    0 IOPS 0 Bytes per second
        Volume-migration-writes                   0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-reads    0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-writes   0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-reads     0 IOPS 0 Bytes per second
```

```
        Exit-protected-maintenance-mode-writes   0 IOPS 0 Bytes per second

Volumes summary:
        0 Bytes all volumes size
        0 Bytes base volumes size
        0 Volumes.
```

remove_storage_pool

Remove a storage pool from a protection domain.

NOTE: Each protection domain must contain at least one storage pool, which can be either the default or a user-defined storage pool. You can only remove all storage pools from a protection domain if they are empty and do not contain any devices.

Syntax

```
scli --remove_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --remove_storage_pool
--protection_domain_name rack_1.1
--storage_pool_name sp_1.1_rack_1.1
```

rename_storage_pool

Name, or rename, a storage pool.

Syntax

```
scli --rename_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --new_name <NAME>
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

--new_name *<NAME>*
New name to associate with the storage pool

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --rename_storage_pool --protection_domain_name rack_1.1 --
storage_pool_name sp_1.1_rack_1.1 --new_name sp_2.2_rack_1.1
```

Storage pool names

Assign the storage pool a meaningful name in order to facilitate future object identification. The name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

reset_scanner_error_counters

Reset background device scanner error counters for the specified storage pool or device. You can use this command to reset counters for comparison errors, read errors, or both counter types.

Syntax

```
scli --reset_scanner_error_counters ((((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) |
(--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) (--device_name <NAME> | --device_path <PATH>))) )
  [--reset_compare_errors]
  [--reset_read_errors]
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<SDS IP ADDRESS>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--device_id *<ID>*

Device ID

--device_name *<NAME>*

Device name

--device_path *<PATH>*

SDS storage device path or file path

--storage_pool_id *<ID>*

Storage pool ID

--reset_read_errors

Reset the read error counter.

--reset_compare_errors

Reset the compare error counter.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

NOTE: If you use the `device_id` parameter in the command, you do not need to also supply the SDS ID or SDS name.

Example

```
scli --reset_scanner_error_counters --protection_domain_name pd120 --storage_pool_name
sp120  --sds_name sds13 --reset_compare_errors
```

set_capacity_alerts_threshold

Set the threshold for triggering capacity usage alerts.

Alerts thresholds are calculated from each storage pool capacity after deducting the defined amount of spare capacity. You can set thresholds for a single storage pool or for all storage pools.

Syntax

```
scli --set_capacity_alerts_threshold --capacity_high_threshold <PERCENT> --
capacity_critical_threshold <PERCENT> (--all_storage_pools | --system_default | (((--
protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>)
|--storage_pool_id <ID>))
```

Parameters

--capacity_high_threshold *<PERCENT>*

Threshold of the non-spare capacity of the Storage Pool that will trigger a high-priority alert, expressed as a percentage. Setting a zero threshold for a storage pool resets it to use the global threshold.

--capacity_critical_threshold *<PERCENT>*

Threshold of the non-spare capacity of the storage pool that triggers a critical-priority alert, expressed as a percentage. Setting a zero threshold for a storage pool resets it to use the global threshold.

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--all_storage_pools

Thresholds apply to all storage pools

--system_default

Thresholds apply to newly created storage pools

Example

```
scli --mdm_ip 192.168.1.200 --set_capacity_alerts_threshold --system_default --
capacity_high_threshold 80 --capacity_critical_threshold 90
```

set_rebalance_mode

Enable or disable rebalancing of the specified storage pool.

CAUTION: Rebalancing is an essential part of the PowerFlex system and should only be disabled temporarily, in special circumstances. Disabling rebalance may cause the system to become unbalanced even if no capacity is added or removed—for example, during recovery from an SDS or device failure.

Syntax

```
scli --set_rebalance_mode (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--enable_rebalance | --
disable_rebalance)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--enable_rebalance

Enable rebalancing of the specified storage pool.

--disable_rebalance

Disable rebalancing of the specified storage pool.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rebalance_mode --protection_domain_name pd10 --storage_pool_name sp20 --
enable_rebalance
```

set_rebalance_policy

Set the rebalance I/O priority policy for a storage pool. This feature affects system performance and should only be modified by advanced users.

Syntax

```
scli --set_rebalance_policy (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --policy <POLICY> (Options)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--policy *<POLICY>*

Policy to use for rebalance I/O priority. Can be one of the following:

no_limit

Rebalance I/Os are not limited.

limit_concurrent_io

Limit the number of allowed concurrent rebalance I/Os to the value defined for `concurrent_io_limit`.

favor_application_io

(Default) Limit the number of allowed concurrent rebalance I/Os to the values defined for `concurrent_io_limit` and `bandwidth_limit`, regardless of user I/Os.

Options

Choose one or several:

--concurrent_io_limit *<LIMIT>*

The maximum number of concurrent rebalance I/Os per device. Default: 1.

--bandwidth_limit <BANDWIDTH>

The maximum bandwidth of rebalance I/Os, in KB/s, per device. The valid range is 1024-1048576. Default: 10,240. This property will take effect only if the policy is set to `favor_application_io`.

--i_am_sure

Skip the safety questions for command execution. (For example: "This could damage the stored data. Are you sure?")

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rebalance_policy --protection_domain_name pd1 --storage_pool_name sp1 --
policy favor_application_io --bandwidth_limit 5120
```

set_rebuild_mode

Enable or disable rebuilds in the specified storage pool.

CAUTION: Rebuilds are an essential part of the PowerFlex system and should only be disabled temporarily, in special circumstances. If rebuilds are disabled, redundancy will not be restored after failures.

Syntax

```
scli --set_rebuild_mode (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--enable_rebuild | --
disable_rebuild)
  [--i_am_sure]
```

Parameters

--protection_domain_id <ID>

Protection domain ID

--protection_domain_name <NAME>

Protection domain name

--storage_pool_name <NAME>

Storage pool name

--storage_pool_id <ID>

Storage pool ID

--enable_rebuild

Enable rebuilds in the specified storage pool.

--disable_rebuild

Disable rebuilds in the specified storage pool.

--i_am_sure

Skip the safety questions for command execution. (For example: "This could damage the stored data. Are you sure?")

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rebuild_mode --protection_domain_name pd10 --storage_pool_name sp20 --
enable_rebuild
```

set_rebuild_policy

Set the rebuild I/O priority policy for a storage pool. This feature affects system performance and should only be modified by advanced users.

Syntax

```
scli --set_rebuild_policy (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --policy <POLICY> (Options)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--policy *<POLICY>*

Policy to use for rebuild I/O priority. Can be one of the following:

no_limit

Rebuild I/Os are not limited.

limit_concurrent_io

(Default) Limit the number of allowed concurrent rebuild I/Os to the value defined for `concurrent_io_limit`.

favor_application_io

Limit the number of allowed concurrent rebuild I/Os to the values defined for `concurrent_io_limit` and `bandwidth_limit`, regardless of user I/Os.

Options

Choose one or several:

--concurrent_io_limit *<LIMIT>*

The maximum number of concurrent rebuild I/Os per device. Default: 1.

--bandwidth_limit *<BANDWIDTH>*

The maximum bandwidth of rebuild I/Os, in KB/s, per device. The valid range is 1024-1048576. Default: 10,240. This property will take effect only if the policy is set to `favor_application_io`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rebuild_policy --protection_domain_name pd1 --storage_pool_name sp1 --policy
favor_application_io --bandwidth_limit 5120
```

set_rebuild_rebalance_parallelism

Set the maximum number of concurrent rebuild and rebalance activities on all SDSs within the specified storage pool.

Syntax

```
scli --set_rebuild_rebalance_parallelism (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--
limit <LIMIT>)
```

Parameters

--protection_domain_id *<ID>*
Protection domain ID

--protection_domain_name *<NAME>*
Protection domain name

--storage_pool_name *<NAME>*
Storage pool name

--storage_pool_id *<ID>*
Storage pool ID

--limit *<LIMIT>*
The maximum number of concurrent rebuild and rebalance activities (1..10) on SDSs in the storage pool.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rebuild_rebalance_parallelism --protection_domain_name pd_18 --
storage_pool_name flash_18 --limit 5
```

set_rmcache_usage

Control the use of read RAM cache in the specified Storage Pool.

When using read RAM cache, you must also enable caching in each SDS in the storage pool using the `enable_sds_rmcache` command. Caching only begins once devices have been added to the SDSs.

NOTE: You can configure read RAM cache for a storage pool and then disable caching on one or more SDSs individually.

Syntax

```
scli --set_rmcache_usage (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--use_rmcache | --
```

```
dont_use_rmcache)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--use_rmcache

Use read RAM cache in the storage pool.

--dont_use_rmcache

Do not use read RAM cache in the storage pool (the default).

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rmcache_usage --protection_domain_name pd10 --storage_pool_name spflash --
use_rmcache
```

set_rmcache_write_handling_mode

Set the RMcache write handling mode of the specified storage pool. Writes typically skip the cache, but in certain circumstances you may want to cache writes, and reads.

Syntax

```
scli --set_rmcache_write_handling_mode (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --
rmcache_write_handling_mode {passthrough | cached}
[--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--rmcache_write_handling_mode {passthrough | cached}

The write-handling mode used by the RMcache. One of:

passthrough

Writes skip the cache and are stored in storage only.

cached

Writes are stored in both cache and storage (the default).

--i_am_sure

Skip the safety questions for command execution. (For example: "This could damage the stored data. Are you sure?")

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_rmcache_write_handling_mode --protection_domain_name pd1 --storage_pool_name
spflash --rmcache_write_handling_mode cached
```

set_storage_pool_external_acceleration_type

Set the external acceleration type for devices in the storage pool.

Syntax

```
scli --set_storage_pool_external_acceleration_type ((((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --
external_acceleration <TYPE>
  [--override_sds_device_specific_external_acceleration | --
keep_sds_device_specific_external_acceleration]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--external_acceleration *<TYPE>*

External acceleration for the device, where TYPE is one of the following:

none

All devices are not accelerated by a non-PowerFlex read or write cache.

read

All devices are accelerated by a non-PowerFlex read cache.

write

All devices are accelerated by a non-PowerFlex write cache.

read_and_write

All devices are accelerated by both non-PowerFlex read and write cache.

--override_sds_device_specific_external_acceleration

Override SDS device-specific external acceleration.

NOTE: The external acceleration type of the device is overridden according to the external acceleration type set for the `--external_acceleration` parameter. If there is a mismatch between the external acceleration type of the storage pool and the external acceleration type of the device, the command fails with informative error message.

--keep_sds_device_specific_external_acceleration

Keep SDS device-specific external acceleration.

Example

```
scli --set_storage_pool_external_acceleration_type --protection_domain_name
pd1 --storage_pool_name sp2 --external_acceleration read_and_write --
override_sds_device_specific_external_acceleration
```

set_storage_pool_media_type

Set the media type of the devices in the storage pool.

Syntax

```
scli --set_storage_pool_media_type (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --
media_type <TYPE>
  [--override_sds_device_media_type]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--media_type *<TYPE>*

Media type of the device where TYPE is one of the following:

HDD

All devices in this storage pool are HDDs.

SSD

All devices in this storage pool are SSDs.

transitional

Media type is defined per device

--override_sds_device_media_type

Override SDS device media type

Example

```
scli --set_storage_pool_media_type --protection_domain_name pd2 --storage_pool_name sp2
--media_type ssd --override_sds_device_media_type
```

set_checksum_mode

Inflight checksum protection mode can be used to validate data reads and writes in storage pools, in order to protect data from data corruption. Modify the checksum protection policy of a storage pool.

Syntax

```
scli --set_checksum_mode (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) (--
enable_inflight_checksum | --disable_inflight_checksum)
```

Parameters

--protection_domain_id *<ID>*
: Protection domain ID

--protection_domain_name *<NAME>*
: Protection domain name

--storage_pool_name *<NAME>*
: Storage pool name

--storage_pool_id *<ID>*
: Storage pool ID

--enable_inflight_checksum
: Enable inflight checksum for the storage pool.

--disable_inflight_checksum
: Disable inflight checksum for the storage pool.

Example

```
scli --set_checksum_mode --protection_domain_name PD_1 --storage_pool_name SP_2 --
enable_inflight_checksum
```

set_protected_maintenance_mode_policy

Set the protected maintenance mode I/O priority policy for a storage pool. This feature affects system performance, and only advanced users should modify it.

Syntax

```
scli --set_protected_maintenance_mode_policy (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --
policy <POLICY> (Options)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--policy *<POLICY>*

Policy to use for protected maintenance mode I/O priority. Can be one of the following:

no_limit

Protected maintenance mode I/Os are not limited.

limit_concurrent_io

(Default) Limit the number of allowed concurrent protected maintenance mode I/Os to the value defined for `concurrent_io_limit`.

favor_application_io

Limit the number of allowed concurrent protected maintenance mode I/Os to the values defined for `concurrent_io_limit` and `bandwidth_limit`, regardless of user I/Os.

Options

Choose one or several:

--concurrent_io_limit *<LIMIT>*

The maximum number of concurrent protected maintenance mode I/Os per device. Default: 1.

--bandwidth_limit *<BANDWIDTH>*

The maximum bandwidth of protected maintenance mode I/Os, in KB/s, per device. The valid range is 1024-1048576. Default: 10,240. This property will take effect only if the policy is set to `favor_application_io`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --set_protected_maintenance_mode_policy --protection_domain_name pd1 --
storage_pool_name sp1 --policy favor_application_io --bandwidth_limit 5120
```

test_inaccessible_data

After identifying where there is inaccessible data using the `--query_inaccessible_user_address_space` command, manually clear the inaccessible data from the storage pool. You can then test the inaccessible data, which triggers a rebuild operation, and the data is accessible again.

Syntax

```
scli --test_inaccessible_data (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--device_id *<ID>*
Device ID

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or filepath

Example

```
scli --mdm_ip 192.168.1.200 --test_inaccessible_data --device_id fceff00700040001
```

34

System commands

This section contains commands for managing the system.

abort_upgrade

Stop an upgrade that is in process.

Syntax

```
scli --abort_upgrade
```

Parameters

None.

Example

```
scli --abort_upgrade
```

add_certificate

Add a certificate authority (CA) signed certificate to the trusted certificates list of the CLI. CLI automatically trusts all applied certificates that the CA has signed.

Syntax

```
scli ---add_certificate --certificate_file <FILE>
```

Parameters

--certificate_file *<FILE>*

File name of the root or trusted certificate

Example

```
scli --add_certificate --certificate_file CAcert-cacert.pem
```

add_standby_mdm

Add a standby MDM to the system. This MDM can later be added to the cluster.

Syntax

```
scli --add_standby_mdm --mdm_role {manager    | tb} --new_mdm_ip <IP>
  [--new_mdm_port <PORT>]
  [--new_mdm_management_ip <IP>]
  [--new_mdm_virtual_ip_interface <INTF>]
  [--new_mdm_name <NAME>]
  [--allow_asymmetric_ips]
  [--force_clean]
  [--i_am_sure]
```

Designate the standby MDM's role, either manager or tiebreaker (default). You can give the standby a unique name.

Parameters

--mdm_role {manager | tb}

Designates whether the MDM role is a manager (primary MDM of the cluster) or will only serve as a tiebreaker in the cluster primary MDM election process

--new_mdm_ip *<IP>*

Comma-separated list of up to four IP addresses assigned to the new MDM. Can contain both IPv4 and IPv6 addresses. Omit the space after each comma.

--new_mdm_port *<PORT>*

Port of the new MDM. Default: 9011.

--new_mdm_management_ip *<IP>*

Comma-separated list of up to four IP addresses used to manage the MDM. Can contain both IPv4 and IPv6 addresses. Omit the space after each comma.

--new_mdm_virtual_ip_interface *<INTF>*

Comma-separated list of interface names to be used for the MDM virtual IP addresses. The order of interfaces must match the order of virtual IP addresses assigned to the cluster. Omit the space after each comma.

--new_mdm_name *<NAME>*

Name of the new MDM

--allow_asymmetric_ips

Allow the added node to have a different number of IP addresses from the primary node.

--force_clean

Clean the previous configuration of the MDM.

--i_am_sure

Skip the safety questions for command execution, and give preemptive approval.

Example

```
scli --add_standby_mdm --mdm_role manager --new_mdm_ip 192.168.1.153 --
new_mdm_management_ip 10.103.110.153 --new_mdm_virtual_ip_interface eth4
```

allow_commands_during_upgrade

Allows cluster commands to be sent during an upgrade for this session.

Syntax

```
scli --allow_commands_during_upgrade
```

Parameters

None.

Example

```
scli --allow_commands_during_upgrade
```

create_mdm_cluster

Create an MDM cluster from the selected primary MDM.

This command does not require the user to be logged in before running it. When the command is executed, the MDM cluster operates in single mode.

Syntax

```
scli --create_mdm_cluster --primary_mdm_ip <IP>
  [--primary_mdm_management_ip <IP>]
  [--cluster_virtual_ip <IP>]
  [--primary_mdm_virtual_ip_interface <INTF>]
  [--primary_mdm_name <NAME>]
  [--accept_license]
  [--disable_client_secure_communication]
  [--approve_certificate_once]
  [--approve_certificate]
```

Parameters

--primary_mdm_ip*<IP>*

Comma- separated list of IP addresses assigned to the primary MDM, to be used for MDM cluster internal control communications. Omit the space after each comma.

--primary_mdm_management_ip *<IP>*

Comma-separated list of IP addresses, used to manage the MDM. Omit the space after each comma.

--cluster_virtual_ip *<IP>*

Comma-separated list of virtual IP addresses to be used for the cluster. Omit the space after each comma.

--primary_mdm_virtual_ip_interface *<INTF>*

Comma-separated list of interface names to be used for the MDM virtual IP addresses. The order of interfaces must match the order of the virtual IP addresses assigned to the cluster. Omit the space after each comma.

--primary_mdm_name*<NAME>*

MDM name. Each MDM name must be unique. Ensure that each server on which the MDM is installed has a unique hostname.

--accept_license

Accept the license agreement.

--disable_client_secure_communication

Create the cluster without management client secure communications.

--approve_certificate_once

One-time approval of the MDM's certificate (without adding the certificate to the truststore).

--approve_certificate

Automatic approval of the MDM's certificate for the next commands (adds the certificate to the truststore).

Example

```
scli --create_mdm_cluster --primary_mdm_ip 192.168.1.152 --cluster_virtual_ip
192.168.100.152 --primary_mdm_virtual_ip_interface eth4
```

finalize_upgrade

Finalize the upgrade process.

Syntax

```
scli --finalize_upgrade
```

Parameters

None.

Example

```
scli --finalize_upgrade
```

generate_certificate

Generate a new certificate and private key for the SDS or SDR or SDT. The certificate and key are signed by the system.

Syntax

```
scli --generate_certificate ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) | (--sdt_id <ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port <PORT>]))
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--sdr_id *<ID>*

SDR ID

--sdr_port <PORT>

Port assigned to the SDR

--sdr_name *<NAME>*

SDR name

--sdt_id *<ID>*

SDT ID

--sdr_ip *<IP>*

SDR IP address

--sdt_name *<NAME>*

SDT name

--sdt_ip *<IP>*

SDT IP address

--sdt_port *<PORT>*

Port assigned to the SDT

generate_mdm_certificate

Create a new self-signed certificate and private key for the specfied MDM.

NOTE: Running this command on the primary MDM causes the MDM to restart. This may cause the MDM to switch ownership.

Syntax

```
scli --generate_mdm_certificate (--target_mdm_id <ID> | --target_mdm_ip <IP> [--
target_mdm_port <PORT>] | --target_mdm_name <NAME>)
```

Parameters

--target_mdm_id *<ID>*

ID of the MDM for which the certificate and private key are required

--target_mdm_ip *<IP>*

IP address of the MDM for which the certificate and private key are required

--target_mdm_port *<PORT>*

Port of the MDM for which the certificate and private key are required

--target_mdm_name *<NAME>*

Name of the MDM for which the certificate and private key are required

Example

```
scli --generate_mdm_certificate --target_mdm_ip 198.168.1.2 --target_mdm_port 9011
```

generate_mdm_csr_file

Create a certificate signing request (CSR) file for the specified MDM.

The file is used to obtain security certificates from a certificate authority and is saved in the specified MDM directory:

- Linux: `/opt/emc/scaleio/mdm/cfg`
- Windows: `C:\Program Files\emc\scaleio\mdm\cfg`

Syntax

```
scli --generate_mdm_csr_file (--target_mdm_id <ID> | --target_mdm_ip <IP> [--
target_mdm_port <PORT>] | --target_mdm_name <NAME>)
  [OPTIONS]
```

Parameters

--target_mdm_id *<ID>*

ID of the MDM for which the certificate and private key are required

--target_mdm_ip *<IP>*

IP address of the MDM for which the certificate and private key are required

--target_mdm_port *<PORT>*

Port of the MDM for which the certificate and private key are required

--target_mdm_name *<NAME>*

Name of the MDM for which the certificate and private key are required

--common_name *<NAME>*

Common name to appear in the CSR. The default is the host name of the machine.

Options

NOTE: If you use any of these options, the default values will be disabled for all of them.

--country *<NAME>*

Country to appear in the CSR

--state *<NAME>*

State to appear in the CSR

--location *<NAME>*

Location to appear in the CSR

--organization *<NAME>*

Organization to appear in the CSR

--organizational_unit *<NAME>*

Organizational unit to appear in the CSR

--email_address *<EMAIL>*

Email address to appear in the CSR

Example

```
scli --generate_mdm_csr_file --target_mdm_ip 192.168.1.2 --target_mdm_port 9011 --
country australia --state victoria --location melbourne --organization example_pty_ltd
--organizational_unit sales --email_address sales@example.com.au
```

modify_cluster_virtual_ips

Set or modify the virtual IP address of an MDM cluster.

NOTE: After changing or adding a virtual IP address, you are required to update the SDC with the new virtual IP address.

Notes

For data networks using IPv6, if you plan to implement a floating virtual IP address for the MDM (when the MDM is installed on a Linux server), disable the IPv6 DAD setting on the server's relevant interfaces, using the command: `sysctl net.ipv6.conf.<interface_name>.dad_transmits=0`

Syntax

```
scli --modify_cluster_virtual_ips (--cluster_virtual_ip <IP> | --clear_all)
```

Parameters

--cluster_virtual_ip *<IP>*

Comma-separated list of IP addresses to be used for the cluster. Omit the space after each comma.

--clear_all

Clear all virtual IP addresses.

Example

```
scli --modify_cluster_virtual_ips --cluster_virtual_ip 192.168.100.152,192.168.100.153
```

modify_management_ip

Modify an MDM node's management IP addresses.

If you are using this command to add additional management IP addresses, you must enter all the existing IP addresses in the command, and add the new ones at the end. Up to four IP addresses are supported.

Syntax

```
scli --modify_management_ip (--new_mdm_management_ip <IP> | --clear_all) (--
target_mdm_id <ID> | --target_mdm_ip <IP> [--target_mdm_port <PORT>] | --target_mdm_name
<NAME>)
  [--allow_duplicate_management_ips]
  [--i_am_sure]
```

Parameters

--new_mdm_management_ip *<IP>*
Comma-separated list of IP address to be used to manage the MDM. Omit the space after each comma.

--clear_all
Clear all management IP addresses.

--target_mdm_id *<ID>*
MDM ID

--target_mdm_ip *<IP>*
MDM IP address

--target_mdm_port *<PORT>*
MDM port

--target_mdm_name *<NAME>*
MDM name

--allow_duplicate_management_ips
Allow duplicate management IP addresses.

--i_am_sure
Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --modify_management_ip --target_mdm_id 0x34e8f1df6c84a410 --new_mdm_management_ip
fd00::192:168:1:17,10.103.110.17
```

Implementing the changes

After modifying MDM IP addresses, ensure that you implement the change in all components and interfaces that are mapped to the MDM:

1. PowerFlex Manager: In the **Login** window, log in using the new IP address, then clear the old IP address from the history if it is now obsolete.
2. REST Gateway and OpenStack: In the REST Gateway, modify the mapping to the new management IP address.
3. On each SDC, perform the following locally, in the command line with root access permissions:
   - If for any reason, the management IP address is not updated in the SDC, run:

     ```
     /opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
     ```

   - If the SDC is mapped to more than one PowerFlex system, run:

     ```
     /opt/emc/scaleio/sdc/bin/drv_cfg --add_mdm
     ```

4. vSphere: Using the vSphere web plug-in interface, unregister PowerFlex and reregister it with the new IP address.

modify_spare_policy

Modify the current spare capacity reservation policy.

To ensure data protection during server failures, PowerFlex reserves 10 percent of its capacity by default, instead of allowing this capacity to be used for volume allocation. To ensure full system protection in the event of a node failure, the spare capacity must be at least equal to the amount of capacity in the node containing the maximum capacity or the maximum Fault Set capacity.

If all nodes contain equal capacity, you should set the capacity value to at least 1/N of the total capacity (where N is the number of SDS nodes).

Syntax

```
scli --modify_spare_policy (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) --spare_percentage
<PERCENT>
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--spare_percentage *<PERCENT>*

Percentage of the total capacity to be set aside as spare

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --modify_spare_policy
--protection_domain_name rack_1.1 --storage_pool_name --spare_percentage 10
```

modify_virtual_ip_interfaces

Set the virtual IP interfaces of a node in the MDM cluster.

Syntax

```
scli --modify_virtual_ip_interfaces (--new_mdm_virtual_ip_interface <INTF> | --
clear_all) (--target_mdm_id <ID> | --target_mdm_ip <IP> [--target_mdm_port <PORT>] | --
target_mdm_name <NAME>)
```

Parameters

--new_mdm_virtual_ip_interface *<INTF>*

Comma-separated list of interface names to be used for the MDM virtual IP interfaces. Omit the space after each comma. The order of interfaces must match the order of virtual IPs assigned to the cluster.

--clear_all

Clear all virtual IP interfaces.

--target_mdm_id *<ID>*
MDM ID

--target_mdm_ip *<IP>*
MDM IP address

--target_mdm_port *<PORT>*
MDM port

--target_mdm_name *<NAME>*
MDM name

Example

```
scli --modify_virtual_ip_interfaces --target_mdm_ip 192.168.1.152 --
new_mdm_virtual_ip_interface eth4
```

query_all

Retrieve general information about the system.

Syntax

```
scli --query_all
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all
```

Sample output

```
# scli --query_all
System Info:
        Product: Dell PowerFlex Version: R3_6.0.270
        ID: 1df6301000964d0f
        Manager ID: 0000000000000000

License info:
        Installation ID: 1e14a1273652dda9
        SWID:
        Maximum capacity: Unlimited
        Usage time left: 84 days *** Non-Production License ***
        Enterprise features: Enabled
        The system was activated 6 days ago

System settings:
        Capacity alert thresholds: High: 80, Critical: 90
        MDM restricted SDC mode: NONE
        Management Clients secure communication: Enabled
        TLS version: TLSv1.2
        CLI preemptive login banner acceptance via shell: Enabled
        CLI login banner acceptance by password: Enabled
```

```
        User authentication method: Native
        MDM connection authentication: Enabled
        Defragmentation: Enabled

SDC-SDS connectivity status: All connected

SDC-SDR connectivity status: All connected

SDR-SDS connectivity status: All connected

Query all returned 2 Protection Domains:
Protection Domain pd1 (Id: fcc9af5500000000) has 1 storage pools, 0 acceleration pools,
1 Fault Sets, 4 SDS nodes and 7 volumes
        Free net capacity: 231.5 GB (237022 MB), Volume allocation limit: 2.5 TB (2560
GB)
        Operational state is Active
        I/O flow control is enabled
        Fine granularity default number of concurrent writes: 1000
        FGL metadata cache is disabled
        FGL default metadata cache size: 0 MB
        Replication capacity utilization is in Normal level

Storage Pool sp1 (Id: 3df1cd9600000000) has 7 volumes and 231.5 GB (237022 MB) free net
capacity. 2.5 TB (2560 GB) volume allocation limit.
        The number of parallel rebuild/rebalance jobs: 2
        Rebuild is enabled and using Limit-Concurrent-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1
        Rebalance is enabled and using Favor-Application-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        VTree Migration is enabled and using Favor-Application-IO policy with the
following parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        Protected Maintenance Mode is enabled and using Limit-Concurrent-IO policy with
the following parameters:
          Number of concurrent IOs per device: 1
        Fragmentation: Enabled
        Background device scanner: Enabled, Read Error Action: report and fix, Compare
Error Action: report and fix, Bandwidth Limit 3072 KBps per device
        Zero padding: Enabled
        Spare policy: 56% out of total
        Inflight Checksum: Disabled
        Doesn't use RAM Read Cache
        Capacity alert thresholds: High: 80, Critical: 90
        Media Type: HDD, External Acceleration Type: NONE, Inflight Requests Factor:
115, Inflight Bandwidth Factor: 115
        Data layout: Medium granularity
        Replication Capacity Max Allowed Ratio: 10%
        Replication Capacity Max Allowed: 27.0 GB (27648 MB)
        Replication Allocated Capacity: 24.0 GB (24576 MB)
        Replication Used Capacity: 5.8 GB (5926 MB)
        Number of devices with inaccessible capacity: 0
        Medium granularity persistent checksum:
                Persistent checksum is enabled
                State is PROTECTED
                Checksum builder limit is: 3072 KBps
                Validation on read is disabled

SDR-SDS connectivity status: All connected

Protection Domain pd2 (Id: fcc9af5600000001) has 1 storage pools, 1 acceleration pools,
0 Fault Sets, 1 SDS nodes and 0 volumes
        Free net capacity: 22.3 GB (22875 MB), Volume allocation limit: 0 Bytes
        Operational state is Active
        I/O flow control is enabled
        Fine granularity default number of concurrent writes: 1000
        FGL metadata cache is disabled
        FGL default metadata cache size: 0 MB
        Replication capacity utilization is in Normal level
```

```
Storage Pool sp2 (Id: 3df1cd9700000001) has 0 volumes and 22.3 GB (22875 MB) free net
capacity. 0 Bytes volume allocation limit.
        The number of parallel rebuild/rebalance jobs: 2
        Rebuild is enabled and using Limit-Concurrent-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1
        Rebalance is enabled and using Favor-Application-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        VTree Migration is enabled and using Favor-Application-IO policy with the
following parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        Protected Maintenance Mode is enabled and using Limit-Concurrent-IO policy with
the following parameters:
          Number of concurrent IOs per device: 1
        Fragmentation: Enabled
        Background device scanner: Enabled, Read Error Action: report and fix, Compare
Error Action: report and fix, Bandwidth Limit 3072 KBps per device
        Zero padding: Enabled
        Spare policy: 70% out of total
        Inflight Checksum: Disabled
        Capacity alert thresholds: High: 80, Critical: 90
        Media Type: SSD, External Acceleration Type: NONE, Inflight Requests Factor:
115, Inflight Bandwidth Factor: 115
        Data layout: Medium granularity
        Replication Capacity Max Allowed Ratio: 10%
        Replication Capacity Max Allowed: 3.0 GB (3072 MB)
        Replication Allocated Capacity: 0 Bytes
        Replication Used Capacity: 0 Bytes
        Number of devices with inaccessible capacity: 0
        Medium granularity persistent checksum:
                Persistent checksum is enabled
                State is PROTECTED
                Checksum builder limit is: 3072 KBps
                Validation on read is disabled

SDR-SDS connectivity status: All connected

SDS Summary:
        Total 5 SDS Nodes
        5 SDS nodes have membership state 'Joined'
        5 SDS nodes have connection state 'Connected'

Physical data:
        1.3 TB (1340 GB) total capacity
                61.3 GB (62756 MB) in-use capacity
                        61.3 GB (62756 MB) user data
                        0 Bytes metadata
                        0 Bytes fine granularity spares
                507.6 GB (519794 MB) unused capacity
                771.4 GB (789950 MB) spare capacity
                0 Bytes decreased capacity
        Capacity Health:
                61.3 GB (62756 MB) protected capacity
                0 Bytes failed capacity
                0 Bytes degraded-failed capacity
                0 Bytes degraded-healthy capacity
                0 Bytes unreachable-unused capacity
                0 Bytes semi-protected capacity
                0 Bytes in-maintenance capacity
                0 Bytes temporary capacity

User data:
        30.6 GB (31378 MB) total user data
                30.6 GB (31378 MB) medium granularity user data
                0 Bytes fine granularity user data
                        0 Bytes compressed data (Compression ratio: Not applicable)
                        0 Bytes uncompressed data
        0 Bytes trimmed capacity
        30.6 GB (31378 MB) user data without trimmed capacity
        1.0:1 total compression ratio
```

```
        2.1:1 thin and snapshot saving
        2.1:1 overall saving
        Data Movement:
                0 Bytes rebalance capacity
                0 Bytes forward-rebuild capacity
                0 Bytes backward-rebuild capacity
                0 Bytes enter protected maintenance mode capacity
                0 Bytes exit protected maintenance mode capacity
                30.6 GB (31378 MB) at-rest capacity
                0 Bytes total moving capacity
                        0 Bytes active moving capacity
                                0 Bytes active rebalance capacity
                                0 Bytes active forward-rebuild capacity
                                0 Bytes active backward-rebuild capacity
                                0 Bytes active enter protected maintenance mode capacity
                                0 Bytes active exit protected maintenance mode capacity
                        0 Bytes pending moving capacity
                                0 Bytes pending rebalance capacity
                                0 Bytes pending forward-rebuild capacity
                                0 Bytes pending backward-rebuild capacity
                                0 Bytes pending enter protected maintenance mode capacity
                                0 Bytes pending exit protected maintenance mode capacity

Bandwidth:
        Primary-reads                               348 IOPS 182.9 MB (187289 KB) per-second
        Primary-writes                              556 IOPS 273.2 MB (279737 KB) per-second
        Secondary-reads                             0 IOPS 0 Bytes per second
        Secondary-writes                            557 IOPS 271.8 MB (278283 KB) per-second
        Backward-rebuild-reads                      0 IOPS 0 Bytes per second
        Backward-rebuild-writes                     0 IOPS 0 Bytes per second
        Forward-rebuild-reads                       0 IOPS 0 Bytes per second
        Forward-rebuild-writes                      0 IOPS 0 Bytes per second
        Rebalance-reads                             0 IOPS 0 Bytes per second
        Rebalance-writes                            0 IOPS 0 Bytes per second
        Volume-migration-reads                      0 IOPS 0 Bytes per second
        Volume-migration-writes                     0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-reads      0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-writes     0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-reads       0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-writes      0 IOPS 0 Bytes per second

Volumes summary:
        6 thin-provisioned volumes. Total user data size: 30.6 GB (31378 MB)
        1 snapshot. Total user data size: 0 Bytes
        64.0 GB (65536 MB) all volumes size
        56.0 GB (57344 MB) base volumes size
        6 volumes are not mapped
        1 snapshot policy
        1 volume are replicated
        3 replication journal volumes
        3 user volumes

SDR Summary:
        Total 3 SDR Nodes
        3 SDR nodes have membership state 'Joined'
        3 SDR nodes have connection state 'Connected'

Replication Peer System Summary:
        Total 1 Replication Peer System Nodes
        1 Replication Peer System nodes have membership state 'Joined'
```

Output description

The following tables explain the output for `query_all`.

Table 10. System information and description

System information	Description
Product	The PowerFlex version of the system.

Table 10. System information and description (continued)

System information	Description
ID	The system ID number.
Manager ID	The ID of the management application that controls the MDM.

Table 11. License information

License information	Description
Installation ID	The Installation ID number, which is required for licensing purposes.
SWID	The SWID number, which is required for licensing purposes.
Maximum capacity	The maximum capacity permitted by the system's license.
Usage time left	The amount of time left until the license expires.
Enterprise features	Indicates whether enterprise features are enabled.
The system was activated *n* days ago	The number of days since the system was activated .

Table 12. System settings

System settings	Description
Volumes are/are not obfuscated	The obfuscation setting for all volumes in the system. Default: not obfuscated.
Capacity alert thresholds	The alert thresholds, as percentages, configured for capacity usage.
Thick volume reservation	The amount of capacity, as a percentage, which is reserved for thick volumes.
MDM restricted SDC mode	Indicates whether users on remote clients are restricted to read-only access: ● enabled: read-only access ● disabled: full read/write access

Table 13. Protection domain details

Protection domain	Description
Protection domain summary	The following items are displayed separately for each protection domain: ● Name ● ID ● Number of storage pools in the protection domain ● Number of fault sets in the protection domain ● Number of SDS nodes in the protection domain ● Number of volumes associated with the protection domain ● Amount of capacity in the protection domain that is available for volume allocation
Operational state	Indicates the current state of the protection domain, such as active, inactive, and so on.

Table 14. Storage pool details

Storage pool	Description
Storage pool summary	Lists the following information for each storage pool: ● Name ● ID

Table 14. Storage pool details (continued)

Storage pool	Description
	• Number of volumes associated with the storage pool • Amount of capacity in the storage pool that is available for volume allocation
The number of parallel rebuild/rebalance jobs	The number of parallel rebuild and rebalance jobs existing in the system
Rebuild is enabled/disabled	Indicates whether the rebuild feature is enabled or disabled. (In normal conditions, rebuild should be enabled.) When enabled, the rebuild policy that is configured in the system is displayed, along with the relevant parameters.
Rebalance is enabled/disabled	Indicates whether the rebalance feature is enabled or disabled. (In normal production conditions, rebalance should be enabled.) When enabled, the rebalance policy configured in the system is displayed, along with the relevant parameters.
Background device scanner	Displays the mode and the bandwidth limit per device.
Zero padding is enabled/disabled	Indicates whether zero padding is enabled or disabled.
Spare policy	The percentage of total capacity reserved as spare.
Read RAM cache	Indicates whether the read RAM cache feature is used in the storage pool. If the read RAM cache is used, it displays the write handling mode currently in use (passthrough or cached).
Capacity alert thresholds	The alert thresholds, as percentages, configured for capacity usage. Thresholds may be configured for the entire system, or per storage pool.

Table 15. SDS details

SDS summary	Description
Total	The total number of SDS nodes.
n SDS nodes have membership state 'Joined'	The number of SDS nodes that are connected to the cluster, can receive I/O, and hold primary and secondary copies of data. It takes some time from actual disconnection until the SDS is disconnected from the cluster, or the reverse. There are also up-pending and down-pending states.
n SDS nodes have connection state 'Connected'	The number of SDS nodes currently connected to the MDM.
x TB (*y* GB) total capacity	The total amount of available raw storage. This does not represent the total capacity available for volume allocation.
x TB (*y* GB) unused capacity	The quantity of raw capacity in the system that can be earmarked for specific purposes, such as Spare, or used for new volume creation.
x Bytes snapshots capacity	The quantity of capacity used for storing snapshots.
x TB (*y* GB) in-use capacity	The total quantity of healthy, degraded, and failed capacity.
x Bytes thin capacity	The quantity of capacity currently needed for storage purposes.
x TB (*y* GB) protected capacity	The quantity of capacity that is fully protected (primary and secondary copies of the data exist).
x Bytes failed capacity	The quantity of capacity that is not available at all (neither primary, nor secondary copies).

Table 15. SDS details (continued)

SDS summary	Description
x Bytes degraded-failed capacity	The quantity of degraded-failed capacity. When an SDS fails, all its capacity is defined as degraded-failed, and the secondary copies (which are spread across the protection domain) are defined as degraded-healthy.
x Bytes degraded-healthy capacity	The quantity of degraded-healthy capacity. When an SDS fails, all of its capacity is defined as degraded-failed, and the secondary copies (which are spread across the protection domain) are defined as degraded-healthy.
x Bytes unreachable-unused capacity	The quantity of capacity not configured for any use type in the system that is currently unavailable.
x Bytes active rebalance capacity	The quantity of capacity that is currently being migrated to a different location for load balancing purposes.
x Bytes pending rebalance capacity	The quantity of capacity that is waiting in the job queue for migration to a different location for load balancing purposes.
x Bytes active forward-rebuild capacity	The quantity of capacity for which one copy of data exists and a second copy is being created.
x Bytes pending forward-rebuild capacity	The quantity of capacity for which one copy of data exists, and a job for the creation of a second copy is waiting in the job queue.
x Bytes active backward-rebuild capacity	The quantity of capacity for which one copy of data went offline and came back online, and changes are currently being synchronized in that copy.
x Bytes pending backward-rebuild capacity	The quantity of capacity for which one copy of data went offline and came back online, and changes are waiting in the job queue to be synchronized in that copy.
x Bytes rebalance capacity	The total quantity of capacity that is either currently rebalancing or is pending rebalance.
x Bytes forward-rebuild capacity	The total quantity of degraded capacity that is either currently in forward rebuild state or is pending forward rebuild.
x Bytes backward-rebuild capacity	The total quantity of degraded capacity that is either currently in backward rebuild state or is pending backward rebuild.
x Bytes active moving capacity	The quantity of capacity that is currently being migrated from one location to another.
x Bytes pending moving capacity	The quantity of capacity that is waiting in the job queue for migration from one location to another.
x Bytes total moving capacity	The total quantity of active and pending moving (migrating) capacity.
x TB (*y* GB) spare capacity	The quantity of capacity that is reserved for system use when recovery from failure is required. This capacity cannot be used for storage purposes.
x TB (*y* GB) at-rest capacity	The quantity of capacity that is fully protected and not in a rebuild or rebalance state.
x Bytes decreased capacity	The quantity of decreased capacity that was deducted from devices (using the **Set Device Capacity Limit** GUI command or `modify_sds_device_capacity` CLI command) and cannot be used for any purpose
Primary-reads *x* IOPS *y* bytes per second	Number of primary data copy read IOPS, and read bandwidth.
Primary-writes *x* IOPS *y* bytes per second	Number of primary data copy write IOPS, and write bandwidth.

Table 15. SDS details (continued)

SDS summary	Description
Secondary-reads *x* IOPS *y* bytes per second	Number of secondary data copy (protection) read IOPS, and read bandwidth.
Secondary-writes *x* IOPS *y* bytes per second	Number of secondary copy (protection) write IOPS, and write bandwidth.
Backward-rebuild-reads *x* IOPS *y* bytes per second	Number of backward rebuild read IOPS, and read bandwidth.
Backward-rebuild-writes *x* IOPS *y* bytes per second	Number of backward rebuild write IOPS, and write bandwidth.
Forward-rebuild-reads *x* IOPS *y* bytes per second	Number of forward rebuild read IOPS, and read bandwidth.
Forward-rebuild-writes *x* IOPS *y* bytes per second	Number of forward rebuild write IOPS, and write bandwidth.
Rebalance-reads *x* IOPS *y* bytes per second	Number of rebalance read IOPS, and read bandwidth.
Rebalance-writes *x* IOPS *y* bytes per second	Number of rebalance write IOPS, and write bandwidth.

Table 16. Volumes summary

Volumes summary	Description
n thick-provisioned volumes	The number of thick-provisioned volumes, and the total size in TB and GB.
n volumes mapped to all SDC nodes	The number of volumes mapped to SDC nodes.

query_cluster

Retrieve MDM cluster information.

Syntax

```
scli --query_cluster [--show_certificate_info]
```

Parameters

--show_certificate_info

Display certificates for all the MDMs

Example

```
scli --mdm_ip 192.168.1.200 --query_cluster --show_certificate_info
```

Sample output

The following is sample output for a five-node cluster:

```
scli --query_cluster
Cluster:
    Mode: 5_node, State: Normal, Active: 5/5, Replicas: 3/3
    Virtual IP Addresses: N/A
Primary MDM:
    Name: MDM_116, ID: 0x155c8db3025d5f01
        IP Addresses: 10.55.137.116, Port: 9011, Virtual IP interfaces: N/A
```

```
        Certificate Info:
            Subject:    /GN=MDM/CN=vm-centos8u1-3.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Issuer:     /GN=MDM/CN=vm-centos8u1-3.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Valid From: Jan 30 09:34:16 2021 GMT
            Valid To:   Jan 29 10:34:16 2031 GMT
            Thumbprint: 2E:68:A4:48:A3:BC:DF:77:D9:8B:1C:77:01:99:04:5B:1D:F9:77:B6
Secondary MDMs:
    Name: MDM_114, ID: 0x75c68cb264723f00
        IP Addresses: 10.55.137.114, Port: 9011, Virtual IP interfaces: N/A
        Certificate Info:
            Subject:    /GN=MDM/CN=vm-centos8u1-1.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Issuer:     /GN=MDM/CN=vm-centos8u1-1.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Valid From: Jan 30 09:34:16 2021 GMT
            Valid To:   Jan 29 10:34:16 2031 GMT
            Thumbprint: F7:B6:37:3C:45:EF:98:5B:64:57:5E:D3:7D:A1:B7:AD:E4:7B:5A:EA
    Name: MDM_115, ID: 0x7386c66b551fdf02
        IP Addresses: 10.55.137.115, Port: 9011, Virtual IP interfaces: N/A
        Certificate Info:
            Subject:    /GN=MDM/CN=localhost/L=Hopkinton/ST=Massachusetts/C=US/O=EMC/
OU=ASD
            Issuer:     /GN=MDM/CN=localhost/L=Hopkinton/ST=Massachusetts/C=US/O=EMC/
OU=ASD
            Valid From: Jan 30 09:34:18 2021 GMT
            Valid To:   Jan 29 10:34:18 2031 GMT
            Thumbprint: A8:FB:1E:A2:6C:92:32:6C:42:FB:A0:A3:50:1D:6F:90:06:E2:1B:A0
```

Possible values for items in the sample output

Table 17. Possible values for items in sample output

Item	Possible values
Mode	Cluster mode: • 1_node • 3_node • 5_node
State	State of the cluster: • Normal • Degraded (one or more cluster members are down or inactive)
Active	The number of cluster members that are active. If all members are in normal state, this number equals the number of cluster members.
Replicas	The number of MDM repository holders (the primary MDM, plus the number of secondary MDMs in normal state)
Secondary MDM status	• Normal • Error • Disconnected • Not synchronized
Tiebreaker status	• Normal • Error • Disconnected
Standby MDM	Role (manager or tiebreaker)

query_performance_parameters

Retrieve performance parameters. The output displays the configurations of performance-related parameters for the performance profile that is currently active. When no parameters are added to the command, the name of the active profile is shown.

Syntax

```
scli --query_performance_parameters
  [--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_ip <IP>] | --all_sds]
[--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_name <NAME> | --
all_sdc] [--peer_system_id <ID> | --peer_system_name <NAME> | --peer_system_ip <IP> [--
peer_system_port <PORT>] | --peer_system_object_id <ID> | --all_peer_system] [--sdr_id
<ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>] | --all_sdr] [--print_all]
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--all_sds
Query SDS performance parameters from all SDSs.

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC Global Unique Identifier

--sdc_ip *<IP>*
SDC IP address

--all_sdc
Query SDC performance parameters from all SDCs.

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system Name

--peer_system_ip *<IP>*
Replication peer system IP address

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

--all_peer_system
Query replication peer system performance parameters from all replication peer systems.

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
SDR IP address

--sdr_port *<PORT>*
Port assigned to the SDR

--all_sdr
Query SDR performance parameters from all SDRs.

--print_all
Print all performance parameters, including those with active profile default values.

Example

```
scli --query_performance_parameters --print_all
```

Sample output

```
MDM Performance Parameters:
        Active profile: high_performance
                                                                  Current Value
Profile Value
        MDM_NUMBER_SDC_RECEIVE_UMT                                        5
5
        MDM_NUMBER_SDS_RECEIVE_UMT                                       20
20
        MDM_NUMBER_SDS_SEND_UMT                                          10
10
        MDM_NUMBER_SDS_KEEPALIVE_RECEIVE_UMT                             10
10
        MDM_SDS_CAPACITY_COUNTERS_UPDATE_INTERVAL                         1
1
        MDM_SDS_CAPACITY_COUNTERS_POLLING_INTERVAL                        5
5
        MDM_SDS_VOLUME_SIZE_POLLING_INTERVAL                             15
15
        MDM_SDS_VOLUME_SIZE_POLLING_RETRY_INTERVAL                        5
5
        MDM_NUMBER_SDS_TASKS_UMT                                      1536
1536
        MDM_INITIAL_SDS_SNAPSHOT_CAPACITY                             1024
1024
        MDM_SDS_SNAPSHOT_CAPACITY_CHUNK_SIZE                          5120
5120
        MDM_SDS_SNAPSHOT_USED_CAPACITY_THRESHOLD                        50
50
        MDM_SDS_SNAPSHOT_FREE_CAPACITY_THRESHOLD                       200
200
        MDM_NET_ALLOC_RCV_BUFFER_WAIT_MS                               500
500
        MDM_NET_BREAK_DO_IO_LOOP                                         5
5
        MDM_FGL_INITIAL_SNAPSHOT_CAPACITY                             5120
5120
        MDM_FGL_SNAPSHOT_CAPACITY_CHUNK_SIZE                          5120
5120
        MDM_FGL_SNAPSHOT_USED_CAPACITY_THRESHOLD                        50
50
        MDM_FGL_SNAPSHOT_FREE_CAPACITY_THRESHOLD                       200
200
        MDM_RENAME_DISABLE_MODE_ON_SNAP_DELETION                         0
```

```
        MDM_PORT_FLAP_FEATURE_ENABLED                                 1
1
        MDM_DISQUALIFICATION_EPOCH_SEC                              120
120
        MDM_DISCONNECTIONS_ERT                                        3
3
        MDM_DISCONNECTIONS_ERTF_MS                                60000
60000
        MDM_HIGH_LATENCY_ERT                                         50
50
        MDM_HIGH_LATENCY_ERTF_MS                                  60000
60000
        MDM_PACKET_LOSS_ERT                                          50
50
        MDM_PACKET_LOSS_ERTF_MS                                   60000
60000
        RPL_MDM_PORT_FLAP_FEATURE_ENABLED                             0
0
        RPL_MDM_DISQUALIFICATION_EPOCH_SEC                          120
120
        RPL_MDM_DISCONNECTIONS_ERT                                    3
3
        RPL_MDM_DISCONNECTIONS_ERTF_MS                            60000
60000
        RPL_MDM_HIGH_LATENCY_ERT                                     50
50
        RPL_MDM_HIGH_LATENCY_ERTF_MS                              60000
60000
        RPL_MDM_PACKET_LOSS_ERT                                      50
50
        RPL_MDM_PACKET_LOSS_ERTF_MS                               60000
60000
        RPL_MDM_CAP_MGR_JOURNAL_VOLUME_ALLOC_SIZE_GB                  8
8
        RPL_MDM_SLEEP_TIME_FOR_CE_ANALYZER_MS                       500
500
        RPL_MDM_SYNCED_OPERATION_TIMEOUT_SECONDS                     30
30
        RPL_MDM_CAP_MGR_MONITOR_TIMEOUT_MS                        1000
1000
        RPL_MDM_NUM_CG_TASKS_UMT                                      1
1
        RPL_MDM_NUM_PAIR_TASKS_UMT                                    1
1
        RPL_MDM_NUM_SDR_RECEIVE_UMT                                  10
10
        RPL_MDM_NUM_SDR_SEND_UMT                                     10
10
        RPL_MDM_NUM_SDR_KEEPALIVE_RECEIVE_UMT                        10
10
        RPL_MDM_SDR_COUNTERS_POLLING_INTERVAL_SECONDS                 1
1
        RPL_MDM_CAP_MGR_INCREASE_THRESHOLD_SIZE_GB                    4
4
        RPL_MDM_CAP_MGR_DECREASE_CAP_TIMEOUT_MS                    2000
2000
        RPL_MDM_RPO_DRIFT_UNTIL_BARRIER                              64
64
        RPL_MDM_NUM_SDR_TASKS_UMT                                   512
512
        RPL_MDM_NUM_PEER_MDM_TASKS_UMT                              126
126
        RPL_MDM_NUM_PEER_MDM_RECEIVE_UMT                             10
10
        RPL_MDM_NUM_PEER_MDM_SEND_UMT                                10
10
        RPL_MDM_CURRENT_BARRIER_MAX_NUM_IOS_THRESHOLD          1999000000
1999000000
        RPL_MDM_CURRENT_BARRIER_MAX_SIZE_MB_THRESHOLD             65536
65536
        RPL_MDM_SDR_SESSION_CAPACITY_LOW_WATERMARK_MB             20480
20480
```

```
        RPL_MDM_SDR_SESSION_CAPACITY_HIGH_WATERMARK_MB                4096
4096
        RPL_MDM_CAP_MGR_CRITICAL_CAPACITY_THRESHOLD_MB                 32
32
        RPL_MDM_CAP_MGR_SWITCH_SESSIONS_INTERVAL_MS                 120000
120000
        RPL_MDM_CAP_MGR_MINOR_UTILIZATION_THRESHOLD                    75
75
        RPL_MDM_CAP_MGR_MAJOR_UTILIZATION_THRESHOLD                    80
80
        RPL_MDM_CAP_MGR_VERY_HIGH_UTILIZATION_THRESHOLD                90
90
        RPL_MDM_CAP_MGR_CRITICAL_UTILIZATION_THRESHOLD                 95
95
```

query_properties

Retrieve any set of properties on any set of objects of the specified type.

The properties that can be queried depend on the type of the object that is being queried. For example, SDS has different attributes than volume.

Several preset combinations are available for use with this command. A preset is a predefined set of properties, usually with some logical relationship, which specifies a group of commonly queried properties. The available presets are also dependent on the object type. All object types have at least one preset called `all`, which queries all the properties for that type.

Syntax

```
scli --query_properties --object_type <TYPE> (--object_id <ID> | --all_objects) (--
properties <PROPS> | --preset <PRESET>)
  [--group_by_property]
  [--display_property_options]
  [--flat_list]
```

Parameters

`--object_type <TYPE>`

Object type to query. One of:

SYSTEM

PROTECTION_DOMAIN

STORAGE_POOL

FAULT_SET

SDS

HOST

VOLUME

DEVICE

VTREE

MDM

SP_SDS

ACCELERATION_POOL

SNAPSHOT_POLICY

SDR

PEER_MDM

REPLICATION_PAIR

REPLICATION_CONSISTENCY_GROUP

SDT

NOTE: To view detailed presets and properties options for a type, enter `scli --query_properties --object_type <TYPE>`, where *<TYPE>* is one of the above options.

--object_id *<ID>*

Comma-separated list of object ID used for querying one or more specific objects. Omit the space after each comma.

--all_objects

Query all objects of the specified type.

--properties *<PROPS>*

Query a comma-separated list of properties. Omit the space after each comma. See the scli help output for a list of properties that can be queried per object type

--preset *<PRESET>*

Query a predefined set of properties. Omit the space after each comma. See the scli help output for the list of presets that can be used per object type.

--group_by_property

Group results by property type, rather than by object (the default).

--display_property_options

Display all available properties grouped by object type.

--flat_list

Print results as a flat list rather than by group (better for searching).

NOTE: Use `scli --query_properties --object_type <TYPE>` with one of the valid types above, to get detailed presets and properties options.

Example

```
scli --query_properties --object_type SDS --all_objects --properties
IPS,PORT,PROTECTION_DOMAIN_ID,RMCACHE_ENABLED
```

Sample output

```
# scli --query_properties --object_type SDS --all_objects --properties
IPS,PORT,PROTECTION_DOMAIN_ID,RMCACHE_ENABLED
SDS 5fcafa4300000001:
        IPS                                             10.55.105.40,10.76.30.11
        PORT                                            7072
        PROTECTION_DOMAIN_ID                            a080539500000000
        RMCACHE_ENABLED                                 Yes

SDS 5fcafa4600000004:
        IPS                                             10.55.105.43,10.76.30.15
        PORT                                            7072
        PROTECTION_DOMAIN_ID                            a080539500000000
        RMCACHE_ENABLED                                 Yes
```

Object presets

The following table provides a list of presets that can be used per object type.

Table 18. Object type and presets

Object type	ALL	CAPACITY	RPL_JOURNAL_CAPACITY	RPL_BW_COUNTERS	IO	RAM_CACHE	PERFORMANCE_PARAMETERS	RF_CACHE
SYSTEM	Yes	Yes	No	No	No	Yes	Yes	Yes
PROTECTION_DOMAIN	Yes	Yes	No	No	No	Yes	Yes	No
STORAGE_POOL	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
FAULT_SET	Yes	Yes	No	No	Yes	Yes	No	Yes
SDS	Yes	Yes	No	No	Yes	Yes	Yes	Yes
SDC	Yes	No	No	No	Yes	No	Yes	No
VOLUME	Yes	No	No	No	Yes	Yes	No	No
DEVICE	Yes	Yes	No	No	Yes	No	No	Yes
VTREE	Yes	Yes	No	No	No	No	No	No
MDM	Yes	No	No	No	No	No	No	No
SP_SDS	Yes	Yes	No	No	Yes	No	No	Yes
ACCELERATION_POOL	Yes	No	No	No	No	No	No	No
SNAPSHOT_POLICY	Yes	No	No	No	No	No	No	No
SDR	Yes	No	Yes	Yes	No	No	Yes	No
PEER_MDM	Yes	No	No	No	No	No	Yes	No
REPLICATION_PAIR	Yes	No	No	No	No	No	No	No
REPLICATION_CONSISTENCY_GROUP	Yes	No	No	No	No	No	No	No

query_remote_read_only_limit_state

Query the MDM's remote access restriction state. If the state is enabled, remote users may only issue read-only commands to the MDM. If the state is disabled, all command types may be issued to the MDM by remote clients.

Syntax

```
scli --query_remote_read_only_limit_state
```

Parameters

None.

Example

```
scli --query_remote_read_only_limit_state
```

query_system_limits

Retrieve information about the system limits.

Syntax

```
scli --query_system_limits
```

Parameters

None.

Example

```
scli --query_system_limits
```

Sample output

```
System limits (45 entries)
  Number of protections domains:  Max: 256
  Number of storage pools:  Max: 1024
  Number of fault sets:  Max: 16384
  Number of SDSs:  Max: 512
  Number of SDCs:  Max: 1024
  Number of volume trees:  Max: 131072
  Number of volumes:  Max: 131072
  Number of snapshots:  Max: 133120
  Number of system users:  Max: 256
  Number of LDAP services:  Max: 8
  Number of sessions:  Max: 128
  Number of rsyslog clients:  Max: 16
  Number of storage pools per protection domain:  Max: 64
  Number of SDSs per protection domain:  Max: 128
  Number of fault sets per protection domain:  Max: 64
  Number of SDSs per fault set:  Max: 64
  Number of devices per SDS:  Max: 64
  Number of devices per storage pool:  Max: 300
  Number of volumes per medium-granularity volume tree:  Max: 127
  Number of volumes per fine-granularity volume tree:  Max: 127
  Number of snapshots per medium-granularity volume tree:  Max: 254
  Number of snapshots per fine-granularity volume tree:  Max: 254
  Number of volumes mapped to a single SDC:  Max: 8192
  Number of volumes per protection domain:  Max: 32768
  Number of LDAP groups per LDAP service:  Max: 8
  Number of IPs per MDM:  Min: 1  Max: 4
  Number of IPs per SDS/SDC:  Min: 1  Max: 8
  Single volume size in GB:  Min: 8  Max: 1048576
  Total size of all volumes in Storage Pool in GB:  Max: 4194303
  Total size of all volumes in System (GB):  Max: 16777216
  Number of VTrees per protection domain:  Max: 32768
  Number of acceleration devices in acceleration pool:  Max: 512
  Number of acceleration pools per protections domain:  Max: 32
  Number of acceleration devices per acceleration pool that belong to the same SDS:
Max: 8
  Capacity per medium granularity device in GB:  Min: 90
  Capacity per fine granularity device in GB:  Min: 128
  Capacity per SDS in GB:  Max: 131072
  Number of Replication Consistency Groups:  Max: 1024
  Number of Replication Destination Systems:  Max: 1
  Total Number of SDRs per system:  Max: 128
```

```
  Total Number of Replication Pairs in system:  Max: 32768
  Total Number of concurrent Move Pairs operations:  Max: 512
  Single replication volume size in GB:  Min: 8  Max: 65536
  Number of mappings between volumes and SDCs:  Max: 262143
```

query_upgrade

Retrieve information about the upgrade process.

Syntax

```
scli --query_upgrade
   [--show_sdc_info]
```

Parameters

--show_sdc_info

Include SDC information

Example

```
scli --query_upgrade --show_sdc_info
```

refresh_mdm_cluster_capabilities

Refresh the exposed cluster node capabilities.

This command is usually used to instruct the cluster nodes to check whether LDAP and SSL libraries are installed.

Syntax

```
scli --refresh_mdm_cluster_capabilities
```

Parameters

Example

```
scli --refresh_mdm_cluster_capabilities --mdm_port 6611
```

remove_standby_mdm

Remove a standby MDM or standby tiebreaker, that is not part of the cluster, from the system.

Syntax

```
scli --remove_standby_mdm (--remove_mdm_id <ID> | --remove_mdm_ip <IP> [--
remove_mdm_port <PORT>] | --remove_mdm_name <NAME>)
```

Parameters

--remove_mdm_id *<ID>*
ID of the MDM to remove

--remove_mdm_ip *<IP>*
IP address of the MDM to remove

--remove_mdm_port *<PORT>*
Port associated with the MDM to remove

--remove_mdm_name *<NAME>*
Name of the MDM to remove

Example

```
scli --remove_standby_mdm --remove_mdm_name mdm153
```

rename_mdm

Rename an MDM cluster node.

Syntax

```
scli --rename_mdm (--target_mdm_id <ID> | --target_mdm_ip <IP> [--target_mdm_port
<PORT>] | --target_mdm_name <NAME>) --new_mdm_name <NAME>
```

Parameters

--target_mdm_id *<ID>*
ID of the MDM to be renamed

--target_mdm_ip *<IP>*
IP address of the MDM to be renamed

--target_mdm_port *<PORT>*
Port of the MDM to be renamed

--target_mdm_name *<NAME>*
Current name of the MDM to be renamed

--new_mdm_name *<NAME>*
New MDM name. The MDM name must be unique. If a new name is not provided, the old name is deleted.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --rename_mdm --new_mdm_name mdm3 --target_mdm_name mdm2
```

rename_system

Name or rename the system.

To determine the current system name, use the `query_cluster` or `query_all` command.

Syntax

```
scli --rename_system --new_name <NAME>
```

Parameters

--new_name *<NAME>*

New name to associate with the system.

Example

```
scli --rename_system --new_name sio-prod-sys2
```

query_all_system_networks

Retrieve information about all system networks.

query_system_network

Retrieve information about a system network.

Syntax

```
scli --query_system_network (--network_id <ID> | --network_name <NAME>)
```

Parameters

--network_id *<ID>*

System network ID

--network_name *<NAME>*

System network name

add_system_network

Add a system network to the system.

Syntax

```
scli --add_system_network --network_ip <IP> (--network_mask <MASK> | --network_prefix
<PREFIX>) [--network_name <NAME>] [(--host_group_id <ID> | --host_group_name <NAME>)]
```

Parameters

--network_ip *<ID>*

The new network IP

--network_mask *<MASK>*

The new network mask

--network_prefix *<PREFIX>*

The new network prefix

--network_name *<NAME>*

A name to be assigned to the system network

--network_prefix *<PREFIX>*

The new network prefix

--host_group_id *<ID>*

A comma separated list of host group IDs that have access to the system network

--host_group_name *<NAME>*

A comma separated list of host group names that have access to the system network

rename_system_network

Name or rename a system network.

Syntax

```
scli --rename_system_network (--network_id <ID> | --network_name <NAME>) --new_name
<NAME>
```

Parameters

--network_id *<ID>*

System network ID

--network_name *<NAME>*

System network name

--new_name *<NAME>*

A new name to be assigned to the system network

remove_system_network

Remove a system network from the system.

Syntax

```
scli --remove_system_network (--network_id <ID> | --network_name <NAME>)
```

Parameters

--network_id *<ID>*
System network ID

--network_name *<NAME>*
System network name

replace_cluster_mdm

Replace members of the MDM cluster. Up to half (rounded down) of the cluster members can be replaced at one time.

Syntax

```
scli --replace_cluster_mdm ((--add_secondary_mdm_id <ID> |
        --add_secondary_mdm_ip <IP> [--add_secondary_mdm_port <PORT>] |
        --add_secondary_mdm_name <NAME>)| (--remove_secondary_mdm_id <ID> |
        --remove_secondary_mdm_ip <IP> [--remove_secondary_mdm_port <PORT>] |
        --remove_secondary_mdm_name <NAME>)) ((--add_tb_id <ID> | --add_tb_ip <IP>
        [--add_tb_port <PORT>] | --add_tb_name <NAME>) | (--remove_tb_id <ID> | --
remove_tb_ip
        <IP> [--remove_tb_port <PORT>] | --remove_tb_name <NAME>)) [--allow_leave_failed]
        [--i_am_sure]
```

Parameters

--add_secondary_mdm_id *<ID>*
Comma-separated list of secondary MDM IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_secondary_mdm_ip *<IP>*
Comma-separated list of secondary MDM IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_secondary_mdm_port *<PORT>*
Comma-separated list of secondary MDM ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_secondary_mdm_name *<NAME>*
Comma-separated list of secondary MDM names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_secondary_mdm_id *<ID>*
Comma-separated list of secondary MDM IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_secondary_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_secondary_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_secondary_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--add_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--allow_leave_failed

Allow leaving currently failed MDMs in the cluster.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --replace_cluster_mdm --add_secondary_mdm_name mdm33 --remove_secondary_mdm_name
mdm31 --add_tb_name tb44 --remove_tb_name tb41
```

replace_mdm_security_files

Replace the security certificate and key of the MDM. The replacement certificate can be either a self-generated and self-signed certificate, or a certificate generated and signed by an external certificate authority. Security permissions are required in order to run this command.

NOTE: Running the command on the primary MDM causes the MDM to restart. It may also cause a brief, single point of failure period due to an MDM switch over.

Syntax

```
scli --replace_mdm_security_files (--target_mdm_id <ID> | --target_mdm_ip <IP> [--
target_mdm_port <PORT>] | --target_mdm_name <NAME>)
```

Parameters

--target_mdm_id *<ID>*
MDM ID

--target_mdm_ip *<IP>*
MDM IP address

--target_mdm_port *<PORT>*
MDM port

--target_mdm_name *<NAME>*
MDM name

Example

```
scli --replace_mdm_security_files --target_mdm_name mdm153
```

set_component_authentication_properties

Enable or disable component authentication properties.

Syntax

```
scli --set_component_authentication_properties (--use_authentication | --
dont_use_authentication)
```

Parameters

None.

--use_authentication
Use authentication between system components

--dont_use_authentication
Do not use authentication between system components

Example

```
scli --set_component_authentication_properties --use_authentication
```

set_management_client_communication

Enable or disable secure communications between the MDM and the management client.

Syntax

```
scli --set_management_client_communication (--enable_client_secure_communication | --
disable_client_secure_communication)
```

Parameters

--enable_client_secure_communication

Enable the use of security when communicating with management clients.

--disable_client_secure_communication

Disable the use of security when communicating with management clients.

Example

```
scli --set_management_client_communication --enable_client_secure_communication
```

set_performance_parameters

Apply performance profiles to system components. You can apply separate profiles for SDSs, SDCs, the MDM cluster, replication peer systems, and SDRs.

When a container is provided in the command (protection domain, fault set, all SDSs, all SDCs), the command configures all objects in the container. Objects that are added later will not be configured.

NOTE: If you change the performance profile of an SDS located on an SVM, you must afterwards perform manual memory allocation on the SVM.

Syntax

```
scli --set_performance_parameters ([--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>] | --protection_domain_id <ID> | --protection_domain_name
<NAME> | --fault_set_id <ID> | ((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --fault_set_name <NAME>) | --all_sds] [--sdc_id <ID> | --sdc_name <NAME>
| --sdc_guid <GUID> | --sdc_ip <IP> | --all_sdc] [(((--protection_domain_id <ID>
| --protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id
<ID>)] [--peer_system_id <ID> | --peer_system_name <NAME> | --peer_system_ip <IP>
[--peer_system_port <PORT>] | --peer_system_object_id <ID> | --all_peer_system] [--
sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>] | --all_sdr] [--
apply_to_mdm]) (--profile <PROFILE>)
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--fault_set_id *<ID>*

Fault set ID

--fault_set_name *<NAME>*

Name of the new fault set

--all_sds

Set SDS-related parameters on all SDSs.

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC global unique identifier

--sdc_ip *<IP>*

SDC IP address

--all_sdc

Set SDS-related parameters on all SDCs.

--peer_system_id *<ID>*

Replication peer system ID

--peer_system_name *<NAME>*

Replication peer system name

--peer_system_ip *<IP>*

Replication peer system IP address

--peer_system_port *<PORT>*

Port assigned to the replication peer system

--peer_system_object_id *<ID>*

Replication peer system internal object ID

--all_peer_system

Set Replication peer system related parameters to all replication peer systems.

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

SDR IP address

--sdr_port *<PORT>*

SDR port

--all_sdr

Set SDR-related parameters on all SDRs.

--apply_to_mdm

Apply the performance profile to the MDM. This flag is only applicable when setting the profile.

--profile {compact | high_performance}

Set the performance profile as compact or high_performance. The high performance profile configures a predefined set of parameters for very high performance use cases.

NOTE: For a complete list of parameters controlled by the profiles, see the *PowerFlex Performance Fine-Tuning Technical Notes*.

Example

```
scli --set_performance_parameters --all_sds --all_sdc --profile high_performance
```

start_upgrade

Start the upgrade process.

Syntax

```
scli --start_upgrade
```

Parameters

None.

Example

```
scli --start_upgrade
```

switch_cluster_mode

Switch the MDM cluster from one operating mode or configuration to another.

Syntax

```
scli --switch_cluster_mode --cluster_mode <MODE> ((--add_secondary_mdm_id <ID> |
        --add_secondary_mdm_ip <IP> [--add_secondary_mdm_port <PORT>] |
        --add_secondary_mdm_name <NAME>)| (--remove_secondary_mdm_id <ID> |
        --remove_secondary_mdm_ip <IP> [--remove_secondary_mdm_port <PORT>] |
        --remove_secondary_mdm_name <NAME>)) ((--add_tb_id <ID> | --add_tb_ip <IP>
        [--add_tb_port <PORT>] | --add_tb_name <NAME>) | (--remove_tb_id <ID> | --
remove_tb_ip
        <IP> [--remove_tb_port <PORT>] | --remove_tb_name <NAME>)) [--allow_leave_failed]
        [--i_am_sure]
```

Parameters

--cluster_mode *<MODE>*

Cluster operation mode. Can be one of the following:

1_node

The MDM cluster will operate in single-node mode.

NOTE: Single-node mode should only be used temporarily during maintenance activities. Single-node mode does not provide failover protection for the MDM.

3_node

The MDM cluster will operate in 3-node mode.

5_node

The MDM cluster will operate in 5-node mode.

--add_secondary_mdm_id *<ID>*

Comma-separated list of secondary MDM IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_secondary_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_secondary_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_secondary_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_secondary_mdm_id *<ID>*

Comma-separated list of secondary MDM IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_secondary_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_secondary_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_secondary_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--add_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--allow_leave_failed

Allow leaving currently failed MDMs in the cluster.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --switch_cluster_mode --cluster_mode 5_node --add_secondary_mdm_ip 192.168.1.164 --
add_tb_ip 192.168.1.134
```

switch_mdm_ownership

Switch MDM cluster ownership from the current primary MDM to a different MDM.

Syntax

```
 scli --switch_mdm_ownership (--new_primary_mdm_id <ID> | --new_primary_mdm_ip
       <IP> [--new_primary_mdm_port <PORT>]| --new_primary_mdm_name <NAME>)
```

Parameters

--new_primary_mdm_id *<ID>*

MDM ID. Must be an MDM with manager role.

--new_primary_mdm_ip *<IP>*

MDM IP. Must be an MDM with manager role.

--new_primary_mdm_port *<PORT>*

MDM port. Must be an MDM with manager role.

--new_primary_mdm_name *<NAME>*

MDM name. Must be an MDM with manager role.

Example

```
scli --switch_mdm_ownership --new_primary_mdm_ip 192.168.1.164
```

generate_mdm_mno_csr

Generate a certificate signing request (CSR) file for the given address. The file is saved at <generated CSR path> and also locally in the configuration directory of the MDM.

Syntax

```
scli --generate_mdm_mno_csr --csr_file_path <generated CSR path>
```

Parameters

--target_mdm_id *<ID>*
MDM ID

--target_mdm_ip *<IP>*
MDM IP address

--target_mdm_port *<PORT>*
MDM port

--target_mdm_name *<NAME>*
MDM name

replace_mdm_mno_certificate

Replace the old MDM certificate with a new certificate at a given address.

Syntax

```
scli --replace_mdm_mno_certificate (--target_mdm_id <ID> | --target_mdm_ip
<IP> [--target_mdm_port <PORT>] | --target_mdm_name <NAME>) --certificate_file
<MDM_CERT_FILENAME>
```

Parameters

--target_mdm_id *<ID>*
MDM ID

--target_mdm_ip *<IP>*
MDM IP address

--target_mdm_port *<PORT>*
MDM port

--target_mdm_name *<NAME>*
MDM name

query_all_network_sets

Retrieve information about all network sets.

query_network_set

Retrieve information about a network set.

Syntax

```
scli --query_network_set (--network_set_id <ID> | --network_set_name <NAME>)
```

Parameters

--network_set_id *<ID>*
Network set ID

--network_set_name *<NAME>*
Network set name

modify_network_set

Modify the list of system networks of an existing network set.

Syntax

```
scli --modify_network_set (--network_set_id <ID> | --network_set_name <NAME>) [--
network_id <ID> | --network_name <NAME>]
```

Parameters

--network_set_id *<ID>*
Network set ID

--network_set_name *<NAME>*
Network set name

--network_id *<ID>*
A comma separated list of system network IDs to be replaced with the old one

--network_name *<NAME>*
A comma separated list of system network names to be replaced with the old one

rename_network_set

Name or rename a network set.

Syntax

```
scli --rename_network_set (--network_set_id <ID> | --network_set_name <NAME>) --new_name
<NAME>
```

Parameters

--network_set_id *<ID>*
Network set ID

--network_set_name *<NAME>*
Network set name

--new_name *<NAME>*
A new name to be assigned to the network set

remove_network_set

Remove a network set from the system.

Syntax

```
scli --remove_network_set (--network_set_id <ID> | --network_set_name <NAME>)
```

Parameters

--network_set_id *<ID>*
Network set ID

--network_set_name *<NAME>*
Network set name

35

Volume commands

This section contains commands for managing volumes.

add_volume

Add a volume to the PowerFlex system.

You can create a volume when the requested capacity is available and system capacity is balanced. To start allocating volumes, at least three SDS nodes must be configured in the system.

The created volume cannot be used until it is mapped to at least one SDC.

Syntax

```
scli --add_volume (((--protection_domain_id <ID> | --protection_domain_name <NAME>) --
storage_pool_name <NAME>) | --storage_pool_id <ID>) --size_gb <SIZE>
  [--volume_name <NAME>]
  [--compression_method none | normal]
  [--thin_provisioned | --thick_provisioned | <BLANK>]
  [--use_rmcache | --dont_use_rmcache | <BLANK>]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--size_gb *<SIZE>*

Volume size, in GB. Basic allocation granularity is 8 GB.

--volume_name *<NAME>*

Name to be associated with the added volume

--compression_method {none | normal}

Indicates the compression method to be used, relevant only for fine granularity data layouts.

--thin_provisioned | --thick_provisioned | <BLANK>

Specifies whether the specified volume will be thin provisioned or thick provisioned (the default). If left blank, the default of the storage pool is used (thick provisioned for medium granularity, thin provisioned for fine granularity).

--use_rmcache | --dont_use_rmcache | <BLANK>

Specifies whether the specified volume will use read RAM cache for caching. Default: `--use_rmcache`. If left blank, use the value of the storage pool.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the protection domain ID or name, and the storage pool name.

Example

```
scli --mdm_ip 192.168.1.200 --add_volume --size_gb 8 --volume_name vol_1 --
protection_domain_name pd1 --storage_pool_name sp1
```

Volume names

Assign each volume a meaningful name associated with its operational role. When a name has not been defined, the system may display default system-defined names that use the volume's ID.

Each volume name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex Manager.

clear_scsi2_reservation

Clear SCSI2 reservation from an SDC.

Syntax

```
scli --clear_scsi2_reservation (--volume_ID <ID> | --volume_name <NAME>) [--wait] [--
i_am_sure]
```

Parameters

--volume_id *<ID>*

ID of the volume where SCSI2 reservation from the SDC will be reset

--volume_name *<NAME>*

Name of the volume where SCSI2 reservation from the SDC will be reset

--wait

Wait for the command to be executed. If specified, the return code is the actual return code of the command.

--i_am_sure

Skip the safety questions for command execution.

Example

```
scli --clear_scsi2_reservation --volume_name public --sdc_ip 10.76.60.233 --wait
```

clear_scsi3_reservation

Clear SCSI3 reservations from all SDCs.

Syntax

```
scli --clear_scsi3_reservation (--volume_id <ID> | --volume_name <NAME>) [--wait] [--
i_am_sure]
```

Parameters

--volume_id ***<ID>***

ID of the volume for which SCSI3 reservation will be reset

--volume_name ***<NAME>***

Name of the volume for which SCSI3 reservation will be reset

--wait

Wait for the command to be executed. If specified, the return code is the actual return code of the command.

--i_am_sure

Skip the safety questions for command execution.

Example

```
scli --clear_scsi3_reservation --volume_name temp1 --i_am_sure
```

map_volume_to_host

Map a volume to SDC or to NVMe host.

Syntax

```
scli --map_volume_to_host (--volume_id <ID> | --volume_name <NAME>) (--host_id <ID>
| --host_name <NAME> | --nvme_host_nqn <NQN> | --sdc_guid <GUID> | --sdc_ip <IP>) [--
access_mode <ACCESS_MODE>] [--allow_multi_map]
```

Add the attribute, `--volume class` to map volumes to host:

```
scli --map_volume_to_host --volume_id <ID> --sdc_guid <GUID> --volume_class nascluster
```

For example:

```
scli --map_volume_to_host --volume_id 2f362c3800000100 --sdc_guid 01D16FC2-342B-4CC2-
A050-DCD97B4E0GH7 --volume_class nascluster
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe host NQN

--sdc_guid *<GUID>*

SDC global unique identifier

--sdc_ip *<IP>*

SDC IP address

--access_mode *<ACCESS_MODE>*

Mapping access mode. Can be one of the following: read_write (default), read_only, no_access.

--allow_multi_map

Allow this volume to be mapped to more than one host.

unmap_volume_from_host

Unmap a volume from one host or all hosts.

Syntax

```
scli --unmap_volume_from_host (--volume_id <ID> | --volume_name <NAME>) (--host_id <ID>
| --host_name <NAME> | --nvme_host_nqn <NQN> | --sdc_guid <GUID> | --sdc_ip <IP> | --
all_host) [--i_am_sure]
```

Add the attributes `--volume class` and `--sdc_guid` to unmap volumes from the host:

```
scli --unmap_volume_from_host --volume_id <ID> --sdc_guid <GUID> --volume_class
nascluster
```

For example:

```
scli --unmap_volume_from_host --volume_id 2f362c3800000100 --sdc_guid 01D16FC2-342B-4CC2-
A050-DCD97B4E0GH7 --volume_class nascluster
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--host_id *<ID>*

Host ID

--host_name *<NAME>*

Host name

--nvme_host_nqn *<NQN>*

NVMe Host NQN

--sdc_guid *<GUID>*

SDC global unique identifier

--sdc_ip *<IP>*

SDC IP address

--all_host

Unmap the volume from all hosts.

--i_am_sure

Preemptive approval

modify_volume_capacity

Increase the capacity of a volume. You can increase (but not decrease) a volume capacity at any time, as long as there is enough capacity for the volume size to grow.

Syntax

```
scli --modify_volume_capacity (--volume_id <ID> | --volume_name <NAME>) --size_gb <SIZE>
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--size_gb *<SIZE>*

New volume size, in GB. Basic allocation granularity is 8 GB.

Example

```
scli --mdm_ip 192.168.1.200 --modify_volume_capacity
--volume_name vol_1 --size_gb 150000
```

overwrite_volume_content

Overwrite the destination volume content with the content of the source volume. Both volumes must belong to the same V-Tree.

Syntax

```
scli --overwrite_volume_content (--source_vol_id <ID> | --source_vol_name <NAME>) (--
destination_vol_id <ID> | --destination_vol_name <NAME>)
[--overwrite_locked_auto_snapshot]
```

Parameters

--source_vol_id *<ID>*

Source volume ID

--source_vol_name *<NAME>*

Source volume name

--destination_vol_id *<ID>*

Destination volume ID

--destination_vol_name *<NAME>*

Destination volume name

--overwrite_locked_auto_snapshot

Use this flag to overwrite volume content when the destination volume is a locked auto snapshot

Example

```
scli --overwrite_volume_content --source_vol_name vol1 --destination_vol_name vol2
```

query_all_volumes

Retrieve information about all volumes in the system. You can optionally filter the query results by protection domain, storage pool, or snapshot consistency group ID.

Syntax

```
scli --query_all_volumes
   [(((--protection_domain_id <ID> | --protection_domain_name <NAME>) --
storage_pool_name <NAME>) | --storage_pool_id <ID>) | --scg_id <ID>]
```

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--scg_id *<ID>*

Snapshot consistency group ID

Example

```
scli --mdm_ip 192.168.1.200 --query_all_volumes --protection_domain_id dc65bd9900000000
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
Storage Pool 3df1cd9600000000 Name: sp1
 Volume ID: f6351e3700000003 Name: vol1_114 Size: 16.0 GB (16384 MB) Mapped to 1 SDC
Thin-provisioned  intercepted for replication
 Volume ID: f6351e3800000004 Name: vol2_114 Size: 8.0 GB (8192 MB) Not mapped Thin-
provisioned
 Volume ID: f6351e3900000005 Name: vol3_pol1 Size: 8.0 GB (8192 MB) Not mapped Thin-
provisioned
 Volume ID: f6683c2700000002 Name: N/A Size: 8.0 GB (8192 MB) Not mapped Auto Snapshot
of vol3_pol1

Protection Domain fcc9af5600000001 Name: pd2
Storage Pool 3df1cd9700000001 Name: sp2
 <No volumes defined>
```

```
Protection Domain fcc9d66500000002 Name: rrr
 <No volumes defined>
Query-all-volumes returned 4 volumes
```

query_volume

Retrieve information about the specified volume.

Syntax

```
scli --query_volume (--volume_id <ID> | --volume_name <NAME>)
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

Example

```
scli --query_volume --volume_id f6351e3900000005
```

Sample output

```
>> Volume ID: f6351e3900000005 Name: vol3_pol1
   Provisioning: Thin
   Size: 8.0 GB (8192 MB)
   Total base user data: 0 Bytes
   Storage Pool 3df1cd9600000000 Name: sp1
   Protection Domain fcc9af5500000000 Name: pd1
   Creation time: 2021-01-31 13:00:40
   Number of Direct Snapshots: 2
   Reads:  0 IOPS 0 Bytes per second
   Writes: 0 IOPS 0 Bytes per second
   Data layout: Medium granularity
   Mapping:
      SDC ID: c27ab52600000001 IP: 10.55.137.119 Name: N/A Access mode: READ_WRITE
   No SCSI reservations
   The volume is a source volume of Snapshot Policy: aaa4278700000000
   VTree migration info:
      Migration status: Paused
      Pause Reason: No space in destination Storage Pool
      Source:      Storage Pool: sp1 ID: 3df1cd9600000000 Protection Domain: pd1 ID:
fcc9af5500000000
      Destination: Storage Pool: sp2 ID: 3df1cd9700000001 Protection Domain: pd2 ID:
fcc9af5600000001
      Conversion type: No conversion
      Queue position: 1
      Progress percentage : 0%
   Doesn't use RAM Read Cache
   Replication info:
      Replication state: Not being Replicated
```

query_volume_sdc_meters

Query the IOPS, bandwidth, and latency counters for volume mappings. You can either specify a volume, an SDC, or both. For volume only, display metrics for all mappings of the specified volume. For SDC only, display metrics for all mappings of volumes to the specified SDC. For both volume and SDC, display metrics for a specific mapping.

Syntax

```
scli --query_volume_sdc_meters [--volume_id <ID> | --volume_name <NAME>] [--sdc_id <ID>
| --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>]
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC global unique identifier

--sdc_ip *<IP>*
SDC IP address

Example

```
scli --query_volume_sdc_meters --volume_id f6351e3700000003
```

Sample output

```
Information on 1 mapping:

Volume: f6351e3700000003 SDC: c27ab52600000001
Reads:  426 IOPS 212.8 MB (217907 KB) per-second
        213 IOPS 22.033 ms average latency
Writes: 450 IOPS 224.6 MB (230000 KB) per-second
        225 IOPS 261.137 ms average latency
```

query_volume_sdt_meters

Query the IOPS, bandwidth, and latency counters for volume mappings. You can either specify a volume, an SDT, or both. For volume only, display metrics for all mappings of the specified volume. For SDT only, display metrics for all mappings of volumes to the specifed SDT. For both volume and SDT, display metrics for a specific mapping.

Syntax

```
scli --query_volume_sdt_meters [(--volume_id <ID> | --volume_name <NAME>)] [(--sdt_id
<ID> | --sdt_name <NAME> | --sdt_ip <IP> [--sdt_port <PORT>])]
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--sdt_id *<ID>*
SDC ID

--sdt_name *<NAME>*
SDC name

--sdt_ip *<IP>*
SDC IP address

--sdt_port *<PORT>*
Port assigned to the SDT

remove_volume

Remove a PowerFlex volume.

NOTE: Removing a volume erases all the data on the corresponding volume.

Before removing a volume, you must ensure that it is not mapped to any SDCs. If the volume is mapped to SDCs, unmap the volume using the `unmap_volume_from_sdc` command or the PowerFlex Manager, or the vSphere plug-in (where applicable) before removing it.

Syntax

```
scli --remove_volume (--volume_id <ID> | --volume_name <NAME>)
{--remove_entire_snapshot_tree | --remove_with_descendant_snapshots | --
remove_descendant_snapshots_only}
  [--i_am_sure]
```

Parameters

--volume_id *<ID>*
ID of the volume to remove

--volume_name *<NAME>*
Name of the volume to remove

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

[Options]

Additional optional parameters:

--remove_entire_snapshot_tree

Remove the entire volume tree (V-Tree) attached to the specified volume, including parents and siblings.

--remove_with_descendant_snapshots

Remove the specified volume, and all volumes that were created as snapshots of the specified volume or one of its descendants.

--remove_descendant_snapshots_only

Remove snapshots created from the specified volume, but do not remove the specified volume.

NOTE: Before removing a V-Tree, you must unmap every volume in the V-Tree.

Example

```
scli --mdm_ip 192.168.1.200 --remove_volume --volume_name vol_1
```

rename_volume

Changes the name of a volume. You can use this command at any time.

Syntax

```
scli --rename_volume --volume_id <ID>| --volume_name <NAME> --new_name <NAME>
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--new_name *<NAME>*

New name to assign to the volume

Example

```
scli --mdm_ip 192.168.1.200 --rename_volume
--volume_name vol1 --new_name vol_new_1
```

set_volume_rmcache_usage

Control the use of RMcache in the specified volume.

You can only configure one volume at a time. If you want to ensure that all I/Os for this volume are cached, the relevant storage pool should be configured to use RMcache, and the relevant SDSs should all have RMcache enabled.

NOTE: You can also enable RMcache at the volume level during initial creation of the volume using the `--add_volume` command.

Syntax

```
scli --set_volume_rmcache_usage (--volume_id <ID>| --volume_name <NAME>)(--use_rmcache |
--dont_use_rmcache)
  [--i_am_sure]
```

Parameters

`--volume_id <ID>`

Volume ID

`--volume_name <NAME>`

Volume name

`--use_rmcache`

Use read RAM cache on the volume.

`--dont_use_rmcache`

Do not use RMcache on the volume. This is the default setting.

`--i_am_sure`

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_volume_rmcache_usage --volume_name vol_10 --use_rmcache
```

set_volume_mapping_access_mode

Set the mapping access mode for the specified volume.

Syntax

```
scli --set_volume_mapping_access_mode (--volume_id <ID> | --volume_name <NAME>) (--
host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --sdc_guid <GUID> | --sdc_ip
<IP> | --all_host) --access_mode <ACCESS_MODE>
```

Parameters

`--volume_id <ID>`

Volume ID

`--volume_name <NAME>`

Volume name

`--host_id <ID>`

Host ID

`--host_name <NAME>`

host name

--nvme_host_nqn *<NQN>*
NVMe host NQN

--sdc_guid *<GUID>*
SDC GUID

--sdc_ip *<IP>*
SDC IP address

--all_host
Use this flag to apply the volume access mode to all hosts

--access_mode {read_write | read_only | no_access}
The access mode

set_volume_access_mode_limit

Set the highest mapping access mode allowed for a volume.

Syntax

```
scli --set_volume_access_mode_limit (--volume_id <ID> | --volume_name <NAME>) --
access_mode_limit <ACCESS_MODE>
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--access_mode_limit {read_write | read_only}
Access mode limit.

Example

```
scli --mdm_ip 192.168.1.200 set_volume_access_mode_limit --volume_id a4b5862000000000 --
access_mode_limit read_only
```

query_auto_provisioned_destination_volume_suffix

Query the suffix added to auto-provisioned destination volumes.

Syntax

```
scli --query_auto_provisioned_destination_volume_suffix
```

modify_auto_provisioned_destination_volume_suffix

Modify the suffix added to auto-provisioned destination volumes.

Syntax

```
scli --modify_auto_provisioned_destination_volume_suffix --
destination_volume_name_suffix <suffix>
```

Parameters

--destination_volume_name_suffix <suffix>

The new suffix. The suffix value is mandatory but an empty string ("") is permitted.

add_snapshot_policy

Create a new snapshot policy to enable you to define the number of snapshots taken at a given time for one or more volumes.

Volumes snapshots are taken according to the rule defined in the snapshot policy. In a multi-level structure, define the time interval in between two rounds of snapshots as well as the number of snapshots to retain. For example, take snapshots every x minutes/hours/days/weeks. There are one to six levels, with the first level having the most frequent snapshots.

Syntax strings

```
scli --add_snapshot_policy [--snapshot_policy_name <NAME>] --snapshot_creation_cadence
<CADENCE> --number_of_snapshots_per_retention_level  <LEVELS> [--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>] [--
paused] [--read_only_snapshots] [--secure_snapshots [--i_am_sure]]
```

Parameters

--snapshot_policy_name *<NAME>*

Snapshot policy name

--snapshot_creation_cadence *<CADENCE>*

The interval in minutes between two snapshots in the policy (minimum is 5).

--number_of_snapshots_per_retention_level *<LEVELS>*

A comma-separated list which represents the number of snapshots per retention level. There are one to six levels, and the first level has the most frequent snapshots. There can be up to 60 snapshots in all levels (two consecutive levels overlap by one).

--paused

Indicates that the snapshot policy should pause on creation.

--read_only_snapshots

Specify all mappings of the created auto snapshots to be limited to read-only access.

--secure_snapshots

Define all auto snapshots created as secure. It will not be possible to edit or remove them before the retention period is complete.

--i_am_sure

Preemptive approval for the creating the snapshots as secure.

--replication_consistency_group_id <ID>

Replication consistency group ID

--replication_consistency_group_name <NAME>

Replication consistency group name

When one of `--replication_consistency_group_id` or `--replication_consistency_group_name` is specified, the snapshot policy applies to that

replication consistency group rather than volumes. Thus, volumes cannot be added to this snapshot policy.

add_source_volume_to_snapshot_policy

Add a source volume to the snapshot policy specified by ID or name.

You can add multiple source volumes to a snapshot policy, but a specific volume can only be the source volume of a single policy. Only one volume per V-Tree may be used as a source volume of a policy.

Syntax

```
scli --add_source_volume_to_snapshot_policy (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>) (--source_vol_id <ID> | --source_vol_name <NAME>)
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Name of the snapshot policy

--source_vol_id *<ID>*

Source volume ID

--source_vol_name *<NAME>*

Source volume name

Example

```
scli --add_source_volume_to_snapshot_policy --snapshot_policy_name snap_pol_1 --
source_vol_id a7e798d100000000
```

align_snapshot_policy_to_current_time

Align the specified snapshot policy's cadence cycles to current time.

Syntax

```
scli --align_snapshot_policy_to_current_time (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>)
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Snapshot policy name

Example

```
scli --mdm_ip 192.168.1.200 --align_snapshot_policy_to_current_time --
snapshot_policy_name SNP1
```

lock_auto_snapshot

Lock an auto snapshot to prevent removal by the snapshot policy.

Syntax

```
scli --lock_auto_snapshot (--auto_snapshot_id <ID> | --auto_snapshot_name <NAME>
```

Parameters

--auto_snapshot_id *<ID>*
Auto snapshot ID

--auto_snapshot_name *<NAME>*
Auto snapshot name

Example

```
scli --lock_auto_snapshot --auto_snapshot_name auto_snapshot_1
```

modify_snapshot_policy

Modify the number of snapshots taken per minute and/or the number of snapshots per retention level for a specific policy

Syntax

```
scli --modify_snapshot_policy (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>) ((--snapshot_creation_cadence <CADENCE> --
number_of_snapshots_per_retention_level <LEVELS>) | ([--set_read_only_snapshots] [--
set_secure_snapshots] [--apply_to_existing_snapshots])) [--i_am_sure]
```

Parameters

--snapshot_policy_id *<ID>*
Snapshot policy ID

--snapshot_policy_name *<NAME>*
Name of the snapshot policy

--snapshot_creation_cadence *<CADENCE>*
The interval in minutes between two snapshots in the policy

--number_of_snapshots_per_retention_level *<LEVELS>*

A comma-separated list which represents the number of snapshots per retention level. There are six levels, and the first level has the most frequent snapshots. There can be up to 60 snapshots total in all levels (note that two consecutive levels overlap by one).

--set_read_only_snapshots

All mappings of the created auto snapshots will be limited to read-only access

--set_secure_snapshots

The auto snapshots will be created as secure. They cannot be edited or removed prior to their policy expiration time

--apply_to_existing_snapshots

Applies the `--set_read_only_snapshots` and `--set_secure_snapshots` flags to the policy's existing auto snapshots (excluding locked snapshots)

--i_am_sure

Skip the safety questions for command execution.

Example

```
scli --modify_snapshot_policy --snapshot_policy_name SP_pol1 --snapshot_creation_cadence
2 --numer of snapshots_per_retention_level 5 --set_read_only_snapshots --i_am_sure
```

pause_snapshot_policy

Pause a snapshot policy so that no auto snapshots will be created or removed.

Syntax

```
scli --pause_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Name of the snapshot policy

Example

```
scli --pause_snapshot_policy --snapshot_policy_name snap_pol_1
```

query_all_snapshot_policies

Retrieve information about all snapshot policies.

Syntax

```
scli --query_all_snapshot_policies
```

Parameters

None.

Example

```
scli --query_all_snapshot_policies
```

Sample output

```
Query all snapshot policies returned 3 policies
Snapshot Policy policy1 (Id: 36a39c6200000000)
        Cadence: 30 minutes
        Number of snapshots per retention level: 1 (read-only)
        State: Active
        Time of last auto snapshot: N/A
        Next run time: 2020-11-12 09:56:17
        Number of source volumes: 0
        Number of auto snapshots: 0

Snapshot Policy policy2_secure (Id: 36a3c37200000001)
        Cadence: 5 minutes
        Number of snapshots per retention level: 1 (secure, read-only)
        State: Active
        Time of last auto snapshot: 2020-11-12 09:47:18
        Next run time: 2020-11-12 09:52:17
        Number of source volumes: 1
        Number of auto snapshots: 1

Snapshot Policy policy_read_write (Id: 36a3ea8200000002)
        Cadence: 5 minutes
        Number of snapshots per retention level: 1
        State: Active
        Time of last auto snapshot: 2020-11-12 09:44:18
        Next run time: 2020-11-12 09:49:17
        Number of source volumes: 1
        Number of auto snapshots: 1
```

query_snapshot_policy

Retrieve information about a snapshot policy.

Syntax

```
scli --query_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>)
[--show_auto_snapshots]
```

Parameters

--snapshot_policy_id *<ID>*
: Snapshot policy ID

--snapshot_policy_name *<NAME>*
: Name of the snapshot policy

--show_auto_snapshots
: Indicate whether to display the full list of auto snapshots

Example

```
scli --query_snapshot_policy --snapshot_policy_name policy1 --show_auto_snapshots
```

Sample output

```
Snapshot Policy policy1 (Id: aaa4278700000000)
        Cadence: 5 minutes
        Number of snapshots per retention level: 1
        State: Active
        Time of last auto snapshot: 2021-02-10 13:39:13
        Next run time: 2021-02-10 13:44:13
        Number of auto snapshots: 2

Source volume information (total 2 volumes):
         1: Volume ID: f6351e3700000003 Name: vol1_114 Provisioning: Thin Size: 16.0 GB
(16384 MB)
         2: Volume ID: f6351e3900000005 Name: vol3_pol1 Provisioning: Thin Size: 8.0 GB
(8192 MB)
```

remove_snapshot_policy

Remove the snapshot policy specified by name or ID. If this is a replication snapshot policy, specify what should be done with the auto snapshots.

Syntax

```
scli --remove_snapshot_policy (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>) (--detach_auto_snapshots | (--remove_auto_snapshots [--
detach_locked_auto_snapshots]))
```

Parameters

--snapshot_policy_id *<ID>*
: Snapshot policy ID

--snapshot_policy_name *<NAME>*
: Name of the snapshot policy

--detach_auto_snapshots
: Detach all associated auto snapshots

--remove_auto_snapshots
: Remove all associated auto snapshots

--detach_locked_auto_snapshots
: When the remove option is specified, lock is detached and not removed.

Example

```
scli --remove_snapshot_policy --snapshot_policy_name SP_pol_1
```

remove_source_volume_from_snapshot_policy

Remove a source volume from the specified snapshot policy.

When removing the source volume from the policy, you must choose how to handle the snapshots created by the policy (auto snapshots). Your selection depends on if there are locked auto snapshots.

- If the source volume has no auto snapshots, you don't need to use the `--detach_auto_snapshots` or `--remove_auto_snapshots` options.
- If the source volume has auto snapshots but none of them are locked, you can choose to detach all snapshots with the `--detach_auto_snapshots` flag. They become regular snapshots as if they were created manually. If you use the `--remove_auto_snapshots`, the snapshots are deleted.
- If the source volume has locked auto snapshots, you can use `--detach_auto_snapshots` to turn the auto snapshots into regular snapshots, as if they were created manually. If you use `--remove_auto_snapshots`, the auto snapshots that are not locked are removed, while the auto snapshots that are locked are detached.

Syntax

```
scli --remove_source_volume_from_snapshot_policy (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>) (--source_vol_id <ID> | --source_vol_name <NAME>) (--
detach_auto_snapshots | (--remove_auto_snapshots [--detach_locked_auto_snapshots]))
```

Parameters

`--snapshot_policy_id <ID>`
: Snapshot policy ID

`--snapshot_policy_name <NAME>`
: Name of the snapshot policy

`--source_vol_id <ID>`
: Source volume ID

`--source_vol_name <NAME>`
: Source volume name

`--detach_auto_snapshots`
: Detach all associated auto snapshots

`--remove_auto_snapshots`
: Remove all associated auto snapshots

`-- detach_locked_auto_snapshots`
: When the `remove_policy_snapshots` option is specified, any locked snapshots will not be removed, but rather detached.

Example

```
scli --remove_source_volume_from_snapshot_policy --snapshot_policy_name snap_pol_1 --
source_vol_id a7e798d100000000 --remove_auto_snapshots --detach_locked_auto_snapshots
```

rename_snapshot_policy

Name or rename the snapshot policy specified by name or ID.

Syntax

```
scli --rename_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name
<NAME>) --new_name <NAME>
```

Parameters

--snapshot_policy_id *<ID>*
Snapshot policy ID

--snapshot_policy_name *<NAME>*
Name of the snapshot policy

--new_name *<NAME>*
New name of the snapshot policy

Example

```
scli --rename_snapshot_policy --snapshot_policy_name SP_pol_1 --new_name snap_pol_1
```

resume_snapshot_policy

Resume a paused snapshot policy so that auto snapshots will once again be created and removed.

Syntax

```
scli --resume_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>)
```

Parameters

--snapshot_policy_id *<ID>*
Snapshot policy ID

--snapshot_policy_name *<NAME>*
Name of the snapshot policy

Example

```
scli --resume_snapshot_policy --snapshot_policy_name snap_pol_1
```

unlock_auto_snapshot

Unlock an auto snapshot to allow the snapshot policy to remove it, if and when required.

Syntax

```
scli --unlock_auto_snapshot (--auto_snapshot_id <ID> | --auto_snapshot_name <NAME>) [--
auto_snapshot_will_be_removed]
```

Parameters

--auto_snapshot_id *<ID>*

Auto snapshot ID

--auto_snapshot_name *<NAME>*

Auto snapshot name

--auto_snapshot_will_be_removed

Acknowledge that if the auto snapshot has expired while it was locked, unlocking it will cause it to be removed.

Example

```
scli --unlock_auto_snapshot --auto_snapshot_name auto_snap_1 --
auto_snapshot_will_be_removed
```

36

V-Tree commands

This section contains commands for managing V-Trees.

migrate_vtree

Migrate an entire V-Tree (a volume and all its snapshots) from one storage pool to another storage pool, within the same protection domain or between two different protection domains.

During V-Tree migration, you can run other tasks such as creating snapshots, deleting snapshots, and entering maintenance mode.

When a V-Tree migration is requested, the MDM first estimates whether the destination storage pool has enough capacity for the migration to complete successfully. The MDM bases the estimation on its information of the current capacity of the V-Tree. If there is insufficient capacity at the destination based on that estimate, migration does not start. (You can use the `--ignore_destination_capacity` option to force the migration even if there is insufficient capacity at the destination, with the intention to increase the capacity as required during the migration.) The MDM does not reserve the estimated capacity at the destination (since the capacity of the source volume can grow during migration and the reserved capacity does not guarantee success). The MDM does not hold onto source capacity once it has been migrated, but releases it immediately.

V-Tree migration can take a long time, depending on the size of the V-Tree and the system workload. During migration, the V-Tree is fully available for user I/O. V-Tree migration is done volume block by volume block. When a single block has completed its migration, the capacity of the block at the source becomes available, and it becomes active in the destination storage pool. During migration, the V-Tree has some of its blocks active in the source storage pool and the remaining blocks active in the destination storage pool.

When migrating from a medium granularity storage pool to a fine granularity storage pool, volumes must be zero padded.

NOTE: You cannot create snapshots when migrating from a medium granularity storage pool to a fine granularity storage pool.

Syntax

```
scli --migrate_vtree (--volume_id <ID> | --volume_name <NAME>) (((--
destination_protection_domain_id <ID> | --destination_protection_domain_name <NAME>) --
destination_storage_pool_name <NAME>) | --destination_storage_pool_id <ID>)
  [--migration_queue_position {head_of_queue | tail_of_queue}]
  [ --convert_to_thin_provisioned]
  [ --convert_to_thick_provisioned]
  [compression_method {none | normal}
  [--ignore_destination_capacity]
```

Parameters

`--volume_id <ID>`
Volume ID of the V-Tree to be migrated

`--volume_name <NAME>`
Volume name of the V-Tree to be migrated

`--destination_protection_domain_id <ID>`
ID of the destination protection domain

`--destination_protection_domain_name <NAME>`
Name of the destination protection domain

--destination_storage_pool_name *<NAME>*

Name of the destination storage pool

--destination_storage_pool_id *<ID>*

ID of the destination storage pool

NOTE: The migrateVTree command is also used for rollback volume migration while using the `destination_storage_pool_id`.

--migration_queue_position {head_of_queue | tail_of_queue}

Define whether this migration should be placed at the head or the tail of the migration queue. By default, it is placed at the tail or it retains its position if the V-Tree is already in migration.

--convert_to_thin_provisioned

Convert a thick-provisioned V-Tree to be thin-provisioned.

NOTE: SDCs with a version earlier than v3.0 do not fully support converting a thick-provisioned V-Tree to a thin-provisioned V-Tree during migration; after migration, the V-Tree is thin-provisioned, but the SDC is unable to trim it. These volumes can be trimmed by unmapping and then remapping them, or by rebooting the SDC. The SDC version does not affect capacity allocation and a V-Tree converted from thick to thin provisioning is reduced in size accordingly in the system.

--convert_to_thick_provisioned

Convert a thin-provisioned V-Tree to be thick-provisioned.

compression_method {none | normal}

Indicate the compression method to be used, relevant only for fine granularity data layouts. By default, the compression is according to the settings of the destination storage pool.

--ignore_destination_capacity

Force migration, even if currently there is not enough capacity at the destination.

Example

```
scli --set_vtree_migration_queue_position --volume_name vol1_test --
migration_queue_position head_of_queue
```

pause_vtree_migration

Pause migration of a V-Tree (a volume and all its snapshots), either gracefully or forcefully. Once paused, you can choose to resume the V-Tree migration, or to roll back the migration and have all volume blocks returned to the original storage pool.

Syntax

```
scli --pause_vtree_migration (--volume_id <ID> | --volume_name <NAME>)
  [--pause_graceful]
  [--pause_forceful]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--pause_graceful

The pause is handled gracefully, allowing all data currently being moved to complete the migration.

--pause_forceful

The pause is handled forcefully, aborting unfinished data and leaving it at the source, if possible.

Example

```
scli --pause_vtree_migration --volume_name vol1_test --pause_ graceful
```

query_all_vtree_migrations

Retrieve the current status of all V-Tree migrations.

Syntax

```
scli --query_all_vtree_migrations
```

Parameters

None.

Example

```
scli --query_all_vtree_migrations
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
Storage Pool 3df1cd9600000000 Name: sp1
VTree ID: 40e7f23600000005 (Name: vol3_pol1 Volume ID: f6351e3900000005)
VTree migration info:
        Migration status: Pending retry
        Pause Reason: No space in destination Storage Pool
        Source:      Storage Pool: sp1 ID: 3df1cd9600000000 Protection Domain: pd1 ID:
fcc9af5500000000
        Destination: Storage Pool: sp2 ID: 3df1cd9700000001 Protection Domain: pd2 ID:
fcc9af5600000001
        Conversion type: No conversion
        Queue position: 1
        Progress percentage : 0%

Protection Domain fcc9af5600000001 Name: pd2
Storage Pool 3df1cd9700000001 Name: sp2
 <No vtrees defined>

Protection Domain fcc9d66500000002 Name: rrr
 <No vtrees defined>
```

query_vtree

Retrieve information about the entire V-Tree (a volume and all of its snapshots) of the specified volume.

Syntax

```
scli --query_vtree (--vtree_id <ID> | --volume_id <ID> | --volume_name
```

Parameters

--vtree_id *<ID>*
V-Tree ID

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

Example

```
scli --query_vtree --volume_id f6351e3800000004
```

Sample output

```
VTree ID: 40e7f23500000004 Name: vol2_114
        Storage Pool 3df1cd9600000000 Name: sp1
        Protection Domain fcc9af5500000000 Name: pd1
        Data layout: Medium granularity
        Provisioning: Thin
        Total capacity in use: 0 Bytes
        Total user data: 0 Bytes
                Total base user data: 0 Bytes
                Total snapshots user data: 0 Bytes
        Number of Volumes: 2
        Migration status: Not in migration

>> Volume ID: f6351e3800000004 Name: vol2_114
   Provisioning: Thin
   Size: 8.0 GB (8192 MB)
   Total base user data: 0 Bytes
   Creation time: 2021-01-31 12:59:42
   Number of Direct Snapshots: 1
   Volume is not mapped
   No SCSI reservations
   The volume is a source volume of Snapshot Policy: aaa4278700000000
   Doesn't use RAM Read Cache
   Replication info:
      Replication state: Not being Replicated

   >> Volume ID: f66afb540000000a Name: N/A
      Auto Snapshot of vol2_114
      Size: 8.0 GB (8192 MB)
      Creation time: 2021-02-10 17:56:45
      SCG ID: 49645e1c00000003
      Volume is not mapped
      No SCSI reservations
      The volume is an auto snapshot of Snapshot Policy: aaa4278700000000
         Auto snapshot is unlocked
         Retention levels: 1
```

```
         Expected deletion time: 2021-02-10 18:02:46
      Doesn't use RAM Read Cache
      Replication info:
         Replication state: Not being Replicated
```

query_vtree_migration

Retrieve the current status of a V-Tree migration.

Syntax

```
scli --query_vtree_migration (--volume_id <ID> | --volume_name <NAME>)
```

Parameters

None.

--volume_id *<ID>*

Volume ID on VTree to query migration of

--volume_name *<NAME>*

Volume name on VTree to query migration of

Example

```
scli --query_vtree_migration --volume_name vol3_pol1
```

Sample output

```
VTree ID: 40e7f23600000005 Name: vol3_pol1
        Storage Pool 3df1cd9600000000 Name: sp1
        Protection Domain fcc9af5500000000 Name: pd1
        Data layout: Medium granularity
        Provisioning: Thin
        Total capacity in use: 0 Bytes
        Total user data: 0 Bytes
                Total base user data: 0 Bytes
                Total snapshots user data: 0 Bytes
        Number of Volumes: 3
        VTree migration info:
                Migration status: Paused
                Pause Reason: No space in destination Storage Pool
                Source:      Storage Pool: sp1 ID: 3df1cd9600000000 Protection Domain:
pd1 ID: fcc9af5500000000
                Destination: Storage Pool: sp2 ID: 3df1cd9700000001 Protection Domain:
pd2 ID: fcc9af5600000001
                Conversion type: No conversion
                Queue position: 1
                Progress percentage : 0%
```

resume_vtree_migration

Resume a paused V-Tree migration.

Syntax

```
scli --resume_vtree_migration (--volume_id <ID> | --volume_name <NAME>)
```

Parameters

None.

--volume_id *<ID>*

Volume ID of the V-Tree

--volume_name *<NAME>*

Volume name

Example

```
scli --resume_vtree_migration --volume_name vol1_test
```

set_vtree_migration_policy

Set the I/O priority policy for V-Tree migration for a specific storage pool. This feature affects system performance and should only be modified by advanced users.

Syntax

```
scli --set_vtree_migration_policy (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>)
--policy {optimize_migration | limit_concurrent_io | favor_application_io |
dynamic_bandwidth_throttling} (Options)
  [--i_am_sure]
```

Note about V-Tree migration throttling

During V-Tree migration, the source V-Tree is copied to two destination copies. This means that for each device in the source storage pool there are *X* MB/sec of reads, while in the destination storage pool there are 2*X* MB/sec of writes. If you set the `bandwidth_limit` to *X* MB/sec, it remains that at the source, while it increases to 2*X* MB/sec at the destination.

Parameters

--protection_domain_id *<ID>*

Protection domain ID

--protection_domain_name *<NAME>*

Protection domain name

--storage_pool_name *<NAME>*

Storage pool name

--storage_pool_id *<ID>*

Storage pool ID

--policy *<POLICY>*

Policy to use for I/O priority of V-Tree migration. Can be one of the following:

limit_concurrent_io

Limit the number of allowed concurrent V-Tree migration I/Os to the value defined for `concurrent_io_limit`.

favor_application_io

(Default) Limit the number of allowed concurrent V-Tree migration I/Os to the values defined for `concurrent_io_limit` and `bandwidth_limit`, regardless of user I/Os.

Options

Choose one or several:

--concurrent_io_limit *<LIMIT>*

The maximum number of concurrent V-Tree migration I/Os per device. Valid range: 1-20.

--bandwidth_limit *<BANDWIDTH>*

The maximum bandwidth of V-Tree migration I/Os, in KB/s, per device. The valid range is 1024-25600 (25 MB). This property is effective only if the policy is set to `favor_application_io`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_vtree_migration_policy --protection_domain_name pd10 --storage_pool_name sp20
--policy favor_application_io --bandwidth_limit 5120
```

set_vtree_migration_queue_position

Set the position of a V-Tree migration to be at the head or the tail of the migration queue.

Syntax

```
scli --set_vtree_migration_queue_position (--volume_id <ID>| --volume_name <NAME>)  --
migration_queue_position {head_of_queue | tail_of_queue}
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--migration_queue_position

Queue position of the V-Tree migration:

- `head_of_queue`
- `tail_of_queue`

Example

```
scli --set_vtree_migration_queue_position --volume_name vol1_test --
migration_queue_position head_of_queue
```